

Cadence SKILL Language User Guide

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Cadence SKILL Language

Cadence® SKILL is a high-level, interactive programming language based on the popular artificial intelligence language, Lisp. Because SKILL supports a more conventional C-like syntax, novice users can learn to use it quickly, and expert programmers can access the full power of the Lisp language. SKILL is as easy to use as a calculator as well as being a powerful programming language whose applications are virtually unlimited.

SKILL brings you a functional interface to the underlying subsystems. SKILL lets you quickly and easily customize existing CAD applications and helps you develop new applications. SKILL has functions to access each Cadence tool using an application programming interface. These functions are common to all of the Cadence tools used in either a graphic or nongraphic environment. Once you master these basic functions, you need to learn only a few new functions to access any tool in the Cadence environment.

SKILL is the command language of the Cadence environment. Whenever you use forms, menus, and bindkeys, the Cadence software triggers SKILL functions to complete your task. For example, in most cases SKILL functions can:

- Open a design window
- Zoom in by a factor of 2
- Place an instance or a rectangle in a design

Other SKILL functions compute or retrieve data from the Cadence Framework environment or from designs. For example, SKILL functions can retrieve the bounding box of the current window or retrieve a list of all the shapes on a layer.

You can enter SKILL functions directly on a command line to bypass the graphic user interface.

Licensing Requirements

SKILL uses **Cadence Design Framework II** license (License Number 111), which is checked out at the launch of the `skill` executable or the workbench.

SKILL Code Examples

The SKILL APIs are explained with illustrative code examples. You can copy these examples and paste them directly into the Command Interpreter Window (CIW) or use the code in non-graphical SKILL mode.

Sample SKILL Code

The following code sample shows the syntax of a SKILL API that accepts three arguments.

axlGetRunStatus

```
axlGetRunStatus(  
    t_sessionName      ← Required argument  
    [ ?optionName t_optionName ] ← Optional keyword argument  
    [ ?historyName t_historyName ] ← Optional keyword argument  
)  
=> l_statusValues      ← Return value
```

The first argument *t_sessionName* is a required argument, where *t* signifies the data type of the argument. The second and third arguments *?optionName t_optionName* and *?historyName t_historyName* are optional keyword arguments (identified by a question mark), which are specified in name-value pairs and can be placed in any order during the function call.

The return value is the value that the SKILL API returns after evaluating the expression. In this case, it is a list of status values, *l_statusValues*.

Example

```
axlSession=axlGetWindowSession( hiGetCurrentWindow()  
=> "session0"  
axlGetRunStatus("session0" ?historyName "Interactive.10" ?optionName "tests")  
=> (1 2)
```


Return value Value of the session name argument Question mark and argument name before the value of the keyword argument Question mark and argument name before the value of the keyword argument

Relationship Between SKILL and Lisp

If you are new either to SKILL or to programming in general, start with [Terms Used in Cadence SKILL Language](#).

Programming Notation

In SKILL, function calls can be written in either of the following notations:

- Algebraic notation used by most programming languages: `func (arg1 arg2 ...)`
- Prefix notation used by the Lisp programming language: `(func arg1 arg2 ...)`

For comparison, here is a SKILL program written first in algebraic notation, then the same program, also implemented in SKILL, using a Lisp style of programming.

```
procedure( fibonacci(n)
    if( (n == 1 || n == 2) then
        1
    else fibonacci(n-1) + fibonacci(n-2)
    )
)
```

Here is the same program implemented in SKILL using a Lisp style of programming.

```
(defun fibonacci (n)
  (cond
    ((or (equal n 1) (equal n 2)) 1)
    (t (plus (fibonacci (difference n 1))
              (fibonacci (difference n 2)))))
  )
)
```

Data Manipulation

Because programs in SKILL are represented as lists, just as they are in Lisp, they can be manipulated like data. You can dynamically create, modify, or selectively evaluate function definitions and expressions. This ability to manipulate data is one of the primary reasons why Lisp is the language of choice for artificial intelligence applications. Because it takes full advantage of the “program is data” concept of Lisp, SKILL can be used to write flexible and powerful applications.

Many SKILL list manipulation functions are available. These functions operate, in most cases, similar to functions of the same name in Lisp, and the Franz Lisp dialect in particular.

SKILL supports a special notation for list construction from templates. This notation is borrowed from Common Lisp and allows selective evaluation within a quoted form. Selective

evaluation eliminates the long sequences of calls to `list` and `append`. See [Cadence SKILL Language](#).

Characters

Unlike many other programming languages, including Common Lisp, SKILL does not have a separate character data type. Characters are instead represented by single character symbols. The character “A,” for example, is the symbol “A.” Unprintable characters can be referred to using the escape sequences.

Terms Used in Cadence SKILL Language

This topic introduces new terms and takes a general look at the Cadence Framework environment. Listed below are some important Cadence SKILL Language terms you need to know before you start using the tool.

Term	Definition
output	The destination of SKILL output can be an xterm screen, a design window, a file, or in many cases, the Command Interpreter Window (CIW).
CIW	Command Interpreter Window: The start-up working window for many Cadence applications, which contains an input line, an output area, a menu banner with commands to launch various tools, and prompts for general information. The output area of the CIW is the destination of many of the examples used in this manual.
SKILL interpreter	The SKILL interpreter executes SKILL programs within the Cadence environment. The interpreter translates a SKILL program’s text source code into internal data structures, which it actively consults during the execution of the program.
compiler	A compiler translates source code into a target representation, which might be machine instructions or an intermediate instruction set.
evaluation	The process whereby the SKILL interpreter determines the value of a SKILL expression.
SKILL expression	An invocation of a SKILL function, often by means of an operator supplying required parameters.

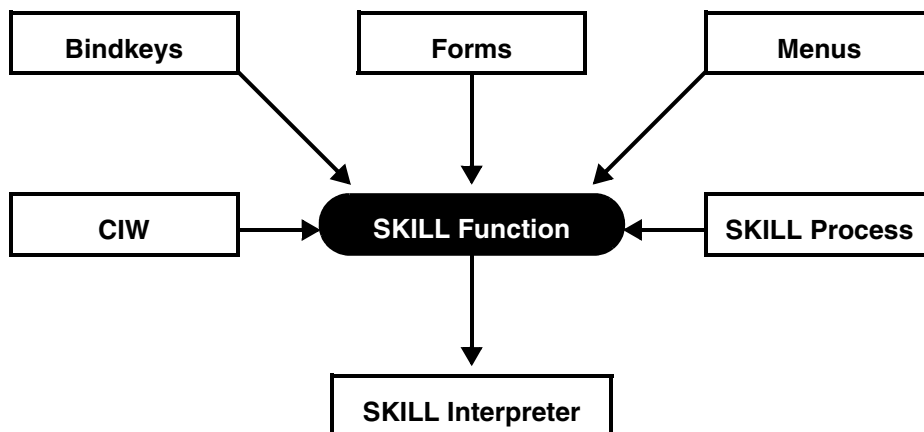
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Term	Definition
SKILL function	A named, parameterizable body of one or more SKILL expressions. You can invoke any SKILL function from the command input line available in the application by using its name and providing appropriate parameters.
SKILL procedure	This term is used interchangeably with the term SKILL function.

Methods to Invoke a SKILL Function

There are many ways to submit a SKILL function to the SKILL interpreter for evaluation. In many applications, whenever you use forms, menus, and bindkeys, the Cadence software triggers corresponding SKILL functions to complete your task. Normally, you do not need to be aware of SKILL functions or any syntax issues.



Term	Definition
Bindkeys	A bindkey associates a SKILL function with a keyboard event. When you cause the keyboard event, the Cadence software sends the SKILL function to the SKILL interpreter for evaluation.
Forms	Some functions require you to provide data by filling out fields in a pop-up form.
Menus	When you choose an item in a menu, the system sends an associated SKILL function to the SKILL interpreter for evaluation.

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Term	Definition
CIW	You can directly enter a SKILL function into the CIW for immediate evaluation.
SKILL Process	You can launch a separate UNIX process that can submit SKILL functions directly to the SKILL interpreter.

You can submit a collection of SKILL functions for evaluation by loading a SKILL source code file.

Alternative Ways to Invoke a Function

In addition to calling a function by stating its name and arguments in a pair of parentheses, as shown below, you can use two other syntax forms to invoke SKILL functions.

```
strcat( "Mary" " had" " a" " little" " lamb" )
```

- You can place the left parenthesis to the left of the function name (Lisp syntax).

```
( strcat "Mary" " had" " a" " little" " lamb" )  
=> "Mary had a little lamb"
```

- You can omit the outermost levels of parenthesis if the SKILL function is the first element at your SKILL prompt, that is, at the top level.

```
strcat "Mary" " had" " a" " little" " lamb"  
=> "Mary had a little lamb"
```

You can use all three syntax forms together. In programming, it is best to be consistent. Cadence recommends the first style noted above.

Cadence SKILL Language Computation and Data

SKILL functions compute or retrieve data from the Cadence Framework environment or from designs. For example, SKILL functions can retrieve the bounding box of the current window or retrieve a list of all the shapes on a layer.

Return Value of a Function

All SKILL functions compute a data value known as the return value of the function. You can

- Assign the return value to a SKILL variable
- Pass the return value to another SKILL function

Any type of data can be a return value. SKILL supports many data types, including integers, text strings, and lists.

Simplest SKILL Data

The simplest SKILL expression is a data item. SKILL data is case sensitive. You can enter data in many familiar ways, including the following.

Sample SKILL Data Items

Data Type	Syntax Example
integer	5
floating point	5.3
text string	"Mary had a little lamb"

Calling a Function

Function names are case sensitive. To call a function, state its name and arguments in a pair of parentheses.

```
strcat( "Mary" " had" " a" " little" " lamb" )  
=> "Mary had a little lamb"
```

- No spaces are allowed between the function name and the left parenthesis.
- Several function calls can be on a single line. Use spaces to separate them.
- You can span multiple lines in the command line or a source code file.

```
strcat(  
    "Mary" " had" " a"  
    " little" " lamb" )  
=> "Mary had a little lamb"
```

- When you enter several function calls on a single line, the system only displays the return result from the final function call.

Operators Are SKILL Functions

SKILL provides many operators. Each operator corresponds to a SKILL function. Here are some examples of useful operators:

Sample SKILL Operators

Operators in Descending Precedence	Underlying Function	Operation
**	expt	arithmetic
*	times	arithmetic
/	quotient	
+	plus	arithmetic
-	difference	
++S, S++	preincrement, postincrement	arithmetic
==	equal	tests for equality
!=	nequal	tests for inequality
=	setq	assignment

The following example shows several function calls using operators on a single line. The calls are separated by spaces. The system displays the return result from the final function call.

```
x = 5 y = 6 x+y  
=> 11
```

Using Variables

You do not need to declare variables in SKILL. SKILL creates a variable the first time it encounters the variable in a session. Variable names can contain

- Alphanumeric characters
- Underscores (_)
- Question marks

The first character of a variable cannot be a digit. Use the assignment operator to store a value in a variable. You enter the variable name to retrieve its value. The `type` SKILL function returns the data type of the variable's current value.

```
lineCount = 4                   => 4  
lineCount                       => 4
```

```
type( lineCount )      => fixnum
lineCount = "abc"      => "abc"
lineCount              => "abc"
type( lineCount )      => string
```

Alternative Ways to Invoke a Function

In addition to calling a function by stating its name and arguments in a pair of parentheses, as shown below, you can use two other syntax forms to invoke SKILL functions.

```
strcat( "Mary" " had" " a" " little" " lamb" )
```

- You can place the left parenthesis to the left of the function name (Lisp syntax).

```
( strcat "Mary" " had" " a" " little" " lamb" )
=> "Mary had a little lamb"
```

- You can omit the outermost levels of parenthesis if the SKILL function is the first element at your SKILL prompt, that is, at the top level.

```
strcat "Mary" " had" " a" " little" " lamb"
=> "Mary had a little lamb"
```

You can use all three syntax forms together. In programming, it is best to be consistent. Cadence recommends the first style noted above.

Solution to Some Common SKILL Problems

Here are three of the most common SKILL problems.

System Doesn't Respond

If you type in a SKILL function and press Return but nothing happens, you most likely have one of these problems.

- Unbalanced parentheses
- Unbalanced string quotes
- The wrong log file filter set

You might have entered more left parentheses than right parentheses. The following steps trigger a system response in most cases.

1. Type a closing right bracket (`]`) character. This character closes all outstanding right parentheses.

2. If you still don't get a response, type a double quote (") character followed by a right bracket (]) character.

In most cases, the system then responds.

If, however, the system responds but subsequently gives erroneous syntax errors for valid input, type an asterisk (*) character followed by a right bracket (]) character.

Inappropriate Space Characters

Do not put any space between the function name and the left parenthesis. Notice that the following error messages do not identify the extra space as the cause of the problem.

- Trying to use the `strcat` function to concatenate several strings.

```
strcat ( "Mary" " had" " a" " little" " lamb")  
Message: *Error* eval: not a function - "Mary"
```

- Trying to make an assignment to a variable.

```
greeting = strcat ( "happy" " birthday" )  
Message: *Error* eval: unbound variable - strcat
```

Data Type Mismatches

An error occurs when you pass inappropriate data to a SKILL function. The error message includes a type template that indicates the expected type of the offending argument.

```
strcat( "Mary had a" 5 )  
Message: *Error* strcat: argument #2 should be either  
a string or a symbol (type template = "S") - 5
```

Here are the characters used in type templates for some common data types.

Some Common Data Types

Data Type	Character in Type Template
integer number	x
floating point number	f
symbol or character string	S
character string (text)	t
any data type (general)	g

SKILL Lists

A SKILL list is an ordered collection of SKILL data objects. The list data structure is central to SKILL and is used in many ways.

The elements of a list can be of any data type, including variables and other lists. A list can contain any number of objects (or be empty). The empty list can be represented either by empty parentheses “()” or the special atom `nil`. The list must be enclosed in parentheses. Lists can contain other lists to form arbitrarily complex data structures. Here are some examples:

Sample Lists

List	Explanation
(1 2 3)	A list containing the integer constants 1, 2, and 3
(1)	A list containing the single element 1
()	An empty list (same as the special atom <i>nil</i>)
(1 (2 3) 4)	A list containing another list as its second element

SKILL displays a list with parentheses surrounding the members of the list. The following example stores a list in the variable `shapeTypeList`, then retrieves the variable's value.

```
shapeTypeList = '( "rect" "polygon" "rect" "line" )  
shapeTypeList => ( "rect" "polygon" "rect" "line" )
```

SKILL provides an extensive set of functions for creating and manipulating lists. Many SKILL functions return lists. SKILL can use multiple lines to display lists. SKILL stores the appropriate integer value in the `_itemsperline` global variable.

Building Lists

There are several main ways to build a list.

- Specify all the elements of the list literally with the single quote (') operator.
- Specify all the elements as evaluated arguments to the `list` function.
- Add an element to an existing list with the `cons` function.
- Merge two lists with the `append` function.
- Add a list using the back quote command.

Both the `cons` and `append` functions allocate a new list and return it to you. You should store the return result in a variable. Otherwise, you cannot refer to the list later.

The following functions allow you to construct new lists and work with existing lists in different ways.

Making a list from given elements

The single quote (`'`) operator builds a list using the arguments exactly as they are presented. The values of `a` and `b` are irrelevant. The `list` function fetches the values of variables for inclusion in a list.

```
'( 1 2 3 )      => ( 1 2 3 )
a = 1           => 1
b = 2           => 2
list( a b 3 )   => ( 1 2 3 )
```

Adding an element to the front of a list (`cons`)

You should store the return result from `cons` in a variable. Otherwise, you cannot refer to the list later. Commonly, you store the result back into the variable containing the target list.

```
result = '( 2 3 )      => ( 2 3 )
result = cons( 1 result ) => ( 1 2 3 )
```

Merging two lists (`append`)

You should store the return result from `append` in a variable. Otherwise, you cannot refer to the list later.

```
oneList = '( 4 5 6 )      => ( 4 5 6 )
aList = '( 1 2 3 )        => ( 1 2 3 )
bList = append( oneList aList ) => ( 4 5 6 1 2 3 )
```

Adding a list using the back quote command

When a single quote (`'`) is used to create a list, it does not evaluate the variables inside the list but considers them as symbols. A list command cannot be used inside an expression following a `'` single quote. For example, this is valid:

```
list(' (1 2) list(3 4) )
```

However, this is not valid:

```
' (list(1 2) ' (5 6) )
```

Use the back quote (```) inside the list command to form another list, using commas before variable names so that they are evaluated. For example:

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```
metallWidth=1.0
keepOutputs = `((0 0) (5 0) (5 ,metallWidth) (0 ,metallWidth) (0 0))
((0 0)
  (5 0)
  (5 1.0)
  (0 1.0)
  (0 0)
)
```

Common Questions and Answers

People often feel that `nil`, and the `cons` and `append` functions are difficult to understand. Let's look at some typical questions.

Typical Questions

Question	Answer
What's the difference between <code>nil</code> and <code>'(nil)</code> ?	<code>nil</code> is a list containing nothing. Its length is 0. <code>'(nil)</code> builds a list containing a single element. The length is 1.
How can I add an element to the end of a list?	Use the <code>list</code> function to build a list containing the individual elements. Use the <code>append</code> function to merge it to the first list. More efficient ways to add an element to the end of a list are discussed in a later chapter.
Can I reverse the order of the arguments to the <code>cons</code> function? Will the results be the same?	You might think that reversing the elements to the <code>cons</code> function will put the element on the end of the list. However, this is not the case.
What's the difference between <code>cons</code> and <code>append</code> ?	The <code>cons</code> function requires only that its second argument be a list. The length of the resulting list is one more than the length of the original list. The <code>append</code> function requires that both its arguments be lists. The length of the resulting list is the sum of the lengths of the two argument lists. <code>append</code> is slower than <code>cons</code> for large lists.

Accessing Lists

Lists are stored internally as a series of branching decision points. Think of the left branch as pointing to the first element and the right branch as pointing to the rest of the list (further

branch decision points). The `car` function follows the left branch and the `cdr` function follows the right branch.

Retrieving the first element of a list (`car`)

`car` returns the first element of a list. `car` was a machine language instruction on the first machine to run Lisp. `car` stands for contents of the address register.

```
numbers = '( 1 2 3 )      => ( 1 2 3 )
car( numbers )            => 1
```

Retrieving the tail of the list (`cdr`)

`cdr` returns a list minus the first element. `cdr` was a machine language instruction on the first machine to run Lisp. `cdr` stands for contents of the decrement register.

```
numbers = '( 1 2 3 )      => ( 1 2 3 )
cdr( numbers )            => ( 2 3 )
```

Retrieving an element given an index (`nth`)

`nth` assumes a zero-based index. `nth(0 numbers)` is the same as `car( numbers)`.

```
numbers = '( 1 2 3 )      => ( 1 2 3 )
nth( 1 numbers )          => 2
```

Determining if a data object is in a list (`member`)

The `member` function cannot search all levels in a hierarchical list. It only looks at the top-level elements. Internally the `member` function follows right branches until it locates a branch point whose left branch dead ends in the element.

```
numbers = '( 1 2 3 )      => ( 1 2 3 )
member( 4 numbers )       => nil
member( 2 numbers )       => ( 2 3 )
```

Counting the elements in a list (`length`)

`length` determines the length of a list, array, or association table.

```
numbers = '( 1 2 3 )      => ( 1 2 3 )
length( numbers )         => 3
```


Modifying Lists

The following functions operate on variables without changing their value or creating new variables.

Coordinates

An xy coordinate is represented by a two-element list. The colon (:) binary operator builds a coordinate from an x value and a y value.

```
xValue = 300
yValue = 400
aCoordinate = xValue:yValue => ( 300 400 )
```

The functions `xCoord` and `yCoord` access the x coordinate and the y coordinate.

```
xCoord( aCoordinate ) => 300
yCoord( aCoordinate ) => 400
```

- You can use the single quote (') operator or `list` function to build a coordinate list.
- You can use the `car` function to access the x coordinate and `car( cdr ( ... ) )` to access the y coordinate.

Bounding Boxes

A bounding box is represented by a list of the lower-left and upper-right coordinates. Use the `list` function to build a bounding box that contains

- Coordinates specified with the binary operator (:).

```
bBox = list( 300:400 500:450 )
```

- Coordinates specified by variables.

```
lowerLeft      = 300:400
upperRight     = 500:450
bBox           = list( lowerLeft upperRight )
```

You can use the single quote (') operator to build the bounding box if the coordinates are specified by literal lists.

```
bBox = ' ( ( 300 400 ) ( 500 450 ) )
```

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Bounding boxes provide a good example of working with the `car` and `cdr` functions. Use any combination of four `a`'s (each `a` executes another `car`) or `d`'s (each `d` executes another `cdr`).

Using `car` and `cdr` with Bounding Boxes

Functions	Meaning	Example	Expression
<code>car</code>	<code>car(...)</code>	lower left corner	<code>ll = car(bBox)</code>
<code>cadr</code>	<code>car(cdr(...))</code>	upper right corner	<code>ur = cadr(bBox)</code>
<code>caar</code>	<code>car(car(...))</code>	x-coord of lower left corner	<code>llx = caar(bBox)</code>
<code>cadar</code>	<code>car(cdr(car(...)))</code>	y-coord of lower left corner	<code>lly = cadar(bBox)</code>
<code>caadr</code>	<code>car(car(cdr(...)))</code>	x-coord of upper right corner	<code>urx = caadr(bBox)</code>
<code>cadadr</code>	<code>car(cdr(car(cdr(...]</code>	y-coord of upper right corner	<code>ury = cadadr(bBox)</code>

File Input/Output

This section introduces how to

- Display values using default formats and application-specific formats
- Write UNIX text files
- Read UNIX text files

Displaying SKILL Data

Display data using

- The `print` and `println` functions
- The `printf` function

The `print` and `println` Functions

The SKILL interpreter has a default display format for each kind of data. The `print` and `println` functions use this format to display data.

Sample Display Formats

Data Type	Example of the Default Format
integer	5
floating point	1.3
text string	"Mary learned SKILL"
variable	bBox
list	(1 2 3)

The `print` and `println` functions display a single data value. `println` is the same as `print` followed by a newline character.

```
for( i 1 3 print( "hello" ))      ;Prints hello three times.
"hello""hello""hello"

for( i 1 3 println( "hello" ))   ;Prints hello three times.
"hello"
"hello"
"hello"
```

The `printf` Function

The `printf` function writes formatted output. This example displays a line in a report.

```
printf(
    "\n%-15s %-15s %-10d %-10d %-10d %-10d"
    layerName purpose
    rectCount labelCount lineCount miscCount
)
```

The first argument is a conversion control string containing directives.

```
%[-][width][.precision]conversion_code
[-]                = left_justify
[width]            = minimum number of character positions
[.precision]       = number of characters to be printed
conversion_code
    d - decimal(integer)
    f - floating point
    s - string or symbol
    c - character
    n - numeric
    L - list (Ignores width and precision fields.)
```

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P - point list (Ignores width and precision fields.)
B - Bounding box list (Ignores width and precision.)

The %L directive specifies the default format. This directive is a convenient way to intersperse application-specific formats with default formats. The `printf` function returns `t`. For more information on directives, see [Formatted Output](#) on page 166.

```
aList = '(1 2 3)
printf( "\nThis is a list: %L" aList ) => t
This is a list: (1 2 3)
aList = nil
printf( "\nThis is a list: %L" aList ) => t
This is a list: nil
```

If the conversion control directive is inappropriate for the data item, `printf` displays an error.

```
printf( "%d %d" 5 nil )
Message: *Error* fprintf/sprintf:
        format spec. incompatible with data - nil
```

Writing Data to a File

To write text data to a file

1. Use the `outfile` function to obtain an output port on a file.
2. Use an optional output port parameter to the `print` and `println` functions and/or use a required port parameter to the `fprintf` function.
3. Close the output port with the `close` function.

Both `print` and `println` accept an optional second argument, which should be an output port associated with the target file. Use the `outfile` function to obtain an output port for a file. Once you are finished writing data to the file, use the `close` function to release the port. The following code

```
myPort = outfile( "/tmp/myFile" )
for( i 1 3
    println( list( "Number:" i) myPort )
)
close( myPort )
```

writes this data to the file `/tmp/myFile`.

```
("Number:" 1)
("Number:" 2)
("Number:" 3)
```

Notice how SKILL displays a port:

```
myPort = outfile( "/tmp/myFile" )
port:"/tmp/myFile"
```

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Use a full path with the `outfile` function. Keep in mind that `outfile` returns `nil` if you don't have write access to the file or if it can't be created in the directory specified in the path. The `print` and `println` functions display an error if the port argument is `nil`. Notice that the type template uses a `p` character to indicate a port is expected.

```
println( "Hello" nil )
Message: *Error* println: argument #2 should be an I/O port
        (type template = "gp") - nil
```

Unlike the `print` and `println` functions, the `printf` function does not accept an optional port argument. Use the `fprintf` function to write formatted data to a file. Its first argument should be an output port associated with the file. The following code

```
myPort = outfile( "/tmp/myFile" )
for( i 1 3
    fprintf( myPort "Number: %d\n" i )
)
close( myPort )
```

writes this data to the file `/tmp/myFile`.

```
Number: 1
Number: 2
Number: 3
```

Reading Data from a File

To read a text file

1. Use the `infile` function to obtain an input port.
2. Use the `gets` function to read the file a line at a time and/or use the `fscanf` function to convert text fields upon input.
3. Close the input port with the `close` function.

Use the `infile` function to obtain an input port on a file. The `gets` function reads the next line from the file. This example prints every line in the `~/ .cshrc` file.

```
inPort = infile( "~/ .cshrc" )
when( inPort
    while( gets( nextLine inPort )
        println( nextLine )
    )
    close( inPort )
)
```

The `fscanf` function reads data from a file according to format directives. This example prints every word in `~/ .cshrc`.

```
inPort = infile( "~/ .cshrc" )
when( inPort
```

```
while( fscanf( inPort "%s" word )
      println( word )
    )
  close( inPort )
)
```

The `gets` function reads the next line from the file. The arguments of `gets` are the variable that will receive the next line and the input port. `gets` returns the text string or `nil` when the end of file is reached.

The `fscanf` function reads data from a file according to conversion control directives. The arguments of `fscanf` are

- Input port
- Conversion control string
- Variable(s) that will receive the matching data values

`fscanf` returns the number of data items matched. Format directives commonly found include the following.

Some Common Format Directives

Format Specification	Data Type	Scans Input Port
%d	integer	for next integer
%f	floating point	for next floating point
%s	text string	for next text string
%e	floating point	for next floating point
%g	floating point	for next floating point

For common output format specifications, see [Formatted Output](#) on page 166.

Flow of Control

This section introduces you to

- Relational Operators: `<`, `<=`, `>`, `>=`, `==`, `!=`
- Logical Operators: `!`, `&&`, `||`
- Branching: `if`, `when`, `unless`
- Multi-way Branching: `case`

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■ Iteration: for, foreach

Iteration refers to repeatedly executing a collection of SKILL expressions by changing - usually incrementing or decrementing - the value of one or more loop variables.

Relational Operators

Use the following operators to compare data values. SKILL generates an error if the data types are inappropriate. These operators all return `t` or `nil`.

Sample Relational Operators

Operator	Arguments	Function	Example	Return Value
<	numeric	lessp	3 < 5	t
			3 < 2	nil
<=	numeric	leqp	3 <= 4	t
>	numeric	greaterp	5 > 3	t
>=	numeric	geqp	4 >= 3	t
==	numeric	equal	3.0 == 3	t
	string list		"abc" == "ABc"	nil
!=	numeric string list	nequal	"abc" != "ABc"	t

It is helpful to know the function name because error messages mention the function (`greaterp` below) instead of the operator (`>`).

```
1 > "abc"
Message: *Error* greaterp: can't handle (1 > "abc")
```

Logical Operators

SKILL considers `nil` as FALSE and any other value as TRUE. The `and` (`&&`) and `or` (`||`) operators only evaluate their second argument if they need to determine the return result.

Sample Logical Operators

Operator	Arguments	Function	Example	Return Value
&&	general	and	3 && 5	5
			5 && 3	3
			t && nil	nil
			nil && t	nil
	general	or	3 5	3
			5 3	5
			t nil	t
			nil t	t

The `&&` and `||` operators return the value last computed. Consequently, both `&&` and `||` operators can be used to avoid cumbersome `if` or `when` expressions.

Using &&

When SKILL creates a variable, it gives the variable a value of `unbound` to indicate that the variable has not been initialized yet. Use the `boundp` function to determine whether a variable is bound. The `boundp` function

- Returns `t` if the variable is bound to a value.
- Returns `nil` if it is not bound to a value.

Suppose you want to return the value of a variable `trMessages`. If `trMessages` is `unbound`, retrieving the value causes an error. Instead, use the expression

```
boundp( 'trMessages ) && trMessages
```

Using ||

Suppose you have a default name, such as `noName`. Suppose you have a variable, such as `userName`. To use the default name if `userName` is `nil`, use the following expression

```
userName || "noName"
```


The if Function

Use the `if` function to selectively evaluate two groups of one or more expressions. The condition in an `if` expression evaluates to `nil` or `non-nil`. The return value of the `if` expression is the value last computed.

```
if( shapeType == "rect"
    then
        println( "Shape is a rectangle" )
        ++rectCount
    else
        println( "Shape is not a rectangle" )
        ++miscCount
)
```

SKILL does most of its error checking during execution. Error messages involving `if` expressions can be obscure. Be sure to

- Be aware of the placement of the parentheses: `if( ... then ... else ... )`.
- Avoid white space immediately after the `if` keyword.
- Use `then` and `else` when appropriate to your logic.

Consider the error message when you accidentally put white space after the `if` keyword.

```
shapeType = "rect"
if ( shapeType == "rect"
    then
        println( "Shape is a rectangle" )
        ++rectCount
    else
        println( "Shape is not a rectangle" )
        ++miscCount
)
```

Message: *Error* if: too few arguments (at least 2 expected, 1 given)...

Consider the error message when you accidentally drop the `then` keyword, but include an `else` keyword, and the condition returns `nil`.

```
shapeType = "label"
if( shapeType == "rect"
    println( "Shape is a rectangle" )
    ++rectCount
    else
        println( "Shape is not a rectangle" )
        ++miscCount
)
```

Message: *Error* if: too many arguments ...

The when and unless Functions

Use the `when` function whenever you have only `then` expressions.

```
when( shapeType == "rect"
      println( "Shape is a rectangle" )
      ++rectCount
    ) ; when

when( shapeType == "ellipse"
      println( "Shape is an ellipse" )
      ++ellipseCount
    ) ; when
```

Use the `unless` function to avoid negating a condition. Some users find this less confusing.

```
unless( shapeType == "rect" || shapeType == "line"
        println( "Shape is miscellaneous" )
        ++miscCount
      ) ; unless
```

The `when` and `unless` functions both return the last value evaluated within their body or `nil`.

The case Function

The `case` function offers branching based on a numeric or string value.

```
case( shapeType
      ( "rect"
        ++rectCount
        println( "Shape is a rectangle" )
      )
      ( "line"
        ++lineCount
        println( "Shape is a line" )
      )
      ( "label"
        ++labelCount
        println( "Shape is a label" )
      )
      ( t
        ++miscCount
        println( "Shape is miscellaneous" )
      )
    ) ; case
```

The optional value `t` acts as a -all and should be handled last. The `case` function returns the value of the last expression evaluated. In this example:

- The value of the variable `shapeType` is compared against the values `rect`, `line`, and `label`. If SKILL finds a match, the several expressions in that arm are evaluated.
- If no match is found, the final arm is evaluated.

- When an arm's target value is a list, SKILL searches the list for the candidate value. If SKILL finds the candidate value, all the expressions in the arm are evaluated.

```
case( shapeType
      ( "rect"
        ++rectCount
        println( "Shape is a rectangle" )
      )
      ( ( "label" "line" )
        ++labelOrLineCount
        println( "Shape is a line or a label" )
      )
      ( t
        ++miscCount
        println( "Shape is miscellaneous" )
      )
    ) ; case
```

The for Function

The index in a `for` expression is saved before the `for` loop and is restored to its saved value after the `for` loop is exited. SKILL does most of its error checking during execution. Error messages involving `for` expressions can be obscure. Be sure to

- Be aware of the placement of the parentheses: `for( ... )`.
- Avoid white space immediately after the `for` keyword.

The example below adds the integers from one to five to an intermediate `sum`. `i` is a variable used as a counter for the loop and as the value to add to `sum`. Counting begins with one and ends with the completion of the fifth loop. `i` increases by one for each iteration through the loop.

```
sum = 0
for( i 1 5
      sum = sum + i
      println( sum )
    )
=> t
```

SKILL prints the value of `sum` with a carriage return for each pass through the loop:

```
1
3
6
10
15
```

The `for` function always returns `t`.

The foreach Function

The `foreach` function is useful for performing operations on each element in a list. Use the `foreach` function to evaluate one or more expressions for each element of a list of values.

```
rectCount = lineCount = polygonCount = 0
shapeTypeList = '( "rect" "polygon" "rect" "line" )
foreach( shapeType shapeTypeList
  case( shapeType
    ( "rect"          ++rectCount )
    ( "line"         ++lineCount )
    ( "polygon"      ++polygonCount )
    ( t              ++miscCount )
  ) ; case
) ; foreach
=> ( "rect" "polygon" "rect" "line" )
```

When evaluating a `foreach` expression, SKILL determines the list of values and repeatedly assigns successive elements to the index variable, evaluating each expression in the `foreach` body. The `foreach` expression returns the list of values over which it iterates.

In the example:

- The variable `shapeType` is the index variable. Before entering the `foreach` loop, SKILL saves the current value of `shapeType`. SKILL restores the saved value after completing the `foreach` loop.
- The variable `shapeTypeList` contains the list of values. SKILL successively assigns the values in `shapeTypeList` to `shapeType`, evaluating the body of the `foreach` loop once for each separate value.
- The body of the `foreach` loop is a single case expression.
- The return value of the `foreach` loop is the list contained in variable `shapeTypeList`.

If you have executed the example above, you can examine the effect of the iterations by typing the name of the counter:

```
rectCount      => 2
lineCount      => 1
polygonCount   => 1
```

Developing a SKILL Function

Developing a SKILL function includes the following tasks.

- Grouping several SKILL statements into a single SKILL statement
- Declaring a SKILL function with the `procedure` function
- Defining function parameters
- Maintaining your source code
- Loading your SKILL source code
- Redefining a SKILL function

Grouping SKILL Statements

Sometimes it is convenient to group several SKILL statements into a single SKILL statement. Use braces `{` and `}` to group a collection of SKILL statements into a single SKILL statement. The return value of the single statement is the return value of the last SKILL statement in the group. You can assign this return value to a variable.

This example computes the pixel height of `bBox` and assigns it to the `bBoxHeight` variable:

```
bBoxHeight = {  
    bBox = list( 100:150 250:400)  
    ll = car( bBox )  
    ur = cadr( bBox )  
    lly = yCoord( ll )  
    ury = yCoord( ur )  
    ury - lly }
```

- The `ll` and `ur` variables hold the lower-left and upper-right coordinates of the bounding box.
- The `xCoord` and `yCoord` functions return the `x` and `y` coordinate of a point.
- The `ury - lly` expression computes the height. This last statement in the group determines the return value of the group.
- The return value is assigned to the `bBoxHeight` variable.

You can declare the variables `ll`, `ur`, `ury`, and `lly` to be local variables. Use the `prog` or `let` functions to define a collection of local variables for a group of several statements. However, defining global variables should be avoided, especially for novices, because it makes harder to debug side effects.

Declaring a SKILL Function

To refer to the group of statements by name, use the `procedure` declaration to associate a name with the group. The group of statements and the name make up a SKILL function.

- The name is known as the function name.
- The group of statements is the function body.
- To execute the group of statements, mention the function name followed immediately by `()`.

The `ComputeBBoxHeight` function example below computes the pixel height of `bBox`.

```
procedure( ComputeBBoxHeight( )  
    bBox = list( 100:150 250:400)  
    ll    = car( bBox )  
    ur    = cadr( bBox )  
    lly   = yCoord( ll )  
    ury   = yCoord( ur )  
    ury - lly  
    ) ; procedure
```

```
bBoxHeight = ComputeBBoxHeight()
```

Defining Function Parameters

To make your function more versatile, you can identify certain variables in the function body as formal parameters.

When you invoke your function, you supply a parameter value for each formal parameter.

In the following example, the `bBox` is the parameter.

```
procedure( ComputeBBoxHeight( bBox )  
    ll    = car( bBox )  
    ur    = cadr( bBox )  
    lly   = yCoord( ll )  
    ury   = yCoord( ur )  
    ury - lly  
    ) ; procedure
```

To execute your function, you must provide a value for the parameter.

```
bBox = list( 100:150 250:400)  
bBoxHeight = ComputeBBoxHeight( bBox )
```

Selecting Prefixes for Your Functions

With only a few exceptions, the SKILL functions in this manual do not use a prefix identifier. Many examples in this manual use a “tr” prefix to indicate they are created for training

purposes. If you look in other SKILL manuals, you will notice that functions for tools are usually grouped with identifiable, unique prefixes.

For example, functions used for technology file administration are all prefixed with “tc”. These prefixes vary across Cadence tools, but all use lowercase letters. It is recommended that you establish a unique prefix using uppercase letters for your own functions.

Maintaining SKILL Source Code

The Cadence environment makes it easy to invoke an editor of your choice. Set the SKILL variable `editor` to a UNIX command line able to launch your editor.

```
editor = "xterm -e vi"
```

The `ed` function invokes an editor of your choice. If you optionally use the `ed1` function, the system loads your file when you quit the editor.

```
ed( "myFile.il")
```

Alternatively, you can use an editor independent of the Cadence environment.

Loading Your SKILL Source Code

The `load` function

- Evaluates each expression in a source code file]
- Is typically used to define a collection of functions
- Returns `t` if all expressions evaluate without errors
- Aborts if there are any errors, any expression following the offending expression is not evaluated

Note: If an error occurs while loading the source code, an error message displays listing the line number on which the error occurred.

Giving a Relative Path

When you pass a relative path to the `load` function, the system resolves it in terms of a list of directories called the SKILL path. You usually establish the SKILL path in your initialization file by using the `setSkillPath` or `getSkillPath` functions.

- The `setSkillPath` function sets the path to a list of directories.
- The `getSkillPath` function returns a list of directories in search order.

Entering a Function

Sometimes you need to define a function without saving the source code file. Using the mouse in your editor, select and paste the function into the command input line.

Setting the SKILL Path

Use the `setSkillPath` function in conjunction with the `prependInstallPath` and `getSkillPath` functions to set the SKILL path.

```
trSamplesPath = list(  
    prependInstallPath( "etc/context" )  
    prependInstallPath( "local" )  
    prependInstallPath( "samples/local" )  
)
```

Use the `prependInstallPath` function to make a path relative to the installation directory. The function prepends `your_install_dir /tools/dfII` to the path. Assuming your installation path is `/cds/9401` `trSamplesPath` is now:

```
("cds/9401/tools.sun4/dfII/etc/context"  
"/cds/9401/tools.sun4/dfII/local"  
"/cds/9401/tools.sun4/dfII/samples/local")
```

Assuming your SKILL path is `("." "~")`, you can set a new SKILL path using `setSkillPath`.

```
setSkillPath( append( trSamplesPath getSkillPath() ) )
```

The return value of `setSkillPath` indicates a path that could not be located, not the SKILL path.

```
("cds/9401/tools.sun4/dfII/samples/local")
```

The SKILL path is

```
("cds/9401/tools.sun4/dfII/etc/context"  
"/cds/9401/tools.sun4/dfII/local"  
"/cds/9401/tools.sun4/dfII/samples/local" "." "~")
```

For more information on the SKILL path, see [Directory Paths](#) on page 152.

Redefining a SKILL Function

Users should be safeguarded against inadvertently redefining functions. Yet, while developing SKILL code, you often need to redefine functions.

The SKILL interpreter has an internal switch called `writeProtect` to prevent the virtual memory definition of a function from being altered during a session.

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By default `writeProtect` is set to `nil`. SKILL functions defined while `writeProtect` is `t` cannot be redefined during the same session. Typically, you set the `writeProtect` switch in your initialization file.

```
sstatus( writeProtect t )      ;sets it to t
sstatus( writeProtect nil )    ;sets it to nil
```

This example tries to redefine `trReciprocal` to prevent division by 0.

```
sstatus( writeProtect t ) => t
procedure( trReciprocal( x ) 1.0 / x ) => trReciprocal
procedure( trReciprocal( x ) when( x != 0.0 1.0 / x ))
*Error* def: function name is write protected
        and cannot be redefined - trReciprocal
```

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Language Characteristics

This chapter explains the basic structure and syntax of the Cadence® SKILL language. The best way to learn SKILL, of course, is by using it to accomplish a real task. Before you begin using SKILL, you should study this chapter.

Programming experience is helpful for those who want to program extensively in SKILL. References to the C programming language in this text make it easier for C programmers to learn SKILL. This text does not require you to be an experienced programmer.

Experienced C programmers must remember that SKILL is not C even though the syntax appears familiar.

For more information, see the following topics:

- [Naming Conventions](#) on page 52
- [Function Calls](#) on page 55
- [SKILL Syntax](#) on page 55
- [Data Characteristics](#) on page 61

Naming Conventions

This section describes Cadence naming conventions for functions, variables, and their arguments.



Caution

To avoid conflict with Cadence-supplied functions and variables, customers should begin their function and variable names with uppercase letters.

Names of Functions

If you look in SKILL API reference manuals, you will notice that functions for tools are usually grouped with identifiable, unique prefixes. These prefixes vary across Cadence tools, but all use lowercase letters.

The recommended naming scheme is to

- Use casing to separate code that is developed within Cadence from that developed outside.
- Use a group prefix to separate code developed within Cadence.
- Add an underscore (_) prefix to an argument in a function definition to indicate to SKILL Lint that it is intentionally unused within the body of the function.

Cadence internal developers should name functions with

- An optional underscore (_) prefix to indicate private functions. Cadence customers should not use private functions. See [Cadence-Private Functions](#) on page 53.
- A prefix that has two to five lowercase characters that signify the code package.
- The lowercase character in the prefix can be one of the following:

Name Type	Character	Meaning
Bit Field Constant	b	Bit-field constant
Constants	c	Enumerated constant
Errors	e	Name of a structure describing an error

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Name Type	Character	Meaning
Internal Functions	i	An internal function, these functions should not be called directly by application programs
Functions	f	Occasionally used as a function indicator.
Macros	m	Rarely used macro indicator.
Global Variables	v	Public global variable

- The root name itself starting with an uppercase character.

Cadence-Private Functions



Functions beginning with an underscore are considered Cadence-private, internal functions and are not supported.

Cadence-private functions are undocumented, unsupported functions that are used internally by Cadence engineering. These Cadence-private functions are subject to change at any time, without notice, because they are not intended for public use.

Names of Variables



Cadence customers can avoid naming conflicts with Cadence-supplied variables by beginning the first letter of their variable names with uppercase letters.

You should not set the following Cadence internal variables without a full understanding of potential consequences.

Variable	Meaning
<code>stdin</code>	Standard input port.
<code>stdout</code>	Standard output port.

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Variable	Meaning
<code>piport</code>	Standard input port from which user input is read.
<code>poport</code>	Standard output port, which is the default for print statements.
<code>ptport</code>	Standard output port for tracing results.
<code>errport</code>	Standard output port for error messages.
<code>printlength</code>	Controls the number of elements in a list that are printed.
<code>printlevel</code>	Controls the depth of elements in a list which are printed.
<code>tracelength</code>	Controls the number of elements in a list that are printed.
<code>tracelevel</code>	Controls the depth of elements in a list that are printed during tracing.
<code>pprintresult</code>	Controls the pretty printing of the results of values returned by the top level.
<code>editor</code>	Controls the default text editor.
<code>gcdisable</code>	Toggles enabling of garbage collection. Use cautiously. Internal variable used in memory management for debugging purposes only.
<code>_gcprint</code>	Toggles the printing of garbage collection messages. Internal variable used in memory management for debugging purposes only.
<code>echoInput</code>	Controls the echoing of expressions or all contents of files being loaded (via <code>load()/loadi()</code>), for example <code>sstatus(echoInput t)</code>
<code>roport</code>	Controls printing messages to CIW.

You might see internal variable names when you are using the debugging tools, especially if you are dumping the stack.

Function Calls

SKILL is a functional programming language, which means that computation is both expressed and performed as a series of function calls.

- Every operator in SKILL corresponds to a predefined function. The operator “+,” for example, corresponds to the function `plus`.
- You can add two numbers together either by calling the function, for example, `plus(x y)`, or by using the `infix` expression `x+y`.

In SKILL, even control statements are implemented by calls to special functions; the special functions then evaluate their arguments in a specific order.

Function calls can be written in either of the following notations.

- Algebraic notation used by most programming languages, that is, `func( arg1 arg2 ... )`.
- Prefix notation used by the Lisp programming language, that is; `(func arg1 arg2 ...)`.

Remember that there must be no white space between the function name and the left parenthesis in the algebraic notation.

The functional programming concepts implemented in SKILL are reflected in the basic syntax of the language. SKILL programs are written as sequences of nested function calls. To utilize SKILL fully, you must first have a good grasp of the underlying concepts of functional programming.

SKILL Syntax

This section describes SKILL syntax, which includes the use of special characters, comments, spaces, parentheses, and other notation.

Special Characters

Certain characters are special in SKILL. These include the `infix` operators such as less than (<), colon (:), and assignment (=). The table below lists these special characters and their meaning in SKILL.

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Note: All non-alphanumeric characters (other than ‘_’ and ‘?’) must be preceded (“escaped”) by a backslash (\) when you use them in the name of a symbol.

Special Characters in SKILL

Character	Name	Meaning
\	backslash	Escape for special characters
()	parentheses	Grouping of list elements, function calls
[]	brackets	Array index, super right bracket
{ }	braces	Grouping of expressions using <code>progn</code>
'	single quote	Quoting the expression to prevent its evaluation
"	double quote	String delimiter
,	comma	Optional delimiter between list elements; also used within the scope of a backquoted expression to force the evaluation of the expression
;	semicolon	Line-style comment character
:	colon	Bit field delimiter, range operator
.	period	<code>getq</code> operator
+, -, *, /	arithmetic	For arithmetic operators; the <code>/*</code> and <code>*/</code> combinations are also used as comment delimiters
!, ^, &,	logical	For logical operators
<, >, =	relational	For relational and assignment operators; <code><</code> and <code>></code> are also used in the specification of bit fields
#	pound sign	Signals special parsing if it appears in the first column
@	“at” sign	If first character, implies reserved word; also used with comma to force evaluation and list splicing in the context of a backquoted expression
?	question mark	If first character, implies keyword parameter
`	backquote	Quoting the expression prevents its evaluation, with support for the comma (,) and comma-at (,@) operators to allow evaluation within backquoted forms
%	percent sign	Used as a scaling character for numbers
\$	—	Reserved for future use

Comments

SKILL permits two different styles of comments. One style is block-oriented, where comments are delimited by `/*` and `*/`. For example:

```
/* This is a block of (C style) comments
comment line 2
comment line 3 etc.
*/
```

The other style is line-oriented where the semicolon (`;`) indicates that the rest of the input line is a comment. For example:

```
a=3 ;Assign value
b=2 ;In the next line, add two numbers
a+b
=>3
2
5
```

There may be times, when you want to break a long comment line for aesthetic reasons. In that case, you can use a backslash (`\`). A backslash allows the comment to continue on to the next line without the need to repeat the comment character (`;`) at the start of that following line. For example:

```
a=3 ;Assign value
b=2 ;In the next \
line add two numbers
a+b
=>3
2
5
```

Note: If the backslash in the comment line is followed by a command, the command will be considered as part of the comment and will not be executed. For example:

```
a=3 ;Assign value
b=2 ;In the next line add two numbers\
a+b
=>3
2
```

In this example, the sum of the two numbers, *a* and *b*, is not displayed.

In such a case, place `' '` immediately after the backslash to avoid disrupting the flow of code. This is shown in the example below.

```
a=3
b=2 ;To add two numbers\' '
a+b
=>3
2
5
```

For simplicity, line-oriented comments are recommended. Block-oriented comments cannot be nested because the first `*/` encountered terminates the whole comment.

White Space

White space sometimes takes on semantic significance and a few syntactic restrictions must therefore be observed. See [Solution to Some Common SKILL Problems](#) on page 27.

Write function calls so the name of a function is immediately followed by a left parenthesis; no white space is allowed between the function name and the parenthesis. For example:

`f(a b c)` and `g()` are legal function calls, but
`f (a b c)` and `g ()` are illegal.

The unary minus operator must immediately precede the expression it applies to. No white space is allowed between the operator and its operand. For example:

`-1`, `-a`, and `-(a*b)` are legal constructs, but
`- 1`, `- a`, and `- (a*b)` are illegal.

The binary minus (subtract) operator should either be surrounded by white space on both sides or there should be no white space on either side. To avoid ambiguity, one or the other method should be used consistently. For example:

`a - b` and `a-b` are legal constructs for binary minus, but `a -b` is illegal.

White Space Characters

The white space characters in SKILL are usually the same as the C standard white space characters: `\t \f \r \n \v`. Respectively, they are space, tab, form feed, carriage return, new line, and vertical tab.

The default break characters used by the SKILL language function `parseString()`, when the optional second argument is not specified, are the white space characters: `/t /f /r /n /v`.

Parentheses

There is a subtle point about SKILL syntax that C programmers, in particular, must be careful to note.

Parentheses in C

In C, the relational expression given to a conditional statement such as `if`, `while`, and `switch` must be enclosed by an outer set of parentheses for purely syntactical reasons, even if that expression consists of only a single Boolean variable. In C, an `if` statement might look like:

```
if (done) i=0; else i=1;
```

Parentheses in SKILL

In SKILL, however, parentheses are used for calling functions, delimiting multiple expressions, and controlling the order of evaluation. You can write function calls in prefix notation

```
(fn2 arg1 arg2) or (fn0)
```

as well as in the more conventional algebraic form:

```
fn2(arg1 arg2) or fn0()
```

The use of syntactically redundant parentheses causes variables, constants, or expressions to be interpreted as the names of functions that need to be further evaluated. Therefore,

- Never enclose a constant or a variable in parentheses by itself. For example, `(1)`, `(x)`.
- For arithmetic expressions involving `infix` operators, you can use as many parentheses as necessary to force a particular order of evaluation, but never put a pair of parentheses immediately outside another pair of parentheses, for example, `((a + b))`; the expression delimited by the inner pair of parentheses would be interpreted as the name of a function.

For example, because `if` evaluates its first argument as a logical expression, a variable containing the logical condition to be tested should be written without any surrounding parentheses; the variable by itself is the logical expression. This is written in SKILL as:

```
if( done then i = 0 else i = 1)
```

Super Right Bracket

When you are entering deeply nested expressions, it often becomes tedious to match up each left parenthesis with a right parenthesis at the end of the expression. The right bracket `]` can be used as a super right parenthesis to close off all open parentheses that are still pending. It is a convenient shorthand notation for interactive input, but it is not recommended for use in programs. For example:

```
f1( f2( f3( f4( x ) ) ) ) )
```

can also be written as

```
f1( f2( f3( f4( x ]
```

Backquote, Comma, and Comma-At

SKILL supports a special notation for list construction from templates. This notation allows selective evaluation within a quoted form. This selective evaluation eliminates long sequences of calls to `list` and `append`.

In absence of commas and the comma-at (`,`, `@`) construction, backquote functions in the same way as single quote. However, if a comma appears inside a backquoted form, the expression that immediately follows the comma is evaluated, and the result of evaluation replaces the original form.

Commas are still acceptable as argument list separators in all contexts except within the scope of a backquote. This means that the backquote comma syntax does not have any implications for SKILL code created before the `backquote comma` facility was implemented. For example:

```
y = 1
'(x y z)          => (x y z)
'(x ,y z)         => (x 1 z)
```

The comma-at construction causes evaluation just as the comma does, but the results of evaluation must be a list, and the elements of the list, rather than the list itself, replace the original form. For example:

```
x = 1
y = '(a b c)
'(.x ,y z)          => (1 (a b c) z)
'(.x ,@y z)         => (1 a b c z)
```

Here's an example of a simple macro implemented with backquote:

```
defmacro(myWhen (@rest body)
.....'if( ,car( body) progn( ,@cdr(body))))
```

The expression

```
a = 2
b = 7
myWhen( eq( a b ) printf( "The same\n" ) list( a b ))
```

expands to

```
if( eq( a b ) progn( printf( "The same\n" ) list( a b )))
```

Line Continuation

SKILL places no restrictions on how many characters can be placed on an input line, even though SKILL does impose an 8191 character limit on the strings being input. The parser reads as many lines as needed from the input until it has read in a complete form (that is, expression). If there are parentheses that have not yet been closed or binary *infix* operators whose right sides have not yet been given, the parser treats carriage returns (that is, newlines) just like spaces.

Because SKILL reads its input on a form-by-form basis, it is rarely necessary to “continue” an input line. There might be times, however, when you want to break up a long line for aesthetic reasons. In that case, you can tell the parser to ignore a carriage return in the input line by preceding it immediately with a backslash (\) character.

```
string = "This is \  
a test."  
=> "This is a test."
```

SKILL always considers a backslash at the end of a line as a continuation regardless of whether the line is a command line, a command string, or a comment.

Length of Input Lists

The SKILL parser imposes a limit of 6000 on the number of elements that can be in a list being read in. Internally and on output, there is no limit to how many elements a list can contain.

Data Characteristics

This section describes the following basic data characteristics:

- Data Types
- Numbers
- Strings
- Atoms
- Escape Sequences
- Symbols
- Characters

These other SKILL data characteristics are discussed in the chapters indicated:

- Lists - see [“Advanced List Operations”](#) on page 179
- Property Lists, Defstructs, and Arrays - see [“Data Structures”](#) on page 87
- Type Predicates - see [“Arithmetic and Logical Expressions”](#) on page 121

Data Types

SKILL supports several data types, including integer and floating-point numbers, character strings, arrays, and a highly flexible linked list structure for representing aggregates of data.

For symbolic computation, SKILL has data types for dealing with symbols and functions.

For input/output, SKILL has a data type for representing I/O ports. The table below lists all the data types supported by SKILL with their internal names and single-character mnemonic abbreviations.

Data Types Supported by SKILL

Data Type	Internal Name	Single Character Mnemonic
array	array	a
Cadence database object	dbobject	d
floating-point number	flonum	f
any data type	general	g
linked list	list	l
integer or floating point number		n
user-defined type		o
I/O port	port	p
defstruct		r
relative object design (ROD) object	rodObj	R
symbol	symbol	s
symbol or character string		S
character string (text)	string	t
function object		u
window type		w

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Data Types Supported by SKILL

Data Type	Internal Name	Single Character Mnemonic
integer number	fixnum	x
binary function	binary	y

Note: In a SKILL function, the last type template character is propagated to any remaining arguments. Therefore, for the case where few template characters are supplied, or if an @rest argument is given, the last type template character is used for the remainder of the arguments except for the @aux arguments.

Numbers

SKILL supports the following numeric data types:

- Integers
- Floating-point
- Scaling factors

Integers

Integers (`fixnum`'s) in SKILL are stored internally as 32-bit wide numbers. Unless they are preceded by one of the prefixes listed in the table below, integers are interpreted as decimal numbers. Binary numbers are prefixed with "0b," octal numbers with a leading "0," and hexadecimal numbers with "0x." If a number with one of these prefixes is not a valid binary, octal, or hexadecimal number, respectively, it is treated as a decimal.

Prefixes for Binary/Octal/Hexadecimal Integers

Radix	Prefix	Examples [value in decimal]
binary	0b or 0B	0b0011 [3] 0b0010 [2]
octal	0	077 [63] 011 [9] 067 => 55 (octal value) 099 => 99 (decimal)

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Language Characteristics

Prefixes for Binary/Octal/Hexadecimal Integers

Radix	Prefix	Examples [value in decimal]
hexadecimal	0x or 0X	0x3f [63] 0xff [255]

Note: If very large numeric values are used (that is, greater than 2147483647 or less than -2147483647), integer overflow might occur. The parser would then return the `INT_MAX` value and display a warning message.

Floating-Point Numbers

You use the same syntax for floating-point numbers in SKILL as you do in most programming languages. You can have an integer followed by a decimal point, a fraction, and an optionally signed exponent preceded by “e” or “E”. Either the integer or the fraction must be present to avoid ambiguity. The following examples illustrate correct syntax:

10.0, 10., 0.5, .5, 1e10, 1e-10, 1.1e10

Scaling Factors

SKILL provides a set of scaling factors that can be added on at the end of a decimal number (integer or floating point) to achieve the desired scaling.

- Scaling factors must appear immediately after the numbers they affect; spaces are not allowed between a number and its scaling factor.
- Only the first nonnumeric character that appears after a number is significant; other characters following the scaling factor are ignored.
- If the number being scaled is an integer, SKILL tries to keep it an integer; the scaling factor must be representable as an integer (that is, the scaling factor is an integral multiplier and the result does not exceed the maximum value that can be represented as an integer). Otherwise a floating-point number is returned. The scaling factors are listed in the following table.

Scaling Factors

Character	Name	Multiplier	Examples
Y	Yotta	10^{24}	10Y [10e+25]
Z	Zetta	10^{21}	10Z [10e+22]

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Language Characteristics

Scaling Factors

Character	Name	Multiplier	Examples
E	Exa	10^{18}	10E [10e+19]
P	Peta	10^{15}	10P [10e+16]
T	Tera	10^{12}	10T [10e+13]
G	Giga	10^9	10G [10,000,000,000]
M	Mega	10^6	10M [10,000,000]
k or K	kilo	10^3	10k [10,000]
%	percent	10^{-2}	5% [0.05]
m	milli	10^{-3}	5m [5.0e-3]
u	micro	10^{-6}	1.2u [1.2e-6]
n	nano	10^{-9}	1.2n [1.2e-9]
p	pico	10^{-12}	1.2p [1.2e-12]
f	femto	10^{-15}	1.2f [1.2e-15]
a	atto	10^{-18}	1.2a [1.2e-18]
z	zepto	10^{-21}	1.2z [1.2e-21]
y	yocto	10^{-24}	1.2y [1.2e-24]

Strings

Strings are sequences of characters, for example, “abc” or “123.” A string is marked off by double quotes, just as in the C language; the empty string is represented as “ ”. The SKILL parser limits the length of input strings to a maximum of 8191 characters. There is, however, no limit to the length of strings created during program execution. Strings of >8191 characters can be created by applications and used in SKILL if they are not given as arguments to SKILL string manipulation functions.

You specify

- Printable characters (except a double quote) as part of a string without preceding them with the backslash (\) escape character
- Unprintable characters and the double quote itself by preceding them with the backslash (\) escape character as in the C language

Atoms

An `atom` is any data object that is not an aggregate of other data objects. In other words, `atom` is a generic term covering data objects of all scalar data types. Built into SKILL are several special atoms that are fundamental to the language.

- `nil`: The `nil` atom represents both a false logical condition and an empty list.
- `t`: The symbol `t` represents a true logical condition.

Both `nil` and `t` always evaluate to themselves and must never be used as the name of a variable.

- `unbound`: To make sure you do not use the value of an uninitialized variable, SKILL sets the value of all symbols and array elements initially to `unbound` so that such an error can be detected.

Escape Sequences

The table below lists all the escape sequences supported by SKILL. Any unprintable character can be represented by listing its ASCII code in two or three octal digits after the backslash. For example, BELL can be represented as `\007` and Control-c can be represented as `\003`.

Escape Sequences

Character	Escape Sequence
new-line (line feed)	<code>\n</code>
horizontal tab	<code>\t</code>
vertical tab	<code>\v</code>
backspace	<code>\b</code>
carriage return	<code>\r</code>
form feed	<code>\f</code>
backslash	<code>\\</code>
double quote	<code>\"</code>
ASCII code ddd (octal)	<code>\ddd</code>

Symbols

Symbols in SKILL correspond to variables in C. In SKILL, we often use the terms “symbol” and “variable” interchangeably even though symbols in SKILL are used for other things as well. Each symbol has the following components:

- Print name, which is limited to 255 characters
- Value, which is `unbound` if a value has not been assigned
- Function binding
- Property list ([The Property List of a Symbol](#) on page 91.)

Symbol names can contain alphanumeric characters (a-z, A-Z, 0-9), the underscore (`_`) character, and the question mark (`?`). If the first character of a symbol is a digit, it must be preceded by the backslash character. Other printable characters can be used in symbol names by preceding each special character with a backslash (`\`). The following examples

```
Var0
Var_Name
\*name\+
```

are all legal names for symbols. The internal name for the last symbol is `*name+`.

You can assign values to variables using the equals sign (`=`) assignment operator. You do not need to declare a variable before assigning a value to it, but you cannot use an unassigned variable, that is, an `unbound` variable. Variables are untyped, which means that the same variable name can store any data type.

It is not advisable to give symbols both a value and a function binding, such as

```
myTest = 3 ;assigns the value of 3 to myTest.
procedure( myTest( x y ) x+y ) ;declares the function myTest.
```

Characters

SKILL represents characters by symbols instead of using a separate data type. For example, the function `getc` returns the symbol representing a character. To verify this, read the characters one by one in the string “abc”.

```
myPort = instring( "abc" )=> port:"*string*"
char = getc( myPort ) => a ;;; the first character
type( char ) => symbol ;;; is represented by the symbol a.
getc( myPort ) => b ;;; the next character
getc( myPort ) => c ;;; the next character
getc( myPort ) => nil ;;; end of string
close( myPort )
```

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- Use the `%c` format with the `printf` function to print a single character. For example:

```
char = 'A
printf("Character = %c\n" char ) => t
```

This function prints the following:

```
Character = A
```

- Use extreme care when referring to symbols that represent certain characters. Certain characters must be escaped if they are the initial character in a symbol name. Specifically, you must escape any ASCII character other than an alphabetic character [a-z, A-Z], the underline character (`_`) or the question mark character (`?`).
- You can use a character's ASCII octal value to represent any character other than NULL. Be aware that if the octal code you use as a symbol name defines a printable character then SKILL uses that character as a print name for the symbol. For example,

```
char = '\120 char => P
printf( "Char = %c\n" '\120 ) => t
```

and prints

```
Char = P
```

- As an alternative to ASCII octal values, you can refer to the unprintable characters listed in the [Escape Sequences](#) on page 66. For example, the formfeed character is represented by `\f` and the newline character is represented by `\n`.

The table below lists all the ASCII characters and their corresponding symbol. The only ASCII character that cannot be handled by SKILL is NULL (ASCII code 0) because the null character always terminates a string in UNIX.

Symbols for ASCII Characters

Character(s)	SKILL Symbol Name(s)
a, b, ..., z	a, b, ..., z
A, B, ..., Z	A, B, ..., Z
?, _	?, _
0, 1, ..., 9	\0, \1, ..., \9
^A, ^B, ..., ^Z, ... (octal codes 001-037)	\001, \002, ..., \032, ... (in octal)
<space>	\<space> (backslash followed by a space)
! . ; : ,	\! \. \; \: \,
() [] { }	\(\) \[\] \{ ...

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Symbols for ASCII Characters

Character(s)	SKILL Symbol Name(s)
" # % & + - *	\ " \# \% ...
< = > @ / \ ^ ~	\< \= \> ...
DEL	\177

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Language Characteristics

Creating Functions in SKILL

“[Cadence SKILL Language](#)” on page 19 introduces you to developing a Cadence® SKILL language function. This chapter introduces you to constructs for defining a function and defining local and global variables.

This chapter covers the following topics:

- [Terms and Definitions](#) on page 72
- [Kinds of Functions](#) on page 73
- [Syntax Functions for Defining Functions](#) on page 73
- [Defining Parameters](#) on page 76
- [Type Checking](#) on page 80
- [Local Variables](#) on page 81
- [Global Variables](#) on page 82
- [Redefining Existing Functions](#) on page 84
- [Physical Limits for Functions](#) on page 85

See also “[Advanced Topics](#)” on page 213 for more information.

Terms and Definitions

function, procedure	In SKILL, the terms procedure and function are used interchangeably to refer to a parameterized body of code that can be executed with parameters bound to the formal parameters. SKILL can represent a function as both a hierarchical list and as a function object.
argument, parameter	The terms argument and parameter are used interchangeably. The arguments in a function call correspond to the formal arguments in the declaration of the function.
byte-code	A generic term for the machine code for a “virtual” machine.
virtual machine	A machine that is not physically built, but is emulated in software instead.
function object	The set of byte-code instructions that implement a function’s algorithm. SKILL programs can treat function objects on a basic level like other data types: compare for equality, assigning to a variable, and pass to a function.
function body	The collection of SKILL expressions that define the function’s algorithm.
compilation	The generation of byte-code that implements the function’s algorithm.
compile time	SKILL compiles function definitions when you load source code. Top-level expressions are compiled and then executed.
run time	The time during which SKILL evaluates a function object.

Kinds of Functions

SKILL has several different kinds of functions, classified by the internal names of `lambda`, `nlambda`, and `macro`. SKILL follows different steps when evaluating these functions.

- Most of the functions you will define are `lambda` functions. SKILL executes a `lambda` function after evaluating the parameters and binding the results to the formal parameters.
- You will probably not need to define an `nlambda` function. However, several built-in SKILL functions are `nlambda` functions.

An `nlambda` function should be declared to have a single formal argument. When evaluating an `nlambda` function, SKILL collects all the argument expressions unevaluated into a list and binds that list to the single formal argument. The body of the `nlambda` can selectively evaluate the elements of the argument list.

- It is not likely that you will write many macros. A `macro` function allows you to adapt the normal SKILL function call syntax to the needs of your application. Unlike `lambda` and `nlambda` functions, SKILL evaluates a `macro` at compile-time. When compiling source code, if SKILL encounters a `macro` function call, it evaluates the function call immediately and the last expression computed is compiled in the current function object.

Syntax Functions for Defining Functions

SKILL supports the following syntax functions for defining functions. You should use the `procedure` function or the `defun` function in most cases.

procedure

The `procedure` function is the most general and is easiest to use and understand. Anything that can be done with the other function definition functions can be done with a `procedure` and possibly a `quote` in the call.

The `procedure` function provides the standard method of defining functions. Its return value is the symbol with the name of the function. For example:

```
procedure( trAdd( x y )
  printf( "Adding %d and %d ... %d \n" x y x+y )
  x+y
) => trAdd
trAdd( 6 7 ) => 13
```

lambda

The `lambda` function defines a function without a name. Its return value is a function object that can be stored in a variable. For example:

```
trAddWithMessageFun = lambda( ( x y )
    printf( "Adding %d and %d ... %d \n" x y x+y )
    x+y
) => funobj:0x1814b90
```

You can subsequently pass a function object to the *apply* function together with an argument list. For example:

```
apply( trAddWithMessageFun '( 4 5 ) ) => 9
```

The use of `lambda` can render code difficult to understand. Often the function being defined is required at some other point in the program and so a procedural definition is better. However, the `lambda` structure can be useful when defining special purpose functions and for passing very small functions to functions such as `sort`. For example, to sort a list `signalList` of disembodied property list objects by a property named `strength`, do the following:

```
signalList = '(
    ( nil strength 1.5 )
    ( nil strength 0.4 )
    ( nil strength 2.5 )
)

sort( signalList
    lambda( ( a b ) a->strength <= b->strength )
)
```

Refer to [“Declaring a Function Object \(lambda\)”](#) on page 223 for further details.

nprocedure

Do not use the `nprocedure` function in new code that you write. It is only included in the system for compatibility with prior releases.

- To allow your function to accept an indeterminate number of arguments, use the `@rest` option with the `procedure` function.
- To allow your function to receive arguments unevaluated, use the `defmacro` function.

defmacro

The `defmacro` function provides a means for you to define a `macro` function. You can use a macro to design your own customized SKILL syntax. Your macro is responsible for

translating your custom syntax at compile time into a SKILL expression to be compiled and subsequently executed.

Refer to [“Macros”](#) on page 224 for further discussion and examples.

mprocedure

The `mprocedure` function is a more primitive alternative to the `defmacro` function. The `mprocedure` function has a single argument. The entire custom syntax is passed to the `mprocedure` function unevaluated.

Do not use the `mprocedure` function in new code. It is only included in the system for compatibility with prior releases. Use the `defmacro` function instead. If you need to receive an indeterminate number of unevaluated arguments, use an `@rest` argument.

Refer to [“Macros”](#) on page 224 for further discussion and examples.

defglofun

The `defglofun` function defines a function which is global within a lexical scope.

Summary of Syntax Functions

The following table summarizes each syntax function for declaring a function. You should think twice about using anything other than `procedure`.

Comparison of Syntax Functions

Syntax Function	Function Type	Argument Evaluation	Execution
procedure	lambda	The arguments are evaluated and bound to the corresponding formal arguments.	The expressions in the body are evaluated at run time. The last value computed is returned.
defmacro	macro	The arguments are bound unevaluated to the corresponding formal arguments.	The expressions in the body are evaluated at compile time. The last value computed is compiled.

Comparison of Syntax Functions

Syntax Function	Function Type	Argument Evaluation	Execution
mprocedure	macro	The entire function call is bound to the single formal argument.	The expressions in the body are evaluated at compile time. The last value computed is compiled.
nprocedure	nlambda	All arguments are gathered unevaluated into a list and bound to the single formal argument.	The expressions in the body are evaluated at run time. The last value computed is returned.

Defining Parameters

You can declare how parameters are to be passed to your function by adding the at (@) options in the formal argument list. The available @ options are `@rest`, `@optional`, and `@key`. You can use these options in `procedure`, `lambda`, and `defmacro` argument lists.

@rest Option

The `@rest` option allows an arbitrary number of parameters to be passed to a function in a list. The name of the parameter following `@rest` is arbitrary, although `args` is a good choice.

The following example illustrates the benefits of an `@rest` argument.

```
procedure( trTrace( fun @rest args )
  let( ( result )
    printf( "\nCalling %s passing %L" fun args )
    result = apply( fun args )
    printf( "\nReturning from %s with %L\n" fun result )
    result
  ) ; let
) ; procedure
```

For example, invoking the `trTrace` function passing `plus` and `1, 2, 3` returns 6.

```
trTrace( 'plus 1 2 3 ) => 6
```

and displays the following output in the CIW.

```
Calling plus passing (1 2 3)
Returning from plus with 6
```

- The `trTrace` function calls the `fun` function and passes the arguments it received.

- The `apply` function calls a given function with the given argument list. `trTrace` passes the `@rest` argument list directly to the `apply` function.

The `trTrace` function must accept an arbitrary number of arguments. The number of arguments passed can vary from call to call.

Another benefit of `@rest` is that it puts the arguments into a single list. The `trTrace` function would be less convenient to use if the caller had to put `fun`'s arguments into a list.

If in a function that specifies keyword arguments and passes arguments using `@rest` option, you use the same keyword argument more than once, SKILL uses the first value of the keyword encountered in the function. Any value that does not match with either required or keyword arguments are passed to the `@rest` argument.

The following example illustrates the scenario where the same keyword argument (`x`) is specified twice.

```
defun( test (a @key x y @rest z)
  printf("a=%L x=%L y=%L z=%L\n" a x y z))

(test 0 ?x 1 ?x 2)
=> a=0 x=1 y=nil z=(?x 2)
```

@optional Option

The `@optional` option gives you another way to specify a flexible number of arguments. With `@optional`, each argument on the argument list is matched up with an argument on the formal argument list.

You can provide any optional parameter with a default value. Specify the default value using a default form. The default form is a two-member list. The first member of this list is the optional parameter's name. The second member is the default value.

The default value is assigned only if no value is assigned when the function is called. If the procedure does not specify a default value for an argument, `nil` is assigned.

If you place `@optional` in the argument list of a procedure definition, any parameter following it is considered optional.

The `trBuildBBox` function builds a bounding box.

```
procedure( trBuildBBox( height width @optional
  ( xCoord 0 ) ( yCoord 0 ) )
  list(
    xCoord:yCoord ;;; lower left
    xCoord+width:yCoord+height ) ;;; upper right
  ) ; procedure
```

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Both `height` and `width` must be specified when this function is called. However, the coordinates of the box are declared as optional parameters. If only two parameters are specified, the optional parameters are given their default values. For `xCoord` and `yCoord`, those values are 0.

Examine the following calls to `trBuildBBox` and their return values:

```
trBuildBBox( 1 2 )      => ((0 0) (2 1))
trBuildBBox( 1 2 4 )    => ((4 0) (6 1))
trBuildBBox( 1 2 4 10)  => ((4 10) (6 11))
```

@key Option

`@optional` relies on order to determine what arguments are assigned to each formal argument. The `@key` option lets you specify the expected arguments in any order.

For example, examine the following generalization of the `trBuildBBox` function. Notice that within the body of the function, the syntax for referring to the parameters is the same as for ordinary parameters:

```
procedure( trBuildBBox(
    @key ( height 0 ) ( width 0 ) ( xCoord 0 ) ( yCoord 0 ) )

    list(
        xCoord:yCoord ;;; lower left
        xCoord+width:yCoord+height ) ;;; upper right
    ) ; procedure

trBuildBBox()                => ((0 0) (0 0))
trBuildBBox( ?height 10 )    => ((0 0) (0 10))
trBuildBBox( ?width 5 ?xCoord 10 ) => ((10 0) (15 0))
```

Default Values of Arguments

Each optional argument, whether optional or mandatory (see the arguments defined with `@key`), can be provided with an expression that is used to compute the default value of that optional argument. This is done using the SKILL or SKILL ++ environment in which the function is defined. The expression is evaluated for each call where the given optional parameter is not provided a value, and the optional argument set to the resulting value.

For instance:

```
(inSkill
  (procedure (testSkill a @optional (b (car a)))
    b))
(inSkill (testSkill '(1 2 3)))
*Error* car: argument #1 should be a list (type template = "1") - unbound
```

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Creating Functions in SKILL

```
(inScheme
  (procedure (testScheme a @optional (b (car a)))
    b))
(inScheme (testScheme '(1 2 3)))
1
```

In SKILL, an expression of an optional argument is evaluated in an environment that does not include the bindings of the other function parameters and cannot refer to them.

In SKILL++, an expression can refer to the values of other function parameters. Furthermore, in SKILL++ the expression can refer to parameters that occur later in the parameter list:

```
(inScheme
  (procedure (testScheme2 @key (b (car a)) a)
    b))
(inScheme (testScheme2 ?a '(1 2 3)))
1
```

@aux Option

The @aux option provides a way to declare auxiliary variables that are local to the function body. These variables are specified as variable name-value pairs. If the value is specified, the variable is bound to `t`, otherwise it is bound to `nil`.

Note: @aux variables are semantically similar to `letseq`.

The following definitions are valid and the same rules are applicable for `defun`, `define`, `procedure`, `defmethod`, `lambda`, and `defmacro`:

```
(defun myfunction (a b c @key d e @aux f g h)
  ...)
(defun myfunction (a b c @optional d e @aux f g h)
  ...)
(defun myfunction (a b c @rest d @aux f g h)
  ...)
(defun myfunction (a b c @key d e @rest f @aux g h)
  ...)
(defun myfunction (a b c @optional d e @rest f @aux g h)
  ...)
```

After all other parameter specifiers (such as `@rest`, `@optional`, and `@key`) have been evaluated, the symbols following the `@aux` keyword are processed from left to right.

Note: @aux keyword is supported in SKILL++.

Combining Arguments

@key and @optional are mutually exclusive; they cannot appear in the same argument list. Consequently, there are two standard forms that procedure argument lists follow:

```
procedure(functionname([var1 var2 ...]
    [@optional opt1 opt2 ...]
    [@rest r])
    .
    .
)

procedure(functionname([var1 var2 ...]
    [@key key1 key2 ...]
    [@rest r])
    .
    .
)
```

Type Checking

Unlike most conventional languages that perform type checking at compile time, SKILL performs dynamic type checking when functions are executed (not when they are defined). Each SKILL lambda or macro function can have as part of its definition an argument template that defines the types of arguments that the function expects. Type checking is not supported in mprocedure functions.

Type characters are discussed in “[Data Characteristics](#)” on page 61. For type checking purposes, you can use several composite type characters (shown in the table below) representing a union of data types.

Composite Characters for Type Checking

Character	Meaning
s	Symbol or string
n	Number: fixnum, flonum
u	Function: Either a symbol (function name), a function object (SKILL function/binary function/primeop), or a SKILL list
g	Any data type

You specify the argument type template as a string of type characters at the end of a formal argument list. If the template is present, SKILL matches the data type of each argument against the template at the time the function is invoked. For example:


```
procedure( f(x y "nn") x**2 + y**2 )
```

`nn` specifies that `f` accepts two numerical arguments.

```
procedure( comparelength(str len "tx") strlen(str) == len)
```

`tx` specifies that the first argument must be a string and the second must be an integer.

Local Variables

When you write functions, you should make your variables local. You can define local variables using the `let` and `prog` functions:

- [Defining Local Variables Using the `let` Function](#) on page 81
- [Defining Local Variables Using the `prog` Function](#) on page 82

See also [“Initializing Local Variables to Non-nil Values”](#) on page 82.

Defining Local Variables Using the `let` Function

You can use the `let` function to establish temporary values for local variables.

- You can include a list of the local variables followed by one or more SKILL expressions. These variables are initialized to `nil`.
- The SKILL expressions make up the body of the `let` function, which returns the value of the last expression computed within its body.
- The local variables are known only within the `let` statement. The values of the variables are not available outside the `let` statement.

```
procedure( trGetBBoxHeight( bBox )
  let( ( ll ur lly ury )
    ll      = car( bBox )
    lly     = cadr( ll )
    ur      = cadr( bBox )
    ury     = cadr( ur )
    ury - lly
  ) ; let
) ; procedure
```

- The local variables are `ll`, `ur`, `lly`, and `ury`.
- They are initialized to `nil`.
- The return value is `ury - lly`.

Defining Local Variables Using the prog Function

A list of local variables and your SKILL statements make up the arguments to the `prog` function.

```
prog( ( localVariables ) yourSKILLstatements )
```

The `prog` function allows an explicit loop to be written because the `go` function is supported within the `prog`. In addition, `prog` allows you to have multiple return points through use of the `return` function. If you are not using either of these two features, `let` is much simpler and faster (see [“Defining Local Variables Using the let Function”](#) on page 81).

Initializing Local Variables to Non-nil Values

You can use `let` to initialize local variables to non-`nil` values by making a two element list with the local variable and its initial value. You cannot refer to any other local variable in the initialization expression. For example:

```
procedure( trGetBBoxHeight( bBox )
  let( ( ( ll car( bBox ) ) ( ur cadr( bBox ) ) lly ury )
    lly      = cadr( ll )
    ury      = cadr( ur )
    ury - lly
  ) ; let
) ; procedure
```

Declaring dynamic variables (SKILL) inside lexical code (Scheme)

As explained in Chapter 13 ([Contrast Variable Scoping](#)), SKILL and SKILL++ use different scoping rules. However, there are times when you need to declare dynamic variables in lexical scope. You can declare such dynamic variables using the `defdynamic` or `dynamicLet` functions and reference their values using the `dynamic` function. See [Cadence SKILL Language Reference](#) for more information about these functions.

Global Variables

Besides predefined functions that you are not allowed to modify, there are several variable names reserved by various system functions. They are listed in [“Naming Conventions”](#) on page 52.

The use of global variables in SKILL, as with any language, should be kept to a minimum.

Following standard naming conventions and running SKILL Lint can reduce your exposure to problems associated with global variables.

Testing Global Variables

Applications typically initialize one or more global variables. Before an application runs for the first time, it is likely that its global variables are `unbound`. In such circumstances, retrieving the value of such a global variable causes an error.

Use the `boundp` function to check whether a variable is `unbound` before accessing its value. For example:

```
boundp( 'trItems ) && trItems
```

returns `nil` if `trItems` is `unbound` and returns the value of `trItems` otherwise.

Avoiding Name Clashes

Two applications might accidentally access and set the same global value. Use a standard naming scheme to minimize the chance of this problem occurring. SKILL Lint can flag global variables that do not obey your naming scheme. For details, refer to the chapter on SKILL Lint in *Cadence SKILL IDE User Guide*.

Assume that `trApplication1` and `trApplication2` are two application functions that are supposed to be totally independent. In particular, the order in which they are executed should not matter. Assume both rely on a single global variable. To observe what can happen if the two applications were accidentally coded to use the same global variable, consider the following example.

```
procedure( trApplication1()
  when( !boundp( 'sharedGlobal ) ;;; not set
    sharedGlobal = 1
  ) ; when
) ; procedure

procedure( trApplication2()
  when( !boundp( 'sharedGlobal ) ;;; not set
    sharedGlobal = 2
  ) ; when
) ; procedure
```

The order in which you run `trApplication1` and `trApplication2` determines the final value of `sharedGlobal`.

```
sharedGlobal = 'unbound
trApplication1()      => 1
sharedGlobal          => 1
trApplication2()      => nil
sharedGlobal          => 1

sharedGlobal = 'unbound
trApplication2()      => 2
sharedGlobal          => 2
trApplication1()      => nil
sharedGlobal          => 2
```

Name “clashes” can also occur between functions because programmers can be using the same function names. In this case, a subsequent function definition either overwrites a previous one, or, if `writeProtect` is set, the function definition fails with an error.

Naming Scheme

The recommended naming scheme is to

- Use casing to separate code that is developed within Cadence from that developed outside.
- Use a group prefix to separate code developed within Cadence.

All code developed by Cadence Design Systems should name global variables and functions with an optional underscore; up to three lowercase characters that signify the code package; an optional further lowercase character (one of `c`, `i`, or `v`) and then the name itself starting with an uppercase character. For example, `dmiPurgeVersions()` or `hnlCellOutputs`. Any global object created by code developed outside Cadence shall be named starting with an uppercase character, such as `AcmeGlobalForm`. This includes global variables, functions, namespaces, and objects such as classes or tables.

Reducing the Number of Global Variables

One other technique to reduce the number of global variables is to consolidate a collection of related globals into a disembodied property list or a symbol's property list. That symbol becomes the only global.

This technique could even be extended to associate one symbol with an entire software module. The disadvantage of this approach is that long property lists involve an access time penalty.

Redefining Existing Functions

You often need to redefine a function that you are debugging. The procedure defining constructs allow you to redefine existing functions; however, functions that are write protected cannot be redefined.

- A function not being executed can be redefined if the write protection switch was turned off when the function was initially defined. To turn off the `writeProtect` switch, type

```
sstatus( writeProtect nil )
```

- When building contexts, `writeProtect` is always set to `t`.

Aside from debugging, the ability to have multiple definitions for the same function is useful sometimes. For example, within the Open Simulation System (OSS) “default” netlisting functions can be overridden by user-defined functions.

Finally, you should use a standard naming scheme for functions and variables.

Physical Limits for Functions

The following physical limitations exist for functions:

- Total number of required arguments must be less than 65536
- Total number of *keyword/optional* arguments must be less than 255
- Total number of local variables in a `let` must be less than 65536
- Max size of code vector is less than 1GB

By default, code vectors are limited to functions that can compile less than 32KB words. This translates roughly into a limit of 20000 lines of SKILL code per function. The maximum number of arguments limit of 32KB is mostly applicable in the case when functions are defined to take an `@rest` argument or in the case of `apply` called on an argument list longer than 32KB elements.

To remove that limitation you should set the following value:
`setSaveContextVersion(getNativeContextVersion())`

Then, the generated contexts (version:602) will not be compatible with old releases but allow code vector in functions to be greater than 32KB.

SKILL Lint catches argument numbers greater than the limits with the following message:

```
NEXT RELEASE (DEF6): <filename - line number> (<funcname> :  
definition for <funcname> cannot have more than 255 optional arguments.
```

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Creating Functions in SKILL

Data Structures

For information on data structures and related topics, see the following sections:

- [Access Operators](#) on page 88
- [Symbols](#) on page 89
- [Disembodied Property Lists](#) on page 93
- [Strings](#) on page 97
- [Defstructs](#) on page 107
- [Arrays](#) on page 111
- [Association Tables](#) on page 113
- [Association Lists](#) on page 118
- [User-Defined Types](#) on page 119

Access Operators

Several of the data access operators have a generic nature. That is, the syntax of accessing data for different data types can be the same. You can view the arrow operator as being a property accessor and the array reference operator [] as an indexer. The operators described below are used in examples throughout this chapter.

Arrow (->) Operator

The arrow (->) operator can be applied to disembodied property lists, defstructs, association tables, and user types (special application-supplied types) to access property values. The property must always be a symbol and the value of the property can be any valid Cadence® SKILL language type.

Squiggle Arrow (~>) Operator

The squiggle arrow (~>) operator is a generalization of the arrow operator. It works the same way as an arrow operator when applied directly to an object, but it can also accept a list of such objects. It walks the list applying the arrow operator whenever it finds an atomic object.

The underlying functions for ~> operator are the `setSGq` and `getSGq` functions, which set and retrieve the value of an attribute or a property. For example,

```
setSGq(obj value prop)           ; is equivalent to:
obj~>prop=value
a=getSGq(obj prop)               ; is equivalent to:
a=obj~>prop
info=getSGq(cvId objType)        ; is equivalent to:
info=cvId~>objType
setSGq(rect list(10:10 100:120) bBox) ; is equivalent to:
rect~>bBox=list(10:10 100:120)
```

Array Access Syntax []

The array access syntax [] can be used to access

- Elements of an array
- Key-value pairs in an association list

Symbols

Symbols in SKILL correspond to variables in C. In SKILL, the terms “symbol” and “variable” are often used interchangeably even though symbols in SKILL are used for other things as well. Each symbol has the following components:

- Print name
- Value
- Function binding
- Property list

Except for the name slot, all slots can be optionally empty. It is not advisable to give symbols both a value and a function binding.

Creating Symbols

The system creates a symbol whenever it encounters a text reference to the symbol for the first time. When the system creates a new symbol, the value of the symbol is set to *unbound*.

Normally, you do not need to explicitly create symbols. However, the following functions let you create symbols.

Creating a Symbol with a Given Base Name (*gensym*)

Use the *gensym* function to create a symbol with a given base name. The system determines the index appended to the base name to ensure that the symbol is new. The *gensym* function returns the newly created symbol, which has the value *unbound*. For example:

```
gensym( 'net ) => net2
```

Creating a Symbol from Several Strings (*concat*)

Use the *concat* function to create a symbol when you need to build the name from several strings.

The Print Name of a Symbol

Symbol names can contain alphanumeric characters (a-z, A-Z, 0-9), the underscore (`_`) character, and the question mark (`?`). If the first character of a symbol is a digit, it must be

preceded by the backslash character (\). Other printable characters can be used in symbol names by preceding each special character with a backslash.

Retrieving the Print Name of a Symbol (`get_pname`)

Use the `get_pname` function to retrieve the print name of a symbol. This function is most useful in a program that deals with a variable whose value is a symbol. For example:

```
location = 'U235  
get_pname( location ) => "U235"
```

The Value of a Symbol

The value of a symbol can be any type, including the type *symbol*.

Assigning a Symbol's Value

Use the `=` operator to assign a value to a symbol. The `setq` function corresponds to the `=` operator. The following are equivalent.

```
U235 = 100  
setq( U235 100 )
```

Retrieving a Symbol's Value

Refer to the symbol's name to retrieve its value.

```
U235 => 100
```

Using the Quote Operator with a Symbol

If you need to refer to a symbol itself instead of its value, use the quote operator.

```
location = 'U235 => U235
```

Storing a Symbol's Value Indirectly (`set`)

You can assign a value indirectly to a symbol with the `set` function. There is no operator that corresponds to the `set` function. The following assigns 200 to the symbol *U235*.

```
set( location 200 )
```

Retrieving a Symbol's Value Indirectly (*symeval*)

You can retrieve a symbol's value indirectly with the *symeval* function. There is no operator that corresponds to the *symeval* function.

```
symeval( location ) => 200
```

Global and Local Values for a Symbol

Global and local variables and function parameters are handled differently in SKILL than they are in C and Pascal.

SKILL uses symbols for both global and local variables. A symbol's current value is accessible at any time from anywhere. SKILL manages a symbol's value slot transparently as if it were a stack. The current value of a symbol is the top of the stack. Assigning a value to a symbol changes only the top of the stack. Whenever the flow of control enters a *let* or *prog* expression, the system pushes a temporary value onto the value stack of each symbol in the local variable list.

The Function Binding of a Symbol

When you declare a SKILL function, the system uses the function's name to determine a symbol to hold the function definition. The function definition is stored in the function slot.

If you are redefining the function, the same symbol is reused and the previous function definition is discarded.

Unlike the symbol's value slot, the symbol's function slot is not affected when the flow of control enters or exits *let* or *prog* expressions.

The Property List of a Symbol

A *property list* is a list containing property name/value pairs. Each name/value pair is stored as two consecutive elements on a property list. The *property name*, which must be a symbol, comes first. The *property value*, which can be of any data type, comes next.

When a symbol is created, SKILL automatically attaches to it a property list initialized to *nil*. A symbol property list can be accessed in the same way structures or records are accessed in C or Pascal, by using the dot operator and arrow operators.

Setting a Symbol's Property List (setplist)

The *setplist* function sets a symbol's property list. For example:

```
setplist( 'U235 ' ( x 200 y 300 ) ) => ( x 200 y 300 )
```

Retrieving a Symbol's Property List (plist)

The *plist* function returns the property list associated with a symbol.

```
plist( 'U235 ) => ( x 200 y 300 )
```

Using the Dot Operator to Retrieve Symbol Properties

The dot (.) operator gives you a simple way of accessing properties stored on a symbol's property list. The dot operator cannot be nested, and both the left and right sides of the dot operator must be symbols. For example:

```
U235.x => 200
```

The *getqq* function implements the dot operator. For example, the following behave identically.

```
U235.x  
getqq( U235 x )
```

The *qq* suffix informally indicates that both arguments are implicitly quoted.

If you ask for the value of a particular property on a symbol and the property does not exist on the symbol's property list, *nil* is returned.

Using the Dot and Assignment Operators to Store Symbol Properties

If you assign a value to a property that does not exist, that property is created and put on the property list.

Using the arrow operator to Retrieve Symbol Properties

The arrow (->) operator gives you a simple way of indirectly accessing properties stored on a symbol's property list. The *getq* function implements the arrow operator. Both the left and right sides of the -> operator must be symbols. For example:

```
designator = 'U235  
U235.x = 200  
U235.y = 300  
designator->x => 200  
designator->y => 300
```

Using the Arrow Operator and the Assignment Operator to Store Symbol Properties

The arrow (->) operator and the assignment (=) operator work together to provide a simple way of indirectly storing properties on a symbol's property list. For example:

```
designator->x = 250
U235.x => 250
```

The *putpropq* function implements both the arrow operator and the assignment operator. For example, the following behave identically.

```
designator->x = 250
putpropq( designator 250 x )
```

Important Symbol Property List Considerations

Property lists attached to symbols are *globally visible to all applications*. Whenever you pass a symbol to a function, that function can add or alter properties on that symbol's property list.

In the following example, even though the sample property is established within a *let* expression, it is still available outside the *let* expression. In other words, when the flow of control enters and subsequently exits a *let* or *prog* expression, the property lists of local symbols are not affected.

```
x = 0
let( ( x )
    x = 2
    x.example = 5
) ; let
x => 0
plist( 'x ) => (example 5)
```



If you want to use symbol property lists to pass data from one function to another, you must make sure you choose unique names to avoid possible name collisions with other applications. Use setplist with caution because you might inadvertently destroy properties of importance to other applications.

Disembodied Property Lists

A disembodied property list is logically equivalent to a record. Unlike C structures or Pascal records, new fields can be dynamically added or removed. The arrow operator (->) can be used to store and retrieve properties in a disembodied property list.

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A disembodied property list starts with a SKILL data object, usually *nil*, followed by alternating name/value pairs. The property names must satisfy the SKILL symbol syntax to be visible to the arrow operator. The first element of the disembodied list does not have to be *nil*. It can be any SKILL data object.

In the following example, a disembodied property list is used to represent a complex number.

```
procedure( trMakeComplex( @key ( real 0 ) ( imaginary 0 ) )
  let( ( result )
    result = ncons(nil)
    result->real = real
    result->imaginary = imaginary
    result
  ) ; let
) ; procedure

complex1 = trMakeComplex( ?real 2 ?imaginary 3 )
=> (nil imaginary 3 real 2)
complex2 = trMakeComplex( ?real 4 ?imaginary 5 )
=> (nil imaginary 5 real 4)

i = trMakeComplex( ?imaginary 1 )
=> (nil imaginary 1 real 0)

procedure( trComplexAddition( cmplx1 cmplx2 )
  trMakeComplex(
    ?real      cmplx1->real + cmplx2->real
    ?imaginary  cmplx1->imaginary + cmplx2->imaginary
  )
) ; procedure

procedure( trComplexMultiplication( cmplx1 cmplx2 )
  trMakeComplex(
    ?real
      cmplx1->real * cmplx2->real -
      cmplx1->imaginary * cmplx2->imaginary
    ?imaginary
      cmplx1->imaginary * cmplx2->real +
      cmplx1->real * cmplx2->imaginary
  )
) ; procedure

trComplexMultiplication( i i ) => (nil imaginary 0 real -1)
```

In several circumstances using a disembodied property list to represent a record has advantages over using a symbol's property list.

- An appropriate symbol to which you can attach a property list might not be available. For example, no symbol exists for complex numbers in the example above.
- If you create a symbol for each record your application tracks and your application requires many records, SKILL will have a lot of extra symbols to manage.
- It is easier to pass a disembodied property list as a parameter than it is to pass a symbol as a parameter.

You can store a disembodied property list as the value of a symbol without affecting the symbol's property list.

```
setplist( 'x '( example 0.5 ))
x = complex1      => (nil imaginary 3 real 2)
plist( 'x )       => ( example 0.5 )
```

Important Considerations

Be careful when you are assigning disembodied property lists to variables.



Property list functions that modify property lists modify the original list structures directly. If the property list is not to be shared, use the copy function to make a copy of the original property list.

This caution applies in general to assigning lists as values. Internally, SKILL uses pointers to the lists in virtual memory. For example, as a result of the following assignment

```
complex1 = complex2
```

both symbols *complex1* and *complex2* refer to the same list in virtual memory. Using the arrow operator to modify the *real* or *imaginary* properties of *complex2* is reflected in *complex1*.

Notice that

```
complex1 == complex2      => t
eq( complex1 complex2 )  => t
```

To avoid this problem, perform the assignment as follows.

```
complex1 = copy( complex2 )
```

Notice that

```
complex1 == complex2      => t
eq( complex1 complex2 )  => nil
```

Additional Property List Functions

Adding Properties to Symbols or Disembodied Property Lists (putprop)

putprop adds properties to symbols or disembodied property lists. If the property already exists, the old value is replaced with a new one. The *putprop* function is a *lambda* function, which means all of its arguments are evaluated.

```
putprop('s 1+2 'x)    => 3
s.x = 1+2             => 3
```

Both examples are equivalent expressions that set the property *x* on symbol *s* to 3.

Getting the Value of a Named Property in a Property List (get)

get returns the value of a named property in a property list. *get* is used with *putprop*, where *putprop* stores the property and *get* retrieves it.

```
putprop( 'U235 8 'pins )
```

Assigns the property *pins* on the symbol *U235* to a value of 8.

```
get( 'U235 'pins )    => 8
U235.pins             => 8
```

Adding Properties to Symbols or Disembodied Property Lists (defprop)

defprop adds properties to symbols or disembodied property lists, but none of its arguments are evaluated. *defprop* is the same as *putprop* except that none of its arguments are evaluated.

```
defprop(s 3 x)        => 3
```

Sets property *x* on symbol *s* to 3.

```
defprop(s 1+2 x)      => 1+2
```

Sets property *x* on symbol *s* to the unevaluated expression 1+2.

Removing a Property and Restoring a Previous Value (remprop)

remprop removes a property from a property list. The return value is not useful.

```
setplist( 'U235 '( x 100 y 200 ) ) => (x 100 y 200)
```

Sets the property list to *(x 100 y 200)*.

```
putprop( 'U235 8 'pins )    => 8
```

Sets the value of the *pins* property to 8.

```
plist( 'U235 )             => (pins 8 x 100 y 200)
```

Verifies the operation.

```
get( 'U235 'pins )         => 8
```

Retrieves the *pins* property.

```
remprop( 'U235 'x )
```

Removes the *x* property.


```
plist( 'U235 )      => (pins 8 y 200)
```

Strings

A *string* is a specialized one-dimensional array whose elements are characters.

The string functions in this section are patterned after functions of the same name in the C run-time library. Strings can be compared, taken apart, or concatenated.

Concatenating Strings

Concatenating a List of Strings with Separation Characters (*buildString*)

buildString makes a single string from the list of strings. You specify the separation character in the third argument. A null string is permitted. If this argument is omitted, *buildString* provides a separating space as the default.

```
buildString( '("test" "il") ".")  => "test.il"
buildString( '("usr" "mnt") "/" )  => "usr/mnt"
buildString( '("a" "b" "c") )      => "a b c"
buildString( '("a" "b" "c") "" )   => "abc"
```

Concatenating Two or More Input Strings (*strcat*)

strcat creates a new string by concatenating two or more input strings. The input strings are left unchanged.

```
strcat( "l" "ab" "ef" ) => "labef"
```

You are responsible for any separating space.

```
strcat( "a" "b" "c" "d" )      => "abcd"
strcat( "a " "b " "c " "d " )  => "a b c d "
```

Appending a Maximum Number of Characters from Two Input Strings (*strncat*)

strncat is similar to *strcat* except that the third argument indicates the maximum number of characters from *string2* to append to *string1* to create a new string. *string1* and *string2* are left unchanged.

```
strncat( "abcd" "efghi" 2)      => "abcdef"
strncat( "abcd" "efghijk" 5)    => "abcdefghi"
```

Comparing Strings

Comparing Two String or Symbol Names Alphabetically (*alphalessp*)

alphalessp compares two objects, which must be either a string or a symbol, and returns *t* if *arg1* is alphabetically less than the name of *arg2*. *alphalessp* can be used with the *sort* function to sort a list of strings alphabetically. For example:

```
stringList = '( "xyz" "abc" "ghi" )  
sort( stringList 'alphalessp ) => ("abc" "ghi" "xyz")
```

The next example returns a sorted list of all the files in the login directory.

```
sort( getDirFiles( "~" ) 'alphalessp )
```

Comparing Two Strings Alphabetically (*strcmp*)

strcmp compares two strings. To test if two strings are equal or not, you can use the *equal* command. The return values for *strcmp* indicate

Return Value	Meaning
1	string1 is alphabetically greater than string2.
0	string1 is alphabetically equal to string2.
-1	string1 is alphabetically less than string2.

```
strcmp( "abc" "abb" )=> 1  
strcmp( "abc" "abc")=> 0  
strcmp( "abc" "abd")=> -1
```

Comparing Two String or Symbol Names Alphanumerically or Numerically (*alphaNumCmp*)

alphaNumCmp compares two string or symbol names. If the third optional argument is non-*nil* and the first two arguments are strings holding purely numeric values, a numeric comparison is performed on the numeric representation of the strings. The return values indicate

Return Value	Meaning
1	arg1 is alphanumerically greater than arg2.

Return Value	Meaning
0	arg1 is alphanumerically identical to arg2.
-1	arg2 is alphanumerically greater than arg1.

```
alphaNumCmp( "a" "b" )      => -1
alphaNumCmp( "b" "a" )      => 1
alphaNumCmp( "name12" "name12" ) => 0
alphaNumCmp( "name23" "name12" ) => 1
alphaNumCmp( "00.09" "9.0E-2" t) => 0
```

Comparing a Limited Number of Characters (strncmp)

strncmp compares two strings alphabetically, but only up to the maximum number of characters indicated in the third argument. The return values indicate the same as for *strcmp* above.

```
strncmp( "abc" "ab" 3) => 1
strncmp( "abc" "de" 4) => -1
```

Getting Character Information in Strings

Getting the Length of a String in Characters (strlen)

Refer to [“Pattern Matching of Regular Expressions”](#) on page 101 for information on the backslash notation used below.

```
strlen( "abc" ) => 3
strlen( "\\007" ) => 1
```

Indexing with Character Pointers

Getting the Index Character of a String (getchar)

getchar returns an indexed character of a string or the print name if the string is a symbol.

```
getchar("abc" 2) => b
getchar("abc" 4) => nil
```

getchar returns a symbol whose print name is the character, not a string.

SKILL represents an individual character by the symbol whose print name is the string consisting solely of the character. For example:

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```
getchar( "1.2" 2 )      => \.  
type( getchar( "1.2" 2 ) ) => symbol  
get_pname( getchar( "1.2" 2 ) ) => "."
```

If you are familiar with C, you might notice that the *getchar* SKILL function is totally unrelated to the C function of the same name.

Getting the Tail of a String (index, rindex)

index returns the remainder of *string1* beginning with the first occurrence of *string2*.

```
index( "abc" 'b )      => "bc"  
index( "abcdabce" "dab" ) => "dabce"  
index( "abc" "cba" )   => nil  
index( "dandelion" "d") => "dandelion"
```

rindex returns the remainder of *string1* beginning with the last occurrence of *string2*.

```
rindex( "dandelion" "d") => "delion"
```

Getting the Character Index of a String (nindex)

nindex finds the symbol or string, *string2*, in *string1* and returns the character index, starting from one, of the first point at which *string2* matches part of *string1*.

```
nindex( "abc" 'b )      => 2  
nindex( "abcdabce" "dab" ) => 4  
nindex( "abc" "cba" )   => nil
```

Creating Substrings

Copying Substrings (substring)

substring creates a new substring from an input string, starting at an index point (*arg2*) and continuing for a given length (*arg3*).

```
substring("abcdef" 2 4) => "bcde"  
substring("abcdef" 4 2) => "de"
```

Breaking Lists Into Substrings (parseString)

parseString breaks a string into a list of substrings with specified break characters, which are indicated by an optional second argument.

```
parseString( "Now is the time" ) => ("Now" "is" "the" "time")
```

Space is the default break character

```
parseString( "prepend" "e" )      => ("pr" "p" "nd" )
```

e is the break character.

```
parseString( "feed" "e")          => ("f" "d")
```

A sequence of break characters in *t_string* is treated as a single break character.

```
parseString( "~/exp/test.il" ". /") => ("~" "exp" "test" "il")
```

Both . and / are break characters.

```
parseString( "abc de" " ")        => ("a" "b" "c" " " "d" "e")
```

The single space between c and d contributes " " in the return result.

Converting Case

Converting to Upper Case (*upperCase*)

upperCase replaces lowercase alphabetic characters with their uppercase equivalents. If the parameter is a symbol, the name of the symbol is used.

```
upperCase("Hello world!")      => "HELLO WORLD!"
symName = "nameofasymbol"      => "nameofasymbol"
upperCase(symName)              => "NAMEOFASYMBOL"
```

Converting to Lower Case (*lowerCase*)

lowerCase replaces uppercase alphabetic characters with their lowercase equivalents. If the parameter is a symbol, the name of the symbol is used.

```
lowerCase("Hello World!")      => "hello world!"
```

Pattern Matching of Regular Expressions

In many applications, you need to match strings or symbols against a pattern. SKILL provides a number of pattern matching functions that are built on a few primitive C library routines with a corresponding SKILL interface.

A *pattern* used in the pattern matching functions is a string indicating a regular expression.

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Here is a brief summary of the rules for constructing regular expressions in SKILL:

Rules for Constructing Regular Expressions

Synopsis	Meaning
<code>c</code>	Any ordinary character (not a special character listed below) matches itself.
<code>.</code>	A dot matches any character.
<code>\</code>	A backslash when followed by a special character matches that character literally. When followed by one of <code><</code> , <code>></code> , <code>(</code> , <code>)</code> , and <code>1,...,9</code> , it has a special meaning as described below.
<code>[c...]</code>	A nonempty string of characters enclosed in square brackets (called a set) matches one of the characters in the set. If the first character in the set is <code>^</code> , it matches a character not in the set. A shorthand <code>S-E</code> is used to specify a set of characters <code>S</code> up to <code>E</code> , inclusive. The special characters <code>]</code> and <code>-</code> have no special meaning if they appear as the first character in a set.
<code>*</code>	A regular expression in the above form, followed by the closure character <code>*</code> matches zero or more occurrences of that form.
<code>+</code>	Similar to <code>*</code> , except it matches <i>one</i> or more times.
<code>\(...\)</code>	A regular expression wrapped as <code>\(form \)</code> matches whatever <i>form</i> matches, but saves the string matched in a numbered register (starting from one, can be up to nine).
<code>\n</code>	A backslash followed by a digit <i>n</i> matches the contents of the <i>n</i> th register from the current regular expression.
<code>\<...\></code>	A regular expression starting with a <code>\<</code> and/or ending with a <code>\></code> restricts the pattern matching to the beginning and/or the end of a word. A word defined to be a character string can consist of letters, digits, and underscores.
<code>rs</code>	A composite regular expression <i>rs</i> matches the longest match of <i>r</i> followed by a match for <i>s</i> .
<code>^, \$</code>	A <code>^</code> at the beginning of a regular expression matches the beginning of a string. A <code>\$</code> at the end matches the end of a string. Used elsewhere in the pattern, <code>^</code> and <code>\$</code> are treated as ordinary characters.

How Pattern Matching Works

The mechanism for pattern matching

- Compiles a pattern into a form and saves the form internally
- Uses that internal form in every subsequent matching against the targets until the next pattern is supplied

The *rexCompile* function does the first part of the task, that is, the compilation of a pattern. The *rexExecute* function takes care of the second part, that is, matching a target against the previously compiled pattern. Sometimes this two-step interface is too low-level and awkward to use, so functions for higher-level abstraction (such as *rexMatchp*) are also provided in SKILL.

Avoiding Null and Backslash Problems

- A null string ("") is interpreted as no pattern being supplied, which means the previously compiled pattern is still used. If there was no previous pattern, an error is signaled.
- To put a backslash character (\) into a pattern string, you need an extra backslash (\) to escape the backslash character itself.

For example, to match a file name with dotted extension “.il”, the pattern “[a-zA-Z]+\\.il\$” can be used, but “[a-zA-Z]\\.il\$” gives a syntax error. However, if the pattern string is read in from an input function such as *gets* that does not interpret backslash characters specifically, you should *not* add an extra backslash to enter a backslash character.

Pattern Matching Functions

Finding a Pattern Within a String or Symbol (*rexMatchp*)

```
rexMatchp("[0-9]*[.][0-9][0-9]*" "100.001")    => t
rexMatchp("[0-9]*[.][0-9]+" ".001")            => t
rexMatchp("[0-9]*[.][0-9]+" ".")               => nil
rexMatchp("[0-9]*[.][0-9][0-9]*" "10.")        => nil

rexMatchp("[0-9" "100")
=> *Error* rexMatchp: Missing ] - "[0-9"
```

Finding a Pattern within a String, Symbol, or PCRE object (*pcreMatchp*)

```
pcreMatchp(
    g_pattern
    S_subject
    [ x_compOptBits ]
```

```
[ x_execOptBits ]  
)  
=> t | nil
```

Checks to see if a string or symbol matches a given regular expression.

Examples

```
pcreMatchp( "[0-9]*[.][0-9][0-9]*" "100.001" ) => t  
pcreMatchp( "[0-9]*[.][0-9]+" ".001" ) => t  
pcreMatchp( "[0-9]*[.][0-9]+" "." ) => nil  
pcreMatchp( "[0-9]" "100" ) =>  
*Error* pcreCompile: compilation failed at offset 4: missing terminating ] for  
character class nil  
pcreMatchp( "((?i)rah)\\s+\\1" "rah rah" ) => t  
pcreMatchp( "^[0-9]+" "abc\\n123\\nefg"  
pcreGenCompileOptBits(?multiLine t) pcreGenExecOptBits( ?notbol t) )  
=> t
```

Compiling a Regular Expression String Pattern (rexCompile)

rexCompile compiles a regular expression string pattern into an internal representation to be used by succeeding calls to *rexExecute*.

```
rexCompile("^ [a-zA-Z]+") => t  
rexCompile("\\ ([a-z]+\\)\\.\\1") => t  
rexCompile("^\\ ([a-z]*\\)\\.\\1$") => t  
rexCompile("[ab]") => *Error* rexCompile: Missing ] - "[ab"
```

Compiling a Regular Expression String Pattern (pcreCompile)

Compiles a regular expression string pattern (*t_pattern*) into an internal representation that you can use in a *pcreExecute* function call.

```
pcreCompile(  
    t_pattern  
    [ x_options ]  
)  
=> o_comPatObj | nil
```


For information about Perl Compatible Regular Expressions (PCRE), see vendor-specific documentation at <http://www.pcre.org> or by using the `man pcrepattern` command.

Examples

```
comPat1 = pcreCompile( "[12[:^digit:]]" ) => pcreobj@0x27d150
comPat2 = pcreCompile( "((?i)ab)c" ) => pcreobj@0x27d15c
comPat3 = pcreCompile( "\\d{3}" ) => pcreobj@0x27d168
```

Matching Against a Previously Compiled Pattern (`rexExecute`)

rexExecute matches a string or symbol against the previously compiled pattern created by the last *rexCompile* call.

```
rexCompile("^ [a-zA-Z] [a-zA-Z0-9]*")      => t
rexExecute('Cell123')                     => t
rexExecute("123 cells")                   => nil
```

The target "123 cells" does not begin with a-z/A-Z.

```
rexCompile("\\ ([a-z]+\\)\\.\\1")           => t
rexExecute("abc.bc")                     => t
rexExecute("abc.ab")                     => nil

rexCompile("\\ (^ [a-z]+\\)\\.\\1")          => t
rexExecute("abc.bc")                     => nil
```

The caret (^) in the *rexCompile* pattern requires that the pattern must match from the beginning of the input string.

Matching Against a Previously Compiled Pattern (`pcreExecute`)

Matches against a string or symbol (*S_subject*) against a previously compiled pattern set up by the last *pcreCompile* call.

```
pcreExecute(
  o_comPatObj
  S_subject
  [ x_options ]
)
=> t | nil
```

Examples

```
comPat1 = pcreCompile( "[12[:^digit:]]" ) => pcreobj@0x27d150
pcreExecute( comPat1 "abc" )             => t
```

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```
comPat2 = pcreCompile( "(?i)ab)c" ) => pcreobj@0x27d15c
pcreExecute( comPat2 "aBc" )      => t

comPat3 = pcreCompile( "\\d{3}" ) => pcreobj@0x27d168
pcreExecute( comPat3 "789" )      => t
```

Matching a List of Strings or Symbols (rexMatchList)

rexMatchList matches a list of strings or symbols against a regular expression string pattern and returns a list of the strings or symbols that match.

```
rexMatchList("^([a-z][0-9]*)" '(a01 x02 "003" aa01 "abc"))
=> (a01 x02 aa01 "abc")

rexMatchList("^([a-z][0-9][0-9]*)"
'(a001 b002 "003" aa01 "abc"))
=> (a001 b002)

rexMatchList("box[0-9]*" '(square circle "cell9" "123"))
=> nil
```

Creating an Association List Made of Matching Strings (rexMatchAssocList)

rexMatchAssocList returns a new association list created out of those elements of an association list whose key matches a regular expression string pattern.

```
rexMatchAssocList("^([a-z][0-9]*)$"
'((abc "ascii") ("123" "number") (a123 "alphanum")
(a12z "ana")))
=> ((a123 "alphanum"))

rexMatchAssocList("^([a-z]*[0-9]*)$"
'((abc "ascii") ("123" "number") (a123 "alphanum")
(a12z "ana")))
=> ((abc "ascii") ("123" "number") (a123 "alphanum"))
```

Turning Meta-Characters On and Off (rexMagic)

rexMagic turns on or off the special interpretation associated with the meta-characters (^, \$, *, +, \, [,], and so forth) in regular expressions. Users of *vi* will recognize this as equivalent to the *set magic/set nomagic* commands.

```
rexCompile( "^([0-9])+" )      => t
rexExecute( "123abc" )        => t
rexSubstitute( "got: \\0" )    => "got: 123"
rexMagic( nil )                => nil
rexCompile( "^([0-9])+" )      => t;Recompile w/o magic.
rexExecute( "123abc" )        => nil
rexExecute( "***^([0-9])+!*" ) => t
rexSubstitute( "got: \\0" )    => "got: \\0"
rexMagic( t )                  => t
rexSubstitute( "got: \\0" )    => "got: ^([0-9])+"

rexMagic(nil) ;; switch off
rexSubstitute("[&")=> "[&]"
```

Replacing a Substring (rexReplace)

rexReplace replaces the substring(s) in the source string that matched the last regular expression compiled by the replacement string. The third argument tells which occurrence of the matched substring is to be replaced. If it's 0 or negative, all the matched substrings will be replaced. Otherwise only the specified occurrence is replaced. *rexReplace* returns the source string if the specified match is not found

```
rexCompile( "[0-9]+" )=> t
rexReplace( "abc-123-xyz-890-wuv" "(*)" 1)=> "abc- (*)-xyz-890-wuv"
rexReplace( "abc-123-xyz-890-wuv" "(*)" 2)=> "abc-123-xyz- (*)-wuv"
rexReplace( "abc-123-xyz-890-wuv" "(*)" 3)=> "abc-123-xyz-890-wuv"
rexReplace( "abc-123-xyz-890-wuv" "(*)" 0)=> "abc- (*)-xyz- (*)-wuv"

rexCompile( "xyz" )      => t
rexReplace( "xyzzxyzz" "xy" 0) => "xyzyxyz" ; No rescanning!

rexCompile( "^teststr" )
rexReplace( "teststr_a" "bb" 0) => "bb_a"
rexReplace( "teststr_a" "bb&" 0) => "bbteststr_a"
rexReplace( "teststr_a" "[&]" 0) => "[teststr]_a"
```

Defstructs

Defstructs are collections of one or more variables. They can be of different types and grouped together under a single name for easy handling. They are the equivalent of structs in C.

The following template for *defstruct* defines a data structure with the named slots:

```
defstruct( s_name s_slot1 [s_slot2..] ) => t
```

The *defstruct* also creates a constructor function, *make_name*, where *name* is the structure name supplied to *defstruct*. The constructor function takes keyword arguments: one for each slot in the structure. All arguments are symbols and none need to be quoted.

Note: The increment (pre-increment, post-increment) and decrement (pre-decrement, post-decrement) operations can't be used to directly modify the variable values when used within structures. However, you can use these operations with arrays and associated table elements. For example,

```
array[index]++
array[index]--
--array[index]
++array[index]
```

Note: The name of any local or global variable should not coincide with the structure name, as it can hide the structure name and overwrite the *defstruct* master array.

Behavior Is Similar to Disembodied Property Lists

Once created, structures behave just like disembodied property lists, but are more efficient in space utilization and access times. Structures can have new slots added at any time. However, these dynamic slots are less efficient than the statically declared slots, both in access time and space utilization.

Defstructs respond to the following operations, assuming *struct* is an instance built from a constructor function:

```
struct->slot
```

Returns the value associated with a slot of an instance.

```
struct->slot = newval
```

Modifies the value associated with a slot of an instance.

```
struct->?
```

Returns a list of the slot names associated with an instance.

```
struct->??
```

Returns a property list (not a disembodied property list) containing the slot names and values associated with an instance.

Additional Defstruct Functions

Testing a SKILL Object (defstructp)

```
defstructp( g_object [st_name] ) => t / nil
```

defstructp tests a SKILL object, returning *t* if it's a structure instance, otherwise *nil*. The second argument is optional and denotes the name of the structure to test for. The test in this case is stronger and only returns *t* if *g_object* is an instance of defstruct *st_name*. The name can be passed either as a symbol or a string.

Printing the Contents of a Structure (printstruct)

```
printstruct( r_structureInstance ) => t
```

For debugging, the *printstruct* function prints the contents of a structure in an easily readable form. It recursively dumps out any slot value that is also a structure instance. The *printstruct* function dumps out a structure instance.

Beware of Structure Sharing (copy_<name>)

Structures can contain instances of other structures; therefore, you need to be careful about structure sharing. If sharing is not desired, a special copy function can be used to generate a copy of the structure being inserted. The *defstruct* function also creates a function for the given *defstruct* called *copy_<name>*. This function takes one argument, an instance of the *defstruct*. It creates and returns a copy of the instance.

Making a Deep or Recursive Copy (copyDefstructDeep)

```
copyDefstructDeep( r_object ) => r_object
```

Performs a deep or recursive copy on defstructs with other defstructs as sub-elements, making copies of all the defstructs encountered. The various *copy_name* functions are called to create copies for the defstructs encountered in the deep copy.

Accessing Named Slots in SKILL Structures

Slot Access Example

This example defines a *card* structure and allocates an instance of *card*.

```
defstruct( card rank suit faceUp ) => t
```

This structure has three slots: *rank suit faceUp*. Next, allocate an instance of *card* and store a reference to it in *aCard*.

```
aCard = make_card( ?rank 'ace ?suit 'spades )  
=> array[5]:21556040
```

Structure instances are implemented as arrays. Refer to “[Arrays](#)” on page 111.

```
aCard => array[5]:21556040
```

The *type* function returns the structure name.

```
type( aCard ) => card
```

Use the Arrow Operator -> and ~> to Access Slots

```
aCard->rank => ace  
aCard->faceUp = t => t
```

Use ->? to Get a List of the Slot Names

```
aCard->? => ( faceUp suit rank )
```

Use ->?? to Get a List of Slot Names and Values

```
aCard->?? => ( faceUp t suit spades rank ace )
```

Slots can be created dynamically for an instance by referencing them with the -> operator.

If you have a list of instances of defstructs and you wish access the same slot in all the instances in that list use the ~> operator.

```
cardList = list( make_card( ?rank 'ace ?suit 'spades )
                  make_card( ?rank 'king ?suit 'diamonds))

cardList~>rank      => (ace king)
cardList~>faceUp    => (nil nil)
cardList~>faceUp = t => (t t)
cardList~>faceUp    => (t t)
```

Extended defstruct Example

1. Define a structure.

```
defstruct(point x y)           => t
```

2. Define another structure.

```
defstruct(bbox ll ur)         => t
```

3. Make an instance.

```
p1 = make_point(?x 100 ?y 200) => array[4]:xxx
```

4. Make another instance.

```
p2 = make_point(?x 0 ?y 0)     => array[4]:xxxx
```

5. Make a *bbox* instance.

```
b1 = make_bbox()              => array[4]:xxxx
```

6. Set a field in *b1*.

```
b1->ll = p2                    => array[4]:xxxx
```

7. Set the other field.

```
b1->ur = p1                    => array[4]:xxxx
```

8. Look inside and note the recursive printing.

```
printstruct( b1 )              ; Look inside
  Structure of type bbox :      ; Note the recursive printing
    ll :
      Structure of type point:
        x:0
        y:0

    ur:
      Structure of type point:
```

```
        x: 100
    y: 200
b1->ll->x = 12
=> 12

printstruct( p2 )
Structure of type point :
    x: 12
    y: 0

p1->??
=> (y 200 x 100)

b1->??
=> (ur array[4]:xxx ll array[4]:xxx)
```

9. Add a previously undefined slot.

```
b1->color = 'blue
printstruct( b1 )
    Structure of type bbox :
        ll :
            Structure of type point :
                x: 12
                y: 0
        ur:
            Structure of type point :
                x: 100
                y: 200
        color: blue

b1->?
=> (color ll ur)
```

Returns the list of currently defined fields.

Arrays

An *array* represents aggregate data objects in SKILL. Unlike simple data types, you must explicitly create arrays before using them so the necessary storage can be allocated. SKILL arrays allow efficient random indexing into a data structure using familiar syntax.

- Arrays are not typed. Elements of the same array can be different data types.
- SKILL provides run-time array bounds checking.
- Arrays are one dimensional. You can implement higher dimensional arrays using single dimensional arrays. You can create an array of arrays.
- The array bounds are checked with each array access during run-time. An error occurs if the index is outside the array bounds.

Allocating an Array of a Given Size

Use the *declare* function to allocate an array of a given size.

```
declare( week[7] )      => array[7]:9780700
week                   => array[7]:9780700
type( week )           => array
arrayp( week )         => t
days = '(monday tuesday wednesday
          thursday friday saturday sunday)
for( day 0 length(week)-1
    week[day] = nth(day days))
```

- The *declare* function returns the reference to the array storage and stores it as the value of *week*.
- The *type* function returns the symbol *array*.
- The *arrayp* function returns *t*.

Accessing Arrays

When the name of an array appears without an index on the right side of an assignment statement, only the array object is used in the assignment; the values stored in the array are not copied. It is therefore possible for an array to be accessible by different names. Indexes are used to specify elements of an array and always start with 0; that is, the first element of an array is element 0. SKILL normally checks for an out-of-bounds array index with each array access.

```
declare(a[10])
a[0] = 1
a[1] = 2.0
a[2] = a[0] + a[1]
```

Creates an array of 10 elements. *a* is the name of the array, with indexes ranging from 0 to 9. Assigns the integer 1 to element 0, the float 2.0 to element 1, and the float 3.0 to element 2.

```
b = a
```

b now refers to the same array as *a*.

```
declare(c[10])
```

Declares another array of 10 elements.

```
declare(d[2])
```

Declares *d* as an array of 2 elements.

```
d[0] = b
```

d[0] now refers to the array pointed to by *b* and *a*.

`d[1] = c`

`d[1]` is the array referred to by `c`.

`d[0][2]`

Accesses element 2 of the array referred to by `d[0]`.
This is the same element as `a[2]`.

Brackets (`[ ]`) are used to represent array references and are part of the statement syntax. The `declare` function is also an example of an *nlambda* function. The arguments of *nlambda* functions are passed literally (that is, not evaluated). It is up to the called function to evaluate selected arguments when necessary.

Association Tables

An association table is a generalized array, a collection of key/value pairs. The SKILL data types that can be used as keys in a table are integer, float, string, list, symbol, and instances of certain user-defined types. An association table is implemented internally as a hash table.

An association table lets you look up any entry with valid instances of SKILL data types. Data is stored in key/value pairs that can be quickly accessed with syntax for standard array access and various iterative functions. This access is based on a system that uses the SKILL *equal* function to compare keys.

Association tables offer convenience and performance not available in disembodied property lists, arrays, or association lists. Disembodied property lists and association lists are not efficient in situations where their contents can expand greatly. In addition, using symbols for properties or keys in a disembodied property list can be wasteful. A simple conversion process converts disembodied property lists and association lists to association tables. An association table can also be converted to a list of association pairs.

The following sections describe how to create, access, and modify association tables. For more information, see the [Data Structures](#) video on the Cadence Online Support website.”

Initializing Tables

The *makeTable* function defines and initializes the association table. This function takes a single string argument as the table name for printing purposes. An optional second argument provides the default value that is returned when a query to the table yields no match. The *tablep* predicate verifies the data type of a table, and the *length* function determines the number of keys in the table. To refer to and add elements, use the syntax for standard array access.

The following example creates a table and loads it with keys and related values that pair numbers and colors for a color map. The keys can be any of the data type mentioned earlier; they are not restricted to numeric data types.

```
myTable = makeTable("atable1" 0) => table:atable1
tablep(myTable)                    => t
myTable[1] = "blue"                => "blue"
myTable["two"] = '(r e d)'         => (r e d)
myTable["three"] = 'green'         => green
length(myTable)                   => 3
```

If a new pair is added to the table but its key already exists, the new value replaces the existing value in the table.

If a key to be accessed does not exist, the process returns either the default value specified at table creation or the symbol *unbound*, if no default value was specified.

Manipulating Table Data

The *foreach*, *forall*, and *setof* functions scan the contents of an association table and perform iterative programming functions on each key and its associated value. Standard input and output functions are available through the *readTable*, *writeTable*, and *printstruct* functions.

The *append* function appends data from existing disembodied property lists, association lists, or an association table to another existing association table. You specify the association table (created with the *makeTable* function) as the first argument for this function. For the second argument, you specify the disembodied property list, association list, or other association table whose data is to be appended. You can use the *remove* function to remove an entry from an association table.

Testing Whether a Data Value is a Table (tablep)

Use the *tablep* function to test whether a data value is a table.

```
myTable = makeTable("atable1" 0) => table:atable1
tablep(myTable)                  => t
tablep(9)                        => nil
```

Converting the Contents of an Association Table to an Association List (tableToList)

This function eliminates the inefficiency that you gain from referencing data in an association table. Do not use this function for processing data in an association table. Instead, use this function interactively to look at the contents of a table.

```
tableToList(myTable)
=> (("two" (r e d)) ("three" green) (1 "blue"))
```

Writing the Contents of an Association List to a File (`writeTable`)

The *writeTable* function is for writing basic SKILL data types that are stored in an association table. The function cannot write database objects or other user-defined types that might be stored in association tables.

```
writeTable("inventory.log" myTable) => t
```

Appending the Contents of a File to an Existing Association Table (`readTable`)

The file must have been created with the *writeTable* function so that the contents are in a usable format.

```
readTable("inventory.log" myTable) => t
```

Printing the Contents of an Object in a Tabular Format (`printstruct`)

For debugging, the *printstruct* function prints the contents of a structure in an easily readable form. It recursively prints nested structures.

```
printstruct(myTable)
=>      1: "blue"
      "three": green
      "two": (r e d)
```

Removing Contents from a Table (`remove`)

The *remove* function can be used to remove an entry from an association table.

```
remove(g_key o_table)
=> l_result
```

Consider an example with the structure of the type association table `myTable` as follows:

```
"1": "blue"
"3": "yellow"
"0": "red"
"2": "green"
```

Use the *remove* function as follows:

```
remove("1" myTable)
```

When you print the contents of the table again, the entry “1” is removed from the table.

Structure of type association table (table:myTable):

```
"3": "yellow"
"0": "red"
"2": "green"
```

Note: When you use the *remove* function, the removal is destructive, that is, any other reference to the table will also see the changes (see “[Removing Elements from a List](#)” on page 190).

Traversing Association Tables

Use the *foreach* function to visit every key in an association table. For example, use the following function call to print each key/value pair in a table.

```
foreach( key myTable
        println(
            list( key myTable[ key ] )
        )
    )
```

Note: You can also use `myTable[ ' ? ]` or `myTable->?` to get a list of all the available keys in an association table.

You can also use the functions *forall*, *exists* and *setof* to traverse association tables. (These functions are described in detail in “[Advanced List Operations](#)” on page 179)

For example, use the following function call to test if every key/value pair in a table are such that the key is a string and value is an integer.

```
forall( key myTable
        stringp(key) && fixp(myTable[key])
    )
```

To check if there is a single pair that satisfies the above expression, call the following function.

```
exists( key myTable
        stringp(key) && fixp(myTable[key])
    )
```

- To write the entire contents of a table to a file, use *writeTable*.
- To read a file (created using *writeTable*), use *readTable*.
- To view the contents of a table, use *printstruct*.

The *append* function appends data from existing disembodied property lists or association lists to an existing association table. In addition, the *append* function can add data from one association table to another association table as shown in the following example.

Checking if a Key Exists in the Table

The most efficient method to check a key existence in a table, regardless of whether the table was created with a default value or not, is to use:

```
member(key table)
```

Returns `t` if a key is a key in the table. Otherwise, returns `nil`.

The testing for the default value does not let you know if the key is in the table:

```
defaultValue = 123
table = makeTable("TblName" defaultValue)
table[1] = defaultValue

table[1] == defaultValue    => t
table[2] == defaultValue    => t

member(1 table)             => t
member(2 table)             => nil
```

If the table was created without a default value, then checking the value associated with the key in the table (`table[key]`) is unbound is equivalent to using the member function:

```
table = makeTable("TblName")
table[1] = 123

table[1] != 'unbound        => t
table[2] != 'unbound        => nil

member(1 table)             => t
member(2 table)             => nil
```

Implementing Sparse Arrays

A sparse array is an indexed collection, most of whose entries are unused. For large sparse arrays, it is wasteful to allocate the entire array. Instead, you can use an association table for a one-dimensional array. Use integers as the keys. To implement a two-dimensional sparse array, use lists of index pairs as keys.

```
procedure( tr2DSparseArray()
    makeTable( gensym( 'trSparseArray ) )
) ; procedure

trSparseTimesTable = tr2DSparseArray( )
for( i 0 3
    for( j 0 6
        trSparseTimesTable[ list( i j ) ] = i*j
```

```
    ) ; for  
  ) ; for
```

List-Oriented Functions for Association Tables

Several list-oriented functions also work on tables, including iteration.

List-Oriented Functions for Tables

Use this	To do this
Syntax for array access	To store and retrieve entries in a table
<i>makeTable</i> function	To create and initialize the association table. The arguments are the table name (required) and (optional) the default value to return for keys not present in the table. The default is <i>unbound</i> .
<i>foreach</i> function	To execute a collection of SKILL expressions for each key/value pair in a table
<i>setof</i> function	To return a list of keys in a table that satisfy a criterion.
<i>length</i> function	To return the number of key/value pairs in a table
<i>remove</i> function	To remove a key from a table

Association Lists

A list of key/value pairs is a natural means to record associations. An association list is a list of lists. The first element of each list is the key. The key can be an instance of any of SKILL types.

```
assocList = '( ( "A" 1 ) ( "B" 2 ) ( "C" 3 ) )
```

The *assoc* function retrieves the list given the index.

```
assoc( "B" assocList ) => ( "B" 2 )  
assoc( "D" assocList ) => nil
```

Use the *rplaca* function to destructively update an entry. The following replaces the *car* of the *cdr* of the association list entry.

```
rplaca( cdr( assoc( "B" assocList ) ) "two" )  
=> ( "two" )
```

```
assocList => (( "A" 1 ) ( "B" "two" ) ( "C" 3 ))
```

Association lists behave the same way as association tables. For lists with less than ten pairs, it is more efficient to use association lists than association tables. For lists likely to grow beyond ten pairs, it is more efficient to use association tables.

User-Defined Types

User-defined types are special foreign or external data types exported into SKILL by various applications. Their behavior is predetermined by the applications that own them. For example, database and window objects are usually implemented as C-structs and exported into SKILL as user-defined types.

The application that defines the SKILL behavior for the user-defined types provides methods for SKILL to apply in various situations. For example, when you apply the accessor operators `->` or `~>` to a user-defined type, the SKILL engine resolves the operation by calling the accessor method implemented for that type by an application.

There are other methods to support a user-defined type's behavior in SKILL. For example, to test two user-defined types for equality (*equal*), the application exporting the type provides a method to overload the SKILL *equal* function just for that type. The *equal* method takes two arguments and returns *t* or *nil*.

The application exporting the type determines what methods are needed to support the type in SKILL. If the application does not supply a method, SKILL applies a default behavior. In general, to a user, instances of a user-defined type look and feel similar to instances of *defstructs*.

Specific information on user-defined types is supplied by the applications exporting the types. For example, creating instances of user-defined types happens when certain application functions are called, such as *dbOpen*.

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Data Structures

Arithmetic and Logical Expressions

Expressions are Cadence® SKILL language objects that also evaluate to SKILL objects. SKILL performs a computation as a sequence of function evaluations. A SKILL *program* is a sequence of expressions that perform a specified action when evaluated by the SKILL interpreter. The three types of primitive expressions in SKILL are constants, variables, and function calls. You can combine constants, variables, and function calls with *operators* to form arithmetic and logical expressions.

A *constant* is an expression that evaluates to itself. That is, evaluating a constant returns the constant itself. For example:

```
123
10.5
"abc"
```

A *variable* stores values used during the computation and returns its value when evaluated. For example:

```
a
x
init_var
```

When SKILL creates a variable, it gives the variable an initial value of `unbound` (*that-value-which-represents-no-value*). If the interpreter encounters an unbound variable, you get an error message. You must initialize all variables. For example:

```
myVariable = 5
```

You will encounter the unbound variable error message if you misspell a variable because the misspelling creates a new variable.

A *function call* applies the named function to the list of arguments and returns the result when called. For example:

```
f(a b c d)
abs(-123)
exit()
```

Creating Arithmetic and Logical Expressions

You can combine constants, variables, and function calls with *infix* operators such as less than (<), colon (:), and greater than (>) to form arithmetic and logical expressions. For example:

```
1+2  
a*b+c  
x>y
```

You can form arbitrarily complicated expressions by combining any number of primitive expressions (constants, variables, and function calls) and operators.

Role of Parentheses

Parentheses delimit the names of functions from their argument lists and delimit nested expressions. In general, the innermost expression of a nested expression is evaluated first, returning a value used in turn to evaluate the expression enclosing it, and so on until the expression at the top level is evaluated.

Parentheses resemble natural mathematical notation and are used in both arithmetic and control expressions.

Quoting to Prevent Evaluation

Occasionally you might want to prevent expressions from being evaluated. This is done by “quoting” the expression, that is, putting a single quote just before the expression.

Quoting Variables

Quoting is often used with the names of variables and their values. For example, putting a single quote before the variable `a` (that is, `'a`) prevents `a` from being evaluated. Instead of returning the value of `a`, the name of the variable `a` is returned when `'a` is evaluated.

Quoting Lists

You usually specify lists of data within a program by quoting. Quoting is necessary because of the common list representation used for both program and data. For example, evaluating the list `(f a b c)` leads to the interpretation that `f` is the name of a function and `(a b c)` is a list of arguments for `f`. By quoting the same list, `'(f a b c)`, the list is instead treated as a data list containing the four elements `f`, `a`, `b`, and `c`.

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Arithmetic and Logical Expressions

Try It Out

The best way to understand evaluation and quoting is by trying them out in SKILL. An interactive session with the SKILL interpreter will help to clarify and reinforce many of the concepts just described.

Arithmetic and Logical Operators

All arithmetic operators are translated into calls to predefined SKILL functions. These operators are listed in the table below in descending order of operator precedence. The table also lists the names of the functions, which can be called like any other SKILL function.

Error messages report problems using the name of the function. The letters in the Synopsis column refer to data types. Refer to the [Data Types](#) on page 62 for a discussion of data type characters.

Arithmetic and Logical Operators

Type of Operator	Name of Function(s)	Synopsis	Operator
Data Access	arrayref	a[index]	[]
Data Access	setarray	a[index] = expr	
Data Access	bitfield1	x<bit>	<>
Data Access	setqbitfield1	x<bit>=expr	
Data Access	setqbitfield	x<msb:lsb>=expr	
Data Access	quote	'expr	'
Data Access	getqq	g.s	.
Data Access	getq	g->s	->
Data Access	putpropqq	g.s = expr, g->s = expr	~>
Data Access	putpropq	d~>s, d~>s = expr	
Unary	preincrement	++s	++
Unary	postincrement	s++	++
Unary	predecrement	--s	--
Unary	postdecrement	s--	--
Unary	minus	-n	-
Unary	null	!expr	!

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Arithmetic and Logical Expressions

Arithmetic and Logical Operators

Type of Operator	Name of Function(s)	Synopsis	Operator
Unary	bnot	$\sim x$	$\sim$
Binary	expt	$n1 ** n2$	$**$
Binary	times	$n1 * n2$	$*$
Binary	quotient	$n1 / n2$	$/$
Binary	plus	$n1 + n2$	$+$
Binary	difference	$n1 - n2$	$-$
Binary	leftshift	$x1 \ll x2$	$\ll$
Binary	rightshift	$x1 \gg x2$	$\gg$
Binary	lessp	$n1 < n2$	$<$
Binary	greaterp	$n1 > n2$	$>$
Binary	leqp	$n1 \leq n2$	$\leq$
Binary	geqp	$n1 \geq n2$	$\geq$
Binary	equal	$g1 == g2$	$==$
Binary	nequal	$g1 \neq g2$	$\neq$
Binary	band	$x1 \& x2$	$\&$
Binary	bband	$x1 \sim\& x2$	$\sim\&$
Binary	bxor	$x1 \wedge x2$	$\wedge$
Binary	bxnor	$x1 \sim\wedge x2$	$\sim\wedge$
Binary	bor	$x1 \mid x2$	
Binary	bnor	$x1 \sim\mid x2$	$\mid, \sim\mid$
Binary	and	rel. expr && rel. expr	$\&\&$
Binary	or	rel. expr rel. expr	$\parallel$
Binary	range	$g1 : g2$	$:$
Binary	setq	$s = \text{expr}$	$=$
Inplace Operator	plus	$a = a + b$	$+=$
Inplace Operator	minus	$a = a - b$	$-=$
Inplace Operator	divide	$a = a / b$	$/=$

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Arithmetic and Logical Expressions

Arithmetic and Logical Operators

Type of Operator	Name of Function(s)	Synopsis	Operator
Inplace Operator	multiply	$a = a * b$	<code>*=</code>
Inplace Operator	expt	$a = a ** b$	<code>**=</code>
Inplace Operator	bor	$a = a b$	<code> =</code>
Inplace Operator	band	$a = a \& b$	<code>&=</code>
Inplace Operator	bxor	$a = a ^ b$	<code>^=</code>
Inplace Operator	leftshift	$a = a << b$	<code><<=</code>
Inplace Operator	riightshift	$a = a >> b$	<code>>>=</code>

The following table gives more details on some of the arithmetic operators.

More on Arithmetic Operators

Arithmetic Operator	Comments
<code>+</code> , <code>-</code> , <code>*</code> , and <code>/</code>	Perform addition, subtraction, multiplication, and division operations.
Exponentiation operator <code>**</code>	Has the highest precedence among the binary operators.
Shift operators (<code><<</code> , <code>>></code>)	Shift their first arguments left or right by the number of bits specified by their second arguments. Both the left and right shifts are logical (that is, vacated bits are 0-filled).
Preincrement operator (<code>++</code> appearing before the name of a variable)	Takes the name of a variable as its argument, increments its value (which must be a number) by one, stores it back into the variable, and then returns the incremented value.
Postincrement operator (<code>++</code> appearing after the name of a variable)	Takes the name of a variable as its argument, increments its value (which must be a number) by one, and stores it back into the variable. However, it returns the original value stored in the variable as the result of the function call.
Predecrement and postdecrement operators	Similar to pre- and postincrement, but they decrement instead of increment the values of their arguments by one.
Range operator (<code>:</code>)	Evaluates both of its arguments and returns the results as a two-element list. It provides a convenient way of grouping a pair of data values for subsequent processing. For example, <code>1:3</code> returns the list <code>(1 3)</code> .

Predefined Arithmetic Functions

In addition to the basic infix arithmetic operators, several functions are predefined in SKILL.

Predefined Arithmetic Functions

Synopsis	Result
General Functions	
add1(n)	$n + 1$
sub1(n)	$n - 1$
abs(n)	Absolute value of n
exp(n)	e raised to the power n
log(n)	Natural logarithm of n
max(n1 n2 ...)	Maximum of the given arguments
min(n1 n2 ...)	Minimum of the given arguments
mod(x1 x2)	$x1$ modulo $x2$, that is, the integer remainder of dividing $x1$ by $x2$
round(n)	Integer whose value is closest to n
sqrt(n)	Square root of n
sxtd(x w)	Sign-extends the rightmost w bits of x , that is, the bit field $x\langle w-1:0 \rangle$ with $x\langle w-1 \rangle$ as the sign bit
zxted(x w)	Zero-extends the rightmost w bits of x , executes faster than doing $x\langle w-1:0 \rangle$

Trigonometric Functions

sin(n)	sine, argument n is in radians
cos(n)	cosine
tan(n)	tangent
asin(n)	arc sine, result is in radians
acos(n)	arc cosine
atan(n)	arc tangent

Predefined Arithmetic Functions

Synopsis	Result
Random Number Generator	
random(x)	Returns a random integer between 0 and x-1. If random is called with no arguments, it returns an integer that has all of its bits randomly set.
srandom(x)	Sets the initial state of the random number generator to x

Bitwise Logical Operators

The *bnot*, *band*, *bband*, *bxor*, *bxnor*, *bor*, and *bnor* operators all perform bitwise logical operations on their integer arguments.

Bitwise Logical Operators

Meaning	Operator
bitwise AND	&
bitwise inclusive OR	
bitwise exclusive OR	^
left shift	>>
right shift	<<
one's complement (unary)	~

Bit Field Operators

Bit field operators operate on bit fields stored inside 32-bit words. To avoid confusion in naming bits, SKILL uses the uniform convention that the least significant bit in an integer is bit 0, with the bit number increasing as you move left in the direction of the most significant bit.

- You can select bit fields by naming the leftmost and the rightmost bits in the bit field or by just naming the bit if the bit field is only one bit wide.
- You can use either integer constants or integer variables to specify bit positions, but expressions are not allowed.
- All bit fields are treated as unsigned integers.

- Use the `sxtb` function for sign-extending a bit field.

Bit Field Examples

x = 0b01101 => 13

Assigns `x` to 13 in binary.


$$x < 0 \Rightarrow 1$$

The contents of the rightmost bit of `x` is 1.

$$x_{\langle 1 \rangle} \Rightarrow 0$$

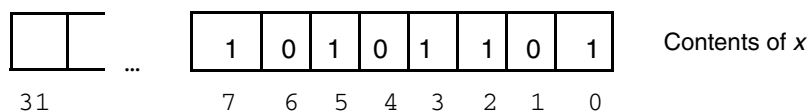
The contents of bit one of x is 0.

$$x \langle 2:0 \rangle \Rightarrow 5$$

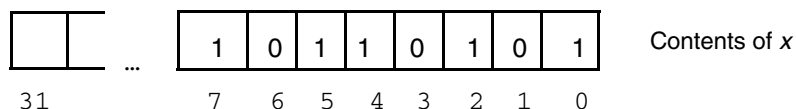
Extracts the contents of the rightmost three bits of `x`. These are `101` or `5` in decimal.

$$x_{\langle 7:4 \rangle} = 0b1010 \Rightarrow 173$$

Stores the bit pattern into the bits 4 through 7.


$$(x + 4) \langle 4 : 3 \rangle \Rightarrow 2$$

Adds 4 to `x`, then extracts the 3rd and 4th bits. 173 plus 4 is 177. SKILL returns the result of the last expression, which is binary 10 or 2 decimal.



Calling Conventions for Bit Field Functions

Because of limitations in the grammar, only integer constants or names of variables are permitted inside the angle brackets. To use the value of an expression to specify a bit position, you must either first assign the value of the expression to a variable or directly call the bit field

functions without using the less than (<), colon (:), and greater than (>) infix operators. The calling conventions for the bit field functions are as follows:

```
bitfield1(  
    x_value  
    x_bitPosition )  
setqbitfield1(  
    s_name  
    x_newvalue  
    x_bitPosition )  
bitfield(  
    x_value  
    x_leftmostBit  
    x_rightmostBit )  
setqbitfield(  
    s_name  
    x_newvalue  
    x_leftmostBit  
    x_rightmostBit )
```

Mixed-Mode Arithmetic

SKILL makes a distinction between integers and floating-point numbers.

- Integer arithmetic is used if all the arguments given to an arithmetic operator are integers.
- Floating-point arithmetic is used if all arguments are floating-point numbers.
- When integers and floating-point numbers are mixed, SKILL uses integer arithmetic until it encounters a floating-point number. SKILL then switches to floating-point arithmetic and returns a floating-point number as the result.

See the following topics for more information:

- [Floating-Point Issues](#) on page 129
- [Integer and Floating-Point Division](#) on page 130
- [Type Conversion Functions \(fix and float\)](#) on page 130
- [Comparing Floating-Point Numbers](#) on page 131

Floating-Point Issues

Because IEEE floating-point numbers use what are essentially binary (base 2) fractions to represent real numbers, some functions that operate on floating-point numbers yield unexpected (and incorrect) results.

There are some floating-point numbers that the computer cannot represent as binary fractions. In these cases, the computer uses a binary (base 2) floating-point number to approximate your decimal (base 10) floating-point number. These approximations can lead to incorrect results for some functions that operate on floating-point numbers. Consider the following:

Decimal fraction	Equivalent	Binary fraction	Result
0.125	$1/10 + 2/100 + 5/1000$	0.001	Exact representation
0.3333...	$3/10 + 3/100 + 3/1000...$	0.010101010101...	Approximation
.1	$1/10$	0.0001100110011...	Approximation

Binary representation can result in a loss of precision. Also, when a program does not save or retrieve the least significant bits of a number, that number loses precision. See also [“Comparing Floating-Point Numbers”](#) on page 131 and [“Type Conversion Functions \(fix and float\)”](#) on page 130.

Note: Some applications use single precision to represent floating-point numbers. SKILL uses double precision.

Integer and Floating-Point Division

The division operator requires special attention because

- Integer division truncates its results
- Floating-point division computes an exact number

Type Conversion Functions (fix and float)

Before you can compare an integer to a floating-point number, you must convert both numbers to the same type (that is, `integer` or `float`).

Note: See also [“Comparing Floating-Point Numbers”](#) on page 131 for more information.

The `fix` and `fix2` functions convert a floating-point number and floating-point calculations, respectively, to an integer. The `float` function converts an integer to a floating-point number. If the argument given to `fix` or `float` is already of the desired type, the argument is returned. See [Cadence SKILL Language Reference](#) for more information about these functions.

Important

Because of the “[Floating-Point Issues](#)” on page 129, you can pass a floating-point argument to the `fix` function that yields an incorrect result. For example:

```
Skill > fix(4.1 * 100)
409

Skill > fix((30 - 18.1) * 10)
118
```

Tip

You can load the following SKILL file (`real_fix.il`) into your SKILL development environment to force correct results in cases like those shown above.

```
procedure(real_fix(number)
let( (num_string fix_num)
sprintf( num_string, "%f" number)
fix_num = atoi(car(parseString(num_string)))
fix_num
)
);
```

Comparing Floating-Point Numbers

Note: Two numbers can only be equal if they are of the same type (either `integer` or `float`) and have identical values. Two floating-point numbers can appear the same when printed while differing internally in their least significant bits.

Simple comparison rarely works for floating-point numbers. For example:

```
if( (a == b) println("same"))
```

You can compare against an acceptable tolerance range with more success as follows (assuming `a` is not zero):

```
if( (abs(a - b)/a < 1e-6) println("same"))
```

You can also use the `fix2` for rounding the result in floating-point calculations. This function returns the largest integer not larger than the given argument.

Caution

Unless two floating-point numbers are assigned identical constants or are generated by exactly the same sequence of computations, it is unlikely SKILL will treat them as equal when compared.

You can think of a loop (such as `for`) as a special case of comparing floating-point numbers. Each time a program increments a floating-point loop variable, the number can lose precision

resulting in a cumulative error. Under these circumstances, your loop might not execute the expected number of times.

Function Overloading

Some applications that are based on SKILL overload the arithmetic and/or bit-level functions with new or modified semantics. While `sqrt(-1)` normally signals an error in SKILL, it returns a valid result as a complex number when some applications are running.

Because the arithmetic and bit-level operators are just simpler syntax for calling their corresponding functions, by overloading these functions with extended semantics for new data types, you can use the same familiar notation in writing expressions involving objects of these new data types.

By overloading the `plus`, `difference`, `times`, and `quotient` functions for a complex-number data type, you can use `+`, `-`, `*`, and `/` in forming arithmetic expressions involving complex numbers as you normally do in writing mathematical formulas.



Caution

This kind of function overloading is done by the individual application. There is no support for user-definable function overloading in SKILL. Refer to the reference manuals of the individual applications for more details about which functions/operators have been overloaded and what semantics to use.

Integer-Only Arithmetic

In addition to standard arithmetic functions that can handle integers and floating-point numbers, SKILL provides several integer-only arithmetic functions that are slightly faster than the standard functions. These functions are named by prepending `x` to the names of the corresponding standard functions: `xdifference`, `xplus`, `xquotient`, and `xtimes`.

When integer mode is turned on using the `sstatus` function, the SKILL parser translates all arithmetic operators into calls on integer-only arithmetic functions. This results in small execution time savings and makes sense only for compute-intensive tasks whose inner loops are dominated by integer arithmetic calculations.

```
sstatus( integermode t)=> t
```

Turns on integer mode.

```
status( integermode )=> t
```

Checks the status of `integermode` and returns `t` if `integermode` is on. The default is off.

The internal variables are typically Boolean switches that accept only the Boolean values of `t` and `nil`. For efficiency and security, system variables are stored as internal variables that can only be set by *status*, rather than as SKILL variables you can set directly. Refer to “[Names of Variables](#)” on page 53 for a discussion of internal variables.

True (non-nil) and False (nil) Conditions

Unlike C or other programming languages that use integers to represent true and false conditions, SKILL uses the nonnumeric special atom `nil` to represent the false condition. The true condition is represented by the special atom `t` or anything other than `nil`.

Relational Operators

The relational operators `lessp` (<), `leqp` (<=), `greaterp` (>), `geqp` (>=), `equal` (==), and `nequal` (!=) operate on numeric arguments and return either `t` or `nil` depending on the results.

Logical Operators

The logical operators `and` (&&), `or` (||), and `null` (!), on the other hand, operate on nonnumeric arguments that represent either the false (`nil`) or true (non-`nil`) conditions.

False/True Conditions Do Not Equal Constants 0 and 1

If you program in C, be especially careful not to interpret the false/true conditions as equivalent to the integer constants 0 and 1.

Testing for Equality and Inequality

You can also use the `equal` (==) and the `nequal` (!=) operators to test for the equality and inequality of nonnumeric atoms.

- Two atoms are equal if they are the same type and have the same value.
- Two lists are equal if they contain the same elements.

Controlling the Order of Evaluation

The binary operators `&&` and `||` are often used to control the order of evaluation.

The `&&` Operator

The `&&` operator evaluates its first argument and, if the result is `nil`, returns `nil` without evaluating the second argument. If the first argument evaluates to non-`nil`, `&&` proceeds to evaluate the second argument and returns that value as the value of the function.

The `||` Operator

The `||` operator also evaluates its first argument to see if the result is non-`nil`. If so, `||` returns that result as its value and the second argument is not evaluated. If the first argument evaluates to `nil`, `||` proceeds to evaluate the second argument and returns that value as the value of the function.

Testing Arithmetic Conditions

In addition to the six infix relational operators, several arithmetic predicate functions are available for efficient testing of arithmetic conditions. These predicates are listed in the table below.

Arithmetic Predicate Functions

Synopsis	Result
<code>minusp(n)</code>	<code>t</code> if <code>n</code> is a negative number, <code>nil</code> otherwise
<code>plusp(n)</code>	<code>t</code> if <code>n</code> is a positive number, <code>nil</code> otherwise
<code>onep(n)</code>	<code>t</code> if <code>n</code> is equal to 1, <code>nil</code> otherwise
<code>zerop(n)</code>	<code>t</code> if <code>n</code> is equal to 0, <code>nil</code> otherwise
<code>evenp(x)</code>	<code>t</code> if <code>x</code> is an even integer, <code>nil</code> otherwise
<code>oddp(x)</code>	<code>t</code> if <code>x</code> is an odd integer, <code>nil</code> otherwise

Differences Between SKILL and C Syntax

Arithmetic and logical expressions in SKILL are the same as in the C programming language with the following minor differences:

- SKILL supports the following exponentiation operator: `**` (two asterisks).
- The function `mod` replaces the modulo operator “`%`” so that you do not have to use “`%`” for this infrequently-used function: Use `mod(i j)` instead of `i % j`.
- The more general `if/then/else` control construct in SKILL makes the conditional expression operators `?` and `:` obsolete.
- The indirection operator `*` and the address operator `&` do not have any meaning in SKILL and are not supported.
- The set of bitwise operators is augmented by `~&` (nand), `~|` (nor), and `~^` (xnor).
- Logical expressions that evaluate to false return the special atom `nil` and those that evaluate to true return any non-`nil` value (usually, the special atom `t`).

SKILL Predicates

The following predicate functions test for a condition:

- [The atom Function](#) on page 135
- [The boundp Function](#) on page 135

For a list of predicate functions for testing the type of a data object, see “[Type Predicates](#)” on page 138.

The atom Function

`atom` checks if an object is an atom. Atoms are all SKILL objects (except nonempty lists). The special symbol `nil` is both an atom and a list.

```
atom( 'hello ) => t
x = '(a b c)
atom( x ) => nil
atom( nil ) => t
```

The boundp Function

`boundp` checks if a symbol is bound (has an assigned value).

```
x = 5 ; Binds x to the value 5.
y = 'unbound ; Unbinds y
boundp( 'x ) => t
boundp( 'y ) => nil
```

```
y = 'x           ; Binds y to the constant x.  
boundp( y ) => t ; Returns t because y evaluates to x, which is bound.
```

Using Predicates Efficiently

Some predicates are faster than others. For example, the `eq`, `neq`, `memq`, and `caseq` functions are faster than their close relatives the `equal`, `nequal`, `member`, and `case` functions.

The `equal`, `nequal`, `member`, and `case` functions compare values while the `eq`, `neq`, `memq`, and `caseq` functions test if the objects are the same. That is, they test whether the objects reside in the same location in virtual memory.

These functions result in quicker tests but might require that you alter your application to take advantage of them.

The eq Function

`eq` checks addresses when testing for equality. The `eq` function returns `t` if both arguments are the same object in virtual memory. You can test for equality between symbols using `eq` more efficiently than using the `==` operator. The following example illustrates the differences between `equal` (`==`) and `eq` for lists.

```
list1 = '( 1 2 3 )      => ( 1 2 3 )  
list2 = '( 1 2 3 )      => ( 1 2 3 )  
list1 == list2           => t  
eq( list1 list2 )        => nil  
  
list3 = cons( 0 list1 )  => ( 0 1 2 3 )  
list4 = cons( 0 list1 )  => ( 0 1 2 3 )  
list3 == list4           => t  
eq( cdr( list3 ) list1 ) => t  
  
aList = '( a b c )      => a b c  
eq( 'a car( aList ) )    => t
```

The equal Function

`equal` tests for equality.

- If the arguments are the same object in virtual memory (that is, they are `eq`), `equal` returns `t`.
- If the arguments are the same type and their contents are equal (for example, strings with identical character sequence), `equal` returns `t`.

- If the arguments are a mixture of fixnums and flonums, `equal` returns `t` if the numbers are identical (for example, `1.0` and `1`).

This test is slower than using `eq` but works for comparing objects other than symbols.

```
x = 'cat
equal( x 'cat )      => t
x == 'dog             => nil; == is the same as equal.
x = "world"
equal( x "world" )   => t
x = '(a b c)
equal( x '(a b c))   => t
```

The neq Function

`neq` checks if two arguments are *not* equal, and returns `t` if they are not. Any two SKILL expressions are tested to see if they point to the same object.

```
a = 'dog
neq( a 'dog )        => nil
neq( a 'cat )        => t
z = '(1 2 3)
neq(z z)             => nil
neq('(1 2 3) z)      => t
```

The nequal Function

`nequal` checks if two arguments are *not* logically equivalent and returns `t` if they are not.

```
x = "cow"
nequal( x "cow" )    => nil
nequal( x "dog" )    => t
z = '(1 2 3)
nequal(z z)          => nil
nequal('(1 2 3) z)   => nil
```

The member and memq Functions

These functions test for list membership. `member` tests using `equal` while `memq` uses `eq` and is therefore faster. These functions return a non-`nil` value if the first argument matches a member of the list passed in as the second argument.

```
x = 'c
memq( x '(a b c d))    => (c d)
memq( x '(a b d))      => nil
x = "c"
member( x '("a" "b" "c" "d")) => ("c" "d")
memq('c '(a b c d c d)) => (c d c d)
memq( concat( x ) '(a b c d)) => (c d)
```

The tailp Function

`tailp` returns the first argument if a list cell `eq` to the first argument is found by `cdr`'ing down the second argument zero or more times, `nil` otherwise. Because `eq` is being used for comparison, the first argument must point to a tail list in the second argument for this predicate to return a non-`nil` value.

```
y = '(b c)
z = cons( 'a y )      => (a b c)
tailp( y z )          => (b c)
tailp( '(b c) z )     => nil
```

`nil` was returned because `'(b c)` is not `eq` the `cdr( z )`.

Type Predicates

Many predicate functions are available for testing the data type of a data object. The suffix `p` on the name of a function usually indicates that it is a predicate function. Type predicates appear in the table below.

Note: `g` (general) can be any data type.

Type Predicates

Function	Value Returned
<code>arrayp(g)</code>	<code>t</code> if <code>g</code> is an array, <code>nil</code> otherwise
<code>bcdp(g)</code>	<code>t</code> if <code>g</code> is a binary function, <code>nil</code> otherwise
<code>dtpr(g)</code>	<code>t</code> if <code>g</code> is a non-empty list, <code>nil</code> otherwise (<code>dtpr(nil)</code> returns <code>nil</code>)
<code>fixp(g)</code>	<code>t</code> if <code>g</code> is a fixnum, <code>nil</code> otherwise
<code>floatp(g)</code>	<code>t</code> if <code>g</code> is a flonum, <code>nil</code> otherwise
<code>listp(g)</code>	<code>t</code> if <code>g</code> is a list, <code>nil</code> otherwise (<code>listp(nil)</code> returns <code>t</code>)
<code>null(g)</code>	<code>t</code> if <code>g</code> is <code>nil</code> , <code>nil</code> otherwise
<code>numberp(g)</code>	<code>t</code> if <code>g</code> is a number (that is, fixnum or flonum), <code>nil</code> otherwise
<code>otherp(g)</code>	<code>t</code> if <code>g</code> is a foreign data pointer, <code>nil</code> otherwise
<code>portp(g)</code>	<code>t</code> if <code>g</code> is an I/O port, <code>nil</code> otherwise
<code>stringp(g)</code>	<code>t</code> if <code>g</code> is a string, <code>nil</code> otherwise
<code>symbolp(g)</code>	<code>t</code> if <code>g</code> is a symbol, <code>nil</code> otherwise
<code>symstrp(g)</code>	<code>t</code> if <code>g</code> is either a symbol or a string, <code>nil</code> otherwise

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Type Predicates

<code>type (g)</code>	a symbol whose name describes the type of g
<code>typep (g)</code>	same as type(g)

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Arithmetic and Logical Expressions

Control Structures

The following are the control structures for the Cadence SKILL Language.

- Control Functions
- Selection Functions
- Declaring Local Variables with prog
- Grouping Functions

Control Functions

The Cadence® SKILL language control functions provide a great deal of functionality familiar to users of a language such as C. These high-level control functions give SKILL additional power over most Lisp languages.

The control functions are also the biggest cause of inefficient code in the SKILL language. One of the inevitabilities of providing so many control structures is that some are more efficient than others and that there is a great deal of overlap between the functions. This means that it is easy for a programmer to choose a structure that works perfectly well for the task in hand, but is not in fact the best structure to use as far as efficiency and (even occasionally) readability are concerned.

A control function is any function that controls the evaluation of expressions given to it as arguments. The order of evaluation can depend on the result of evaluating test conditions, if any, given to the function. In addition to supporting standard control constructs such as `if/while/for`, SKILL makes it easy for you to define control functions of your own. Because control functions in SKILL correspond to “statements” in conventional languages, this manual sometimes uses the terms interchangeably.

Conditional Functions

Conditional functions test for a condition and perform operations when that condition is found.

There are four conditional functions available to the SKILL programmer: `if`, `when`, `unless`, and `cond`. These each have their own distinct characteristics and uses. Because the four functions carry out similar tasks, it is easy for the programmer to choose an inappropriate function. Choose a conditional function according to the following criteria:

- `if`: There are two values to consider, true and false.
- `when`: There are statements to carry out when the test proves true.
- `unless`: There are statements to carry out unless the test proves true.
- `cond`: There is more than one test condition, but only the statements of one test are to be carried out.

The `cond` Function

The `cond` function offers multiway branching.

```
cond(  
  ( condition1 exp11 exp12 ... )  
  ( condition2 exp21 exp22 ... )  
)
```

```
( condition3 exp31 exp32 ... )  
( t expN1 expN2 ... )  
); cond
```

The `cond` function

- Sequentially evaluates the conditions in each branch until it finds one that is non-`nil`. It then executes all the expressions in the branch and exits.
- Returns the last value computed in the branch it executes.

The `cond` function is equivalent to

```
if          condition1      exp11    exp12 ...  
else if     condition2      exp21    exp22 ...  
else if     condition3      exp31    exp32 ...  
...  
else       expN1expN2 ....
```

For example, this version of the `trClassify` function is equivalent to the one using the `prog` and `return` functions in [“The return Function”](#) on page 146.

```
procedure( trClassify( signal )  
  cond(  
    ( !signal nil )  
    ( !numberp( signal ) nil )  
    ( signal >= 0 && signal < 3 'weak )  
    ( signal >= 3 && signal < 10 'moderate )  
    ( signal >= 10 'extreme )  
    ( t 'unexpected )  
  ) ; cond  
) ; procedure
```

Iteration Functions

There are two basic iteration functions available in the SKILL language: `while` and `for`. These are both general functions that have many uses.

The while Function

The `while` function is the more general function because everything that can be done with a `for` can be done with a `while`.

When using the `while` function remember that all parts of the test condition are evaluated on each pass of the loop. This means that if there are parts of the test that do not depend on the contents of the loop, they should be moved outside of the loop. Consider the following code:

```
while( i < length(myList)  
  ...
```

```
        i++
    )
```

If the value of the symbol `myList` does not change within this loop, the value of `length(myList)` is being re-evaluated on each loop for no reason. It would be better to assign the value of `length(myList)` to a variable before starting the `while` loop.

When using a `while` loop, consider whether it would be better to use one of the list iteration and quantifier functions such as `foreach`, `setof`, or `exists`.

The for Function

The main advantage of the `for` function is that it automatically declares the loop variable. This means that the variable does not need to be declared in a local variable section of a structure such as `prog` or `let`. It also means that the variable cannot be used outside the loop, which differs from the case in C. Consider the following code:

```
for( i 1 length(myList)
    evaluateList(i)
)
if( i == 0
    printf("The list was empty!\n")
)
```

The `if` test is incorrect because the variable `i` will be unbound by the time it is executed.

Selection Functions

There are two selection functions in SKILL: `caseq` and `case`. The difference between these functions is the range of values that are allowed within the test conditions.

`caseq` is a considerably faster version of `case`. `caseq` uses the function `eq` rather than `equal` for comparison. The comparators for `caseq` are therefore restricted to being either symbols or small integer constants ($-256 \leq i \leq 255$), or lists containing symbols and small integer constants.

The `caseq` and `case` functions allow lists of elements within the test parts and match if the test value is `eq` or `equal` to one of those elements, as appropriate.

One common fault with the use of the `caseq` function is the misconception that the values in the conditional part of the function are evaluated. Consider the following call to `caseq`:

```
caseq( x
      ('a "a")
      ('b "b")
    )
```


The conditional parts of this, 'a and 'b, are `not` evaluated, so this code equates to

```
caseq( x
      ((quote a) "a")
      ((quote b) "b")
    )
```

That is, if the value of `x` is the symbol `a` or is the symbol `quote`, `caseq` returns the value `a`. This is not what was required.

Be careful when using symbols in these selection functions because the symbol `t` indicates the default case and should not therefore be used. For example, consider the case where a function returns one of the values `t`, `nil`, or `indeterminate`.

It might be tempting to write a function such as

```
caseq( value
      (t                printf("Succeeded.\n"))
      (nil              printf("Failed.\n"))
      (indeterminate    printf("Indeterminate.\n"))
    )
```

But this function will not work because the `t` case is the default and always matches. The correct way to write this test is

```
caseq( value
      (nil              printf("Failed.\n"))
      (indeterminate    printf("Indeterminate.\n"))
      (t                when( eq(value,t)
                             printf("Succeeded.\n")
                           )
    )
) /* caseq */
```

The problem can also be avoided by putting the `t` within parentheses because the default case only matches against a single `t`. This is not recommended because it is an implementation dependency. The SKILL Lint program always warns of dubious uses of the `t` case in a selection function.

Declaring Local Variables with `prog`

All variables that appear in a SKILL program are global to the whole program unless they are explicitly declared as local variables. You declare local variables using the `prog` control construct, which initializes all its local variables to `nil` upon entry and restores their original values (that is, the values of the variables before the `prog` was executed) upon exit from the `prog`.

A symbol's current value is accessible at any time from anywhere. The SKILL interpreter transparently manages a symbol's value slot as if it were a stack.

- The current value of a symbol is the top of the stack.
- Assigning a value to a symbol changes only the top of the stack.

Whenever your program invokes the `prog` function, the system pushes a temporary value onto the value stack of each symbol in the local variable list. When the flow of control exits, the system pops the temporary value off the value stack, restoring the previous value.

The `prog` Function

The `prog` function allows an explicit loop to be written since `go` is supported within the `prog`. In addition, `prog` allows you to have multiple return points through use of the function `return`. If you are not using either of these two features, `let` is much simpler and faster.

If you need to conditionally exit a collection of SKILL statements, use the `prog` function. A list of local variables and your SKILL statements make up the arguments to the `prog` function.

```
prog( ( local variables ) your SKILL statements )
```

The `return` Function

Use the `return` function to force the `prog` to immediately return a value skipping over subsequent statements. If you do not call the `return` function, the `prog` expression returns `nil`.

Example: The `trClassify` function returns either `nil`, `weak`, `moderate`, `extreme`, or `unexpected` depending on `signal`. It does not use any local variables.

```
procedure( trClassify( signal )
  prog( ()
    unless( signal return( nil ))
    unless( numberp( signal ) return( nil ))
    when( signal >= 0 && signal < 3 return( 'weak ))
    when( signal >= 3 && signal < 10 return( 'moderate ))
    when( signal >= 10 return( 'extreme ))
    return( 'unexpected )
  ) ; prog
) ; procedure
```

Use the `prog` function and the `return` function to exit early from a `for` loop. This example finds the first odd integer less than or equal to 10.

```
prog( ( )
  for( i 0 10
    when( oddp( i )
      return( i )
    ) ; when
  ) ; for
) ; prog
```

Grouping Functions

Three main functions allow the grouping of statements where only a single statement would otherwise be allowed. These functions are `prog`, `let`, and `progn`. In addition, the `let` and `prog` functions allow for the declaration of local variables. The `prog` function also allows for the use of `return` statements to jump out from within a piece of code and the `go` function, along with labels, to jump around within the code.

When considering whether to use `prog`, `let`, or `progn`, the function with the least extra functionality should be used at all times because the functions are progressively more expensive in terms of run time. Use the functions as follows:

- If local variables and jumps are not needed, use a `progn`.
- If local variables are needed but not jumps, use a `let`.
- Only if jumps are needed, use a `prog`.

Using `prog`, `return`, and `let`

The `prog` statement should be used only when it is absolutely necessary. Its overuse is one of the biggest causes of inefficiency in all SKILL code. Returning from the middle of a piece of code is not only highly expensive, but can also lead to code that is difficult to read and understand. As with all high level programming languages, the use of `go` (the SKILL 'goto' statement) is highly discouraged. There are cases when it is necessary, but these are few.

A programmer usually uses the `prog` form when a certain amount of error checking must be done at the start of a function, with the rest of the function only being carried out if the error checking succeeds. Consider the following piece of code:

```
procedure(check(arg1 arg2)
  prog( ()
    when(illegal_val(arg1)
      printf("Arg1 in error.\n")
      return()
    ) /* end when */

    when(illegal_val(arg2)
      printf("Arg2 in error.\n")
      return()
    ) /* end when */

    Rest of code ...
  ) /* end prog */
) /* end check */
```

This code is reasonably clear, except that it is easy for someone to miss the `return` statements, and it uses the `prog` form. Consider the following alternative. This code seems to be a longer procedure, but it is clearer, faster, and more maintainable:

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```
procedure(check(arg1 arg2)
  cond(
    ( illegal_val(arg1)
      printf("Arg1 in error.\n")
      nil
    ) /* check arg1 */
    (illegal_val(arg2)
      printf("Arg2 in error.\n")
      nil
    ) /* check arg2 */
    (t
      Rest of code ...
    )
  ) /* end cond */
) /* end check */
```

A separate function could be called from within the `t` condition, which could expect its arguments to be correct. This would, at the small cost of an extra function call, separate the error checking code completely from the main body of the function, thereby making it even easier for programmer maintaining the code to see what is involved in the function, without having to worry about the peripheral interfaces.

Another common mistake with the use of the `prog` and `let` functions is the initialization of the local variables to `nil`. All local variables in a `prog` or `let` are automatically initialized to `nil`. Remember that the `let` function allows local variables to be initialized within the declaration. This saves both time and space, and, as long as care is taken over the layout of the code, can be no less readable:

```
procedure(test())
  let( ((localvar1 initvalue1)
      (localvar2 initvalue2)
      localvar3
    )
    Rest of code ...
  ) /* end let */
) /* end procedure */
```

When setting initial values within the declaration, reference cannot be made to other local variables. For example, the following is wrong:

```
procedure(incorrect(list)
  let( ((listHead      car(list))
      (headval      car(listHead)) /* WRONG!!! */
    )
    Rest of code ...
  )
)
```

The `prog` and `let` functions have different return values.

- The `prog` function returns the value given in a `return` statement or, if it exits without a `return`, returns `nil`.
- The `let` function always returns the value of the last statement.

This means that in converting a `prog` to a `let`, it might be necessary to add an extra `nil` to the end of the function.

Using the `progn` Function

The `progn` function is a simple means of grouping statements where multiple statements are required, but only one is expected.

An example is the `setof` function, which only allows a single statement in the conditional part.

Remember that there is an overhead in using `progn`. It should only be used where there is more than one statement, and only one statement is allowed.

Using the `prog1` and `prog2` Functions

Two minor grouping functions that have roughly the same overhead as the `progn` function are `prog1` and `prog2`.

The `prog1` Function

`prog1` evaluates expressions from left to right and returns the value of the first expression.

```
prog1(  
    x = 5  
    y = 7 )  
=> 5
```

The `prog2` Function

`prog2` evaluates expressions from left to right and returns the value of the second expression.

```
prog2(  
    x = 4  
    p = 12  
    x = 6 )  
=> 12
```

`prog1` and `prog2` are often useful when a local variable would otherwise be needed to hold a temporary variable before that variable is returned. These two functions should be used with caution, because they can detract from the readability of the program, and they are usually only useful where otherwise a `let` would be necessary. For example:

```
procedure(main()  
    let( (status)
```

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```
        initialize()
        /* Main body of program */
        status = analyzeData()
        wrapUp()
/* Return the status */
    status
) /* end let */
) /* end main */
```

This code can be more efficiently (but less clearly) implemented using a `prog2`:

```
procedure (main())
    prog2 (
        initialize()
        /* Main body of program.
        * The value of this will be returned by prog2.
        */
        analyzeData()
        wrapUp()
    ) /* end prog2 */
) /* end main */
```

I/O and File Handling

You can read about I/O and file handling topics in the following sections:

- [File System Interface](#) on page 152
- [Ports](#) on page 162
- [Output](#) on page 164
- [Input](#) on page 169
- [System-Related Functions](#) on page 175

File System Interface

All input and output in the Cadence® SKILL language is defined with respect to the UNIX file system. Writing I/O statements in SKILL requires an understanding of files, directories, and paths.

Files

A file contains data, usually organized in multiple records, and has several attributes such as name, the date the file was created, the last time it was accessed, access permissions, and so on. A device is a file with special attributes.

Directories

A directory has a name, just like a file, but it contains a list of other files. Directories can be nested to as many levels as desired. A directory allows related files to be grouped together. Because of the thousands of files that can exist on a single disk, using directories helps to avoid chaos. Most network-wide file systems are dependent on directories.

Directory Paths

Often, there are several directories you want to search in a particular order by specifying a set of directory paths. You can specify a file name in an absolute sense or in a relative sense.

The following description uses ‘path’ as a generic term where either a file name or a directory name can be used. However, because a directory under UNIX is just a special kind of file, file name is often used as a synonym for path.

Absolute Paths

You can specify the path with a slash character (/). When used as the first character of a name, it represents the system root directory. Intermediate levels of directories can use the slash character again as a separator.

Relative Paths

Any path that does not begin with a slash is a relative name.

If the path begins with a tilde followed by a slash (`~/`), the search path begins in your home directory.

If the tilde is followed by a name, the name is interpreted as a user name. That is, `~jones/file1` specifies a file named `file1` in `jones`' home directory.

If the path begins with a period and a slash (`./`), the search begins with the current working directory.

If the file name begins with two periods and a slash (`../`), the search begins with the parent of the current working directory.

If you are using a function that refers to the SKILL path, refer to the following section.

The SKILL Path

SKILL provides a flexible mechanism for using relative paths. An internal list of paths, referred to as the `SKILL path`, is used in many file-related functions.

Importance of the First Path Character

When a relative path that does not begin with `~` or `./` is given to a function, the paths in the SKILL path are used as directory names and prepended to the given path (with a `/` separator if needed) to form possible paths. The `setSkillPath` and `getSkillPath` functions access and change this internal SKILL path setup.

Path Order when the Same File Name Exists in Multiple Directories

The order of the paths on the SKILL path is important when the same file name is in multiple directories. If a file is opened for input or queried for status, all readable directories in the SKILL path are checked, in order, for the given file name. The first one found is taken to be the intended path.

Path Order when a File is Updated or Written for the First Time

The order of the paths is also important when a file is updated or written for the first time. If you open an output file, all directory paths in the SKILL path are checked, in the order specified, for that file name. If found, the system overwrites the first updateable file in the list. If no updateable file is found, it places a new file of that name in the first writable directory.

Know Your SKILL Path

Having an implicit list of search paths provides a powerful shortcut in many situations, but it can also be a source of possible confusion. When in doubt, double check the current setup of your SKILL path or set it to `nil`.

When you start your system, the SKILL path might be set to a default value. You can use the `setSkillPath` function to make sure it is set up correctly.

Working with the SKILL Path

Setting the Internal SKILL Path (`setSkillPath`)

`setSkillPath` sets the internal SKILL path. You can specify the directory paths either as a string, where each alternate path is separated by spaces, or as a list of strings. The system tests the validity of each directory path as it puts the input into standard form.

- If all directory paths exist, it returns `nil`
- If any path does not exist, it returns a list in which each element is an invalid path

The paths on the SKILL path are always searched for in the path order you specify. Even if a path does not exist (and hence appears in the returned list), it remains on the new SKILL path. The use of the SKILL path in other file-related functions can be effectively disabled by calling `setSkillPath` with `nil` as the argument.

```
setSkillPath('("." "~" "~/cpu/test1"))
=> nil                               ; If "~/cpu/test1" exists.
=> ("~/cpu/test1")                   ; If "~/cpu/test1" does not exist.
```

The same task can be done with the following call that puts all paths in one string.

```
setSkillPath(". ~ ~/cpu/test1")
```

Finding the Current SKILL Path (`getSkillPath`)

`getSkillPath` returns directory paths from the current SKILL path setting. The result is a list where each element is a path component as specified by `setSkillPath`.

```
setSkillPath('("." "~" "~/cpu/test1"))=> nil
getSkillPath()=> (". " "~" "~/cpu/test1")
```

The example below shows how to add a directory to the beginning of your search path (assuming a directory `~/lib`).

```
setSkillPath(cons("~/lib" getSkillPath()))=> nil
getSkillPath()=> ("~/lib" "." "~" "~/cpu/test1")
```

Working with the Installation Path

Finding the Installation Path (`getInstallPath`)

`getInstallPath` returns the system installation path (that is, the root directory where the Cadence products are installed in your file system) as a list of a single string, where

- The path is always returned in absolute format
- The result is always a list of one string

```
getInstallPath() => ("/usr5/cds/4.2")
```

Attaching the Installation Path to a Given Path (`prependInstallPath`)

`prependInstallPath` prepends the Cadence installation path to the given path (possibly adding a slash (/) separator if needed) and returns the resulting path as a string. The typical use of this function is to compute one member of a list passed to `setSkillPath`.

```
getInstallPath()  
=> ("/usr5/cds/4.2")
```

Assume this is your install path.

```
prependInstallPath("etc/context" )  
=> "/usr5/cds/4.2/etc/context"
```

A slash (/) is added.

```
prependInstallPath("/bin" )  
=> "/usr5/cds/4.2/bin"  
  
setSkillPath( list("." prependInstallPath("bin")  
                  prependInstallPath("etc/context")) )  
=> nil
```

Finding the Root of the Hierarchy (`cdsGetInstPath`)

`getInstallPath` returns the root of the `dfII` hierarchy whereas `cdsGetInstPath` returns the root of the hierarchy. `cdsGetInstPath` is more general and is meant to be used by all `dfII` and non-`dfII` applications. For example:

```
getInstallPath() => ("/usr/mnt/hamilton/9304/tools/dfII")  
cdsGetInstPath() => "/usr/mnt/hamilton/9304"
```

Checking File Status

Checking if a File Exists (`isFile`, `isFileName`)

`isFileName` checks if a file exists. The file name can be specified with either an absolute path or a relative path. In the latter case, the current SKILL path is used if it's not `nil`. Only the presence or absence of the name is checked. If found, the name can belong to either a file or a directory. `isFileName` differs from `isFile` in this regard.

```
isFileName("myLib")=> t
```

A directory is just a special kind of file.

```
isFileName("triadc")=> t  
isFileName("triad1")=> nil
```

Result if `triad1` is not in the current working directory.

`isFile` checks if a file exists. `isFile` is identical to `isFileName`, except that directories are not viewed as (regular) files. Uses the current SKILL path for relative paths.

```
isFile("triadc")=> t
```

Checking if a Path Exists and if it is the Name of a Directory (`isDir`)

`isDir` checks if a path exists and if it is the name of a directory. When the path is a relative path, the current SKILL path is used if it's non-`nil`.

```
isDir("myLib")    => t  
isDir("triadc")   => nil
```

Assumes `myLib` is a directory and `triadc` is a file under the current working directory and the SKILL path is `nil`.

```
isDir("test")=> nil
```

Result if `test` does not exist.

Checking if You Have Permission to Read a File or List a Directory (`isReadable`)

`isReadable` checks if you have permission to read the file or list the directory you specify. Uses the current SKILL path for relative paths.

```
isReadable("./")=> t
```

Result if current working directory is readable.

```
isReadable("~/myLib")=> nil
```

Result if `~/myLib` is not readable or does not exist.

Checking for Permission to Write a File or Update a Directory (`isWritable`)

`isWritable` checks if you have permission to write a file or update a directory that you specify. It uses the current SKILL path for relative paths.

```
isWritable("/tmp")           => t
isWritable("~/test/out.1")   => nil
```

Result if `out.1` does not exist or there is no write permission to it.

Checking for Permission to Execute a File or Search a Directory (`isExecutable`)

`isExecutable` checks if you have permission to execute a file or search a directory. A directory is executable if it allows you to name that directory as part of your UNIX path in searching files. It uses the current SKILL path for relative paths.

```
isExecutable("/bin/ls")      => t
isExecutable("/usr/tmp")     => t
isExecutable("attachFiles")  => nil
```

Result if `attachFiles` does not exist or is not executable.

Determining the Number of Bytes in a File (`fileLength`)

`fileLength` determines the number of bytes in a file. A directory is viewed just as a file in this case. `fileLength` uses the current SKILL path if a relative path is given.

```
fileLength("/tmp")           => 1024
```

Return value is system-dependent.

```
fileLength("~/test/out.1")    => 32157
```

This examples assumes the file exists. If the file does not exist, you get an error message, such as

```
*Error* fileLength: no such file or directory - "~/test/out.1"
```

Getting Information About Open Files (`numOpenFiles`)

`numOpenFiles` returns the number of files that are open and the maximum number of files that a process can open. The numbers are returned as a two-element list.

```
numOpenFiles()               => (6 64)
```

Result is system-dependent.

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```
f = infile("/dev/null")      => port: "/dev/null"
numOpenFiles()              => (7 64)
```

One more file is open now.

Working with File Offsets (fileTell, fileSeek)

`fileTell` returns the current offset (from the beginning of the file) in bytes for the file opened on a port.

`fileSeek` sets the position for the next operation to perform on the file opened on a port. The position is specified in bytes. `fileSeek` takes three arguments. The first two are for port and for offset designated in number of bytes to move forward (or backward with a negative argument). The valid values for the third argument are

- 0 Offset from the beginning of the file
- 1 Offset from current position of file pointer
- 2 Offset from the end of the file.

Let the file `test.data` contain the single line of text:

```
0123456789 test xyz
```

```
p = infile("test.data")  => port: "test.data"
fileTell(p)              => 0
for(i 1 10 getc(p))      => t      Skip first 10 characters
fileTell(p)              => 10

fscanf(p "%s" s)          => 1      s = "test" now
fileTell(p)              => 15

fileSeek(p 0 0)           => t
fscanf(p "%d" x)          => 1      x = 123456789 now

fileSeek(p 6 1)           => t
fscanf(p "%s" s)          => 1      s = "xyz" now

fileSeek(p -12 2)         => t
fscanf(p "%d" x)          => 1      x = 89 now
```

Working with Directories

Creating a Directory (createDir)

`createDir` creates a directory. The directory name can be specified with either an absolute or relative path; the SKILL path is used in the latter case. You get an error message if the directory cannot be created because you do not have permission to update the parent directory or a parent directory does not exist.

```
createDir("/usr/tmp/test") => t
createDir("/usr/tmp/test") => nil
```

Directory already exists.

Creating Parent Directories (createDirHier)

`CreateDirHier` creates all directories specified in the given SKILL path that do not already exist. The directory names in the given SKILL path can be specified with either absolute or relative; the SKILL path is used in the latter case. A path that is anchored to the current directory, for example, `./`, `../`, or `../..`, etc., is not considered as a relative path.

The permissions associated with new directories are subject to the file creation mask on systems supporting that concept. If the directory with the specified name already exists, `nil` is returned.

Deleting a Directory (deleteDir)

`deleteDir` deletes a directory. The directory name can be specified with either an absolute or relative path; the SKILL path is used in the latter case. You get an error message if you do not have permission to delete a directory or the directory you want to delete is not empty.

```
createDir("/usr/tmp/test")=> t
deleteDir("/usr/tmp/test")=> t
deleteDir("/usr/bin")
```

If you do not have permission to delete `/bin`, you get an error message about permission violation.

```
deleteDir("~/")
```

Assuming there are some files in `~`, you get an error message that the directory is not empty.

Deleting a File (deleteFile)

`deleteFile` deletes a file. The file name can be specified with either an absolute or relative path; the SKILL path is used in the latter case. If a symbolic link is passed in as the argument, it is the link itself, not the file or directory referenced by the link, that gets removed.

```
deleteFile("~/test/out.1") => t
```

If the file exists and is deleted.

```
deleteFile("~/test/out.2")=> nil
```

If the file does not exist.

```
deleteFile("/bin/ls")
```

If you do not have write permission for `/bin`, signals an error about permission violation.

Creating a Unique File Name (`makeTempFileName`)

`makeTempFileName` appends a string suffix to the last component of a path template such that the resultant composite string does not duplicate any existing file name. (That is, it checks that the file does not exist; the SKILL path is not used in this checking.)

Successive calls to `makeTempFileName` return different results only if the first name returned is used to create a file in the same directory before a second call is made.

- The last component of the resultant path is guaranteed to be no more than 14 characters. The example below requests a “file” with 15 characters

```
makeTempFileName("/tmp/123456789123456")  
=> "/tmp/12345678a08717"
```

- If the original template has a long last component, it is truncated from the end if needed

```
makeTempFileName("/tmp/123456789.123456")  
=> "/tmp/12345678a08717"
```

- Any trailing Xs are removed from the template before the new string suffix is appended

You should follow the convention of placing temporary files in the `/tmp` directory on your system.

```
d = makeTempFileName("/tmp/testXXXX") => "/tmp/testa00324"
```

Trailing Xs are removed.

```
createDir(d)                                => t
```

The name is used this time.

```
makeTempFileName("/tmp/test")               => "/tmp/testb00324"
```

A new name is returned this time.

Listing the Names of All Files and Directories (`getDirFiles`)

`getDirFiles` lists the names of all files and directories (including `.` and `..`) in a directory. Uses the current SKILL path for relative paths.

```
getDirFiles(car(getInstallPath())) =>  
("." ".." "bin" "cdsuser" "etc" "group" "include" "lib" "pvt"  
"samples" "share" "test" "tools" "man" "local" )
```


Expanding the Name of a File to its Full Path (simplifyFilename)

`simplifyFilename` returns the fully expanded name of a file. Tilde expansion is performed, `./` and `../` are compressed, and redundant slashes are removed. Symbolic links are also resolved by default, unless the second (optional) argument `g_dontResolveLinks` is specified to non-nil. If the file you supply is not absolute, the current working directory is prefixed to the returned file name.

```
simplifyFilename("~/test") => "/usr/mnt/user/test"
```

Returns the fully expanded name of `test`, assuming the user's home directory is `/usr/mnt/user`.

Getting the Current Working Directory (getWorkingDir)

`getWorkingDir` returns the current working directory as a string. The result is put into a "`~/`prefixed" form if possible by testing for commonality with the current user's home directory. For example, `~/test` is returned in preference to `/usr/mnt/user1/test`, assuming that the home directory for `user1` is `/usr/mnt/user1` and the current working directory is `/usr1/mnt/user1/test`.

```
getWorkingDir() => "~/project/cpu/layout"
```

Changing the Current Working Directory (changeWorkingDir)

Changes the working directory to the name you supply. The name can be specified with either a relative or absolute path. If you supply a relative path, the `cdpath` shell variable is used to search for the directory, not the SKILL path.

Different error messages are output if the operation fails because the directory does not exist or you do not have search (execute) permission.



Use this function with care: if `“.”` is either part of the SKILL path or the libraryPath, changing the working directory can affect the visibility of SKILL files or design data.

Assume there is a directory `/usr5/design/cpu` with proper permission and there is no `test` directory under `/usr5/design/cpu`.

```
changeWorkingDir( "/usr5/design/cpu") => t  
changeWorkingDir( "test")
```

Signals an error that no such directory exists.

Ports

All input and output in SKILL goes through a data type called a port. A port can be opened either for reading (an input port) or writing (an output port). Ports are analogous to FILE* variables used by the `stdio` library in C. Most implementations of the UNIX operating system impose a strict limit (typically between 30 and 64) on the number of files that can be open at any time.

Your application typically needs to use some of these scarce file descriptors, leaving you with only a few free ports with which to work. You should therefore always close ports that are no longer in use and avoid using an excessive number of ports; you might otherwise run out of ports when your code is moved to a different UNIX machine.

Predefined Ports

Most I/O functions in SKILL accept a port as an optional argument. If a port is not specified, the `piport` and `poport` are used as default ports for input and output respectively. The table below lists the names and the use of the input/output ports predefined in SKILL.



Caution

You can redefine the default values, if necessary, but many internal SKILL functions are hard wired to use particular ports and you should be careful not to assign any of them an illegal value.

The `stdin`, `stdout`, and `stderr` ports are also predefined. These ports correspond to the standard input, standard output, and standard error streams available to every UNIX program.

Predefined Input/Output Ports

Name	Usage
<code>piport</code>	Standard input port, analogous to and initialized to <code>stdin</code> in C
<code>poport</code>	Standard output port, analogous to and initialized to <code>stdout</code> in C
<code>errport</code>	Output port for printing error messages, analogous to and initialized to <code>stderr</code> in C
<code>ptport</code>	Output port for printing trace information; initialized to <code>stdout</code> in C
<code>woport</code>	Output port for printing warning messages; analogous to and initialized to <code>stdout</code> in C.

Opening and Closing Ports

The following functions work with opening and closing ports. Both of the file opening functions use the SKILL path variable.

Opening an Input Port to Read a File (infile)

`infile` opens an input port ready to read a file you specify. The file name can be specified with either an absolute path or a relative path. In the latter case, the current SKILL path is used if it's not *nil*.

```
infile("~/test/input.il")=> port:"~/test/input.il"
```

Result if such a file exists and is readable.

```
infile("myFile") => nil
```

Result if `myFile` does not exist according to the SKILL path or exists but is not readable.

Opening an Output Port to Write a File (outfile)

`outfile` opens an output port ready to write to the file you specify. The file name can be specified with either an absolute path or a relative path.

- If a relative path is given and the current SKILL path setting is not *nil*, all directory paths from the SKILL path are checked, in the order specified, for that file name
- If found, the system overwrites the first updateable file in the list
- If no updateable file is found, it places a new file of that name in the first writable directory

```
p = outfile("out.il" "w") => port:"out.il"
```

Returns the name of the output port ready to write to the file.

```
outfile("/bin/ls") => nil
```

Returns *nil* if the file cannot be opened for writing.

Writing Out All Characters in the Output Buffer of a Port (drain)

`drain` writes out all characters that are in the output buffer of a port. `drain` is analogous to a combination of `fflush` and `fsync` in C. You get an error message if the port to drain is an input port or has been closed.

```
drain() => t  
drain(port) => t
```

Writing Data to a Port in Virtuoso

To write text data to a port in Virtuoso, use the `fprintf` function. For example,

```
fprintf(errport "An error message\n")
```

Prints the text in red in the CIW and tags it with `\e` in the CDS.log file.

```
fprintf(errport "An error message \n")  
An error message  
t
```

Whereas,

```
fprintf(woport "A warning message\n")
```

Prints the text in orange in the CIW and tags it with `\w` in the CDS.log file.

```
fprintf(woport "A warning message \n")  
A warning message  
t
```

Draining, Closing, and Freeing a Port (close)

The port is drained, closed, and freed. When a file is closed, it frees the `FILE*` associated with the port. Do not use this function on `piport`, `poport`, `stdin`, `stdout`, and `stderr`.

```
p = outfile("~/test/myFile") => port:"~/test/myFile"  
close(p)                     => t
```

Output

SKILL provides functions for unformatted and formatted output.

Unformatted Output

Printing the Value of an Expression in the Default Format (print, println)

`print` prints the value of an expression using the default format for the data type of the value (for example, strings are enclosed in double quotes).

```
print("hello")  
"hello"  
=> nil
```

Prints to `poport` and returns `nil`.

`println` prints the value of an expression just like `print`, but a newline character is automatically printed after printing the input value. `println` flushes the output port after printing each newline character.

```
println("Hello World!")
"Hello World!"
=> nil
```

Printing a Newline (\n) Character (newline)

Prints a newline (\n) character. If you do not specify the output port, it defaults to `poport`, the standard output port. The `newline` function flushes the output port after printing each newline character.

```
print("Hello") newline() print("World!")
"Hello"
"World!"
=> nil
```

Printing a List with a Limited Number of Elements and Levels of Nesting (`printlev`)

`printlev` prints a list with a limited number of elements and levels of nesting. Lists are normally printed in their entirety no matter how many elements they have or how deeply nested they are.

Applications can, however, set upper limits on the number of elements and the levels of nesting shown when printing lists using `printlev`. These limits are sometimes necessary to control the volume of interactive output because the SKILL top-level automatically prints the results of expression evaluation. Limits can also protect against infinite looping on circular lists possibly created when novices use the destructive list modification functions, such as `rplaca` or `rplacd`, without a thorough understanding of how they work. `printlev` uses the following syntax:

```
printlev( g_value x_level x_length [p_outputPort] ) => nil
```

Two integer variables, `print length` and `print level` (specified by `x_length` and `x_level`), control the maximum number of elements and the levels of nesting that are printed. List elements beyond the maximum specified by `print length` are abbreviated as "...". Lists nested deeper than the maximum level specified by `print level` are abbreviated as "&.". Both `print length` and `print level` are initialized to `nil` (meaning no limits are imposed) by SKILL, but each application can set its own limits.

The `printlev` function is identical to `print` except that it takes two additional arguments specifying the maximum level and length to use in printing the expression.

```
List = '(1 2 (3 (4 (5))) 6)
printlev(List 100 2)
```

```
(1 2 ...)  
=> nil  
  
printlev(List 3 100)  
(1 2 (3 (4 &)) 6)  
=> nil  
  
printlev(List 3 3 p)  
(1 2 (3 (4 &)) ...)  
=> nil
```

Assumes port p exists. Prints to port p.

Formatted Output

You can precede format characters with a field width specification. For example, %5d prints an integer in a field that is 5 columns wide. If the field width begins with the digit “0”, zero padding is done instead of blank padding. For the format characters `f` and `e`, the width specification can be followed by a period “.” and an integer specifying the precision, that is, the number of digits to print after the decimal point.

Output is right justified within a field by default unless an optional minus sign “-” immediately follows the “%” character, which will then be left justified. To print a percent sign, you must use two percent signs in succession. You must explicitly put “\n” in your format string to print a newline character and “\t” for a tab.

Common Output Format Specifications

Format Specification	Type(s) of Argument	Prints
%d	fixnum	Integer in decimal radix
%o	fixnum	Integer in octal
%x	fixnum	Integer in hexadecimal
%f	flonum	Floating-point number in the style [-]ddd.ddd
%e	flonum	Floating-point number in the style [-]d.ddde[-]ddd
%g	flonum	Floating-point number in style f or e, whichever gives full precision in minimum space
%s	string, symbol	Prints out a string (without quotes) or the print name of a symbol
%c	string, symbol	The first character
%n	fixnum, flonum	Number

Common Output Format Specifications

Format Specification	Type(s) of Argument	Prints
%L	list	Default format for the data type
%P	list	Point
%B	list	Box

For formatted output, SKILL makes available the standard C `stdio` library routines `printf`, `fprintf`, and `sprintf`. SKILL provides a robust interface to these routines. Below is a brief description of each routine in the context of the SKILL runtime environment. If more detailed descriptions are needed for these functions, consult your C programming manual.

Writing Formatted Output to Ooport (`printf`)

`printf` writes formatted output to `poport`. Optional arguments following the format string are printed according to their corresponding format specifications. *printf* is identical to *fprintf* except that it does not take a port argument and the output is written to *poport*.

```
x = 197.9687
printf("The test measures %10.2f.\n" x)
```

Prints the following line to `poport` and returns `t`.

```
The test measures          197.97. => t
```

Writing Formatted Output to a Port (`fprintf`)

`fprintf` writes formatted output to the port given as the first argument. The optional arguments following the format string are printed according to their corresponding format specifications.

```
x = 197.9687
fprintf(p "The test measures %10.2f.\n" x)
```

Prints the following line to port `p` and returns `t`.

```
The test measures          197.97. => t
```

Writing Formatted Output to a String Variable (`sprintf`)

`sprintf` formats the output and puts the resultant string into the variable given as the first argument. If `nil` is specified as the first argument, no assignment is made. The formatted string is returned. Because of internal buffering in `sprintf`, there is a limit to how many

characters `sprintf` can handle, but the limit is large enough (8192 characters) that it should not present any problem.

```
sprintf(s "Memorize %s number %d!" "transaction" 5)
=> "Memorize transaction number 5!"

s
=> "Memorize transaction number 5!"

p = outfile(sprintf(nil "test%d.out" 10))
=> port:"test10.out"
```

Pretty Printing

SKILL provides functions for “pretty printing” function definitions and long data lists with proper indenting to make them more readable and easier to manipulate in text form.

You need the SKILL Development Environment license to pretty print function definitions using the `pp` SKILL function below.

Pretty Printing a Function Definition (`pp`)

`pp` pretty prints a function definition. The function must be a readable interpreted function. (Binary functions cannot be pretty printed.) Each function definition is printed so it can be read back into SKILL. `pp` does not evaluate its first argument but does evaluate the second argument, if given.

```
procedure(fac(n) if(n <= 1 1 n*fac(n-1)))=> fac
pp fac
procedure(fac(n)
  if((n <= 1) 1
    (n * fac(n - 1)))
)
=> nil
```

Defines the factorial function `fac`, then pretty prints it to `poport`.

Pretty Printing Long Data Lists (`pprint`)

`pprint` is identical to `print` except that it tries to pretty print the value whenever possible. (`pprint` does not work the same as the `pp` function. `pp` is an `nlambda` and only takes a function name whereas `pprint` is a `lambda` and takes an arbitrary SKILL object.)

The `pprint` function is useful, for example, when printing out a long list where `print` prints the list on one (possibly huge) line but `pprint` limits the output on a single line and produces a multiple-line printout if necessary. This multiple-line printout makes later input much easier.

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```
pprint '(1 2 3 4 5 6 7 8 9 0 a b c d e f g h i j k)
(1 2 3 4 5
  6 7 8 9 0
  a b c d e
  f g h i j
  k
)
=> nil
```

Input

When describing input functions, this manual often uses the term “form” to refer to a logical unit of input. A form can be an expression, such as source code or a data list that can span multiple input lines. Input functions such as `lineread` read in one input line at a time but continue reading if they do not find a complete form at the end of a line.

You can think of input forms and how the SKILL functions work with them in the following ways.

Input Functions

Input Source	SKILL Evaluated	SKILL Not Evaluated	Application-Specific Formats
File	<code>load</code> <code>loadi</code>	<code>lineread</code>	<code>infile</code> <code>gets</code> <code>getc</code> <code>fscanf</code> <code>close</code>
String	<code>evalstring</code> <code>loadstring</code> <code>errsetstring</code>	<code>linereadstring</code>	<code>instring</code> <code>gets</code> <code>getc</code> <code>fscanf</code> <code>close</code>

SKILL forms read from a file are either evaluated or not evaluated. Input strings can have an application-specific syntax of their own, such as a netlist syntax. It is the programmer’s responsibility to open a port, understand the application-specific syntax, process the input, and then close the port.

Reading and Evaluating SKILL Formats

Reading and Evaluating an Expression Stored in a String (`evalstring`)

`evalstring` reads and evaluates an expression stored in a string. The resulting value is returned. Notice that `evalstring` does not allow the outermost set of parentheses to be omitted, as in the top level. Refer to the [“Top Levels”](#) on page 235 for a discussion of the top level.

```
evalstring("1+2")           => 3
evalstring("cons('a '(b c))") => (a b c)
car '(1 2 3)                 => 1
evalstring("car '(1 2 3)")
```

Signals that `car` is an unbound variable.

Opening a String and Executing its Expressions (`loadstring`)

`loadstring` opens a string for reading, then parses and executes expressions stored in the string just as `load` does in loading a file. `loadstring` is different from `evalstring` in two ways. `loadstring`

- Uses `lineread` mode
- Always returns `t` if it evaluates successfully

```
loadstring "1+2"           => t
loadstring "procedure( f(n) x=x+n )" => t
loadstring "x=10\n f 20\n f 30"  => t
x                               => 60
```

Reading and Evaluating an Expression then Checking for Errors (`errsetstring`)

`errsetstring` reads and evaluates an expression stored in a string. Same as `evalstring` except that it calls `errset` to catch any errors that might occur during the parsing and evaluation.

```
errsetstring("1+2")           => (3)
errsetstring("1+'a")          => nil
```

Returns `nil` because an error occurred.

```
errsetstring("1+'a" t)        => nil
Prints out an error message:
*Error* plus: can't handle (1 + a)
```

Loading Files (`load`, `loadi`)

`load` opens a file, repeatedly calls `lineread` to read in the file, and immediately evaluates each form after it is read in. It closes the file when end of file is reached. Unless errors are discovered, the file is read in quietly. If `load` is interrupted by pressing Control-c, the function skips the rest of the file being loaded.

SKILL has an autoload feature that allows applications to load functions into SKILL on demand. If a function being executed is undefined, SKILL checks if the name of the function (a symbol) has a property called *autoload* attached to it. If the property exists, its value, which must be either a string or a function call that returns a string, is used as the name of a file to load. The file should contain a definition for the function that triggered the autoload. Execution proceeds normally after the function is defined. The whole autoload sequence is functionally transparent. Refer to [“Delivering Products”](#) on page 239

```
load( "testfns.il")           ; Load file testfns.il
fn.autoload = "myfunc.il"    ; Declares an autoload property.
fn(1)
```

`fn` is undefined at this point, so this call triggers an autoload of `myfunc.il`, which contains the definition of `fn`.

```
fn(2)           ; fn is now defined and executes normally.
```

`loadi` is identical to `load`, except that `loadi` ignores errors encountered during the load, prints an error message, and then continues loading.

```
loadi( "testfns.il" )
```

Loads the `testfns.il` file.

```
loadi( "/tmp/test.il")
```

Loads the `test.il` file from the `tmp` directory.

Reading but Not Evaluating SKILL Formats

Parsing the Next Line in the Input Pport into a List (`lineread`)

`lineread` parses the next line in the input port into a list that you can further manipulate. It is used by the interpreter's top level to read in all input and understands only SKILL syntax.

Only one line of input is read in unless there are still open parentheses pending at the end of the first line, or binary *infix* operators whose right-hand argument has not yet been supplied, in which case additional input lines are read until all open parentheses have been closed and all binary *infix* operators satisfied. The symbol `t` is returned if `lineread` reads a blank input line and `nil` is returned at the end of the input file.

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```
lineread(piport)      ; Reads in the next input expression
f 1 2 +               ; First input line of the file being read
3                   ; Second input line
=> f (1 (2 + 3))

lineread(piport)
f(a b c)              ; Another input line of the file
=> ((f a b c))        ; Returns a list of input objects
```

Reading a String into a List (linereadstring)

`linereadstring` executes `lineread` on a string and returns the form read in. Anything after the first form is ignored.

```
linereadstring "abc"           => (abc)
linereadstring "f a b c"       => (f a b c)
linereadstring "x + y"         => ((x + y))
linereadstring "f a b c\n g 1 2 3" => (f a b c)
```

In the last example, only the first form is read in.

Reading Application-Specific Formats

The following input functions are helpful when you must read input from a file that was not written in SKILL-compatible format.

Reading Formatted Input (fscanf)

`fscanf` reads the input from a port according to format specifications in a format string. The results are stored in corresponding variables in the call. `fscanf` can be considered the inverse of the `fprintf` output function. `fscanf` returns the number of input items it successfully matches with its format string. It returns `nil` if it encounters an end of file.

The maximum size of any input string being read as a string variable for `fscanf` is 8K. Also, the function `lineread` is a faster alternative to `fscanf` for reading SKILL objects.

The input formats accepted by `fscanf` are summarized below.

Common Input Format Specifications

Format Specification	Type(s) of Argument	Scans for
%d	fixnum	An integer
%f	flonum	A floating-point number

Common Input Format Specifications

Format Specification	Type(s) of Argument	Scans for
%s	string	A string (delimited by spaces) in the input

```
fscanf( p "%d %f" i d )
```

Scans for an integer and a floating-point number from the input port `p` and stores the values read in the variables `i` and `d`, respectively.

Assume there is a file `testcase` with one line:

```
hello 2 3 world
x = infile("testcase") => port:"testcase"
fscanf( x "%s %d %d %s" a b c d )=> 4
(list a b c d)                => ("hello" 2 3 "world")
```

Reading a Line and Storing it in a Variable (gets)

`gets` reads a line from the input port and stores it as a string in a variable. The string is also returned as the value of `gets`. The terminating newline character of the line becomes the last character in the string. `gets` returns `nil` if EOF is encountered and the variable maintains its last value. Assume the `test1.data` file has the following first two lines:

```
#This is the data for test1
0001 1100 1011 0111
p = infile("test1.data") => port:"test1.data"
gets(s p)                => "#This is the data for test1\n"
gets(s p)                => "0001 1100 1011 0111\n"
s                        => "0001 1100 1011 0111\n"
```

Reading and Returning a Single Character from an Input Port (getc)

`getc` reads a single character from the input port and returns it as the value of `getc`. If the character returned is a non-printable character, its octal value is stored as a symbol. If you are familiar with C, you might notice that the `getc` and `getchar` SKILL functions are totally unrelated. `getc` returns `nil` if EOF is encountered.

The input port arguments for both `gets` and `getc` are optional. If the port is not given, the functions take their input from `piport`. In the following example assume the file `test1.data` has its first line read as:

```
#This is the data for test1
p = infile("test1.data") => port:"test1.data"
getc(p)                 => \#
```

```
getc(p)           => T
getc(p)           => h
```

Reading Application-Specific Formats from Strings

In addition to being able to accept input from the terminal and from text files, SKILL can also take its input directly from strings. Some applications store programs internally as strings and then parse the strings into their corresponding internal SKILL representations as needed.

Because parsing is a relatively expensive operation, you should avoid calling any of the following functions repeatedly on the same string. It is a good practice to convert each string into its internal SKILL representation before using it more than once.

Opening a String for Reading (instring)

Opens a string for reading just as `infile` opens a file. An input port that can be used to read the string is returned.

```
s = "Hello World!"           => "Hello World!"
p = instring(s)              => port:"*string*"
fscanf(p "%s %s" a b) => 2
a                             => "Hello"
b                             => "World!"
close(p)                     => t
```



Always remember to close the port when you are done.

Opening a String for Writing (outstring)

The `outstring` function takes no arguments and returns an opened output port for strings (or an output). After a port is opened, it can be used with functions, such as `fprintf`, `println`, and `close` that write to an output port. You can use the `close` function to close the output port.

You need to use the `getOutstring` function to retrieve the content of the output port (while it is open). For example:

```
s = outstring()
= >port:"*string*"
fprintf(s "Quick brown")
getOutstring(s)
=>"Quick brown"
fprintf(s " fox jumps")
getOutstring(s)
```

```
=> "Quick brown fox jumps"
    fprintf(s " over the lazy dog")
getOutstring(s)
=> "Quick brown fox jumps over the lazy dog"
close(s)
getOutstring(s)
=> nil
```

System-Related Functions

Various SKILL functions are available to interact with and query the system environment.

Executing UNIX Commands

From within SKILL, you can execute individual UNIX commands or invoke the `sh` or `csh` UNIX shell.

Starting the UNIX Bourne-Shell (`sh`, `shell`)

Starts the UNIX Bourne-shell `sh` as a child process to execute a command string. If the `sh` function is called with no arguments, an interactive UNIX shell is invoked that prompts you for UNIX command input (available only in nongraphic applications).

```
sh( "rm /tmp/junk")
```

Removes the `junk` file from the `/tmp` directory and returns `t` if it is removed successfully.

Starting the UNIX C-Shell (`csh`)

Starts the UNIX C-shell `csh` as a child process to execute a command string. Identical to the `sh` function, but invokes the C-shell (`csh`) rather than the Bourne-shell (`sh`).

```
csh( "mkdir ~/tmp" )
```

Creates a directory called `tmp` in your home directory.

System Environment

The following functions find and compare the current time, retrieve the version number of the software you are using, and determine the value of a UNIX environment variable.

Getting the Current Time (getCurrentTime)

`getCurrentTime` returns the current time in the form of a string. The format of the string is month day hour:minute:second year.

```
getCurrentTime( ) => "Jan 26 18:15:18 1993"
```

Comparing Times (compareTime)

`compareTime` compares two string arguments, representing a clock-calendar time. The format of the string is month day hour:minute:second year. The units are seconds.

```
compareTime( "Apr 8 4:21:39 1991" "Apr 16 3:24:36 1991")  
=> -687777.
```

687,777 seconds have occurred between the two dates. For a positive number of seconds, the most recent date needs to be the first argument.

```
compareTime("Apr 16 3:24:36 1991" "Apr 16 3:14:36 1991")  
=> 600
```

600 seconds (10 minutes) have occurred between the two dates.

Getting the Current Version Number of Cadence Software (getVersion)

Returns the version number of software you are using.

```
getVersion()  
=> "cds3 version 4.2.2 Fri Jan 26 20:40:28 PST 1993"
```

Getting the Value of a UNIX Environment Variable (getShellEnvVar)

`getShellEnvVar` returns the value of a UNIX environment variable, if it has been set.

```
getShellEnvVar("SHELL") => "/bin/csh"
```

Returns the current value of the `SHELL` environment variable.

Setting a UNIX Environment Variable (setShellEnvVar)

`setShellEnvVar` sets the value of a UNIX environment variable to a new value.

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```
setShellEnvVar("PWD=/tmp") => t
```

Sets the current working directory to the /tmp directory .

```
getShellEnvVar("PWD")=> "/tmp"
```

Gets the current working directory.

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I/O and File Handling

Advanced List Operations

“Cadence SKILL Language” on page 19 introduces you to building Cadence® SKILL language lists. Information in this chapter helps you perform more advanced list operations.

It is helpful to understand how lists are stored in virtual memory so that you can understand how SKILL functions such as `car` and `cdr` manipulate lists and the issues behind building long lists efficiently (see “Conceptual Background” on page 180).

You can read more about the following topics:

- Summary of List Operations on page 182
- Altering List Cells on page 183
- Accessing Lists on page 187
- Building Lists Efficiently on page 188
- Reorganizing a List on page 191
- Searching Lists on page 192
- Copying Lists on page 193
- Filtering Lists on page 194
- Removing Elements from a List on page 195
- Substituting Elements on page 196
- Transforming Elements of a Filtered List on page 197
- Validating Lists on page 198
- Using Mapping Functions to Traverse Lists on page 198
- List Traversal Case Studies on page 206

Conceptual Background

How Lists Are Stored in Virtual Memory

SKILL functions that manipulate lists and symbols deal with memory pointers. When you assign a list to a variable, the variable is internally assigned a pointer to the head of the list. When a list is taken apart by functions such as `car` and `cdr`, only pointers to various parts of the list are returned and no new list cells are created.

SKILL suppresses your awareness of pointers by how it displays lists and symbols. In general, when SKILL displays a supported data type, it uses a characteristic syntax to suppress irrelevant detail and focus on the essentials of the data. This syntax has implications for list and symbol data types. Instead of displaying memory addresses, SKILL displays

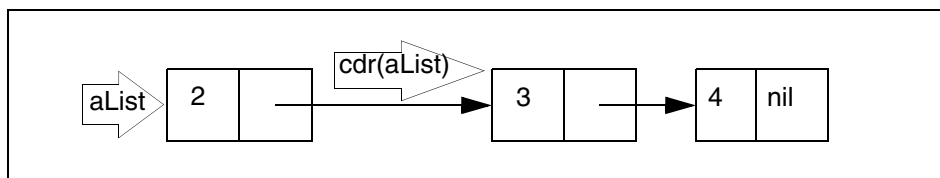
- The elements of a list surrounded by parentheses
- The name of a symbol

A SKILL List as a List Cell

SKILL represents a list by means of a list cell. A list cell occupies two locations in virtual memory.

- The first location holds a reference to the first element in the list.
- The second location holds a reference to the tail of the list, that is, another list cell or `nil`.

The expression `aList = '( 2 3 4 )` allocates the following three list cells.



The `car` function returns the contents of the first location of a list cell.

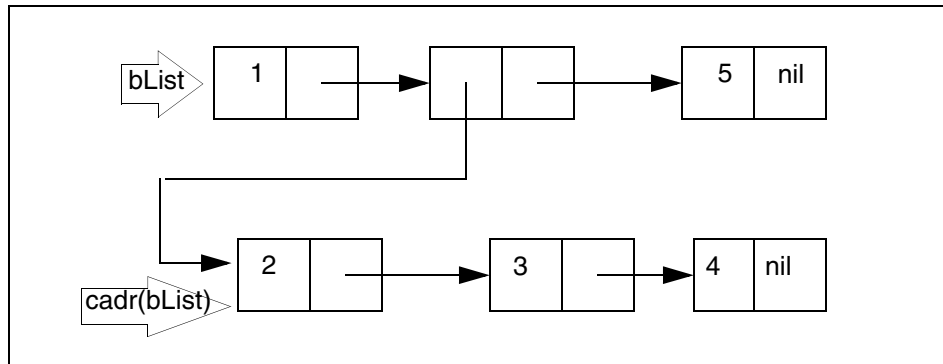
```
car( aList) => 2
```

The `cdr` function returns the contents of the second location of a list cell.

```
cdr( aList) => (3 4)
```

Lists Containing Sublists

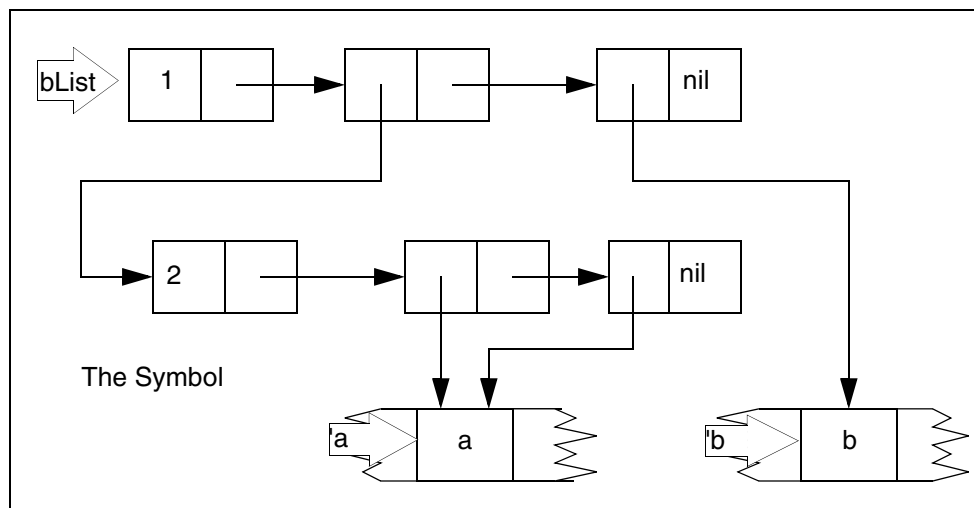
The expression `bList = '( 1 ( 2 3 4 ) 5 )` allocates the following list cells.



`cadr(bList) => (2 3 4)`

Lists Containing Symbols

The expression `bList = '( 1 ( 2 a a ) b )` allocates the following list cells.



Internally the expression `' a` returns the pointer to the symbol `a` in the symbol table.

Destructive versus Non-Destructive Operations

Non-Destructive Operations

The term `non-destructive modification` refers to any operation that allocates a copy of a list that reflects the desired alteration. The original list is not altered. It is your responsibility to update any variables that need to reflect the operation.

Such operations are usually easier for you to implement than destructive operations that do alter the original list. The disadvantage is that making an altered copy of the original list might be significantly time-consuming.

Destructive Operations

The term `destructive list modification` refers to any operation that alters either the `car` or `cdr` of a list cell. Destructive modification functions do not need to create new list structures. They are therefore considerably faster than equivalent nondestructive modification functions.

Depending on the operation, any variable referring to the original list can be affected. Many subtle problems can arise when these functions are used without a thorough understanding of the implications.



You should only use the destructive modification functions described in this chapter with a very good understanding of how the SKILL language represents lists in virtual memory.

Summary of List Operations

The following table summarizes the list operations that are discussed in this chapter. Use the destructive modification functions with great care.

List Operations

Operation	Function	Non-destructive	Destructive
Altering List Cells	<code>rplaca</code> , <code>rplacd</code>		x
Accessing a List	<code>nthelem</code> , <code>nthcdr</code> , <code>last</code>	x	

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List Operations

Operation	Function	Non-destructive	Destructive
Building a List	cons, ncons, xcons, append1	x	
	tconc, nconc, lconc		x
Reorganizing a List	reverse	x	
	sort, sortcar		x
Removing Elements	remove, remq	x	
	remd, remdq		x
Searching Lists	member, memq, exists	x	
Filtering Lists	setof	x	
Substituting	subst	x	
Traversal	mapc, map, mapcar, maplist, mapcan	x	
Traversal	mapcan		x

Altering List Cells

The most fundamental destructive operations concern altering a list cell. You can change either the `car` cell or the `cdr` cell. `rplaca` and `rplacd` are destructive operations. Starting from IC6.1.6, a new function, `setf`, has been introduced for altering a list cell.

The `rplaca` Function

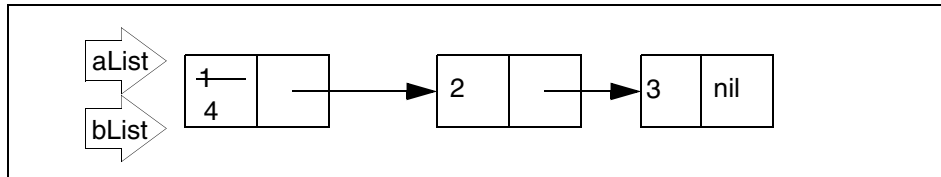
Use the `rplaca` function to replace the first element of a list.

```
aList = '( 1 2 3 ) => ( 1 2 3 )
bList = rplaca( aList 4 ) => ( 4 2 3 )
```

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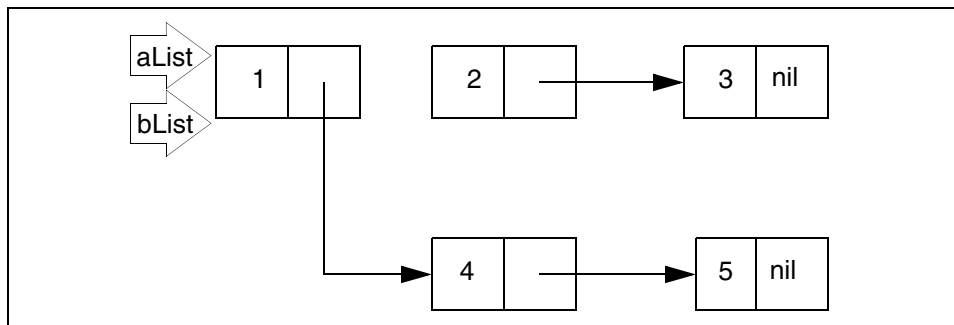
```
aList => ( 4 2 3 )  
eq( aList bList ) => t
```



The rplacd Function

Use the `rplacd` function to replace the tail of a list.

```
aList = '( 1 2 3 ) => ( 1 2 3 )  
bList = rplacd( aList '( 4 5 ) ) => ( 1 4 5 )  
aList => ( 1 4 5 )  
eq( aList bList ) => t
```



Notice that the `rplacd` function returns a list with the desired modifications. An important point to remember about destructive operations is that the modified list is literally the same list in virtual memory as the original list. To verify this fact, use the `eq` function, which returns `t` if both arguments are the same object in virtual memory.

The setf function

The `setf` function assigns a new value to an existing storage location, destroying the value that was previously in that location. Basically, the function expands into an update form that takes the first argument as a *reference location*, evaluates the second argument, and stores the result of evaluating the second argument (new value) in the *reference location*. For example,

```
x = '(a b c d e)  
setf((car x) 42)
```


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Advanced List Operations

displays the following return value:

```
(42 b c d e)
```

A main objective of the `setf` function is to allow you to write your own `setf` expanders/ functions defines as `setf_<expanders>`) so that you can use it with your macros and functions. This means that if you want to write a `setf` function to work with a method, you need to write a `defmethod` form. Also, in SKILL, a generic function must be named by a symbol, so `setf_` is concatenated to the getter function name, such as `setf_GetRandomVal` in the example given below.

```
defgeneric(GetRandomVal (obj))

defclass( Random () ())
defclass( SemiRandom (Random)
  (nextval( @initform nil)))

defmethod( GetRandomVal ((self Random))
  (random))

defmethod( GetRandomVal ((self SemiRandom))
  if( self->nextval
    (progl
      self->nextval
      self->nextval = nil)
    (callNextMethod))
  )

defmethod( setf_GetRandomVal (value (self Random))
  error( "Random objects may not have their random value set"))

defmethod( setf_GetRandomVal ((value fixnum) (self SemiRandom))
  self->nextval = value
  )
```

The list of `setf_` expanders available in SKILL Language are as follows: `setf` helpers: (public functions)

```
setf_car
```

```
setf_cdr
```

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setf_cadr
setf_caar
setf_cdar
setf_cddr
setf_caaar
setf_caadr
setf_cadar
setf_cdaar
setf_caddr
setf_cddar
setf_cdadr
setf_cdddr
setf_last
setf_arrayref
setf_nth
setf_nthelem
setf_nthcdr
setf_xCoord
setf_yCoord
setf_leftEdge
setf_rightEdge
setf_bottomEdge
setf_topEdge
setf_upperRight
setf_lowerLeft
setf_getd
setf_get
setf_getq
setf_getSGq

```
setf_slotValue  
setf_getShellEnvVar
```

Consider an example of using the `setf_cadr` helper function:

```
data = list(1 2 3 4 5)  
setf((cadr data) 42)
```

The return value is:

```
(1 42 3 4 5)
```

Accessing Lists

The following functions are convenient variations and extensions of the `cdr` and `nth` functions introduced in “[Cadence SKILL Language](#)” on page 19.

Selecting an Indexed Element from a List (`nthelem`)

`nthelem` returns an indexed element of a list, assuming a one-based index. Thus `nthelem(1 l_list)` is the same as `car(l_list)`.

```
nthelem( 1 '( a b c ) ) => a  
z = '( 1 2 3 )  
nthelem( 2 z )          => 2
```

Applying `cdr` to a List a Given Number of Times (`nthcdr`)

You supply the iteration count and the list of elements.

```
nthcdr( 3 '( a b c d ) ) => (d)  
z = '( 1 2 3 )  
nthcdr( 2 z )           => ( 3 )
```

Getting the Last List Cell in a List (`last`)

`last` returns the last list cell in a list. The `car` of the last list cell is the last element in the list. The `cdr` of the last list cell is `nil`.

```
last( '(a b c) ) => (c)  
z = '( 1 2 3 )  
last( z )       => (3)
```

Building Lists Efficiently

To build lists efficiently, you must understand how lists are constructed. Using a function like `append` involves searching for the end of a list, which can be unacceptably slow for large lists.

Adding Elements to the Front of a List (`cons`, `xcons`)

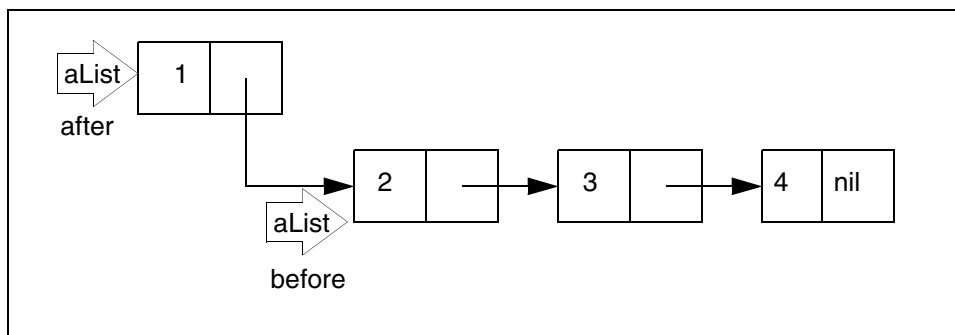
If order is unimportant, the easiest way to build a list is to repeatedly call `cons` to add new elements at the head of the list.

The `cons` function allocates a new list cell consisting of two memory locations. It stores its first argument in the first location and its second argument in the second location.

The `cons` function returns a list whose first element is the one you supplied (1 below) and whose `cdr` is the list you supplied (*aList* below). The expressions

```
aList = '( 2 3 4 )  
aList = cons( 1 aList ) => (1 2 3 4 )
```

allocate the following.



`xcons` accepts the same arguments as the `cons` function, but in reverse order. `xcons` adds an element to the beginning of a list, which can be `nil`.

```
xcons( '( b c ) 'a ) => ( a b c )
```

Building a List with a Given Element (`ncons`)

`ncons` builds a list by adding the `element` you supply to the beginning of an empty list. It is equivalent to `cons( g_element nil )`.

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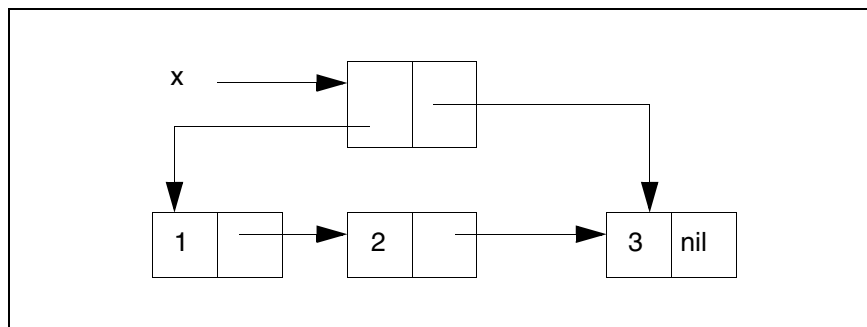
Advanced List Operations

```
ncons( 'a )      => ( a )          ;;; equivalent to cons( 'a nil )
z = '( 1 2 3 )
ncons( z )       => ( ( 1 2 3 ) ) ;;; equivalent to cons( z nil )
```

Adding Elements to the End of a List (tconc)

Because lists are singly linked in only one direction, searching for the end of a list typically requires the traversal of every list cell in the list, which can be slow with a long list containing many list cells. This long traversal poses a problem when you want to build a list by adding elements at the end of a list. If a list must be built by adding new elements at the end of the list, the most efficient way is to use `tconc`.

The `tconc` function creates a list cell (known as a `tconc` structure) whose `car` points to the head of the list being built and whose `cdr` points to the last element of the list.



The `tconc` structure allows subsequent calls to `tconc` to find the end of a list instantly without having to traverse the entire list. For this reason, call `tconc` once to initialize a special list cell and pass this special list cell to subsequent calls on `tconc`. Finally, to obtain the value of the list you have been building, take the `car` of this special list cell. The typical steps required to use `tconc` are as follows:

1. Create the `tconc` structure by calling `tconc` with `nil` as its first argument and the first element of the list being built as the second argument.

```
x = tconc(nil 1 )
```

2. Repeatedly call `tconc` with other elements to be added to the end of the list, each time giving the `tconc` structure as the first argument to `tconc`. There is no need to assign the value returned by `tconc` to a variable because `tconc` modifies the `tconc` structure. For example:

```
tconc(x 2 ), tconc(x 3 ) ...
```

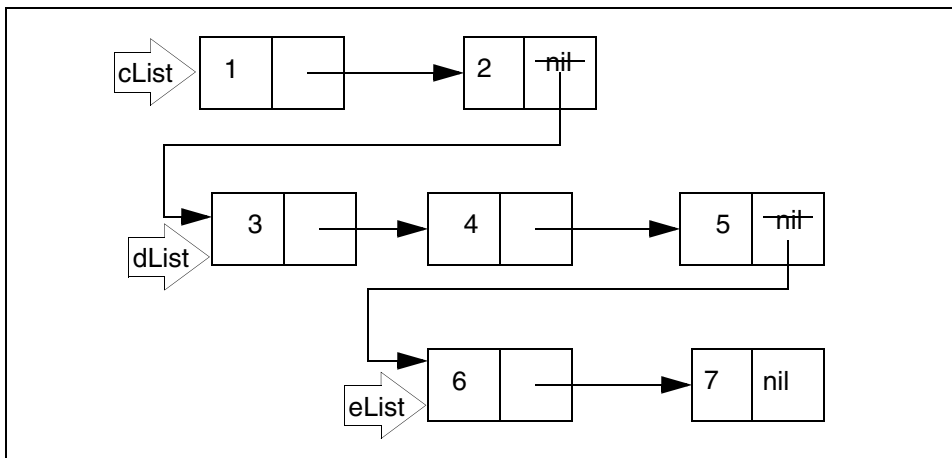
3. After the list has been built, take the `car` of the `tconc` structure to get the value of the list being built. For example:

```
x = car(x)
```

Appending Lists

The nconc Function

Use the `nconc` function to quickly append lists destructively. The `nconc` function takes two or more lists as arguments. Only the last argument list is unaltered.



```

cList = '( 1 2 )
dList = '( 3 4 5 )
eList = '( 6 7 )
nconc( cList dList eList ) => ( 1 2 3 4 5 6 7 )
cList => ( 1 2 3 4 5 6 7 )
dList => ( 3 4 5 6 7 )
eList => ( 6 7 )
  
```

Use the `apply` function and the `nconc` function to append a list of lists.

```

apply( 'nconc '( ( 1 2 ) ( 3 4 5 ) ( 6 7 ) ) )
=> (1 2 3 4 5 6 7 )
  
```

The lconc Function

`lconc` uses a `tconc` structure to efficiently splice lists to the end of lists. The `tconc` structure must initially be created using the `tconc` function. See the example below.

```

x = tconc(nil 1)           ; x is initialized ((1) 1)
lconc(x '(2 3 4))         ; x is now ((1 2 3 4) 4)
lconc(x nil)              ; Nothing is added to x.
lconc(x '(5))             ; x is now ((1 2 3 4 5) 5)
x = car( x )              ; x is now (1 2 3 4 5)
  
```

Reorganizing a List

SKILL provides several functions that reorganize a list. Sometimes the most efficient way to build a list is to reverse it or sort it after you have built it incrementally with the `cons` function. These functions change the sequence of the top-level elements of your list.

Reversing a List

The following is a non-destructive operation.

Reversing the Order of Elements in a List (`reverse`)

`reverse` returns the top-level elements of a list in reverse order.

```
aList = '( 1 2 3 )
aList = reverse( aList ) => ( 3 2 1 )
anotherList = '( 1 2 ( 3 4 5 ) 6 )
reverse( anotherList ) => ( 6 ( 3 4 5 ) 2 1 )
```

Although `reverse( anotherList )` returns the list in reverse order, the value of `anotherList`, the original list, is not modified. It is your responsibility to update any variables that you want to reflect this reversal.

```
anotherList => ( 1 2 ( 3 4 5 ) 6 )
```

Sorting Lists

The following functions are helpful when you must sort lists according to various criteria and locate elements within lists. They are destructive operations.

The `sort` Function

The syntax statement for the `sort` function is

```
sort( l_data u_comparefn ) => l_result
```

`sort` sorts a list of objects (`l_data`) according to the `sort` function (`u_comparefn`) you supply. `u_comparefn( g_x g_y )` returns non-`nil` if `g_x` can precede `g_y` in sorted order and `nil` if `g_y` must precede `g_x`. If `u_comparefn` is `nil`, alphabetical order is used. The algorithm in `sort` is based on recursive merge sort.

```
sort( '(4 3 2 1) 'lessp )      => (1 2 3 4)
sort( '(d b c a) 'alphalessp) => (a b c d)
```

The sortcar Function

`sortcar` is similar to `sort` except that only the `car` of each element in a list is used for comparison by the `sort` function.

```
sortcar( '( (4 four) (3 three) (2 two)) 'lessp )
=> ((2 two) (3 three) (4 four))

sortcar( '( (d 4) (b 2) (c 3) (a 1)) nil )
=> ((a 1) (b 2) (c 3) (d 4))
```

The list is modified in place and no new storage is allocated. Pointers previously pointing to the list might not be pointing at the head of the sorted list.

Searching Lists

SKILL provides several functions for locating elements within a list that satisfy a criterion. The most basic criterion is equality, which in SKILL can mean either `value equality` or `memory address equality`.

The member Function

The `member` function is briefly discussed in [“Cadence SKILL Language”](#) on page 19. It uses the `equal` function as the basis for finding a top-level element in a list. The `member` function

- Returns `nil` if the element is not equal to any top-level element in the list.
- Returns the first tail of the list that starts with the element.

Some examples include

```
member( 3 '( 2 3 4 3 5 )) => (3 4 3 5)
member( 6 '( 2 3 4 3 5 )) => nil
```

The `member` function resembles the `cdr` function in that it internally returns a pointer to a list cell. The `car` of the list cell is equal to the element. You can use the `member` function with the `rplaca` function to destructively substitute one element for the first top-level occurrence of another in a list. For example, find the first occurrence of 3 and replace it with 6:

```
rplaca(
  member( 3 '( 2 3 4 3 5 ))
  6 )
```

SKILL provides a non-destructive `subst` function for substituting an element at all levels of a list. See [“Substituting Elements”](#) on page 196.

The memq Function

The `memq` function is the same as the `member` function except that it uses the `eq` function for finding the element. Because the `eq` function is more efficient than the `equal` function, use `memq` whenever possible based on the nature of the data. For example, if the list to search contains only symbols, then using the `memq` function is more efficient than using the `member` function.

The exists Function

The `exists` function can use an application-specific testing function to locate the first occurrence of an element in a list. The `exists` function generalizes the `member` and `memq` functions, which locate the first occurrence of an element in a list based on equality.

```
exists( x '( 2 4 7 8 ) oddp( x ) ) => ( 7 8 )
exists( x '( 2 4 6 8 ) evenp( x ) ) => ( 2 4 6 8 )
exists( x '(1 2 3 4) (x > 1) )=> (2 3 4)
exists( x '(1 2 3 4) (x > 4) )=> nil
```

The lindex Function

The `lindex` function returns the index number of the given element in a list.

`lindex(l_list g_element)` returns `nil` if the given element is not present in the list.

```
lindex(' (1 2 3 4) 2)  => 2
lindex(' (1 6 4 5) 4)  => 3
lindex(' (1 4 6 7) 3)  => nil
```

Copying Lists

Sometimes it is more efficient to apply a destructive operation to a copy of a list than it is to apply a non-destructive operation. First determine whether a shallow copy of only the top-level elements is sufficient.

The copy Function

`copy` returns a copy of a list. `copy` only duplicates the top-level list cells. All lower-level objects are still shared. You should consider making a copy of any list before using a destructive modification on the list.

```
z = '(1 (2 3) 4)  => (1 (2 3) 4)
x = copy(z)       => (1 (2 3) 4)
equal(z x)        => t
```

z and x have the same value.

```
eq(z x)          => nil
```

z and x are not the same list.

Copying a List Hierarchically

The following function recursively copies a list.

```
procedure( trDeepCopy( arg )
  cond(
    ( !arg nil )
    ( listp( arg );;; argument is a list
      cons(
        trDeepCopy( car( arg ) )
        trDeepCopy( cdr( arg ) )
      )
    )
    ( t arg ) ;;; return the argument itself
  ) ; cond
) ; procedure
```

Filtering Lists

Many list operations can be abstractly considered as making a filtered copy of a list. The filter can be any function that accepts a single argument. If the filter function returns `non-nil`, the element is included in the new list. If the filter function returns `nil`, the element is excluded from the new list.

`setof` makes a filtered copy of the top-level elements of a list, including all elements that satisfy a given criteria. For example, contrast the following two approaches to computing the intersection of two lists.

- One way to proceed is to use the `cons` function as follows:

```
procedure( trIntersect( list1 list2 )
  let( ( result )
    foreach( element1 list1
      when( member( element1 list2 )
        result = cons( element1 result )
      ) ; when
    ) ; foreach
    result
  ) ; let
) ; procedure
```

- The more efficient way is as follows:

```
procedure( trIntersect( list1 list2 )
  setof(
    element list1
```

```

        member( element list2 ) )
    ) ; procedure

```

The criteria is used to decide whether to include each element of the list. The copied element is not transformed.

Removing Elements from a List

SKILL has several functions that remove all top-level occurrences of an element from a list.

The Removal Function

	Non-destructive	Destructive
Uses equal	remove	remd
Uses eq	remq	remdq

Non-Destructive Operations

The remove Function

`remove` returns a copy of an argument with all top-level elements equal to a given SKILL object removed. The `equal` function, which implements the `=` operator, is used to test for equality.

```

aList = '( 1 2 3 4 5 )
remove( 3 aList ) => ( 1 2 4 5 )
aList => ( 1 2 3 4 5 )

```

It is your responsibility to make the appropriate assignment so that `aList` reflects the removal.

```
aList = remove( 3 aList )
```

The element to remove can itself be a list.

```
remove( '( 1 2 ) '( 1 ( 1 2 ) 3 ) ) => ( 1 3 )
```

The remq Function

`remq` returns a copy of an argument list with all top-level elements equal to a given SKILL object removed. The `eq` function is used to test for equality. This function is faster than the `remove` function because the `eq` equality test is faster than the `equal` equality test.

However, the `eq` test is only meaningful for certain data types, such as symbols and lists. The `remq` function is appropriate, for example, when dealing with a list of symbols.

```
remq( 'x ' ( a b x d f x g ) ) => ( a b d f g )
```

The `remq` function does not work on association tables.

Destructive Operations

The `remd` Function

remd removes all top-level elements equal to a given SKILL object from a list. *remd* uses *equal* for comparison. This is a destructive removal.

```
remd( "x" ' ("a" "b" "x" "d" "f")) => ("a" "b" "d" "f")
```

Example

In order to remove the first element from the original list, use the same variable (that holds the original list) to hold the updated list.

```
y = ' ("a" "b" "d" "f")      => ("a" "b" "d" "f")
y=remd("a" y)                => ("b" "d" "f")
y                             => ("b" "d" "f")
```

The `remdq` Function

remdq removes all top-level elements equal to the first argument from a list. *remdq* uses *eq* instead of *equal* for comparison. This is a destructive removal.

```
remdq('x ' (a b x d f x g)) => (a b d f g)
```

Substituting Elements

The `subst` function is a non-destructive operation. It is your responsibility to update any variables that you want to reflect this substitution.

Substituting One Object for Another Object in a List (`subst`)

`subst` returns the result of substituting the `new object` (first argument) for all equal occurrences of the `previous object` (second argument) at all levels in a list.

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```
aList = '( a b c )           => ( a b c )
subst( 'a 'b aList )        => ( a a c )
anotherList = '( a b y ( d y ( e y )))
subst('x 'y anotherList )   => ( a b x ( d x ( e x )))
```

Although `subst('x 'y anotherList )` returns a list reflecting the desired substitutions, the value of `anotherList`, the original list, is not modified.

```
anotherList => ( a b y ( d y ( e y )))
```

Transforming Elements of a Filtered List

Many list operations can be modeled as a filtering pass followed by a transformational pass. For example, suppose you are given a list of integers and you want to build a list of the squares of the odd integers.

Phase I: Filter the odd integers into a list.

You can use the `setof` function here.

```
setof( x '( 1 2 3 4 5 6 ) oddp(x)) => ( 1 3 5 )
```

Phase II: Square each element of the list.

You can use the `mapcar` function together with a function that squares its argument:

```
mapcar(
  lambda( (x) x*x ) ;; square my argument
  '( 1 3 5 )
) => ( 1 9 25 )
```

or use the `foreach` function:

```
foreach( mapcar x '( 1 3 5 ) x*x ) => ( 1 9 25 )
```

The `trListOfSquares` function summarizes this approach.

```
procedure( trListOfSquares( aList )
  let( ( filteredList )
    filteredList =
      setof( element aList oddp( element ) )
    foreach( mapcar element filteredList
      element * element
    ) ; foreach
  ) ; foreach
) ; procedure

trListOfSquares( '( 1 2 3 4 5 6 ) ) => ( 1 9 25 )
```

Validating Lists

A predicate is a function that validates that a single SKILL object satisfies a criterion. SKILL provides many basic predicates. (Refer to “[SKILL Predicates](#)” on page 135 and “[Type Predicates](#)” on page 138.) Predicates return `t` or `nil`.

SKILL provides the `forall` function and the `exists` function so you can check whether all elements or some elements in a list satisfy a criterion. These two functions correspond to quantifiers in mathematical logic.

- `forall` is represented mathematically as $\forall$.
- `exists` is represented mathematically as $\exists$.

The criterion is represented by a SKILL expression with a single argument that you identify as the first argument to the `forall` or `exists` function.

The forall Function

`forall` verifies that an expression remains true for every element in a list. The `forall` function can also be used to verify that an expression remains true for every key/value pair in an association table. (Refer to “[Association Tables](#)” on page 113 for further details.)

```
forall( x '( 2 4 6 8 ) evenp( x ) ) => t
forall( x '( 2 4 7 8 ) evenp( x ) ) => nil
```

The exists Function

`exists` can use an application-specific testing function to locate the first occurrence of an element in a list based on equality. The `exists` function generalizes the `member` and `memq` functions, which locate the first occurrence of an element in a list based on equality.

```
exists( x '( 2 4 7 8 ) oddp( x ) ) => ( 7 8 )
exists( x '( 2 4 6 8 ) evenp( x ) ) => ( 2 4 6 8 )
exists( x '(1 2 3 4) (x > 1) )=> (2 3 4)
exists( x '(1 2 3 4) (x > 4) )=> nil
```

Using Mapping Functions to Traverse Lists

SKILL provides a family of powerful functions for iterating over lists. For historical reasons, these are called mapping functions. The mapping functions are `map`, `mapc`, `mapcar`, `mapcan`, `maplist`, and `mapcon`.

All `map*` functions have the same arguments.

- A function, which must take a single argument
- A list

Using lambda with the map* Functions

It is often convenient to use the `lambda` construct to define a nameless function to be used as the first argument.

For example:

```
mapcar(  
  lambda( ( x ) list( x x**2 ) ) ;;; return pair of x x**x  
  '(0 1 2 3 )  
=> ( (0 0) (1 1) (2 4) (3 9) )
```

Refer to [“Syntax Functions for Defining Functions”](#) on page 73 for further details on the `lambda` construct.

Using the map* Functions with the foreach Function

Alternatively, you can use each mapping function as an option to the `foreach` function. Often, using the `foreach` function results in more understandable code.

For example, the following are equivalent.

```
foreach( mapcar x '( 0 1 2 3 )  
  list( x x**2 ) ;;; build 2 element list of a x and x*x  
)  
=> ( (0 0) (1 1) (2 4) (3 9) )  
  
mapcar(  
  lambda( ( x ) list( x x**2 ) ) ;;; return pair of x x**x  
  '(0 1 2 3 )
```

The relationship between the two usages of the mapping functions is tight. When you use the `foreach` function, SKILL incorporates the expressions within the `foreach` body in a `lambda` function as illustrated below. This `lambda` function is passed to the mapping function. For example, the following are equivalent:

```
foreach( mapcar x aList  
  exp1()  
  exp2()  
  exp3()  
)  
  
mapcar(  
  lambda( ( x )  
    exp1()  
    exp2()  
    exp3()
```

```
    )  
    aList  
  )
```

The following descriptions illustrate both approaches to using each mapping function.

The mapc Function

The `mapc` function is the default mapping function used by the `foreach` macro. When used with the `foreach` function

- The `foreach` function iterates over each element of the argument list.
- At each iteration, the current element is available in the loop variable of the `foreach`.
- The `foreach` function returns the original argument list as a result.

For example:

```
foreach( mapc x '(1 2 3 4 5)  
        println(x)  
)  
mapc(  
    lambda( ( x ) println( x ) )  
    '( 1 2 3 4 5 )  
)
```

displays the following:

```
1  
2  
3  
4  
5
```

The return value is `(1 2 3 4 5)`.

The map Function

The `map` function is useful for processing each list cell because it uses `cdr` to step down the argument list. When used with the `foreach` function

- The `foreach` function iterates over each list cell of its argument.
- At each iteration, the current list cell is available in the loop variable of the `foreach`.
- The `foreach` function returns the original argument list as a result.

For example, suppose you want to make substitutions at the top-level of a list using a look-up table such as the following:

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```
trLookupList = '(
  ( 1 "one" )
  ( 2 "two" )
  ( 3 "three" ))
```

By using the `map` function, you gain access to each successive list cell, allowing you to use the `rplaca` function to make the desired substitution. Notice that the lookup list is an association list so that you can use the `assoc` function to retrieve the appropriate substitution.

```
assoc( 2 trLookupList ) => ( 2 "two" )
assoc( 5 trLookupList ) => nil

procedure( trTopLevelSubst( aList aLookupList )
  let( ( currentElement substValue )
    foreach( map listCell aList
      currentElement = car( listCell )
      substValue = cadr(
        assoc( currentElement aLookupList ) )
      when( substValue
        rplaca( listCell substValue )
      ) ; when
    ) ; foreach
    aList
  ) ; let
  ) ; procedure

trList = '( 1 4 5 3 3 )
trTopLevelSubst( trList trLookupList ) =>
  ("one" 4 5 "three" "three")
```

The mapcar Function

The `mapcar` function is useful for building a list whose elements can be derived one-for-one from the elements of an original list. When used with the `foreach` function

- The `foreach` function iterates over each element of the argument list.
- At each iteration, the current element is available in the loop variable of the `foreach`.
- Each iteration produces a result value.
- These values are returned in a list as the result of the function.

For example:

```
foreach(mapcar x '(1 2 3 4 5)
  println(x)
  x*3
)
```

displays the following:

```
1
2
3
```

4
5

The return value is (3 6 9 12 15) .

The maplist Function

The `maplist` function is useful for processing each list cell because it uses `cdr` to step down the argument list. Like the `mapcar` function, it returns the list of results that it collects for each iteration. When used with the `foreach` function

- The `foreach` function iterates over each list cell of the argument list.
- At each iteration, the current list cell is available in the loop variable of the `foreach`.
- Each iteration produces a result value, and these values are returned in a list as the result of the `foreach` function.

For example, consider the substitution example illustrating the `map` function. In addition to making the substitutions, suppose that the function must display a message giving the number of substitutions made.

By using `maplist` instead of `map` you can build a list of 1s and 0s reflecting the substitutions made. Use the `apply` function with `plus` to add up the numbers to produce the desired count.

```
procedure( trTopLevelSubst( aList aLookUpList )
  let( ( currentElement substValue substCountList )
    substCountList = foreach( maplist listCell aList
      currentElement = car( listCell )
      substValue = cadr(
        assoc( currentElement aLookUpList ) )
      if( substValue
        then
          rplaca( listCell substValue )
          1
        else
          0
        ) ; if
      ) ; foreach
    printf( "There were %d substitutions\n"

    apply( 'plus substCountList )
  )
  aList
) ; let
) ; procedure
trList = '( 1 4 5 3 3 )
trTopLevelSubst( trList trLookUpList ) =>
  ("one" 4 5 "three" "three")
There were 3 substitutions
```

The mapcon Function

Like the `maplist` function, the `mapcon` function is useful for processing each list cell because it uses `cdr` to step down the argument list. It returns a concatenated list of results that it collects after each iteration. When used with the `foreach` function:

- The `foreach` function iterates over each list cell of the argument list.
- At each iteration, the current list cell is available in the loop variable of the `foreach`.
- Each iteration produces a result value, and these values are returned as a concatenated list as the result of the `foreach` function.

An example of the `mapcon` function used with `foreach` is as follows:

```
foreach(mapcon x '(1 2 3 4) list(x))
```

The return value is:

```
((1 2 3 4)
 (2 3 4)
 (3 4)
 (4)
)
```

You can use the `mapcon` function with more than one list argument. In this case, the supplied function argument is applied to successive sublists of the lists. This means that the supplied function is first applied to the lists themselves, and then to the `cdr` of each list, and then to the `cdr` of the `cdr` of each list, and so on.

A `mapcon` function example with a lambda function with one list argument:

```
mapcon((lambda (x)
  (printf "x = %L\n" x)
  (list (car x) (add1 (car x))))) '(1 2 3 4))
```

displays the following

```
x = (1 2 3 4)
x = (2 3 4)
x = (3 4)
x = (4)
```

The return value is: (1 2 2 3 3 4 4 5)

A `mapcon` function example with a lambda function with two list arguments:

```
mapcon((lambda (x y) (printf "x = %L y = %L\n" x y)
  (list (car x) (add1 (car y)))))
```

```
'(1 2 3 4)
'(4 3 2 1)
)
```

displays the following

```
x = (1 2 3 4) y = (4 3 2 1)
x = (2 3 4) y = (3 2 1)
x = (3 4) y = (2 1)
x = (4) y = (1)
```

The return value is: (1 5 2 4 3 3 4 2)

The mapcan Function

Like the `mapcar` function, the `mapcan` function is useful for building a list by transforming the elements of the original list. However, instead of collecting the intermediate results into a list as `mapcar` does, `mapcan` appends them. The intermediate results must be lists. Notice that the resulting list need not be in 1-1 correspondence with the original list.

The `mapcan` function iterates over each element of the argument list. When used with the `foreach` function

- At each iteration, the current element is available in the loop variable of the `foreach`.
- Each iteration produces a result value, which must be a list. These lists are then concatenated by the destructive modification function `nconc`, and the new list is returned as the result of the function as a whole.

For example, flattening a list of lists can be done with:

```
foreach( mapcan x '( ( 1 2 ) ( 3 4 5 ) ( 6 7 ) )
          x
        ) => ( 1 2 3 4 5 6 7 )
mapcan(
  lambda( ( x ) x )
  '( ( 1 2 ) ( 3 4 5 ) ( 6 7 ) )
) => ( 1 2 3 4 5 6 7 )
```

For another example, the following builds a list of squares of the odd integers in the list (1 2 3 4 5 6).

```
foreach( mapcan x '( 1 2 3 4 5 6 )
          when( oddp( x )
                list( x )
              )
        ) => ( 1 3 5 )
mapcan(
  lambda( ( x ) when( oddp( x ) list( x ) ) )
)
```

```
'( 1 2 3 4 5 6 )  
) => ( 1 3 5 )
```

The mapinto Function

Like the `mapcar` function, the `mapinto` function is useful for building a list by transforming the elements of the original list. However, instead of allocating a new list, `mapinto` reuses the preallocated list. The first argument of `mapinto` is a sequence that receives the results of the mapping. It is because of this difference in behavior with the other `map*` functions, `mapinto` cannot be used along with `foreach`.

For example:

```
procedure ( myFunc(x)  
x  
) ;procedure  
mapinto( '("a" "b" "c") 'myFunc '(1 2) )  
=> (1 2 "c")  
mapinto('("a" "b" "c") 'plus '(1 2) '(3 4))  
=> (4 6 "c")
```

Summarizing the List Traversal Operations

The table below summarizes how the mapping functions work. The term `Function` refers to the function you pass to the mapping function. Each table entry is a mapping function that behaves according to the row and column headings. In one case, there is no such mapping function.

Summary of the Mapping Functions

Mapping Function Return Value	Function is Passed Successive Elements	Function is Passed Successive List Cells
Ignores the result of each iteration. Returns the original list.	<code>mapc</code> (default option)	<code>map</code>
Collects the result of each iteration. Returns the list of results.	<code>mapcar</code>	<code>maplist</code>
Collects the result of each iteration. Uses <i>nconc</i> to append the result of each iteration.	<code>mapcan</code>	No such function

List Traversal Case Studies

The most useful mapping functions are `mapc`, `mapcar`, and `mapcan`. A good understanding of these functions simplifies most list handling operations in SKILL.

Handling a List of Strings

Suppose you need to calculate the length of each string in a list of strings so that the lengths are returned in a new list. That is, the function should perform the following transformation:

```
( "sam" "francis" "nick" ) => ( 3 7 4 )
```

Someone not familiar with the `mapcar` option to `foreach` might code this as follows:

```
procedure( string_lengths( stringList )
  let( (result)
    foreach( element stringList
      result = cons( strlen(element) result)
    ) ; foreach
    reverse( result )
  ) ; let
) ; procedure
```

Using the `mapcar` function allows this code to be written as

```
procedure(string_lengths(stringList)
  foreach(mapcar element stringList
    strlen(element)
  ) ; foreach
) ; procedure
```

In fact, the `foreach` function is implemented as a macro that expands to one of the mapping functions, so the same example can be written using the mapping function directly as follows:

```
procedure(string_lengths(stringList)
  mapcar( 'strlen stringList)
) ; procedure
```

Making Every List Element into a Sublist

You can perform this operation with either version of the `trMakeSublists` function.

```
procedure( trMakeSublists( aList )
  foreach( mapcar element aList
    list( element )
  ) ; foreach
) ; procedure

procedure( trMakeSublists( aList )
  mapcar(
    'ncons
    alist
    ;;; return argument in a list
  )
)
```

```
)  
); procedure  
trMakeSublists( '(1 2 3)) => ( ( 1 ) ( 2 ) ( 3 ) )
```

Using mapcan for List Flattening

The `mapcan` function is useful when a new list is derived from an old one, and each member of the old list produces a number of members in the new list. (Using `mapcar` allows only a one-to-one mapping). One application of `mapcan` is in list flattening. Suppose we have a list that contains lists of numbers:

```
x = '( (1 2 3) (4) (5 6 7 8) () (9) )
```

This list can be flattened using the following procedure:

```
procedure(flatten(numberList)  
  foreach( mapcan element numberList  
    element  
  ); foreach  
); procedure  
  
flatten(x) => ( 1 2 3 4 5 6 7 8 9 )
```

This function concatenates all sublists of the argument. Remember that `nconc` is a destructive modification function, so variable `x` no longer holds useful information after the call.

To preserve the value of `x`, each sublist should be copied within the `foreach` function to produce a new sublist that can be harmlessly modified:

```
procedure( flatten( numberList )  
  foreach( mapcan element numberList  
    copy(element)  
  ); foreach  
); procedure  
  
x = '( (1 2 3) (4) (5 6 7 8) () (9) )  
flatten(x) => ( 1 2 3 4 5 6 7 8 9 )  
x => ((1 2 3) (4) (5 6 7 8) nil (9))
```

Flattening a List with Many Levels

By using a type predicate and a recursive step, this procedure can be modified to flatten a list with many levels:

```
procedure(flatten(numberList)  
  foreach(mapcan element numberList  
    if( listp( element )  
      flatten(copy(element)) ;; then  
      ncons(element)  
    ); if  
  ); foreach  
); procedure
```

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```
x = '((1) ((2 (3) 4 ((5)) () 6) ((7 8 ()))) 9)
flatten(x)      => (1 2 3 4 5 6 7 8 9)
x              => ((1) ((2 (3) 4 ((5)) nil 6) ((7 8 nil))) 9)
```

The body of the `foreach` first checks the type of the current list member.

- If the member is a list, the result is a list obtained by flattening a copy of it.
- If the member is not a list, it cannot be flattened further, and the result is a `one-element list` containing the member.

Remember that when using `mapcan`, the result of each iteration must be a list so that the results can be concatenated using `nconc`.

Manipulating an Association List

Mapping functions can be powerful when used together. For example, suppose there is a database of names and extension numbers:

```
thePhoneDB = '(
  ("eric" 2345 6472 8857)
  ("sam" 4563 8857)
  ("julie" 7765 9097 5654 6653)
  ("francis")
  ("pat" 5527)
)
```

The database is stored as a list with one entry for each person. An entry consists of a list of the person's name followed by a list of extensions at which the person can be reached. This type of list is known as an association list. Some people can be reached at several extensions, and some people at none at all. An automated dialing system has been introduced that accepts only name-number pairs: in other words, it requires data in the following format:

```
(("eric" 2345)
 ("eric" 6472)
 ("eric" 8857)
 ("sam" 4563)
 . . . . .)
```

How can the information be transformed from one format to another? Each `person` entry in the original database can produce several entries in the new database, so `mapcan` must be used to traverse person entries. Each `number` in the old database produces a single entry in the new database, so `mapcar` can be used to traverse the numbers.

From this information, a function can be written to translate the database:

```
procedure( translate( phoneDB )
  let( (name)
    foreach( mapcan personEntry phoneDB
      name = car( personEntry )
      foreach( mapcar number cdr( personEntry )
        list( name number )
```



```
        ) ; foreach
      ) ; foreach
    ) ; let
  ) ; procedure
```

To show that this works, consider the innermost `foreach` loop first. This loop is called for each person entry in the database. Suppose that the entry `( "sam" 4563 8857 )` is being processed. In this case, `name` is set to `"sam"` and the `foreach` iterates over the list of numbers `(4563 8857)`.

Because this iteration is a `mapcar`, the result of the `foreach` is a new list with one entry for each number in the old list. Each entry in this new list is a name-number pair constructed by the expression `list(name number)`. Hence, the result of this `foreach` is the list

```
(( "sam" 4563) ( "sam" 8857))
```

The outermost `foreach` concatenates these lists of name-number pairs: it works like the first example of `flatten` given previously. Notice that there is no need to copy the elements before concatenating them (as was the case in `flatten`) because they have just been created.

Using the `exists` Function to Avoid Explicit List Traversal

SKILL provides a large number of list iteration functions. The most basic of these is the `foreach` function, which traverses all elements of a list. This traversal is useful, but often what is required is to traverse just part of a list, to check for a certain condition. In this situation, correct use of the `exists`, `forall`, and `setof` functions can greatly improve your code. Usually, the alternatives are less efficient.

Consider writing a simple function that iterates over all integers in a list and checks whether there is any even integer. There are several approaches.

One approach is to use a `while` loop that explicitly tests a Boolean variable `found`.

```
procedure( contains_even( list )
  let( (found)
    while( list && !found
      when( evenp( car(list) )
        found = t
      )
      list = cdr(list)
    ) /* end while */
    /* Return whether found. */
    found
  ) /* end let */
) /* end contains_pass */

contains_even( '(1 2 3 4) ) => t
contains_even( '(1 7 3 9) ) => nil
contains_even( nil ) => nil
```

Another approach is to jump out of the `while` loop:

```
procedure(contains_even( list )
  prog( ()
    while(list
      if( evenp( car(list) )
        return(t)
      )
      list = cdr(list)
    ) /* end while */
    return(nil)
  ) /* end prog */
) /* end contains_pass */
```

This approach might appear to be better because there is no need for the local variable.

A third approach involves jumping out of a `foreach` loop:

```
procedure(contains_even(list)
  prog( ()
    foreach(element list
      if( evenp(element)
        return(t)
      )
    ) /* end foreach */
    return(nil)
  ) /* end prog */
) /* end contains_pass */
```

But this approach still requires the `prog` to allow the jump out of the `foreach` loop once the element has been found.

A much simpler way of writing this procedure is as follows:

```
procedure(contains_even(list)
  when( exists( element list evenp( element ) )
    t
  ) /* when */
) /* end contains_pass */
```

This approach has neither the `prog` nor any local variables, and is just as intuitive. The `exists` function terminates as soon as a matching list member has been found, so this example is just as efficient in this respect as the previous ones (which had to use `return` to achieve this result). The result of an `exists` is `nil` if no matching entry is found. Otherwise the result is the sublist of the argument that contains the matching member as its head. For example:

```
exists( x '( 1 2 3 4 ) evenp( x ) ) => ( 2 3 4 )
```

Because SKILL considers `nil` to be equivalent to false and non-`nil` to be equivalent to true, the result of an `exists` can be treated as a boolean if desired. If the `contains_even` function was defined to return either `nil` or non-`nil` (rather than `t`), it could be further simplified to

```
procedure( contains_even( list )
    exists( element list evenp( element ))
) /* end contains_pass */
```

As with all the other iterative functions, there is no need to declare the loop variable for `exists` because the loop variable is local to the function and automatically declared.

Commenting List Traversal Code

The examples above demonstrate the power of these mapping functions. Usually, wherever a `foreach` loop is used to build a list, a mapping function can usually be used instead. Because the mapping functions collect all the results and apply a single list building operation, they are faster than the equivalent iterative function and often look more succinct.

However, the mapping functions all look similar, and it is often difficult to see what a piece of code using a mapping function is trying to do. For this reason whenever a mapping function is used, you should write a comment detailing what the function is expected to return.

Cadence SKILL Language User Guide

Advanced List Operations

Advanced Topics

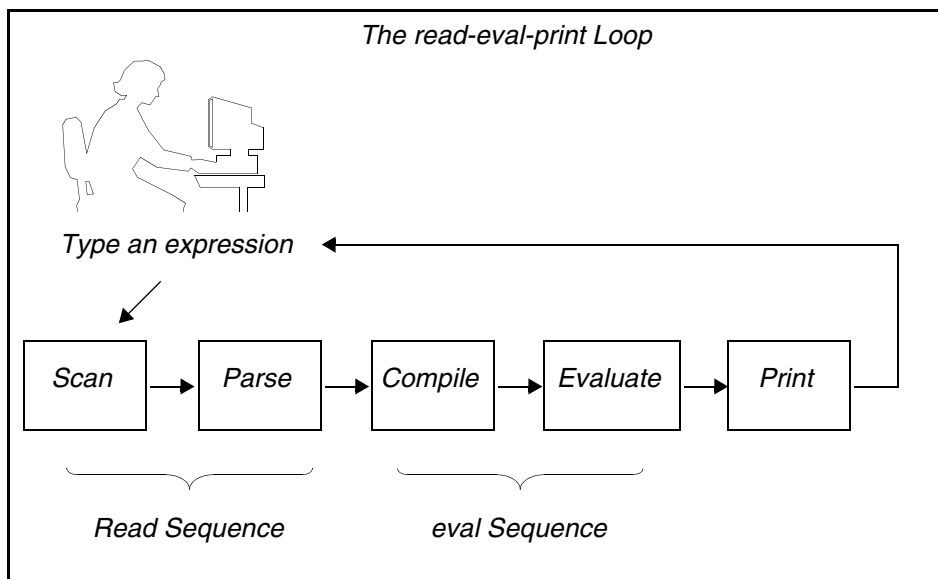
You can find information about advanced topics in the following sections:

- [Cadence SKILL Language Architecture and Implementation](#) on page 214
- [SKILL Namespace](#) on page 215
- [Evaluation](#) on page 219
- [Function Objects](#) on page 221
- [Macros](#) on page 224
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Cadence SKILL Language Architecture and Implementation

When you first encounter the Cadence® SKILL language in an application and begin typing expressions to evaluate, you are encountering what is known as the read-eval-print loop.

The expression you type in is first “read” and converted into a format that can be evaluated. The evaluator then does an “eval” on the output from the “read.” The result of the “eval” is a SKILL data value, which is then printed. The same sequence is repeated when other expressions are entered.



The “read” in this case performs the following tasks.

- The expression is parsed, resulting in the generation of a parse tree.
- The parse tree is then compiled into a body of code, known as a function object, made of a set of instructions that, when executed, results in the desired effect from the expression.

The instructions generated are not those of a particular machine architecture. They are the instructions of an abstract machine. Usually, this set of instructions might be referred to as byte-code or p-code. (p-code was the target instruction set of some of the early Pascal compilers and lent the name to this technique.)

The evaluator executes the byte-code generated by the compiler. In a sense, the evaluator emulates in software the operations of a hardware CPU. This technique of using an abstract

instruction set to be executed by an engine written in software has several advantages for the implementation of an extension language.

- This technique lends itself well to an interpreter-based implementation.
- This technique offers faster performance than direct source interpretation.
- Once SKILL code is compiled into contexts (refer to [Delivering Products](#) on page 239) the context files are faster to load than original source code and are portable from one machine to another.

That is, if the context file is generated on a machine with architecture A from vendor X, it can be copied onto a machine with architecture B from vendor Y and SKILL will load the file without the need for recompilation or translation.

SKILL Namespace

Need for a SKILL Namespace

A SKILL programmer often uses the same name for different purposes when working on a large project with different programmers. Usually this problem is solved by using naming conventions or adding prefixes to function names, such as `leCreateRect`, `dbCreateRect`, `rodCreateRect`, and `geCreateRect`. SKILL provides a language mechanism called *namespace* to separate these symbols, so that a symbol with the same name can exist in several namespaces.

SKILL provides several namespace functions that you can use. For example, you can use namespace functions to create a new namespace, associate symbols with the new or an existing namespace, add or remove symbols to and from a namespace, and also use shadow functions to resolve any name conflicts between symbols within a namespace.

Note: By default all warnings related to SKILL namespaces are suppressed.

Default Namespace

In SKILL, the default namespace is the namespace that is opened at the start of the SKILL interpreter. The symbols from the default namespace can be referenced with their full names, such as `IL::<symbol_name>` or just by their names such as `<symbol_name>` since there is no way to change the current namespace.

Working with a Namespace

Creating a Namespace

You start working with a namespace when you first create a new namespace using the `makeNamespace` function. For example, the following code helps you create an empty namespace, `myNS`:

```
makeNamespace("myNs")
```

You can use the `findNamespace` function to check if a given namespace already exists.

Associating Symbols to a Namespace

Next, you associate new symbols to the namespace (or an existing namespace) by using any of the following methods:

- Use the access operator `:::`
- Use the `addToNamespace` function

Some examples are given below:

```
myNs:::x = 0
```

```
;Creates a new symbol x in the namespace myNs and associates a numerical value 0 to it
```

```
defun( myNs:::y (a) list(a a))
```

```
;Creates a new symbol y in the namespace myNs and associates a function value to it
```

```
(addToNamespace "A" ("a" "b" "c"))
```

```
;Adds symbols a, b, and c to the namespace A
```

Using Export and Import Lists of a Namespace

Finally, to be able to use the symbols associated with your own or other namespaces, you must add them to an export list. An export list is a list of symbols to be imported into the default ("IL") namespace when the `useNamespace()` function is called. When a new namespace is created, its export list is empty. You use the `addToExportList` and the `removeFromExportList` functions to add or remove symbols from the export list of a namespace.

Cadence SKILL Language User Guide

Advanced Topics

You can use the symbols that have been added to the export list by using the "::" and ":::" operators, such as <NAMESPACE>::<SYMBOL>. Consider the following example:

```
makeNamespace("A")
addToExportList(' (A:::abcd) )
defun(A::abcd ()
;;defun a function "abcd" in the namespace:"A"
. . .
. . .
printf("calling A::abcd\n")
)
useNamespace("A") ;; now symbol A::abcd is imported into the default namespace
abcd()
=> "calling A::abcd\n"
getSymbolNamespace('abcd)
=> ns@A ;; this symbol belongs to the namespace:"A"
```

To use symbols that are not available in the export list, you can use only the ":::" operator.

Consider the following examples.

```
myNs:::y(myNs:::x)
;Uses a private function y from the namespace myNs, use the following code

makeNamespace("A")
A::a1
A:::a1
;A::a1 returns an error because a1 is not in the export list of namespace A,
whereas, A:::a1 enables you to access the symbol a1

addToExportList(' (A:::a1) )
A::a1
;Allows you to access a1 from namespace A
```

Note: The `addToNamespace` function can also be used to add symbols to the export list of a namespace. Therefore, this function is suitable for converting the existing code into one that adopts a namespace without the need to modify all the occurrences of that symbol in the code.

Symbols from other namespaces can be imported to the default (or current) namespace by using any of the following methods:

- Implicitly—from the export list by using the `useNamespace` function

In the following example, function `y` can be accessed by its name after the `myNs` namespace is used.

```
useNamespace("myNs" )
=>t
y(6)
=>(6 6)
```

■ Explicitly—by using the `importSymbol` or `shadowImport` functions

In the following example, the explicit import feature is used to gain access to the symbol `x`

```
importSymbol(' (myNs:::x) )
=>t
x = y(x)
=>(0 0)
(eq 'x 'myNs:::x)
=>t
```

To be able to access the imported symbols, you can use the symbol name (without using the `:::` or `:::` operator).

The `unuseNamespace` function is used to remove imported symbols from a namespace.

Resolving Symbol Name Conflicts

Importing operations (that is, using the `importSymbol` or `useNamespace` functions) on a namespace can cause symbol name conflicts. For example, a name conflict can occur if a namespace contains a symbol with the same name as a symbol in the default namespace. Such conflicts can be resolved or avoided. Unhandled name conflicts cause an error and do not allow importing of any symbols.

To resolve a name conflict, you can use the `shadow` or `shadowImport` functions. Using these functions you can determine the symbols that should be used in case of a name conflict.

The `shadow` function is used to protect symbols in the given namespace. This means that the symbols that were *shadowed* cannot be overridden by the import operation. On the other hand, the `shadowImport` function shadows the symbol from the importing namespace and disregards any name conflict. This means that if a symbol of the same name is present then it is removed. See the following example:

```
makeNamespace("ns1")
=>t
ns1:::x = 9; the symbol x is assigned a value
=>9
```

```
importSymbol(' (ns1:::x) )
*error* the symbol 'x' is already exists
shadowImport(' (ns1:::x) ) ;; it takes a list of symbols
=>t
(eq 'ns1:::x 'x)
=>t
```

Nesting Namespaces

Currently, nested namespaces are not permitted.

Note: Procedural interfaces need to be available to create a namespace from a given textual name.

For more information about SKILL namespace functions, see Chapter 15, “[Namespace Functions](#)” in the *SKILL Language Reference Guide*.

Evaluation

SKILL provides functions that invoke the evaluator to execute a SKILL expression. You can therefore store programs as data to be subsequently executed. You can dynamically create, modify, or selectively evaluate function definitions and expressions.

Evaluating an Expression (eval)

`eval` accepts any SKILL expression as an argument. `eval` evaluates an argument and returns its value.

```
eval( '( plus 2 3 ) )=> 5
```

Evaluates the expression `plus(2 3)`.

```
x = 5
eval( 'x )=> 5
```

Evaluates the symbol `x` and returns the value of symbol `x`.

```
eval( list( 'max 2 1 ) ) => 2
```

Evaluates the expression `max(2 1)`.

Getting the Value of a Symbol (`symeval`)

`symeval` returns the value of a symbol. `symeval` is slightly more efficient than `eval` and can be used in place of `eval` when you are sure that the argument being evaluated is indeed a symbol.

```
x = 5
symeval( 'x )=> 5
y = 'unbound
symeval( 'y )=> unbound
```

Returns `unbound` if the symbol is unbound.

Use the `symeval` function to evaluate symbols you encounter in lists. For example, the following `foreach` loop returns `aList` with the symbols replaced by their values.

```
a = 1
b = 2
aList = '( a b 3 4 )
anotherList = foreach( mapcar element aList
    if( symbolp( element )
        then symeval( element )
        else element
    ) ; if
)
=> ( 1 2 3 4 )
```

Applying a Function to an Argument List (`apply`)

`apply` is a function that takes two or more arguments. The first argument must be either a symbol signifying the name of a function or a function object. (Refer to “[Declaring a Function Object \(lambda\)](#)” on page 223.) The rest of the arguments to `apply` are passed as arguments to the function.

`apply` calls the function given as the first argument, passing it the rest of the arguments. `apply` is flexible as to how it takes the arguments to pass to the function. For example, all the calls below have the same effect, that of applying `plus` to the numbers 1, 2, 3, 4, and 5:

```
apply('plus '(1 2 3 4 5) )
=> 15
apply('plus 1 2 3 '(4 5) )
apply('plus 1 2 3 4 5 nil)
=> 15
```

The last argument to `apply` must always be a list.

If the function is a macro, `apply` evaluates it only once, that is, `apply` expands the macro and returns the expanded form, but does not evaluate the expanded form again (as `eval` does).

```
apply('plus (list 1 2) )      ; Apply plus to its arguments.
=> 3

defmacro( sum (@rest nums) `(plus ,@nums)) ; Define a macro.
=> sum

apply('sum '(sum 1 2))
=> (1 + 2)                      ; Returns expanded macro.

eval('(sum 1 2))
=> 3
```

Read-time Evaluation

SKILL provides the `read-time eval` operator `#.` for evaluating expressions at read-time. When this operator is encountered, the expression following the dot is evaluated and the result of this evaluation replaces the `#.` For example, in the code below `get_filename(piport)` is evaluated at read-time and the results are inserted when the entire expression is compiled:

```
procedure(printFileThisWasFrom()
  printf( "The file is:%s\n" #.get_filename(piport) )
)
```

If this function was defined within a file `"code.il"` then after the read-time evaluation, the expression would expand to the following, before being passed to the SKILL evaluator:

```
procedure(printFileThisWasFrom()
  printf( "The file is:%s\n" "code.il" )
)
```

Function Objects

When you use the procedure function to define a function in SKILL, the byte-code compiler generates a block of code known as a function object and places that object on the function property of a symbol.

Subsequently, when SKILL encounters the symbol in a function call, the function object is retrieved and the evaluator executes the instructions.

Function objects can be used in assignment statements and passed as arguments to functions such as `sort` and `mapcar`.

SKILL provides several functions for manipulating function objects.

Retrieving the Function Object for a Symbol (getd)

You can use the `getd` function to retrieve the function object that the procedure function associates with a symbol.

For example:

```
procedure( trAdd( x y )
    printf( "Adding %d and %d ... %d \n" x y x+y )
    x+y
)
=> trAdd
getd( 'trAdd ) => funobj:0x1814bc0
```

If there is no associated function object, `getd` returns `nil`. The following table shows several other possible return values.

The `getd` Function

Function	Return Value	Explanation
<code>trAdd</code>	<code>funobj:0x1814bc0</code>	Application SKILL function.
<code>edit</code>	<code>t</code>	Read-protected SKILL function. A function is read protected when it is loaded from a context or from an encrypted file.
<code>max</code>	<code>lambda:0xf6f25c</code>	Built-in <code>lambda</code> function.
<code>breakpt</code>	<code>nlambda:0xf7a784</code>	Built-in <code>nlambda</code> function.

Assigning a New Function Binding (putd)

The `putd` function binds a function object to a symbol. You can undefine a function by setting its function binding to `nil`. You cannot change the function binding of a write-protected function using `putd`.

For example, you can copy a function definition into another symbol as follows:

```
putd( 'mySqrt getd( 'sqrt ))=> lambda:0x108b8
    Assigns the function mySqrt the same definition as sqrt.

putd( 'newFun
    lambda( ( x y ) x + y )
)
=> funobj:0x17e0b3c

newFun( 5 6 )
=> 11
```

Assigns the symbol `newFun` a function definition that adds its two arguments.

Declaring a Function Object (lambda)

The word `lambda` in SKILL is inherited from Lisp, which in turn inherits it from `lambda calculus`, a mathematical compute engine on which Lisp is based.

The `lambda` function builds a function object. The arguments to the `lambda` function are

- The formal arguments
- The SKILL expressions that make up the function body (these expressions are evaluated when the function object is passed to the `apply` function or the `funcall` function)

Unlike the procedure function, the `lambda` function does not associate the function object with any particular symbol. For example:

```
(lambda (x y) (sqrt (x*x + y*y)))
```

defines an unnamed function capable of computing the length of the diagonal side of a right-angled triangle.

Evaluating a Function Object

Unnamed or anonymous functions are useful in various situations. For example, mapping functions such as `mapcar` require a function as the first argument. You can pass either a symbol or the function object itself.

```
mapcar( 'get_pname '( sin cos tan ))
=> ("sin" "cos" "tan")
mapcar( lambda(( x ) strlen( get_pname(x)) )
        '( sin cos tan ))
=> ( 3 3 3 )
```

A `quote` before a `lambda` construct is not needed. In fact, a `quote` before a `lambda` construct used as a function is slower than one without a `quote` because the construct is compiled every time before it is called. That is, the `quote` prevents the `lambda` construct from being compiled into a function object when the code is loaded. You can save function objects in data structures. For example:

```
var = (lambda (x y) x + y)
=> funobj:0x1eb038
```

The result is a function object stored in the variable `var`. Function objects are first class objects. That is, you can use function objects just like an instance of any other type to pass as an argument to other functions or to assign as a value to variables. You can also use function objects with `apply` or `funcall`. For example:

```
apply(var '(2 8))  
=> 10  
funcall(var 2 8)  
=> 10
```

Efficiently Storing Programs as Data

Whenever possible, store SKILL programs as function objects instead of text strings. Function objects are more efficient because calls to `eval`, `errset`, `evalstring`, or `errsetstring` require the compiler and generate garbage parsing the text strings. On the other hand, unquoted `lambda` expressions are compiled once. Use `apply` or `funcall` to do the evaluation.

Converting Strings to Function Objects (`stringToFunction`)

To convert an expression represented as a string into a function object with zero arguments, use `stringToFunction`. For example:

```
f = stringToFunction("1+2") => funobj:0x220038  
apply(f nil) => 3
```

To convert an expression represented as a list, you can construct a list with `lambda` and `eval` it. Make sure you account for any parameters:

```
expr = '(x + y)  
f = eval( '( lambda ( x y ) ,expr ) ) => funobj:0x33ab00  
apply( f '( 5 6 ) ) => 11
```

You can always construct the expression as a `lambda` construct at the outset to avoid an unnecessary call to `eval`.

Macros

Macros in SKILL are different from macros in C.

- In C, macros are essentially syntactic substitutions of the body of the macro for the call to the macro.
- A macro function allows you to adapt the normal SKILL function call syntax to the needs of your application.

Benefits of Macros

SKILL macros can be used in various situations:

- To gain speed by replacing function calls with in-line code
- To expand constant expressions for readability
- As convenience wrappers on top of existing functions



Caution

In-line expansion increases the size of the code using macros. So, use macros sparingly and only in those situations where they clearly add value to the code.

Macro Expansion

When SKILL encounters a macro function call, it evaluates the function call immediately and the last expression computed is compiled in the current function object. This process is called macro expansion. Macro expansion is inherently recursive: the body of a macro function can refer to other macros including itself.

Macros should be defined before they are referenced. This is the most efficient way to process macros. If a macro is referenced before it is defined, the call is compiled as “unknown” and the evaluator expands it at run-time, incurring a serious penalty in performance for macros.

Redefining Macros

If you are in development mode and you redefine a macro, make sure all code that uses that macro is reloaded or redefined. Otherwise, wherever the macro was expanded, the previous definition continues to be used.

defmacro

To define a macro in SKILL, use `defmacro`.

For example, if you want to check the value of a variable to be a string before calling `printf` and you don't want to write the code to perform the check at every place where `printf` is called, you might consider writing a macro:

```
(defmacro myPrintf (arg) `when((stringp ,arg)
  printf( "%s" ,arg)))
```

As you can see, the macro `myPrintf` returns an expression constructed using backquote, which is substituted for the call to `myPrintf` at the time a function calling `myPrintf` is defined.

mprocedure

The `mprocedure` function is a more primitive facility on which `defmacro` is based. Avoid using `mprocedure` in new code that you are developing. Use `defmacro` instead. While compiling source code, when SKILL encounters an `mprocedure` call, the entire list representing the function call is passed to the `mprocedure` immediately, bound to the single formal argument. The result of the last expression computed within the `mprocedure` is compiled.

Using the Backquote (`) Operator with defmacro

Here is a sample macro that highlights the effect of compile time macro expansion in SKILL. It assumes `isMorning` and `isEvening` are defined.

```
(defmacro myGreeting (_arg)
  let((_res)
    cond( (isMorning())
          (_res= sprintf(nil "Good morning %s" _arg))
          (isEvening())
          (_res= sprintf(nil "Good evening %s" _arg)))
    `println(,_res))
  )
```

Use the utility function `expandMacro` to test the expansion, for example,

```
expandMacro('myGreeting("Sue")) ; using the above definition
=> println("Good morning Sue") ; if isMorning returns t
```

When called, `myGreeting` returns a `println` statement with the desired greeting to be in-line substituted for the call to `myGreeting`. Because the call to `sprintf` inside `myGreeting` is performed outside of the expression returned, the greeting message reflects the time when the code was compiled, and not when it was run.

This is how `myGreeting` should be rewritten to have the greeting message reflect the time the code was run:

```
(defmacro myGreeting (_arg)
  `let((_res)
    cond( (isMorning())
          (_res= sprintf(nil "Good morning %s" ,_arg))
          (isEvening())
          (_res= sprintf(nil "Good evening %s" ,_arg)))
    println(_res))
  )
```

The above, when compiled, substitutes the entire `let` expression in-line for the macro call.

Using an @rest Argument with defmacro

The next macro example shows how a functionality can be implemented efficiently by exploiting in-line expansion. The macro implements `letOrdered`. It differs from regular `let` in that it performs the bindings for the declared local variables in sequence, so an earlier declared variable can be used in expressions evaluated to bind subsequent variable declarations, which is not safe to do in a `let`. For example:

```
(letOrdered ((x 1) (y x+1) ...)
```

guarantees that `x` is first bound to 1 when the binding for `y` is done.

The `letOrdered` macro

```
defmacro(letOrdered (decl @rest body)
  cond(( zerop(length(decl))
        cons('progn body))
    ( t 'let((,car(decl))
              letOrdered(,cdr(decl),@body)))
  )
)
```

is defined recursively. For each variable declaration, it nests `let` statements, thereby guaranteeing that all bindings are performed sequentially. For example:

```
procedure( foo()
  letOrdered(((x 1) (y x+1))
    y+x))
```

expands to

```
procedure(foo()
  let(((x 1))
    let(((y x+1))
      progn( y+x ))))
```

Using @key Arguments with defmacro

The following example illustrates a custom syntax for a special way to build a list from an original list by applying a filter and a transformation. To build a list of the squares of the odd integers in the list

```
( 0 1 2 3 4 5 6 7 8 9 )
```

you write

```
trForeach(
  ?element      x
  ?list         '( 0 1 2 3 4 5 6 7 8 9 )
  ?suchThat     oddp(x)
```

```
?collect      x*x
) => ( 1 9 25 49 81 )
```

instead of the more complicated

```
foreach( mapcar x
  setof(x '(0 1 2 3 4 5 6 7 8 9) oddp(x))
  x * x
) => ( 1 9 25 49 81 )
```

Implementing an easy-to-maintain macro requires knowledge of how to build SKILL expressions dynamically using the backquote (`), comma (,), and comma-at (,@) operators.

The definition for `trForeach` follows.

```
defmacro( trForeach ( @key element list suchThat collect )
  `foreach( mapcar ,element
    setof( ,element ,list ,suchThat )
    ,collect
  ) ; foreach
) ; defmacro
```

Variables

SKILL uses symbols for both global and local variables. In SKILL, global and local variables are handled differently from C and Pascal.

Lexical Scoping

In C and Pascal, a program can refer to a local variable only within certain textually defined regions of the program. This region is called the `lexical scope` of the variable. For example, the lexical scope of a local variable is the body of the function. In particular

- Outside of a function, local variables are inaccessible
- If a function refers to non-local variables, they must be global variables

Dynamic Scoping

SKILL does not rely on lexical scoping rules at all. Instead:

- A symbol's current value is accessible at any time from anywhere within your application
- SKILL transparently manages a symbol's value slot as if it were a stack
- The current value of a symbol is the top of the stack

- Assigning a value to a symbol changes only the top of the stack
- Whenever the flow of control enters a `let` or `prog` expression, the system pushes a temporary value onto the value stack of each symbol in the local variable list (the local variables are normally initialized to `nil`)
- Whenever the flow exits the `let` or `prog` expression, the prior values of the local variables are restored

Dynamic Globals

During the execution of your program, the SKILL programming language does not distinguish between global and local variables.

The term *dynamically scoped variable* refers to a variable used as a local variable in one procedure and as a global in another procedure. Such variables are of concern because any function called from within a `let` or `prog` expression can alter the value of the local variables of that `let` or `prog` expression. For example:

```
procedure( trOne()
  let( ( aGlobal )
    aGlobal = 5 ;;; set the value of trOne's local variable
    trTwo()
    aGlobal      ;;; return the value of trOne's local variable
  ) ;let
) ; procedure

procedure( trTwo()
  printf("\naGlobal: %L" aGlobal )
  aGlobal = 6
  printf("\naGlobal: %L\n" aGlobal )
  aGlobal
) ; procedure
```

The `trOne` function uses the `let` function to define `aGlobal` to be a local variable. However, `aGlobal`'s temporary value is accessible to any function `trOne` calls. In particular, the `trTwo` function changes `aGlobal`.

This change is not intuitively expected and can lead to problems that are difficult to isolate. SKILL Lint reports this type of variable as an "Error Global." It is usually recommended that users should not rely on the dynamic behavior of variable bindings.

Error Handling

SKILL has a robust error handling environment that allows functions to abort their execution and recover from user errors safely. When an error is discovered, you can send an error signal up the calling hierarchy. The error is then caught by the first error catcher that is active. The

default error catcher is the SKILL top level, which catches all errors that were not caught by your own error catchers. In addition, starting from the IC6.1.6 release, SKILL can handle unusual situations or exceptions, which may occur during the execution of your programs by using the `throw` and `catch` functions.

The `errset` Function

The `errset` function catches any errors signalled during the execution of its body. The `errset` function returns a value based on how the error was signalled. If no error is signalled, the value of the last expression computed in the `errset` body is returned in a list.

```
errset( 1+2 ) (3)
```

In the following example, without the `errset` wrapper, the expression

```
1+"text"
```

signals an error and display the messages

```
*Error* plus: can't handle (1 + "text")
```

To trap the error, wrap the expression in an `errset`. Trapping the error causes the `errset` to return `nil`.

```
errset( 1+"text" ) => nil
```

If you pass `t` as the second argument, any error message is displayed.

```
errset( 1+"text" t ) => nil  
*Error* plus: can't handle (1 + "text")
```

Information about the error is placed in the `errset` property of the `errset` symbol. Programs can therefore access this information with the `errset.errset` construct after determining that `errset` returned `nil`.

```
errset( 1+"text" ) => nil  
errset.errset =>  
("plus" 0 t nil  
  ("*Error* plus: can't handle (1 + \"text\")"))
```

When working in the CIW, to ensure that the `errset.errset` variable is not modified internally in the Virtuoso Studio design environment, do not separate `errset` and `errset.errset`. For example, use this construct:

```
errset(sqrt('x)), errset.errset  
=> ("sqrt" 0 t nil      ("*Error* sqrt: cannot handle sqrt(x)"))
```

Using `err` and `errset` Together

Use the `err` function to pass control from the point at which an error is detected to the closest `errset` on the stack. You can control the return value of the `errset` by your argument to the `err` function.

If this error is caught by an `errset`, `nil` is returned by that `errset`. However, if an optional argument is given, that value is returned from the `errset` in a list and can be used to identify which `err` signaled the error. The `err` function never returns.

```
procedure( trDivide( x )
  cond(
    ( !numberp( x ) err() )
    ( zerop( x ) err( 'trDivideByZero ) )
    ( t 1.0/x )
  )
) ; procedure

errset( trDivide( 5 ) )      => ( 0.2 )
errset( trDivide( 0 ) )      => (trDivideByZero)
errset( trDivide( "text" ) ) => nil

errset( err( 'ErrorType) )   => (ErrorType)
errset.errset                => nil
```

The error Function

`error` prints any error messages and then calls `err` flagging the error. The first argument can be a format string that causes the rest of the arguments to print using that format. Here are some examples:

This function:	Prints the following:
<code>error("myFunc" "Bad List")</code>	<code>*Error* myFunc: Bad List</code>
<code>error("bad args - %s %d %L" "name" 100 '(1 2 3))</code>	<code>*Error* bad args - name 100 (1 2 3)</code>
<code>errset(error("test") t)=> nil</code>	<code>*Error* test</code>

The warn Function

`warn` queues a warning message string. After a function returns to the top level, all queued warning messages are printed in the Command Interpreter Window and the system flushes the warning queue. Arguments to `warn` use the same format specification as `sprintf`, `printf`, and `fprintf`.

This function is useful for printing SKILL warning messages in a consistent format. You can also suppress a message with a subsequent call to `getWarn`.

```
arg1 = 'fail
warn( "setSkillPath: first argument must be a string or list of strings - %s\n"
arg1)
=> nil

*WARNING* setSkillPath: first argument must be a string or list of strings - fail
```

The `getWarn` Function

`getWarn` dequeues the most recently queued warning from a previous `warn` function call and returns that warning as its return result.

```
procedure( testWarn( @key ( dequeueWarn nil ) )
  warn("This is warning %d\n" 1 ) ;;; queue a warning
  warn("This is warning %d\n" 2 ) ;;; queue a warning
  warn("This is warning %d\n" 3 ) ;;; queue a warning
  when( dequeueWarn
    getWarn() ;;; return the most recently queued warning
  )
) ; procedure
```

The `testWarn` function prints the warning if `t` is passed in and gets the warning if `nil` is given as an argument.

```
testWarn( ?dequeueWarn nil)
=> nil
*WARNING* This is warning 1
*WARNING* This is warning 2
*WARNING* This is warning 3
```

Returns `nil` and the system prints all the queued warnings.

```
testWarn( ?dequeueWarn t)
=> "This is warning 3\n"
*WARNING* This is warning 1
*WARNING* This is warning 2
```

Returns the dequeued (most recent) warning and the system prints the remaining queued warnings.

The muffleWarnings and getMuffleWarnings Functions

The `muffleWarnings` function can be used when it is not desirable to show warning messages from functions included in a piece of code. In such cases, where the number of messages is dynamic, the `getWarn` function is not feasible.

Wrapping code in `muffleWarnings` dequeues all SKILL warning messages within the enclosed code. You can include the `getMuffleWarnings` function within it to retrieve the muffled warnings, which can then be used to build a new message, if required.

The following example illustrates the use of `muffleWarnings` with function calls that will result in a dynamic number of warning messages.

```
procedure( myTask( task @optional object )
  let((code)
    if( typep( task ) == 'string then
      code = case( task
        ("Open"
          unless( object
            warn( "Object not defined\n" )
          )
          1
        )
        ("Close"
          2
        )
        (t
          warn( "Unknown task specified\n" )
          0
        )
      )
    else
      warn( "Task is expected to a string\n" )
    )
    unless( code
      warn( "Unexpected task result\n" )
    )
    code
  )
)
```

```
myTask( "Open" "Door" )
=> 1
```

```
myTask( "Open" )
*WARNING* Object not defined
=> 1
```

```
myTask( "Look" )
*WARNING* Unknown task specified
*WARNING* Unexpected task result
=> nil
```

```
muffleWarnings(
  myTask( "Open" "Door" )
)
```

```
=> 1
getMuffleWarnings() => nil

muffleWarnings (
    myTask( "Open" )
)
=> 1
getMuffleWarnings() => ("Object not defined\n")

muffleWarnings (
    myTask( "Look" )
)
=> nil
getMuffleWarnings() => ("Unknown task specified\n" "Unexpected task result\n")
```

The results for `getMuffleWarnings` pertain to the preceding `muffleWarnings` command.

The throw and catch Functions

The `throw` and `catch` functions implement the exception mechanism in SKILL. These functions can be used to throw exceptions of various types with the help of `tags` and provide handlers for each `tag`. When a `throw` occurs and a particular `tag` or exception is caught, the SKILL stack is unwound to the initial point of block execution and `catch` block returns the value produced by evaluating `throw` form(s).

The `throw` function should always be defined inside a block. There can also be nested blocks.

The syntax for using the `catch` and `throw` functions is as follows:

```
(s_tag g_form) => g_result
throw(s_tag g_value)
```

You can specify an `s_tag` to be `t`, in which case the function will any condition thrown by the corresponding `throw`.

The steps of execution of the `catch` and `throw` functions is as follows:

1. Evaluate `s_tag` and establish it as a return point.
2. Evaluate the forms and return the results of the last form unless a `throw` occurs.
3. If a `throw` occurs, transfer the control to the block for which `s_tag` corresponds with the `s_tag` argument of the `throw` function. No more forms are evaluated any further.

Top Levels

When you run SKILL or non-graphical applications built on top of SKILL, you are talking to the SKILL top level, which reads your input from the terminal, evaluates the expressions, and prints the results. If an error is encountered during the evaluation of expressions, control is usually passed back to the top level.

When you are talking to the top level, any complete expression that you type (followed by typing the Return key to signal the end of your input) is evaluated immediately. Following the evaluation, the value of the expression is pretty printed.

If only the name of a symbol is typed at the top level, SKILL checks if the variable is `bound`. If so, the value of the variable is printed. Otherwise, the symbol is taken to be the name of a function to call (with no arguments). The following examples show how the outer pair of parentheses can be omitted at the top-level.

```
if (ga > 1) 0 1
if( (a > 1) 0 1 )

loadi "file.ext"
loadi("file.ext")

exit                ; Assuming exit has no variable binding
exit()
```

The default top level uses `lineread`, so it quietly waits for you to complete an expression if there are any open parentheses or any binary `infix` operators that have not yet been assigned a right operand. If SKILL seems to do nothing after you press Return, chances are you have mistyped something and SKILL is waiting for you to complete your expression.

Sometimes typing a super right bracket (]) is all you need to properly terminate your input expression. If you mistype something when entering a form that spans multiple lines, you can cancel your input by pressing *Control-c*. You can also press *Control-c* to interrupt function execution.

Memory Management (Garbage Collection)

In SKILL all memory allocation and deallocation is managed automatically. That is, the developer using SKILL does not have to remember to deallocate unused structures. For example, when you create an array or an instance of a `defstruct` and assign it as a value to a variable declared locally to a procedure, if the structure is no longer in use after the procedure exits, the memory manager reclaims that structure automatically. In fact, reclaimed structures are subsequently recycled. For users programming in SKILL, garbage collection simplifies bookkeeping to the point that most users do not have to worry about storage management at all.

The allocator keeps a pool of memory for each data type and, on demand, it allocates from the various pools and reclaimed structures are returned to the pool. The process of reclaiming unused memory - garbage collection (GC) - is triggered when a memory pool is exhausted.

Garbage collection replenishes the pool by tracking all unused or unreferenced memory and making that memory available for allocation. If garbage collection cannot reclaim sufficient memory, the allocator applies heuristics to expand the memory pools by a factor determined at run time.

Garbage collection is transparent to SKILL users and to users of applications built on top of SKILL. The system might slow down for a brief moment when garbage collection is triggered, but in most cases it should not be noticeable. However, unrestrained use of memory in SKILL applications can result in more time spent in garbage collection than intended.

How to Work with Garbage Collection

The garbage collector uses a heuristic procedure that dynamically determines when and if additional working memory should be allocated from the operating system. The procedure works well in most cases, but because optimal time/space trade-offs can vary from application to application, you might sometimes want to override the default system parameters.

When an application is known to use certain SKILL data types more than others, you can measure the amount of memory pools needed for the session and preallocate that pool. This allocation helps reduce the number of garbage collection cycles triggered in a session. However, because the overhead caused by garbage collection is typically only several percent of total run time, such fine-tuning might not be worthwhile for many applications.

First you need to analyze your memory usage by using `gcsummary`. This function prints a breakdown of memory allocation in a session. See the next section for a sample output. Once you have determined how many instances of a data type you need for the session, you can preallocate memory for that data type by using `needNCells` (described at the end of this section).

For the most part, you do not need to fine tune memory usage. You should first use memory profiling (refer to the chapter about SKILL Profiler in *Cadence SKILL IDE User Guide*) to see if you can track down where the memory is generated and deal with that first. Use the memory tuning technique described in this section as a last resort. Remember, because all memory tuning is global, you can't just tune the memory for your application. All other applications running in the same currently running binary are affected by your tuning.

Note: To troubleshoot performance and memory management issues in your SKILL code, contact Cadence Customer Support.

Printing Summary Statistics

The `gcsummary` function prints a summary of memory allocation and garbage collection statistics in the current SKILL run.

How to Interpret the Summary Report

Column	Contains
Type	Data type names.
Size	Size of each atom representing the data type in bytes.
Allocated	Total number of bytes allocated in the pool for the data type.
Free	Number of bytes that are free and available for allocation.
Static	Memory allocated in static pools that are not subject to GC. This memory is usually generated when contexts are built. When variables are write protected, their contents are shifted to static pools.
GC Count	Number of GC cycles triggered because the pool for this data type was exhausted.

```
***** SUMMARY OF MEMORY ALLOCATION *****
Maximum Process Size (i.e., voMemoryUsed) = 3589448
Total Number of Bytes Allocated by IL = 2366720
Total Number of Static Bytes          = 1605632
```

Type	Size	Allocated	Free	Static	GC count
list	12	339968	42744	1191936	9
fixnum	8	36864	36104	12288	0
flonum	16	4096	2800	20480	0
string	8	90112	75008	32768	0
symbol	28	0	0	303104	0
binary	16	0	0	8192	0
port	60	8192	7680	0	0
array	16	20480	8288	8192	0
TOTALS	--	516096	188848	1576960	9

User Type (ID)	Allocated	Free	GC count
hiField	(20) 8192	7504	0
hiToggleItem	(21) 8192	7900	0
hiMenu	(22) 8192	7524	0
hiMenuItem	(23) 8192	5600	0
TOTALS	-- 32768	28528	0


```
Bytes allocated for :
  arrays      = 38176
  strings     = 43912
  strings(perm) = 68708
  IL stack    = 49140
  (Internal)  = 12288
```

```
TOTAL GC COUNT 9
----- Summary of Symbol Table Statistics -----
Total Number of Symbols = 11201
Hash Buckets Occupied = 4116 out of 4499
Average chain length = 2.721331
```

Allocating Space Manually

The `needNCells` function takes a cell count and allocates the appropriate number of pages to accommodate the cell count. The name of the user type can be passed in as a string or a symbol. However, internal types, like *list* or *fixnum*, must be passed in as symbols. For example:

```
needNCells( 'list 1000 )
```

guarantees there will always be 1000 list cells available in the system.

Exiting SKILL

Normally you exit SKILL indirectly by selecting the `Quit` menu command from the CIW while running the Cadence software in graphic mode, or by typing *Control-d* at the prompt while running in non-graphic mode. However, you can also call the `exit` function to exit a running SKILL application with or without an explicit status code. Both the `Quit` menu command in the CIW and *Control-d* in standalone SKILL trigger a call to `exit`.

Sometimes you might like to do certain cleanup actions before exiting SKILL. You can do this by registering exit-before and/or exit-after functions, using the `regExitBefore` and `regExitAfter` functions. An exit-before function is called before `exit` does anything, and an exit-after function is called after `exit` has performed its bookkeeping tasks and just before it returns control to the operating system. The user-defined exit functions do not take any arguments.

To give you even more control, an exit-before function can return the atom `ignoreExit` to abort the exit call totally. When `exit` is called, first all the registered exit-before functions are called in the reverse order of registration. If any of them returns the special atom `ignoreExit`, the exit request is aborted and it returns `nil` to the caller. After calling the exit-before functions, it does some bookkeeping tasks, calls all the registered exit-after functions in the reverse order of their registration, and finally exits to the operating system.

For compatibility with earlier versions of SKILL, you can still define the functions named `exitbefore` and `exitafter` as one of the exit functions. They are treated as the first registered exit functions (the last being called). To avoid confusing the system setup, do not use these names for other purposes.

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Overview information:

- [Contexts](#) on page 240
- [Autoloading Your Functions](#) on page 256
- [Encrypting and Compressing Files](#) on page 257
- [Protecting Functions and Variables](#) on page 258

Contexts

Note: This information is for users who are writing a large volume of code using the Cadence® SKILL language code and are interested in modularizing the code and possibly autoloading it at run time.

Contexts are binary representations of the internal state of the interpreter. Their primary purpose is to help speed the loading of SKILL files. They are best used when the set of SKILL files to load is large.



A SKILL Development license is required to build contexts using the `saveContext` command.

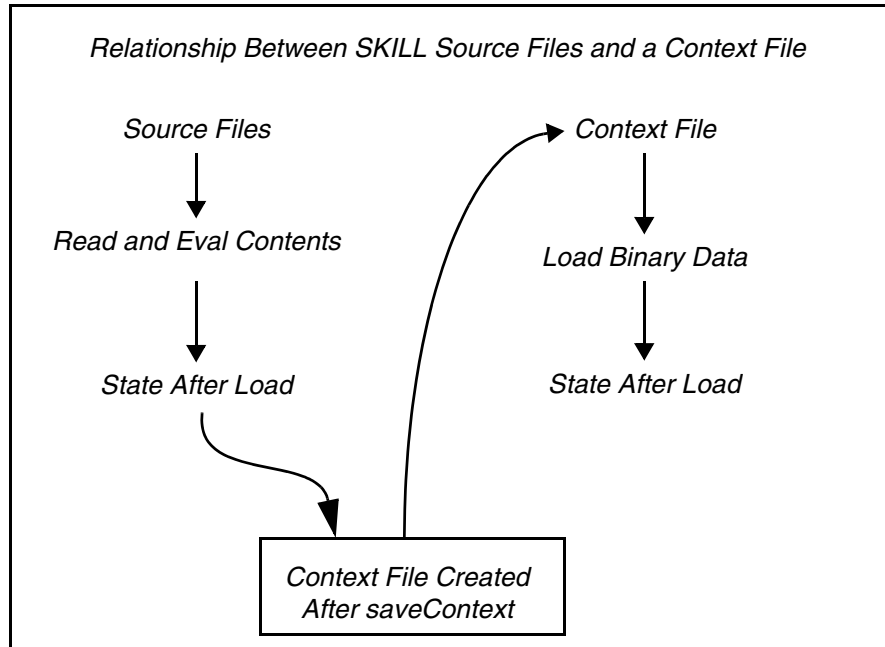
All SKILL-related structures can be saved in a context except those with values meaningless outside of a session. For example, port values, database handles, and window types are meaningful only to current sessions, whereas lists, integers, floats, defstructs, and so forth are transportable from one session to another.

SKILL contexts contain source code and data. Their main purpose is to allow for fast loading and initializing of product code. One way of looking at contexts is to view them as snapshots of the internal state of the interpreter, much like a core file in UNIX is an image of the running process.

Contexts cannot store at save time and retrieve at load time process-dependent structures. For example, file descriptors stored in port structures or process IDs cannot be saved in contexts. All other non process-dependent SKILL structures can be safely stored: Lists, numbers, strings, arrays, and so forth can be saved into and retrieved from a context.

When a SKILL source file is loaded, it is first parsed. The evaluator is called for each complete expression read in. This is how procedures are defined. Context files, on the other hand,

contain binary data that is loaded directly into memory. The binary data stored into a context has been parsed and evaluated before being saved.



Deciding When to Use Contexts

You must decide when it is better to use contexts versus straight or encrypted SKILL code. Some considerations include the following:

- Is the code likely to become a product?

Usually, the first criterion is whether the code is likely to become a product that will be shipped. Contexts offer a good vehicle for productizing code.

- How long does the code take to load?

The second criteria is whether there is enough SKILL code to warrant being in a context. This is difficult to measure. A simple test is to load the source code and see if the time it takes is likely to be unacceptable to a user. For example, if the code is likely to be auto-loaded during the physical manipulation of graphics, the impact of the load and initialization should be minimized. Contexts can help in this case.

The more code and initialization needed at load time, the better it is to use contexts. For small amounts of code (200 lines), contexts might be overkill. To load and initialize 20,000 lines of SKILL code without using contexts takes approximately 30 seconds,

whereas loading the same code using a context takes approximately 4 seconds. In this case, the perceptible impact is high, so using contexts makes sense.

■ Do you need to modularize your code?

Sometimes it is necessary to modularize code according to predefined capabilities. There might be a need to have these capabilities loaded incrementally at run-time. Thus it is not necessary to load all the code at once. Contexts, as snapshots of the interpreter's internal state, can be saved into separate files, even though code that goes into one context might depend on code in another.

The dependencies are resolved at load time. For example, the SKILL code for the schematics editor relies, among other things, on having code for the graphics editor present, but the context for the schematics editor does not contain any of the graphic editor's code or structures.

■ Can you create contexts at integration time?

The process of creating contexts must always be a separate step done at integration time, the time when all C code is compiled and linked and SKILL files are digested to produce context files. During integration and when contexts are being created, the interpreter enters a state that renders all normal use inefficient.

For example, during context creation, the memory management subsystem works in a special mode such that incremental snapshots of the memory can be made and saved into contexts. Context creation and code that is used to create contexts should not be part of the normal function of any product.

Creating Contexts

When you use SKILL contexts for delivering a product, you must develop an organization for those contexts.

Creating the Directory Structure

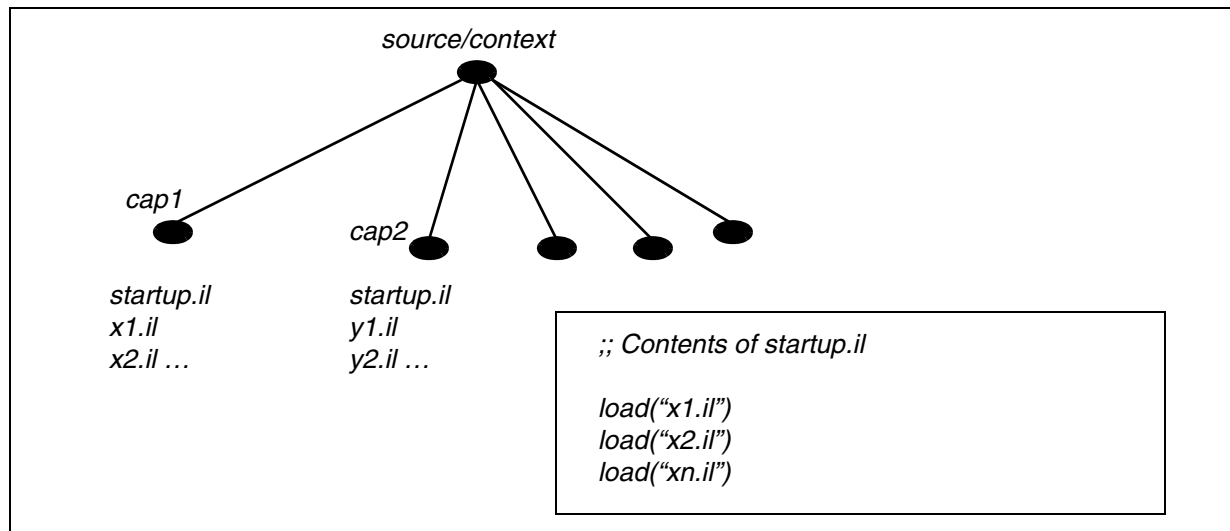
First, group the SKILL code on a product/capability basis. This is best done by designating a special directory, for example, `source/context` and then using make files or simple scripts to copy the code into separate directories under `source/context`.

If the capability names are `cap1` and `cap2`, create two directories under `source/context` named `source/context/cap1` and `source/context/cap2`. Copy the source code for each capability into the respective directory.

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For each capability context directory, create a single file named `startup.il` that uses the SKILL `load` command to load all the files in that directory in the necessary order. The figure below shows a layout of the context directories. To evaluate a command at the top level, use the `loadTopContextForm` command instead of `load`.



The first 12 characters in a context file name must be unique.

How the Process of Building Contexts Works

Using the directory structure described above, the process of building contexts starts by loading code from each directory and generating a binary context corresponding to the state of the interpreter when the files are loaded (in the sequence specified in the `startup.il` file).

The binary context file can go into one of several directories. If auto-loading of the context is necessary, the file can be placed in the `/your_install_dir/tools/dfII/local/context` directory. The autoload mechanism looks in this directory.

Note: 64 bit SKILL-based programs (such as `virtuoso`) require 64 bit context files, which should inturn be created using 64 bit SKILL-based programs. In addition, these 64 bit context files should be placed under the `/64bit` directory. For example, `/your_install_dir/tools/dfII/local/context/64bit`.

Creating Utility Functions

Given the directory structure above, the code for generating the contexts can now be written. First let us define a few utility functions.

Assumptions

Source code is stored under the `/user/source/context` directory. The created contexts are saved under `/user/etc/context`, where `user` is any installation path you choose for keeping sources and contexts. These contexts are hard-coded in the sample code below, but you can pass them as arguments to the functions.

Create `myContextBuild.il`

Put the following code in a separate file. Call it `myContextBuild.il`.

```
procedure( getContext(cxt)
;;-----
;; Given a context name load the context into the session
(let ((ff (strcat "/<user>/etc/context/" cxt ".cxt")))
  (cond ((null (isFile ff)) nil)
        ((null (loadContext ff))
         (printf "Failed to load context %s\n" cxt))
        ((null (callInitProc cxt))
         (printf "Failed to initialize context %s\n"
                 cxt))
        (t (printf "Loading Context %s\n" cxt)))
  )
)

procedure( makeContext(cxt)
;;-----
;; Given a context name create and save the context
;; under "/<user>/etc/context". Assumes user source
;; code is located under /<user>/source/<context>
(let ( (newPath (strcat "/<user>/source/" cxt))
      (oldPath (getSkillPath))
      (fileName (strcat "/<user>/etc/context/" cxt ".cxt"))
      (oldStatus (status writeProtect)))
  )

  (printf "Building context for %s\n" cxt)
  (setSkillPath newPath)
  (sstatus writeProtect t)
  ;; setContext is a function that takes the name of
  ;; a context and indicates to the system that
  ;; whatever is loaded or evaluated from
  ;; the point of this call to the time context
  ;; is saved belongs to the named context.

  (setContext cxt)
```

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```
;; Load all the SKILL files corresponding to the
;; given context. Relies on skill path being set earlier

(loadi "startup.il")

;; After the load save the files into a context
;; file containing all the necessary data to load
;; the context

(saveContext fileName)

    ;; It is important to call the initialization function
    ;; AFTER the context file is saved. This procedure
    ;; may create structures that can't be saved into
    ;; a context file but a subsequent load of SKILL files
    ;; for another context may need the call to have taken
    ;; place.

(callInitProc cxt)

;; Set the SKILL path back to what it was.
(setSkillPath oldPath)
    (unless oldStatus (sstatus writeProtect nil))
))

procedure( buildContext(cxt)
;-----
;; Deletes existing context and prepares to create the context

(progn
    (deleteFile (strcat "</user>/etc/context/" cxt ".cxt"))
    (cond ((isDir cxt "</user>/source/")
        (makeContext cxt))
        (t (printf "Can't find context directory %s\n" cxt)))
    )
)

;; Using getContext, load all the contexts that the code is
;; likely to need. For instance, the most basic
;; of contexts from Cadence are loaded first.

getContext("skillCore")
getContext("dbRead")
getContext ...

;; After loading all the dependencies, build all local
;; contexts

buildContext("cap1")
buildContext("cap2")
buildContext("capn")

exit
;; end of file myContextBuild.il
```

Building the Contexts

At this point everything is ready to build the contexts in a special call to the Cadence executable. Let us assume that the executable is called `cds`. The following line can be typed at the UNIX command line or can be made part of a make file to be called during the normal integration cycle.

```
cds -ilLoadIL myContextBuild.il -nograph
```

The output from the `cds` call can be piped into a file if a record of the build is needed.

ilLoadIL Option

The `-ilLoadIL` option causes the executable to switch into a special mode for context building and to allow itself to first read and evaluate the SKILL file before doing any of the normal operations for initializing the executable. That is why a call to the `exit` command is the last thing the file `myContextBuild.il` performs. It is meaningless to do more with the system after that.

nograph Option

The second option `-nograph` causes the executable to run with all graphics turned off. This option is useful if the integration is done on a machine that does not have X running. This option is not necessary for context building to work.

Initializing Contexts

Certain process-dependent constructs cannot be saved into a context. In the following example, a file port needs to be opened as part of the initialization phase of the code to be loaded. Use

```
myFile = outfile("/tmp/data")
```

The contents of the `myFile` symbol are not saved into a context because they would be meaningless when loaded into a different session. You need to define a special initialization function to initialize variables or structures that can only be initialized in the current session.

You can use the *defInitProc* function to define an initialization for each context. Let us assume a capability called `cap1` that opens a file and starts a child process. If the code for the capability needs to go into a binary context file, you need an initialization function to set up the file and the child process in the current session. The arrangement is as follows:

```
procedure(cap1InitProc()  
;;-----  
;; This procedure initializes two global variables: myFile
```

```
;; and myChild

    myFile = outfile("/tmp/data")
    myChild = ipcBeginProcess(..)
)

;; The next line of code designates the above function
;; to be the initialization code for the context cap1.

(defInitProc "cap1" 'cap1InitProc)

;; end sample code
```

The call to *defInitProc* must be executed when the code is loaded, that is, it should not be included in a procedure but rather it should be placed at the top-level of a file to be executed during the loading of the file.

To summarize the work so far, you have seen how to

- Organize the source code
- Set up code to initialize the context
- Write the SKILL code to drive the building of contexts

Loading Contexts

Context loading can be done in two ways, depending on the need for a particular context.

Loading Contexts at the Start of a Session

If the code in the context is needed at the start of a session, use the *.cdsinit* file to force the loading of the required contexts by adding the following lines to the *.cdsinit* file.

```
loadContext(strcat(prependInstallPath("etc/context/")
                    "cap1.cxt"))
callInitProc("cap1")
```

These two lines are needed for every context that is force-loaded by a call to *loadContext*. The call to *callInitProc* is needed to cause the initialization function for the context to be called. The basic *loadContext* function does not automatically do that.

Loading Contexts on Demand

Another option for contexts is to use the auto-load mechanism. This mechanism forces the context to be loaded on demand. There are two steps to achieving this.

- Generate an auto-load file.

- Load the auto-load file in the `.cdsinit` file.

The context is loaded automatically whenever one of its functions is called. To generate the auto-load file, first isolate all the entry point functions of a certain capability/context. When these functions are known, the contents of the auto-load file should look as follows.

```
;; This is the auto-load file for cap1. func1 through funcN
;; are the entry point functions for the capability cap1.
;; Call this file cap1.al
foreach(x '(func1 func2 func3 ..funcN)
        (putprop x "<user>/etc/context/cap1.cxt" 'autoload))
;; end of auto-load file
```

The call to the *putprop* function causes the symbol name of the function to have the *autoload* property assigned a string value corresponding to the name of the context, with full path, to which the function belongs. This is how the evaluator makes the connection between function name and context file. The autoload mechanism always looks for a context file to autoload under `/your_install_dir/tools/dfII/local/context` directory. If you choose to place the file under this directory, you don't need to give the property a full path; just the name of the context file is enough.

When the function is called during a normal session, and it does not have a function definition, the evaluator force-loads and initializes the context at that point. It is important to stress that the auto-load mechanism automatically calls the initialization function associated with the loaded context. After the context is loaded and the symbol gets a function definition, the evaluator calls the function and continues with the session.

Now that the auto-load file has been created, it can go anywhere the `.cdsinit` file can find it. You can place this file anywhere you choose, provided it is loaded at startup. The entry in the `.cdsinit` file needed to load the auto-load file is as follows.

```
load( "/your_install_dir/tools/dfII/local/context/cap1.al")
```

Working with a Menu-Driven User Interface

Some applications or capabilities have a menu-driven user interface. This implies that potentially the context for a capability does not have to be loaded, but the menus from which the capability is driven have to be put in place. In this case, the auto-load file (`cap.al`) must be augmented with the necessary code to create and insert menus in the appropriate places, such as banners.

When the auto-load file is loaded during the initialization phase of the executable, the desired menus appear and the callbacks for menu entries have the auto-load property set as explained above. The effect is that when menu commands are selected, the callback forces the corresponding context to load and execute the selected function. It is always safer to create and insert menus before adding the auto-load property on the function symbols.

Customizing External Contexts

You might want to customize the loading of externally supplied contexts. Consider the following example. Whenever the schematics context is loaded in an instance, a block of customer code needs to execute to set up special menus or load extra functionality. This can be accomplished using the `defUserInitProc` function. This function takes the following arguments.

- A string denoting the name of the capability and context to customize
- A symbol denoting the user-defined callback function to trigger whenever the named context is loaded.
- An optional `autoInit` argument to automatically initialize the named context.

```
defUserInitProc("schematic", 'mySchematicCustomFunc 'autoInitschematic)
```

The `defUserInitProc` call causes the `mySchematicCustomFunc` function to be called after the context for `schematic` is loaded and initialized.

If multiple `defUserInitProc` calls are defined for the same context, all of them are executed in the order in which they are defined.

For example, the following `defUserInitProc` calls cause the `mySchematicNewFunc` function to be called after the `mySchematicCustomFunc`:

```
defUserInitProc("schematic", 'mySchematicCustomFunc 'autoInitschematic)
defUserInitProc("schematic", 'mySchematicNewFunc)
```

The `defUserInitProc` call is similar to the `defInitProc` function that initializes a context. The context loading mechanism calls the function defined by `defInitProc` first and then calls the function defined by `defUserInitProc`.

The `initfunction` can be a part of the context. To make it a part of the context, you should define it in the context and call `defUserInitProc` between `setContext` and `saveContext` calls to register the function as an `initfunction`.

Note: `loadContext` does not update `initfunction` name for the context if it has already been set by the `defUserInitProc` call.

Potential Problems

Binary context files are built incrementally. This can introduce inter-context warnings, which means that references are made in one context to values outside its own space. Therefore, when the contexts are loaded independently, those values become meaningless. Because the SKILL language is dynamically scoped, excessive use of global data structures and

vague boundaries around the program and data spaces of applications can result in this type of warning. SKILL programmers can practice caution by modularizing their code and by using strict naming conventions for their symbol names.

When inter-context warnings are detected during context building, they are flagged but the context continues to build, leaving the values indicated as crossing context boundaries to be `nil`. The following sample cases of code cause inter-context warnings.

Sharing Lists Across Contexts

Consider the following sample code involving two contexts.

```
;; Code in context 1. This code builds list structures by
;; sharing sub-lists

field = list('nil 'item "hello world" 'item2 22)
form1 = list('xxx field)
form2 = list('yyy field)

;; Code in context 2. Also building list structures but
;; sharing sub-lists from context1

form3 = list('zzz field)

;; end sample code
```

In the three form variables, the `field` structure is shared. However when contexts are saved separately, the list structure for the `field` part of `form3` is not saved with the data in context 2, because it is outside the context's bounds. If the two contexts were created in the sequence given, a warning would be flagged indicating the part of the list in context 2 that is crossing boundaries with context 1. Both contexts would be named and the lists involved displayed.

If each context maintained the lists it needs within the context boundary, this would solve the problem. In this case, if the contents of the variable `field` are needed, a copy should be performed in context 2 to get a copy of the list locally to that context.

The Conditional eq Failure

The context saving algorithms attempt to smooth out the inter-context problems by copying atoms across context boundaries. For instance:

```
;; Code in context 1. Here a function is defined such that
;; when called it constructs and returns a list.

procedure( foo()
  list(42 "hello world" 42)
)
var1 = foo()
```

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```
;; Code in context 2. A call to foo is made and the result is
;; stored in var2

var2 = foo()

(eq (cadr var1) (cadr var2))      ;; Normally returns t
(equal (cadr var1) (cadr var2))  ;; Normally returns t

;; end sample code
```

When separate contexts are generated incrementally for the code above and you load both contexts in a session, the call to *eq* returns *nil* indicating the two strings “hello world” in the two separate lists pointed to by *var1* and *var2* are not the same instance or pointer. That is because the string “hello world” was copied into context 2 when the binary context for context 2 was saved. This allows the list pointed to by *var2* to remain complete. Such copying is only done on atomic structures, such as integers or strings.

This condition is not flagged by any errors at context build time.

Execution During Load

The SKILL code executed during the loading of *startup.il* executes all top-level statements (that is how procedure definitions are done). It was explained earlier that certain data types cannot be saved into a context and have to be regenerated using a context-specific initialization function. There are process-related executions (that is, not necessarily data-related) that have to occur during the context initialization phase. For example, consider the following.

```
;; Assume the function isMorning is a boolean that calls
;; time related internal functions and returns t if the
;; current time is AM.

if ( isMorning()
then greeting = "good morning"
else greeting = "good afternoon")
form->title->value = greeting

;; end of sample
```

When the context is built using the code above, the *isMorning* function is executed and the *greeting* is set correctly. The value of *greeting* and the value of *form->title->value* are “frozen” in the context file. On subsequent loads of the context, the *isMorning* function will not execute so the *greeting* will take the value given to it at the time the context was built. To remedy this situation, the code above should be made part of the initialization function for a context.

Context Building Functions

Saving the Current State of the SKILL Language Interpreter as a Binary File (`saveContext`)

`saveContext` saves the current state of the SKILL language interpreter as a binary file. This function is best used in conjunction with `setContext`. `saveContext` saves all function and variable definitions that occur, usually due to file loading, between the calls to `setContext` and `saveContext`. Those definitions can then be loaded into a future session much faster in the form of a context using the `loadContext` function.

By default all functions defined in a context are read and write protected unless the `writeProtect` system switch was turned off when the function was defined between the calls to `setContext` and `saveContext`. If the full path is not specified, this function uses the SKILL path to determine where to store the context file name.

```
setContext( "myContext")           => t
load("mySkillCode.il")            => t
defInitProc("myContext" 'myInit)  => t
saveContext("myContext.cxt")      => t
```

Context creation fails when the code size is too big inside a construct. The maximum size is 32KB. To avoid a failure, it is recommended that you divide the code into smaller constructs or use the `setSaveContextVersion(getNativeContextVersion())` function to set native save context version and increase the capacity (see [Context Version Functions](#) on page 254). However, contexts created with increased capacity are not compatible with earlier versions of SKILL.

Saving Contexts Incrementally (`setContext`)

`setContext` allows contexts to be saved incrementally, creating micro contexts from a session's SKILL context. To understand this, think of the SKILL interpreter space as linear; the function call `setContext` sets markers along the linear path. Any SKILL files loaded between a *setContext* and a *saveContext* are saved in the file named in the `saveContext` call. This function can be used more than once during a session.

Loading the Context File into the Current Session (`loadContext`)

`loadContext` loads the context file into the current session. You should always fully qualify the file name. The default directories will be searched for the file if the name is not fully qualified. The context file must have been created using the function `saveContext`. For example:

```
loadContext( "/usr/mnt/charb/myContext.cxt" )
```

Loads the `myContext.cxt` context.

Important

In 64-bit environment, 64 bit context files are placed under the `/64bit` directory. If the specified path does not contain `/64bit`, it is automatically prepended to the context file name. For example,

If your current working directory is `/home/user/615/local/context`, `saveContext( "myContext.cxt" )` places the `myContext.cxt` file under `/home/user/615/local/context/64bit`.

Similarly, `loadContext( "/home/usr/615/local/context/myContext.cxt" )` is converted internally to `loadContext( "/home/usr/615/local/context/64bit/myContext.cxt" )`

It is advisable to not load context files supplied by Cadence using `loadContext`. It is better to rely on the autoload mechanism for context files you do not own.

Concatenating the Context Files (UNIX `cat` command)

You can concatenate two or more context files into one using the UNIX `cat` command. The concatenated files can then be loaded together in the order of concatenation. For example.

```
setContext( "myContext" )
defun( myContextFunction ()
    info( "this is myContextFunction\n" )
defUserInitProc( "myContext" 'myContextFunction t)
saveContext( "myContext.cxt" )
setContext( "yourContext" )
defun( yourContextFunction ()
    info( "this is yourContextFunction\n" )
defUserInitProc( "yourContext" 'yourContextFunction t)
saveContext( "yourContext.cxt" )
;;concatenate the context files using the UNIX cat command
system("cat myContext.cxt yourContext.cxt > combinedContext.cxt")
;;load the concatenated context file
loadContext("combinedContext.cxt")
;;The files are loaded in the order of concatenation
=> this is myContextFunction
=>this is yourContextFunction
```

Calling Initialization Functions Associated with a Context (callInitProc)

`callInitProc` takes the same argument as `loadContext` and causes the initialization functions associated with the context to be called. This function need not be used if the loading of the context is happening through the autoload mechanism. Use this function only when calling `loadContext` manually. For example

```
loadContext("myContext")      => t
callInitProc("myContext")    => t
```

Functions defined through `defInitProc` and `defUserInitProc` are called.

Registering an Initialization Function for a Context (defInitProc)

`defInitProc` registers an initialization function associated with a context. The initialization function is called when the context is loaded.

`defInitProc` always returns `t` when set up. The initialization function is not called at this point, but is called when the associated context is loaded.

```
defInitProc("myContext" 'myInitFunc) => t
```

Registering a Function for Contexts You Don't Own (defUserInitProc)

`defUserInitProc` registers a user defined function that the system calls immediately after loading and initializing a context. For instance, this function lets you customize Cadence supplied contexts. Usually, most Cadence-supplied contexts have internally defined an initialization function through the `defInitProc` function. `defUserInitProc` defines a second initialization function, called after the internal initialization function, thereby allowing you to customize on top of Cadence-supplied contexts. The call to `defUserInitProc` is best done in the `.cdsinit` file.

`defUserInitProc` always returns `t` when set up. You can specify the `autoInit` option to automatically initialize contexts. The initialization function is not called at this point, but is called when the associated context is loaded.

```
defUserInitProc( "someContext" 'myInitSomeContext 'autoInitSomeContext) => t
```

Context Version Functions

Contexts created in IC6.1.4 and later releases with a 'native' context version could not be loaded with the contexts created in the earlier releases (IC6.1.3, IC6.1.2, IC6.1.1, or IC5.x.41). So, in IC6.1.4, API to set the context version were introduced to resolve these compatibility issues.

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As described in the table below, a given product release version (6.1.4) can map to a specific SKILL version (31.00), which in turn can have a native context version (602) or a compatible context version (601). To use SKILL contexts across releases, you need to have compatible context versions.

Note: It is possible for two context files to be compatible and still have different number or type of arguments of the referenced functions.

CIC Product Release Version	SKILL Version	Native Context Version	Compatible Context Version
6.1.3	07.02	601	N/A
6.1.4	31.00	602	601

The `saveContext` function builds either a native or a compatible context version in IC 6.1.4 (the default is compatible.)

Note: Compatible context version runs slower than the native context version in IC 6.1.4.

Retrieving the Current `saveContext` Version (`getCurSaveContextVersion`)

`getCurSaveContextVersion` returns the current `saveContext` version (the version which the new context will have.) The possible return values are, 601 for compatible contexts and 602 for native contexts (for IC 6.1.4/CAT 31.00)

```
setSaveContextVersion (getNativeContextVersion ())
601
getCurSaveContextVersion ()
602
```

Resetting the Current `saveContext` Version (`setSaveContextVersion`)

`setSaveContextVersion` resets the current `saveContext` version to `x_newVers` and returns the previous context version. If `x_newVers` has an unsupported value or the function is called between `setContext` and `saveContext`, it returns an error.

```
setSaveContextVersion (getCompatContextVersion ())
601
setSaveContextVersion (0)
*Error* setSaveContextVersion: unsupported context version - 0
```

Retrieving the Native Context Version (getNativeContextVersion)

`getNativeContextVersion` returns the native context version (for IC 6.1.4/CAT 31.00, the native context version is 602).

```
getNativeContextVersion()  
602
```

Retrieving the Compatible Context Version (getCompatContextVersion)

`getCompatContextVersion` returns the compatible context version (for IC 6.1.4/CAT 31.00, the compatible context version is 601).

```
getCompatContextVersion()  
601
```

Note: Do not use the 601 or 602 context version values directly in SKILL functions. Use `getCompatContextVersion`, `getNativeContextVersion`, or `getCurSaveContextVersion` instead to retrieve the values of context versions.

Autoloading Your Functions

Autoloading is the facility through which SKILL code can be loaded dynamically. That is, the code supporting a capability is not loaded until that capability is needed. The load is usually triggered by a call to a function that is undefined in the current session. The evaluator calls on the loader to locate and load the code for the function. You should use autoloading to tune the amount of code loaded in a session to that only needed for that session.

The autoloader follows these rules:

- If there is a property on the function being called with the symbol "autoload" and the value of the property is a string denoting the name of a context (with the extension `.cxt`), the loader looks for the file under `/your_install_dir/tools/dfII/etc/context` (this is where Cadence-supplied contexts are stored) and `/your_install_dir/tools/dfII/local/context`.
- If the value of the autoload property is a string denoting the name of a file with a full path and the file is a context file (`.cxt` extension), the context is loaded. If the extension is anything other than `.cxt`, `loadi` is called with the file name as its argument.
- If the value of the autoload property is an expression, the expression is evaluated. The expression is responsible for taking the necessary steps to define the function triggering the autoload.

Autoloading Your Classes

You can autoload a class definition from your SKILL source file (`.il` or `.ils`) or a context file (`.cxt`) by setting the property `autoloadClass` on a `className` as shown in the following examples:

```
myName.autoloadClass="classX.il"  
myName.autoloadClass="contextX.cxt"
```

The autoload mechanism is triggered when the `autoload` property is set for some class (symbol) and this class is not yet defined then

You need to implement this feature from the `findClass` function. You can also implement nested autoloading to retrieve superclasses for a class being autoloaded. However, this is only possible for SKILL source files, nested autoload for context files is not supported. This means that only the first context file can be autoloaded and the remaining nested context files are skipped.

Encrypting and Compressing Files

Encrypting and compressing files allows distribution of SKILL code in a manner that an end user cannot read the code. This is an alternative method to contexts and is intended for small sets of SKILL code that need protection.

Encrypting a File (encrypt)

You can encrypt SKILL programs and data files. These can subsequently be reloaded using the `load`, `loadi`, or `include` functions. `encrypt` encrypts a file and places the output into another file. If a password is supplied, the same password must be given to the command used to reload the encrypted file.

```
encrypt( "triadb.il" "triadb_enc.il" "option") => t
```

Encrypts the `triadb.il` file into the `triadb_enc.il` file with `option` as the password. Returns `t` if successful.

Reducing the Size of a File (compress)

You can compress SKILL files to remove unnecessary blank spaces and comments from the file. `compress` reduces the size of a source file and places the output into a destination file.

Compression renders the data less readable because indentation and comments are lost. It is not the same as encrypting the file because the representation of destination file is still in ASCII format.

```
compress( "triad.il" "triad_cmp.il") => t
```

Protecting Functions and Variables

The following functions get and set the write-protect status bit on functions and variables. These functions can be used to secure the definitions of functions and the contents of variables when delivering products.

You can write protect a function by setting the `writeProtect` status flag. Once turned on, all subsequent function definitions are write protected.

Explicitly Protecting Functions

Protecting functions on a per-function basis is an alternative to `sstatus` which protects functions on a per-context basis.

Setting the Write-Protect Bit on a Function (`setFnWriteProtect`)

`setFnWriteProtect` sets the write-protect bit on a function.

- If the function has a function value, it can no longer be changed.
- If the function does not have a function value but does have an `autoload` property, the `autoload` is still allowed. This is treated as a special case so that all the desired functions can be write-protected first and autoloaded as needed.

This example defines a function and sets its write protection so it cannot be redefined.

```
procedure( test() println( "Called function test" ))
setFnWriteProtect( 'test' )=> t

procedure( test() println( "Redefine function test" ))
*Error* def: function name already in use and cannot be
    redefined - test

setFnWriteProtect( 'plus' )=> nil
```

Returns `nil` because the `plus` function is already write protected.

Finding the Value of a Function's Write-Protect Bit (`getFnWriteProtect`)

`getFnWriteProtect` returns the value of the write-protect bit on a function. The value is `t` if the function is write-protected or `nil` otherwise.

```
getFnWriteProtect( 'strlen ) => t
```

Protecting Variables

Setting the Write-Protect Bit on a Variable (`setVarWriteProtect`)

`setVarWriteProtect` sets the write-protect on a variable. Use this function only when the variable and its contents are to remain constant.

- If the variable has a value, it can no longer be changed.
- If the variable does not have a value, it cannot be used.
- If the variable holds a list as its value, that list can no longer be changed.

For example, if the list is a disembodied property list, attempting to modify the value of properties will fail:

```
y = 5 ; Initialize the variable y.
setVarWriteProtect( 'y )=> t ; Set y to be write protected.
setVarWriteProtect( 'y )=> nil ; Already write protected.
y = 10 ; y is write protected.
*Error* setq: Variable is protected and cannot be assigned to - y
```

`getVarWriteProtect`

Returns the value of the write-protect on a variable.

```
x = 5
getVarWriteProtect( 'x )=> nil
```

Returns `nil` if the variable `x` is not write protected.

Global Function Protection

Write-protecting code renders the code secure from tampering.

To turn on global protection for functions saved in a context, be sure the following line is in your `makeContext` utility function described earlier:

```
sstatus( writeProtect t)
```

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However, if certain pieces of code need to have write-protection turned off (for example, when a function is user definable), use the following method.

```
;; Sample code showing how to turn off write-protection
;; The following let statement should surround all
;; that is to have write-protection turned off.

let( ((priorStatus (status writeProtect)))

    ;; Turn off write-protection
    (sstatus writeProtect nil)

    ;; Body of code to have no write-protection
    procedure( proc1() ...)
    procedure( proc2() ...)
    ...etc.

    ;; Turn write-protection status back to what it was
    (sstatus writeProtect priorStatus)
) ;; end-of-let
;; end of sample
```

The call to `sstatus` takes only the values `t/nil` as its second argument.

Writing Style

Good style in any language can lead to programs that are understandable, reusable, extensible, efficient, and easy to maintain. These attributes are particularly important to have in programs developed using an extension language like the Cadence® SKILL language because the programs tend to change a lot and the number of contributors can be many.

Useful SKILL programs, or pieces of them, get copied and passed around via e-mail or bulletin boards. Hence, it is important to consider the following when writing SKILL programs:

- Be specific, concise, and most importantly consistent.
- Anticipate a novice reader's questions and document the code accordingly.
- Be conventional. Using fancy techniques that are usually possible in a Lisp-based language, yet usually not well understood, might not be a good idea (unless you are doing it to gain in performance or reduce memory use).
- Avoid too many global states and direct access to them. Use abstractions.
- Make functional interfaces non-modal. That is, try and not predicate your interfaces on a global state or global mode such that a second user of the interface does not experience unpredictable behavior. For example, it is bad style to have a procedure that puts an editor in “insert” mode and uses subsequent calls to insert objects into the design. It is better to have one call that takes a list of objects to be inserted:

GOOD

```
EDinsert(database '(obj1 obj2 obj3))
```

BAD

```
EDstartInsert(database)
EDinsert(obj1)
EDinsert(obj2)
EDinsert(obj3)
EDendInsert(database)
```

As in all programming languages, the layout of code within a file can greatly affect the readability and maintainability of the code. The code examples in this manual use a layout that is both intuitive and easy to understand, but layout is a matter of taste.

The rest of this chapter describes specific situations where coding style issues are important.

- [Code Layout in SKILL Language](#) on page 262
- [Usage of Globals Variables in SKILL](#) on page 266
- [Coding Style Mistakes](#) on page 268
- [Red Flags](#) on page 270

Code Layout in SKILL Language

The readability of the code is the most important aspect of the code layout. The following guidelines can help you to write more readable code.

Comments and Documentation

You can use either of the following methods for commenting in SKILL:

- `;` at the beginning of a line
- `/*...*/` surrounding your comments

Note: Because `/*...*/` comments cannot be nested, some programmers find them easier to see.

While it is typically easier to understand and maintain code that is well-commented, you must also keep your comments consistent with your code. Comments that are misleading because they have not been maintained along with the code can be worse than not having any comments at all.

You should document your data structures by describing the motivation behind them and the effect of each structure on the algorithm. Well-designed and -documented data structures can tell a lot about the nature of the program.

If you are reading someone else's code and find it inadequately documented, write down your questions as comments. They may get answered or will encourage other readers to persevere.

As you develop a program, you should maintain a "to-do" list. For example, you can put a "to-do" comment around a dubious looking piece of code to explain your misgivings so that another developer can track a bug in that code.

Usually, if you find that you need to write convoluted comments about a convoluted algorithm, you should rewrite the algorithm. Strong code can be self-documenting.

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Writing Style

Things to Comment and Document

Cadence suggests that you comment the following items:

Item to comment	What to comment
Code modules	Each code module should have a header containing at least the author, creation date, and change history of the module, plus a general description of the contents.
Procedure definitions	Precede procedure definitions with a block comment detailing the functionality of and interfaces to the procedure. Add a help string inside the procedure (see “ procedure ” on page 73). Help systems extract this information for display. As much as possible, add type templates for procedure arguments to help erroneous use; type information can be extracted by help systems.
Data structures	Describe contents of data structures and impact on algorithms.
Complex conditionals	Comment any test within a conditional function that is nontrivial to indicate the pass/fail conditions.
Mapping functions	Comment complex mapping functions to state what the return values are.
Terminating parentheses	Where a function call extends over several lines, label the closing parenthesis with the function name.

Things Not to Comment and Document

While including any number of comments in your code does not affect performance once it has been read into the SKILL interpreter, Cadence suggests that you do not comment the following in production code:

Item not to comment	Explanation
Long change details	You should use your source code control system (such as RCS or SCCS) to maintain full details of any changes you make and include only a brief outline of your changes in the module header (rather than in the body of the code). Including details in the body of the code can make maintenance more difficult.
PCR details	You should maintain product change request information in the PCR itself.

Function Calls and Brackets

Function calls in SKILL can be written in two distinct ways, with the opening parenthesis either before the function name, as in Lisp, or after it, as in C. Once again, the method chosen is not as important as ensuring that it is chosen consistently. However, putting the parenthesis after the function name does make it easier for a non-Lisp programmer to read the code. It is also easier to distinguish between function names and arguments, and between function calls and other lists.

When a function call extends over more than one line in a file, it is recommended that the closing parenthesis is aligned, on a separate line, with the beginning of the function name. This makes it easier to see where a particular function call finishes and has the added advantage that missing parentheses are easier to see.

Note: Having extra newlines to allow alignment of function arguments or parentheses does not affect the performance of the code once it has been read into the SKILL interpreter.

Avoid Using a Super Right Bracket

Using the super right bracket (`)`) is strongly discouraged except when using the interactive interpreter because

- Missing parentheses can become difficult to locate.
- Inserting another function call around the code containing the super right bracket can be difficult.

- Bracket-matching procedures in editors neither manage nor account for super right bracket.



Tip

One method you can use to make sure that all your parentheses are matching when writing SKILL code is to insert the closing parenthesis immediately after the opening parenthesis of a function call, and then to go back and fill in the arguments.

Brackets in SKILL Are Always Significant

- In C, it is possible to insert brackets where they are not needed and do not affect the functionality of the program.
- In SKILL, the insertion of extra brackets can be, and usually is, incorrect.

The problem usually occurs with infix operators. In SKILL (as in Lisp) every function evaluates to a list whose head is the function name. Thus, code such as

```
a + b
```

is held internally as the list

```
(plus a b)
```

You must understand the relative precedence of the built-in SKILL functions. For example, consider the following code:

```
a = b && c
```

The line of code above is held as the following list:

```
(setq a (and b c))
```

rather than

```
(and (setq a b) c)
```

While this precedence is what you would expect from any language, it might not necessarily be what you want. Consider the following:

```
if(res = func1(arg) && res != no_val then ...)
```

The programmer meant to do one of these:

- Call `func1` and store the result in `res`.
- Check that `res` is not equal to `no_val`.

What happens:

The interpreter evaluates the expression `func1(arg) && (res != no_val)` and assigns the result to `res`.

The programmer needs to write the code as follows to perform the desired function:

```
if((res = func1(arg)) && res != no_val then ...)
```

Using too many parentheses can cause the code to fail. For example, the following statement has too many levels of parentheses:

```
a = ((func1(arg1)) && (func2(arg2)))
```

Parentheses in function calls are only optional for the built-in unary and infix operators, such as `!` and `+`. The following functions are equivalent:

```
!a  
!(a)  
(!a)
```

Commas

Commas between function arguments, and list elements, are optional in SKILL. Programmers from a C programming background will probably want to insert commas, those from a Lisp background probably will not. The general recommendation is that commas should not be used.

Usage of Globals Variables in SKILL

The use of global variables in SKILL, as with any language, should be kept to a minimum. The problems are greater in SKILL however because of its dynamic scoping of variables.

The problem with global variables is that different programmers can be using a variable with the same name. This type of “name clash” can cause problems that are even more difficult to isolate than the “Error Global” problem, because programs can fail because of the order in which they have been run. However, because this problem can be avoided by adopting a standard naming scheme, SKILL Lint will report this type of variable as a “Warning Global”. To illustrate the problem, consider the example code below:

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```
/* *****
 * myShowForm()
 * *****/

procedure( myShowForm()
    /* If we don't already have the form, then create it. */
    unless(boundp('theForm) && theForm
        myBuildForm()
    )
    /* Display the form. */
    hiDisplayForm(theForm)
) /* end myShowForm */

/* *****
 * yourShowForm()
 * *****/

procedure( yourShowForm()
    /* If we don't already have the form, then create it. */
    unless(boundp('theForm) && theForm
        yourBuildForm()
    )
    /* Display the form. */
    hiDisplayForm(theForm)
) /* end yourShowForm */

/* *****
 * myBuildForm()
 * *****/

procedure( myBuildForm()
    hiCreateForm('theForm,
        "My Form"
        "myFormCallback()"
        list( hiCreateStringField(?name 'string1,
            ?prompt "my field")
        ) /* end list */
    ) /* end hiCreateForm */
) /* end myBuildForm */

/* *****
 * yourBuildForm()
 *
 * Build the form.
 * *****/

procedure( yourBuildForm()
    hiCreateForm('theForm,
        "Your Form"
        "yourFormCallback()"
        list( hiCreateStringField(?name 'string1,
            ?prompt "your field")
        ) /* end list */
    ) /* end hiCreateForm */
) /* end myBuildForm */
```

If `myShowForm` is called before `yourShowForm`, the global variable `theForm` will be set to a different value than if `yourShowForm` is called before `myShowForm`.

Coding Style Mistakes

C programmers sometimes make the following mistakes when programming in SKILL. Even though these mistakes have mostly to do with coding style, some of them can have an impact on performance.

- Inefficient Use of Conditionals
- Misusing prog and Conditionals on page 269

Inefficient Use of Conditionals

A common mistake with conditional checks is to use multiple inversions and boolean checks when using De Morgan's Law would result in a simpler and clearer test. For example, the code below shows a common form of check.

```
if( !template || !templateDir
    warn("Invalid templates\n")
) /* end if */
```

This check can be optimized using De Morgan's Law to be:

```
if( !(template && templateDir)
    warn("Invalid templates\n")
) /* end if */
```

Another common mistake is made by many programmers used to having only the standard `if` and `case` conditionals. SKILL provides a rich variety of conditional functions, and appropriate use of these functions can lead to much clearer and faster code. For example, an `unless` could be used in the above check to yield the clearer and more efficient:

```
unless( template && templateDir
    warn("Invalid templates\n")
) /* end unless */
```

Another example is where multiple if-then-else checks are used, such as:

```
if(stringp(layer) then
    layerName = layer
else
    if(fixp(layer) then
        layerNum = layer
    else
        if(listp(layer) then
            layerPurpose = cadr(layer)
        ) /* end if */
    ) /* end if */
) /* end if */
```

This can be more clearly and efficiently implemented using a `cond`:

```
cond(
    (stringp(layer) layerName = layer)
```

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```
        (fixp(layer)      layerNum = layer)
        (listp(layer)     layerPurpose = cadr(layer) )
) /* end cond */
```

When applying multiple tests, either in a nested `if` function or a `cond` function, it is important to consider the order in which the tests will be carried out.

- If one test is more likely to be true than the others then it should go first.
- If all tests are equally likely to be true, then the test that involves the most work should go last.

Misusing `prog` and Conditionals

Often, `prog` statements are used to return from a procedure when an error condition occurs. In the example below, `template` and `templateDir` are both verified, and only if both are correct is the rest of the procedure executed:

```
procedure( EditCallback()
    prog( ( templateFile templateDir fullName )
        templateFile = ReportForm->Template->value
        templateDir   = ReportForm->TemplateDir->value
        /* Check the template directory. */
        if( ((templateDir == "") || (!templateDir)) then
            return(warn("Invalid template directory.\n"))
        )
        /* Check the template file. */
        if( ((templateFile == "") || (!templateFile)) then
            return(warn("Invalid template file name.\n"))
        )
        /* If both are correct then act.*/
        sprintf(fullName "%s/%s" templateDir templateFile)
        if(isFile(fullName) then
            LoadCallback()
        else
            SetUpEnviron()
        ) /* end if */
        return(hiDisplayForm(EditTemplateForm ))
    ) /* end prog */
) /* end EditCallback */
```

Using the fact that a `cond` returns the value of the last statement executed allows us to more efficiently implement this example using a `let`:

```
procedure( EditCallback()
    let((templateFile ReportForm->Template->value)
        (templateDir   ReportForm->TemplateDir->value)
        fullName)
    cond(
        /* Check the template directory. */
```

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```
( (templateDir == "") || (!templateDir)
  warn("Invalid template directory.\n")
)
/* Check the template file. */
( (templateFile == "") || (!templateFile)
  warn("Invalid template file name.\n")
)
/* If both are correct then act. */
(t
  sprintf(fullName "%s/%s" templateDir
          templateFile)
  if(isFile(fullName) then
    LoadCallback()
  else
    SetUpEnviron()
  ) /* if */
  hiDisplayForm(EditTemplateForm )
)
) /* end cond */
) /* end let */
) /* end of EditCallback */
```

Red Flags

The following are situations or functions that require special attention. Their use is often symptomatic of problems with the way interfaces or algorithms are designed. In some cases, their use is legitimate and these comments do not apply.

Use of eval or evalstring

Strictly speaking these calls are inefficient and any code using them is either suffering through using a bad interface from another application or the code itself is badly designed.

Excessive Use of reverse and append

Excessive use of `reverse` and `append` is an indication that algorithms using list structures are badly designed. They are acceptable when prototyping but production code should not suffer their consequences. Both these functions are capable of generating a lot of memory. There are alternatives to these functions and the recommendation is that code using these functions should be rewritten using `tconc`.

Excessive Use of gensym and concat

Symbols are large structures and applications that have to generate symbols at run time may not be designed to use the right data structure. Many times applications use symbols in place of small strings to save on memory because symbols are unique in the system. However, SKILL caches strings so this optimization might not always yield the desired effect.

Overuse of the Functions Combining car and cdr

One function that combines `car` and `cdr` is `cdaddr`. Using such functions, which are not as intuitive as calls to `nthelem` and `nthcdr`, can lead to programmer and reader errors.

Use of eval Inside Macros

Calling `eval` inside a macro means you are determining the value of an entity at compile time as opposed to evaluating it at run time, which may result in undesirable behavior. In general, macros should message expressions and return expressions (see [“Macros”](#) on page 224).

Misuse of prog and return in SKILL++ mode

Misuse of `prog` and `return` in SKILL++ might corrupt SKILL++ environment frames and can lead to a fatal programming error. The following example demonstrates misuse that you should avoid (using `prog` and `return` in a SKILL++ `.ils` file). Immediately following this example is an example of what you should do instead.

```
procedure( myFunc(namelist keys "ll")
  let( (selectedlist)
    foreach( name namelist
      prog( ()
        foreach( key keys
          when( rexMatchp(key name)
            return(t)
          )
        )
      return(nil)
    )
    selectedlist = cons(name selectedlist)
  )
)
```

Instead, do this:

```
procedure( myFunc(namelist keys "ll")
  let( ()
    setof( name namelist
      setof(key keys rexMatchp(key name))
    )
  )
)
```

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)
)

SKILL Code Optimization

Before you embark on optimizing your SKILL code, your SKILL program should work. You might waste time if you try to optimize the performance of a program that has not yet met its functional requirements.

Note: In this chapter, you can find information about techniques for modifying your programs to run faster. You will not learn how to create better algorithms: You need to design algorithmic performance into your programs from the outset.

When optimizing your SKILL code, you should do the following:

- Focus your efforts

Before optimizing, you should determine what you need to optimize. If 80% of the time is spent in 20% of the code, you should focus your efforts on optimizing that small portion of the program where there is a performance bottle-neck.

Always measure the benefit gained after you modify your code. You should leave well-written and well-structured code untouched if the changes do not yield significant performance improvements.

- Use profiling tools to measure code performance

Important

You should expect reduced execution speed when profiling and gathering performance data because these operations are necessarily intrusive in nature. The measurements you gather should take this into account. When profiling, you should think of code performance in terms of percentages rather than absolute values. See “Printing Summary Statistics” on page 237 for information on the how to obtain and interpret a summary of memory allocation.

You can optimize your SKILL code for time spent and memory used.

- Time: You can use the SKILL Profiler to find out where most of the time is spent in your program. You can gather global statistical information about time spent in functions called in a given session. The profiler takes a sample of the runtime stack

at pre-specified intervals and presents timing information as call graphs identifying the critical paths. You can use this information to direct your effort.

You can use `measureTime` to evaluate specific expressions and get timing results. You can also add your own more deterministic instrumentation to the code to collect relevant information about your algorithms and structures.

- ❑ **Memory:** You can use the SKILL Profiler to track memory usage in your program. SKILL has an automatic memory manager and programs can be written to generate excessive amounts of memory. Before you optimize for time, make sure to profile memory usage to see if that is where you need to spend your effort. A good indicator that memory usage needs optimizing is if the function `gc` (garbage collection) appears high among functions profiled for time.

Optimizing Techniques

You can try the following techniques to optimize your code. Not all techniques are effective in all circumstances. You should experiment and measure performance gain.

- [Macros](#)
- [Caching](#)
- [Mapping and Qualifying](#)
- [Write Protection](#)
- [Minimizing Memory](#)

Macros

In many situations, you can improve the performance of your code by replacing function calls with [macros](#). However, because macros are in-line expanded, they can grow the size of the code at runtime. So there is a balance as to what kind of functions can be written as macros.

For example, functions small in size that are likely to be used a lot are good candidates. Expressions that can be reduced at compile time leaving smaller subexpressions to be evaluated at run time are also good candidates.

Caching

Caching is a technique used to save the results of costly computations in a fast access cache structure. You have to balance the benefit of time saved versus amount of memory used for the cache structure.

A good data structure to use for caches is the association table. For example, if you called a compute-intensive function, say factorial or fibonacci, many times in a session, you might consider caching the results so a second call to the function with same argument will run faster. To do this, you first need to create an assoc table and store it as a property on the symbol for the function, for example:

The fibonacci function described in ["Fibonacci Function"](#) on page 388.

```
myFib.cache = makeTable("fibonacciTable" nil)
procedure( myFib(num)
  (let ((res myFib.cache[num]))
    cond( !res myFib.cache[num]= fibonacci(num))
      ( t res)
  ))
```

You could write the function `myFib` as a macro:

```
defmacro( myFib (num)
  'or (myFib.cache[,num]
    myFib.cache[,num] = fibonacci(,num)
  ))
```

For example, compare the following timing numbers (in seconds).

```
          fibonacci(25) myFib(25)
1st call 23 23
2nd call 23 0.009
```

Mapping and Qualifying

Mapping functions (`map`, `mapcar`, `maplist`, and so forth) have been described in detail in [“Advanced List Operations”](#) on page 179. When manipulating lists, you can achieve significant performance gains if you use `map*` functions instead of loops that directly manipulate the lists. The same argument applies to qualifiers like `foreach`, `setof` and `exists`. The qualifiers are generic in nature in that they apply to lists as well as to association tables.

Consider the following two examples performing the same operation of adding consecutive numbers in two equal length lists and returning a list of the results. The example using `mapcar` is at least twice as fast.

```
L1 = '(1 2 3 4 5 ... 100)
L2 = '(100 99 98 .. 1)

mapcar(lambda((x y) x+y) L1 L2)

(let (res)
  while(
    car(L1)
    res=cons(car(L1)+car(L2) res)
    L1 = cdr(L1)
    L2 = cdr(L2)
  ) res
)
```

Write Protection

Ensuring that all of your functions and data structures that are static in nature are write protected reduces the amount of work the garbage collector has to perform whenever it is triggered. Usually, all functions can be write protected because they are not likely to be over-written at run time. Some data structures, like lookup tables, whose contents are not likely to be modified at run time, can also be write protected.

It is highly recommended that SKILL code destined for production be packaged in contexts and that all contexts are built with the status flag `writeProtect` set to `t` (see [“Protecting](#)

[Functions and Variables](#)” on page 258). To set write protection on global variables use `setVarWriteProtect`.

When SKILL memory is write protected, the garbage collector does not touch it, thus considerably reducing the amount paging and work done at run time.

Minimizing Memory

The way memory is used in SKILL is by `consing` (the basic list building operation), creating strings, arrays, and so forth, or by generating instances of defstructs and user types. The goal of memory optimizing is to reduce the overall amount of memory used. This reduction

- Saves on run-time page swapping that an operation has to perform when physical memory resources are scarce
- Reduces the work load on the garbage collector

Run the SKILL Profiler in Memory Mode

To start optimizing memory usage, use the SKILL profiler in memory mode to discover the functions responsible for the largest amount of memory used. From there start tweaking the functions. You should be aware of the nature of the utilities and library calls you use.

For example, removing elements from lists using `remove` makes a copy of the original list minus the element removed. You should check to see if that is the desired behavior. If you can use a destructive remove instead, such as `remd`, you can cut down memory use considerably on this particular operation. Experiment.

Ways to Minimize

When generating large data structures in memory from information on disk, you should ask yourself whether you need to generate the whole image in memory if all of it is not likely to be used. Use the technique known as lazy evaluation to expand your structures on demand rather than at start-up.

For example, you can embed `lambda` constructs in your data structures to retrieve information on demand (see [“Declaring a Function Object \(lambda\)”](#) on page 223 for a description of `lambda` constructs). Experiment.

If you know from the outset the amount of memory your application is likely to need at run time, you can preallocate that memory to reduce the number of times the garbage collector is triggered. There is a delicate balance you have to make here between total memory allocated and garbage collection.

Remember, preallocating memory is **not** a substitute for fine tuning memory use in your program.

Only preallocate memory when you have tuned the program and determined the minimum amount of memory needed to run efficiently. Read “[Memory Management \(Garbage Collection\)](#)” on page 235 to better understand the nature of automatic memory management.

Tail-Call Optimization

When a function is called, its return address is pushed to the call stack. Each recursive call to the function results in a new stack frame being pushed to the call stack. Since the call stack has limited memory, recursive calls to a function can result in stack overflow. SKILL uses tail-call optimization to keep runtime stack-overflow errors in check.

If the last thing a function does before it returns is call another function, rather than allocating a new stack frame for the called function, tail-call optimization allows you to reuse the stack frame of the calling function. By using tail-call optimization, you can eliminate all intermediate return calls and implement recursion as a simple iteration.

To implement tail-call optimization in SKILL, set the `optimizeTailCall` variable to `t`, that is `sstatus(optimizeTailCall t)`

Note: Tail-call optimization is applicable to SKILL code with lexical scoping (`scheme/.ils` files). The `.il` files (dynamic scoping) use the runtime stack to store the function call's arguments, and therefore, the tail-call optimization is not applicable for `.il` files. In addition, tail call optimization does not work in the debug mode. Ensure that `debugMode` switch is set to `nil` before compiling the function and setting `optimizeTailCall` switch.

You can run the following examples and verify how tail-recursion is eliminated when you use tail-call optimization:

Example 1:

```
toplevel('ils) ;; in SCHEME mode
sstatus(debugMode nil) ;; turn off debugMode
tracef t ;; set trace = on
defun(test ()
  let( ()
    --x
    if(x > 0 then
      test()
    )
  )
)
```

```
) ;; recursive call to test()
ILS-<2>x = 4
```

Calling test() without setting optimizeTailCall:

```
test()
  (3 > 0)
  greaterp --> t
  test()
    (2 > 0)
    greaterp --> t
    test()
      (1 > 0)
      greaterp --> t
      test()
```

Calling test() after setting optimizeTailCall:

```
ILS-<2> sstatus(optimizeTailCall t)
ILS-<2> x = 4 ;; set x
ILS-<2> test()
  test()
    (3 > 0)
    greaterp --> t
  test()
    (2 > 0)
    greaterp --> t
  test()
    (1 > 0)
    greaterp --> t
  test()
    (0 > 0)
    greaterp --> nil
```

When you call `test()` by setting `optimizeTailCall`, the stack-overflow errors are eliminated as the existing stack frame is reused.

Example 2:

```
toplevel('ils) ;; in SCHEME mode
ILS-<2> defun(a (x)
  if(x > 0
    b(x)
  )
```

```
)  
a  
ILS-<2> defun(b (x)  
  a(--x)  
)  
b  
;this implements a simple delay:  
ILS-<2> a(1000000)  
*Error* b: Runtime Stack Overflow!  
ILS-<2> sstatus optimizeTailCall t  
a(1000000)  
=>nil ; no stack overflow
```

Benefits of using tail-call optimization:

- Functions execute faster since recursive push and pop operations are reduced.
- Stack-overflow errors are eliminated.

Limitations of using tail-call optimization:

- Debugging and tracing of optimized code is difficult as the intermediate return calls are eliminated.

General Optimizing Tips

See the following sections for general optimizing tips:

- [Element Comparison](#)
- [List Accessing](#) on page 283
- [List Building](#) on page 283
- [List Searching](#) on page 286
- [List Sorting](#) on page 287
- [Element Removal and Replacing](#) on page 287
- [Alternatives to Lists](#) on page 287

Element Comparison

In SKILL, there are two basic functions used to compare values. These functions are `eq` and `equal` (also known as the infix operator, `==`).

It is important to understand the difference between these two functions, because both are useful in particular circumstances. There are several functions in SKILL which have alternative implementations depending on whether the user wants to compare using the `eq` or `equal` function. For example, the two functions `memq` and `member` are used to search a list for an object. `memq` uses the `eq` function for comparison and `member` uses the `equal` function.

The `eq` function is far stricter in its comparison than the `equal` function. There are objects which `equal` would consider to be the same, but which `eq` considers to be different.

You can compare the following objects using `eq`:

- SKILL symbols
- Small integers ($-2^{**29} \leq i \leq 2^{**28}$)
- List objects (NOT their contents)
- Any pointers
- Characters (NOT strings)
- Ports

The important things that cannot be reliably compared by `eq` are strings and lists, unless they are identical objects referenced by the same pointer. In many situations SKILL tries to optimize memory use by caching certain objects and reusing them. For example, there is a string caching mechanism that saves SKILL from generating the same string multiple times. Code and data segments in static (write-protected) memory are also cached so they are reused within the static space.

The following are some examples of the more unexpected differences between the comparison operations:

```
x = '(1 2 3)
y = '(1 2 3)
eq(x y)           => nil
equal(x y)        => t

s1 = "string"
s2 = "string"
s3 = s2
eq(s1 s2)         => t      ; unreliable
equal(s1 s2)      => t      ; reliable
```

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```
eq(s3 s2)           => t      ; reliable
equal(s3 s2)        => t      ; reliable
eq(12345 12345)      => nil
l = '(12345)
eq(car(l) car(l))    => t
l = '("string")
eq(car(l) car(l))    => t
```

To understand more about `equal` and `eq`, keep in mind how they are implemented. The following are pseudo-code definitions of the two functions (they are implemented in C):

```
eq(A B)
  if A is the same object as B
  then t
  else nil
end eq

equal(A B)
  if eq(A B)
  then t
  else
    if the contents of A and B are 'equal'
    then t
    else nil
end equal
```

Suppose the two functions are used to compare two SKILL objects, A and B. If A and B are in fact the same object then `eq` will immediately return `t`. Because the first thing that `equal` does is call `eq`, it too will immediately return `t` in this case. Now suppose that A and B represent distinct objects. In this case, `eq` will immediately return `nil`. `equal`, however, goes on to try to establish if the contents of the two objects are the same, (for example, if the objects are lists, `equal` compares each element of the two lists for equality) and this process involves a large overhead. To summarize this behavior:

```
eq('a 'a)           => t      (fast)
equal('a 'a)         => t      (fast)
eq('a 'b)            => nil    (fast)
equal('a 'b)         => nil    (SLOW)
```

If in doubt about which of `eq` and `equal` to use, observe the following rules:

- If the objects are simple (symbols, small integers, pointers, characters), use `eq` because `eq` is faster than `equal`.
- If the objects are compound or complex (lists or strings, for example), consider what functionality is needed. To test whether two strings contain the same characters, use `equal`; To test whether two strings are in fact the same object, use `eq`.

Some common tests can be more efficiently implemented by using the built-in SKILL functions, or by using the fact that in SKILL, any non-`nil` object is true, not just `t`. Examples of some simple transformations are:

Original Test	Improved Test
<code>a == 0</code>	<code>zerop(a)</code>
<code>a == 1</code>	<code>onep(a)</code>
<code>strcmp(a,"teststring") == 0</code>	<code>a == "teststring"</code>
<code>a != nil</code>	<code>a</code>
<code>a == nil</code>	<code>!a</code>
<code>!null(a)</code>	<code>a</code>

For `a != nil`, providing a Boolean value is all that is required, for example,
`if( a != nil )` can be coded as `if( a )`

For `!null(a)`, providing a Boolean value is all that is required, for example,
`if( !null(a) )` can be coded as `if( a )`

List Accessing

The basic list accessing operations of SKILL (`car`, `cdr` and so forth) are fast, and their performance is predictable. The `nth` and `nthcdr` functions are significantly faster than the equivalent number of basic operations (because they avoid procedure call overhead) and should be used if lists are long. The operation that should be used with the most care is the `last` operation. This function must traverse the entire list to find the last element, so a large overhead is incurred for long lists.

List Building

There are two main methods of building lists: iteratively, either as part of a program's operation or when all the elements are the same type and non-iteratively when the format and number of elements are known. List building is an area that is open to abuse, and it is important that the processes involved are clearly understood.

Iterative List Creation

The standard function for adding an element to the start of a list is the `cons` function, which is efficient and has predictable performance. New users often find `cons` difficult to use because elements are added to the `front` of the list, giving a result that is “back to front”. There are several methods of producing a list in the “right” order, which vary in efficiency:

Use `append1` to add each element to the end of the list rather than to the start.

This is the most inefficient method, and lists should never be iteratively created in this way. As each element is added, the `append1` function makes a copy of the original list, with the new element on the end. If a list of n elements is built using this method, then on the order of n^2 list cells are created, and most of them are promptly discarded again.

Use `nconc` to add each element to the end of the list rather than to the start.

This method is also inefficient, and lists should never be iteratively created in this way. Because `nconc` is a “destructive” append, only n list cells need to be created to form the list. However, on the order of n^2 list cells must be traversed to build the list.

Use `cons` to build the list backwards, and then use `reverse` to turn the list around.

This is a much more efficient and easily understood method. To create a list of n elements, $2n$ list cells are created, but half of them are immediately discarded.

Use the `tconc` structure and function to build the list in the right order.

This is the most efficient method in terms of storage requirements. To create a list of n elements, only $n+1$ list cells are created. Because creation of list cells is relatively time consuming, this means that using `tconc` to build a long list is faster than using `cons` and `reverse`. A slight disadvantage is that the code is less intuitive.

In general, if the code is not time critical, it might be better to build the list backwards using `cons` and then apply `reverse`. If the code is in a time critical part of the program, then the `tconc` method should be used, along with some detailed commenting. From a memory usage point of view, if the list being built is long, then it is better to use the `tconc` method to prevent the garbage collector being called unnecessarily.

To build lists that are derived from existing lists, it is far better to use the mapping functions, possibly coupled with the `foreach` function. This is discussed in detail later.

Non-Iterative List Creation

To build lists of known format, use one of the `list`, `quote`, or `backquote` functions as follows:

- Use `list` when all elements of the result must be evaluated (in other words, none of the list members are known constants).
- Use `backquote` when the list contains a mixture of known constants and evaluated entries.
- Use `quote` when the list consists entirely of known constants.

For example, suppose we have the three variables, `a`, `b`, and `c` such that:

```
a = 1
b = "a string"
c = '(A B C D E)
```

If we wanted to make a disembodied property list (dpl) of symbol-value pairs using these variables then we could use `list` or a mixture of `quote` and `backquote`, as follows:

```
/* These are identical in function. */
dpList1 = list('a a 'b b 'c c)
dpList2 = '(a ,a b ,b c ,c)
```

In the second case, some items are preceded with commas (with no space after the comma) and some are not. Those `not` preceded with commas are treated as literals, as if this was a normal quoted list. Those preceded by commas are evaluated, as if the list was declared using `list`.

If this was the only use of `backquote`, it would be of little use. However, there are two useful extra features. The first is that you can splice in entire lists by using the `,@` (comma-at) specifier, as follows:

```
'(a ,a b ,b c ,@c) => (a 1 b "a string" c A B C D E)
```

Here, the five elements `A`, `B`, `C`, `D`, and `E` have been used in place of the placeholder `,@c`. This cannot be done easily using `list`. The second useful feature is that the expansion can descend hierarchically. Suppose that instead of a dpl we wanted to create an assoc list. To do this using just `list`, would require the following:

```
list( list('a a) list('b b) list('c c) )
=> ((a 1) (b "a string") (c (A B C D E)))
```

Using `backquote` simplifies this to:

```
'((a ,a) (b ,b) (c ,c))
=> ((a 1) (b "a string") (c (A B C D E)))
```

The last element can still be flattened using `,@` if required:

```
'((a ,a) (b ,b) (c ,@c))  
=> ((a 1) (b "a string") (c A B C D E))
```

Note: Care should be taken with lists defined using `quote`. These lists form part of the program code, and if edited using the destructive list operations the program code will be changed. This is particularly important when building `tconc` lists. It is tempting to initialize a `tconc` structure to `'(nil nil)`, but this is wrong. If this is done, then as each element is added the program itself is being modified.

Consider the following naive list copy:

```
procedure(copylist(inputList)  
    let( ((tc '(nil nil)))  
        foreach(element inputList  
            tconc(tc element)  
        )  
        car(tc)  
    ) /* let */  
 ) /* copylist */
```

This works the first time it is called, but the list assigned to `tc` in the variable declaration part is being modified as part of the program, so the next time the function is called, the copied list will be appended to the list that was copied in the first call:

```
copylist('(1 2 3)) => (1 2 3)  
copylist('(a b c)) => (1 2 3 a b c)
```

The list should be initialized by using either `list(nil nil)` or `tconc(nil nil)`. The second method makes it more obvious that the variable is being initialized as a `tconc` structure, but in this case the return value would be `cdar(tc)` rather than `car(tc)`.

List Searching

There are two methods for searching a list, depending on its structure. For a simple list, the functions `memq` and `member` can be used to search the list. The `memq` function is faster because it uses the `eq` function for comparison and is therefore preferred whenever the list contains elements for which the `eq` function is suitable.

If the list is an `assoc` list, that is, it is a list of key-value pairs, then the functions `assoc` and `assq` should be used to do the searching. Again, `assq` is faster because it uses the `eq` function for the comparison. It is therefore worthwhile, when building `assoc` lists, trying to ensure that the key elements are suitable for use with the `eq` function. In particular, when building an `assoc` list that would normally have keys that are strings, it may be worthwhile using the `concat` function to turn these strings into symbols, and then using those symbols as the keys in the list. This will then allow the `assq` rather than the `assoc` function to be used.

There are, however, two disadvantages with this method. The first disadvantage is that symbols in SKILL use memory and are not garbage collected (they are persistent), so creating many symbols uses up memory. Garbage collection is also slowed because the speed of this is directly related to the number of symbols. The second disadvantage is that the `concat` function is itself slow, so the overhead of this might outweigh any gains from using `assq` instead of `assoc`.

List Sorting

The `sort` and `sortcar` functions for lists are based on a recursive merge sort and are thus reasonably efficient. The list is sorted in-place, with no new list elements created, thus the list returned replaces the one passed as argument, and the one passed as argument should no longer be used.

Element Removal and Replacing

Two non-destructive functions are provided for removal of elements from a list, `remq` and `remove`. The equivalent but destructive functions are `remd` and `remdq`. These functions remove all elements from the list which match a given value, using the `eq` and `equal` function respectively. The `remq` function is faster than the `remove` function.

The function `subst` is provided for replacement of all elements of a list matching a particular value with another value. This function uses `equal` and so should be used sparingly.

It should be noted that the non-destructive functions return a `copy` of the original list with the matching elements removed. This means that these functions should not be used within a loop in order to remove a large number of elements. If a number of different elements must be removed from a list, then it is more efficient to generate a new list by traversing the old one just once, selecting only the required elements for the new list.

Alternatives to Lists

In many cases, there are faster and more compact alternatives to list structures. For example, if you need a property list that is likely to remain small in contents and most of the properties are known, consider using a `defstruct`.

If you need an `assoc` or property list whose contents are likely to be large (in the order of tens at least), then consider using `assoc` tables. `Assoc` tables offer much faster access time and for a large set of key-value pairs memory usage is more efficient. `Assoc` tables are not ordered (they are implemented as hash tables).

Miscellaneous Comparative Timings

This section gives comparative timings for various pieces of SKILL code to further demonstrate and reinforce the comments made in the previous sections. The examples are listed in an order that matches the structure of the preceding sections.

The timings were ascertained using the SKILL `profile` command and are expressed in ratios of the first example. In producing the timings, every effort has been made to compare like with like.

Element Comparison

The difference in speed between the `eq` and `equal` functions can be demonstrated using the following functions:

```
procedure(equal_test()  
    equal('fred 'joe)  
    equal('fred 'fred)  
) /* end equal_test */  
  
procedure(eq_test()  
    eq('fred 'joe)  
    eq('fred 'fred)  
) /* end eq_test */
```

Each procedure has two comparisons, one failing and one succeeding. The comparative times for these were:

```
equal_test 1.00  
eq_test    0.92
```

Further tests demonstrated that it is when the symbols are `not` equal that the `eq` function gains over the `equal` function. This means that if the test is expected to succeed on most occasions, there is little difference between the two functions.

List Building

As noted, there are several methods for building a list. The following examples attempt to build a list of the first 50 integers. The last example builds the list in descending order; the others build the list in ascending order.

```
procedure(list1()  
    let( (returnList)  
        for(i 1 50  
            returnList = append1(returnList i)  
        )  
        /* return */  
        returnList  
    ) /* end let */  
) /* end list1 */
```


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```
procedure(list2())
let( ((returnList list(nil nil))
      )
      for(i 1 50
            tconc(returnList i)
          )
      car(returnList)
    ) /* end let */
) /* end list2 */

procedure(list3())
let( (returnList)
      for(i 1 50
            returnList = cons(i returnList)
          )
      reverse(returnList)
    ) /* end let */
) /* end list3 */

procedure(list4())
let( (returnList)
      for(i 1 50
            returnList = cons(i returnList)
          )
      returnList
    ) /* end let */
) /* end list4 */
```

The outcome of this test depends on the length of the list being built. These examples use a medium length list, and the results of running these examples are:

```
list1 1.00
list2 0.14
list3 0.11
list4 0.10
```

These results demonstrate that with this size of list there is little difference between using the `tconc` method and the `cons` and `reverse` method. In fact, there will be little difference between these methods for any length of list because they both have to carry out the same basic functions. The only difference is that the `reverse` method must find twice as many list elements as the `tconc` method. This gives a greater chance of the garbage collector being called, which might cause the program to slow down, especially for large lists.

Mapping Functions

To demonstrate the relative speeds of the mapping functions, consider the following implementations of a function that picks every even integer out of a list of integers, returning the list of even integers in the same order as the originals.

```
procedure(map1(intList)
let( (res)
      while(intList
            when(evenp(car(intList))
                  res = cons(car(intList) res)
                )
            intList = cdr(intList)
          )
    )
```

```
        ) /* end while */
        reverse(res)
    ) /* end let */
) /* end map1 */

procedure(map2(intList)
    let( (res)
        foreach(i intList
            when(evenp(i)
                res = cons(i res)
            )
        ) /* end foreach */
        reverse(res)
    ) /* end let */
) /* end map2 */

procedure(map3(intList)
    foreach(mapcan i intList
        when(evenp(i)
            ncons(i)
        )
    ) /* end foreach */
) /* end map3 */

procedure(map4(intList)
    mapcan((lambda (i) when(evenp(i) ncons(i)))
        intList
    )
) /* end map4 */
```

The relative timings for these are:

map1	1.00
map2	0.68
map3	0.64
map4	0.29

This shows that the version using a `lambda` function along with the basic mapping function is the fastest. However, this is the least readable of these functions and should only be used with a great deal of caution, and a large number of comments.

Data Structures

It is difficult to give meaningful comparisons between the data structure functions because they are all suitable for different tasks. The following two examples attempt to compare the time taken to access and change one element of a data structure stored as an array, simple list, assoc list, defstruct and property list.

```
elem_number = 9
elem_symbol = 'j
procedure(access1(array)
    array[elem_number]
) /* end access1 */

procedure(access2(list)
    nth(elem_number list)
) /* end access2 */
```

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```
procedure(access3(assoc_list)
  cadr(assq(elem_symbol assoc_list))
) /* end access3 */

procedure(access4(dstruct)
  get(dstruct elem_symbol)
) /* end access4 */

procedure(access5(plist)
  get(plist elem_symbol)
) /* end access5 */

procedure(access6(assocTable)
  assocTable[elem_symbol]
) /* end access */
```

The comparative timings for these are:

```
access1      1.00
access2      1.3
access3      1.47
access4      1.08
access5      1.55
access6      1.11

procedure(set1(array val)
  array[elem_number] = val
) /* end set1 */

procedure(set2(list val)
  rplaca(nthcdr(elem_number list) val)
) /* end set2 */

procedure(set3(assoc_list val)
  rplaca(cdr(assq(elem_symbol assoc_list)) val)
) /* end set3 */

procedure(set4(dstruct val)
  putprop(dstruct val elem_symbol)
) /* end set4 */

procedure(set5(plist val)
  putprop(plist val elem_symbol)
) /* end set5 */

procedure(set6(assocTable val)
  assocTable[elem_symbol] = val
) /* end set6 */
```

The comparative timings for these are:

```
set1      1.00
set2      1.14
set3      1.09
set4      0.93
set5      1.71
set6      1.2
```

No comment will be made about the readability of these functions because access procedures should be made available for all data structure access anyway. It is clear from these results that there is little to be gained, in terms of speed from the choice of data structure, although arrays do seem to be fastest overall. When choosing data structures it is

more important to consider the other factors mentioned in the data structures section of this document.

Because the structures used contained a small number of elements, the experiment is naturally biased. You can repeat this experiment using `measureTime` to find out how effective your data structure accesses are for a given volume of data. For example, for sets of hundreds of elements, association tables will be significantly faster to access than property lists and assoc lists. For large sets it would be impractical to use arrays or defstructs.

About SKILL++ and SKILL

Cadence® SKILL++ is the second generation extension language for Cadence software. SKILL++ combines the ease-of-use of the SKILL environment with the power of the Scheme programming language. The major power brought in from Scheme is its use of lexical scoping and functions with lexically-closed environments called closures:

- Lexical scoping makes reliable and modular programming more easily achievable, because you have total control over the use and reference of variables for any code without worrying about accidental corruptions caused by the use of the same variable name in a remote place.
- Closures are powerful entities that only exist in the more advanced programming languages. They encapsulate the code and related data into a single unit, with total control on the exported interface. Many modern programming idioms and paradigms, such as message-passing and objects with inheritance, can be implemented elegantly using closures.

Other advantages of SKILL++ include the following:

- SKILL++ provides environments as first-class objects. With closures and first-class environments, you can create your own module or package systems. You are no longer restricted to SKILL's single flat name space model. Instead, you can organize the code into a hierarchy of name spaces.
- In addition to Scheme semantics, SKILL++ includes an object layer that makes explicit object-oriented style programming possible. The object layer supports classes, generic functions, methods, and multiple inheritance.
- Because SKILL++ and SKILL can coexist harmoniously in the same environment, backward compatibility and interoperability are not an issue. All existing SKILL code can still run without any changes, and all or part of any SKILL package can be migrated to SKILL++. Code developed in SKILL++ and SKILL can call each other and share the same data structures transparently.

Background Information about SKILL and Scheme

SKILL was originally based on a flavor of Lisp called “Franz Lisp.” Franz Lisp and all other flavors of Lisp were eventually superseded by an ANSI standard for Lisp called “Common Lisp.” The semantics and nature of SKILL make it ideal for scripting and fast prototyping. But the lack of modularity and good data abstraction, or its general openness, make it harder to apply modern software engineering principles especially for large endeavors. Since its inception, SKILL has been used for writing very large systems within the Cadence tools environments. To this end Cadence chose to offer Scheme within the SKILL environment.

Scheme is a Lisp-like language developed originally for teaching computer science at the Massachusetts Institute of Technology and is now a popular language in computer science and sometimes EE curriculums. Scheme is a modern language whose semantics empower engineers to develop sound software systems. There is an IEEE standard for Scheme. Scheme was also the choice of the CAD Framework Initiative (CFI) for an extension language base. Cadence supplies major Scheme functionality as part of the SKILL environment. Benefits include the following:

- New programs written in Scheme can coexist and call procedures in existing programs written in SKILL without paying penalties in performance or functionality.
- Scheme and SKILL will share the run-time environment so structures allocated in Scheme programs can be passed without modification to SKILL programs and visa-versa.
- Suppliers and consumers can choose to migrate to Scheme independently of each other. For example, a Cadence-supplied layer can choose to remain written in SKILL while users of that layer can switch to using Scheme. The converse is also true.

SKILL++ Relation to IEEE and CFI Standard Scheme

CFI has chosen IEEE standard Scheme as the base of their proposed CAD Framework extension language. Because the intended use was as a CAD tool extension language, which must be embeddable within large applications in a mixed language environment, CFI relaxed the requirement for the support of full “call-with-current-continuation” and the full numeric tower (only numbers equivalent to C's long and double are required).

For the same reason, CFI added a few extensions such as exception handling and functions for evaluating Scheme code, as well as a specification on the foreign function interface APIs, to their proposal.

SKILL++ is designed with IEEE Scheme and CFI Scheme compliance in mind, but due to its SKILL heritage and compatibility, it is not fully compliant with either standard.

The following sections describe the differences between SKILL++ and the standard Scheme language.

Syntax Differences

SKILL++ uses the same familiar SKILL syntax with the following restrictions and extensions.

Restrictions

- Because most of the special characters are used as infix symbols in SKILL++ and SKILL, they cannot be used as regular name constituents. However, many standard Scheme functions and syntax forms have been systematically renamed for ease of use under SKILL++'s syntax, for example

```
pair?           ==> pairp
list->vector     ==> listToVector
make-vector     ==> makeVector
set!            ==> setq
let*            ==> letseq (for "sequential let")
```

- Except for vector literals (such as `#(1 2 3)`), the `#...` syntax is not supported, so use `t` for `#t`, `nil` for `#f`, and use single character symbols for character literals and so forth.

Extensions

- Like SKILL, but unlike standard Scheme, SKILL++ symbols are case-sensitive.
- SKILL++ inherited all the SKILL syntactic features, such as infix notation, optional/keyword arguments with default values, and many powerful looping special forms (such as `for`, `foreach`, `setof`).
- SKILL++ code can define macros using the `mprocedure/defmacro` as well as use existing macros defined in SKILL.

Semantic Differences

SKILL++ adopts the standard Scheme semantics with the following restrictions and extensions.

Restrictions

- The atom `nil` is the same as the empty list `'()`, as well as the `false` value. Standard Scheme uses `#f` for the only false value and treats the empty list as a true value.

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- The `cons` cells in SKILL++ are like SKILL's, that is, their `cdr` slot can only be either `nil` or another `cons` cell. In standard Scheme, the `cdr` slot of a `cons` cell can hold any value.
- The SKILL++ `map` function and the SKILL `map` function share the same name and implementation, but behave differently from standard Scheme's `map` function. To get the behavior of Scheme's `map`, use `mapcar` in SKILL++.
- Strings in SKILL++ and SKILL are immutable, so there is no support for functions like Scheme's `string-set!`.
- The `character` type is not supported yet. As in SKILL, symbols of one character can be used as characters.
- No “eof” object. The `lineread` function returns `nil` on end-of-file.

Extensions

- Environments are treated as first class objects. The `theEnvironment` form can be used to get the enclosing lexical environment, and bindings in an environment can be easily accessed. This provides a powerful encapsulation tool.
- SKILL++ inherits the SKILL set of powerful data structures, such as `defstruct` and association tables, as well as all the functions and many special forms of SKILL.
- Support for transparent cross-language (SKILL <-> SKILL++) mixed programming.

Syntax Options

SKILL++ adopts the same SKILL syntax, which means SKILL++ programs can be written in the familiar infix syntax and the general Lisp syntax.

If syntax is not an issue for you and you are comfortable with the infix notation, continue to use that.

If you are concerned that the knowledge of SKILL++ you build by programming in the infix syntax will not be useful if you were to program in a Scheme environment (without SKILL), then use the Lisp syntax for programming in SKILL++. The syntactic differences between SKILL++ using Lisp syntax and standard Scheme are:

- In SKILL++, you may not use any special characters (such as `+`, `-`, `/`, `*`, `%`, `!`, `$`, `&`, and so forth) in identifiers because most of these characters are used as infix operators.
- As a general convention, Scheme functions ending with an exclamation point (`!`) are provided either without the exclamation point or as the equivalent SKILL function. For example, Scheme `set!` is SKILL++ `setq`. Scheme functions using `->` are provided

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using `To` as part of the function name. For example, `list->vector` becomes `listToVector`. See “[Scheme/SKILL++ Equivalents](#)” in the *Cadence SKILL Language Reference* for a complete list of name mappings.

- Scheme's dotted pairs are not available in SKILL++. Use simple lists instead.
- You can use `=>` and `...` as identifiers.

Compliance Disclaimer

Cadence-supplied Scheme is not fully IEEE compliant for the following reason:

- Scheme was not originally designed as an extension language. Features in Scheme that cannot be used safely in conjunction with a system written in C/C++ are omitted, such as “`call/cc`” and non-null terminated strings.
- Cadence puts a high value on making the system fully backward compatible for SKILL programs and procedural interfaces. As a result, the empty list `nil` is a Boolean `true` in the Scheme standard while Cadence's SKILL and SKILL++ treat `nil` as a Boolean `false`. Without this treatment of `nil`, the migration of existing SKILL programs to Scheme would require many existing procedural interfaces written in SKILL to change.

In general, SKILL++ is implemented to support both the Lisp syntax and the more familiar SKILL infix syntax for writing Scheme programs, as well as to provide a smooth path for migrating SKILL code to Scheme.

References

You can read the following for more information about Scheme:

Structure and Interpretation of Computer Programs, Harold Abelson, Gerald Sussman, Julie Sussman, McGraw Hill, 1985.

Scheme and the Art of Programming, G. Springer & D. Friedman, McGraw Hill, 1989.

An Introduction to Scheme, J. Smith, Prentice Hall, 1988.

“Draft Standard for the Scheme Programming Language,” P1178/D5, October 1, 1990. IEEE working paper.

Extension Language Environment

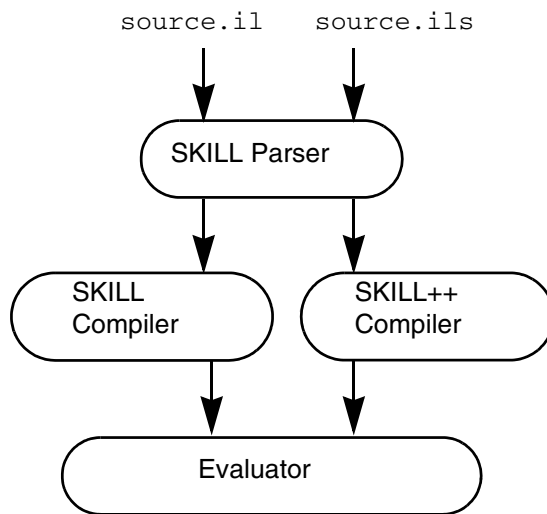
Cadence's extension language environment supports two integrated extension languages, SKILL and SKILL++. Your application can consist of source code written in either language. Programs in either language can call each other's functions and share data.

- You can maintain existing SKILL applications with no change whatsoever.
- Using SKILL++, you can hide private functions and private data so that you can design and implement your application with reusable components.

You should consider developing new applications using the SKILL++ language, in conjunction with SKILL procedural interfaces as necessary.

For source code files, the file extension indicates the language:

`.il` for SKILL, `.ils` for SKILL++.



By default, the session accepts SKILL expressions. Interactively, you can invoke a top level for either SKILL or SKILL++ as follows:

Language	Command
SKILL	<code>toplevel 'il</code>
SKILL++	<code>toplevel 'ils</code>

Advanced programmers can use the `eval` function to evaluate an expression using either SKILL semantics or SKILL++ semantics.

Contrast Variable Scoping

Variables associate an identifier with a memory location. A referencing environment is the collection of identifiers and their associated memory locations. The scope of a variable refers to the part of your program within which the variable refers to the same location.

During your program's execution, the referencing environment changes according to certain scoping rules. Even though the syntax of both languages is identical, SKILL and SKILL++ use different scoping rules and partition a program into pieces differently.

SKILL++ Uses Lexical Scoping

SKILL++ uses lexical scoping. Because lexical scoping relies solely on the static layout of the source code, the programmer can confidently determine how reusing a program affects the scope of variables.

The lexical scoping rule is only concerned with the source code of your program. The phrase 'a part of your program' means a block of text associated with a SKILL++ expression, such as `let`, `letrec`, `letseq`, and `lambda` expressions. You can nest blocks of text in the source code.

SKILL Uses Dynamic Scoping

SKILL uses dynamic scoping. Modifying a SKILL program can sometimes unintentionally disrupt the scope of a variable. The probability of introducing subtle bugs is higher.

The dynamic scoping rule is only concerned with the flow of control of your program. In SKILL, the phrase 'a part of your program' means a period of time during execution of an expression. Usually, the dynamic scoping and lexical scoping rules agree. In these cases, identical expressions in both SKILL and SKILL++ return the same value.

Lexical versus Dynamic Scoping

It is important that the scope of variables not be unintentionally disrupted when a programmer modifies or otherwise reuses a program. Subtle bugs can result when modification or reuse in another setting changes the scope of a variable.

Lexical scoping makes it possible for you to inspect the source code to determine the effect on the scope of variables. Dynamic scoping—which relies on the execution history of your program—can make it difficult to write reusable, modular code by preventing confident reuse of existing code.

Example 1: Sometimes the Scoping Rules Agree

The following example has line numbers added for reference only.

```
1: let ( (x)
2:     x = 3
3:     x
4: )
```

In both SKILL and SKILL++, the `let` expression establishes a scope for `x` and the expression returns 3.

- In SKILL++, the scope of `x` is the block of text comprising line 2 and line 3. The `x` in line 2 and the `x` in line 3 refer to the same memory location.
- In SKILL, the scope of `x` begins when the flow of control enters the `let` expression and ends when the flow of control exits the `let` expression.

Example 2: When Dynamic and Lexical Scoping Disagree

In example 1, the two scoping rules agree and the `let` expression returns the same value in both SKILL and SKILL++. Example 2 illustrates a case in which dynamic and lexical scoping disagree. Notice that the following extends the first example by merely inserting a function call to the `A` function between two references to `x`. However, the `A` function assigns a value to `x`.

```
1: procedure( A() x = 5 )
2: let( ( x )
3:     x = 3
4:     A()
5:     x
6: )
```

Consider the `x` in line 1.

- In SKILL++, the `x` in line 1 and the `x` in line 5 refer to different locations because the `x` in line 1 is outside the block of text determined by the `let` expression. The `let` expression returns 3.
- In SKILL, because line 1 executes during the execution of the `let` expression, the `x` in line 1 and the `x` in line 5 refer to the same location. The `let` expression returns 5.

Example 3: Calling Sequence Effects on Memory Location

In SKILL, dynamic scoping dictates that the memory location affected depends on the function calling sequence. In the code below, function `B` updates the global variable `x`. Yet, when called from the function `A`, function `B` alters function `A`'s local variable `x` instead. Function `B` updates the `x` that is local to the `let` expression in `A`.

- In SKILL, function `A` returns 6.
- In SKILL++, function `A` returns 5.

```
procedure( A()
  let( ( x )
    x = 5          ;;; set the value of A's local variable x
    B()
    x              ;;; return the value of A's local variable x
  ) ;let
) ; procedure

procedure( B()
  let( ( y z )
    x = 6
    z
  )
) ; procedure
```

See “[Dynamic Scoping](#)” on page 228 for guidelines concerning the use of dynamic scoping.

Difference in Symbol Usage

SKILL and SKILL++ share the same symbol table. Each symbol in the symbol table is visible to both languages.

How SKILL Uses Symbols

SKILL has a data structure called a symbol. A symbol has a name which uniquely identifies it and three associated memory locations. For more information, see “[Symbols](#)” on page 89.

The Value Slot

SKILL uses symbols for variables. A variable is bound to the value slot of the symbol with the same name. For example, `x = 5` stores the value 5 in the value slot of the symbol `x`. The `symeval` function returns the contents of the value slot. For example, `symeval( 'x )` returns the value 5.

The Function Slot

SKILL uses the function slot of a symbol to store function objects. SKILL evaluates a function call, such as

```
fun( 1 2 3 )
```

by fetching the function object stored in the function slot of the symbol `fun`. Dynamic scoping does not affect the function slot at all.

The Property List

Dynamic scoping does not affect the property list at all. See [“Important Symbol Property List Considerations”](#) on page 93.

Summary

You can call the `set`, `symeval`, `getd`, `putd`, `get`, and `putprop` SKILL functions to access the three slots of a symbol. The following table summarizes the SKILL operations that affect the three slots of a symbol.

SKILL Construct	Value Slot	Function Slot	Property List
assignment operator (=)	x		
<code>set</code> , <code>symeval</code>	x		
<code>let</code> and <code>prog</code> constructs	x		
procedure declaration		x	
<code>getd</code> , <code>putd</code>		x	
function call		x	
<code>get</code> , <code>putprop</code>			x

How SKILL++ Uses Symbols

Normally, each SKILL++ global variable is bound to the function slot of the symbol with the same name. In this way, SKILL and SKILL++ can share functions transparently.

Sometimes your SKILL++ program needs to access a SKILL global variable. You can use the `importSkillVar` function to change the binding of the SKILL++ global from the function slot of a symbol to the value slot of the symbol.

Difference in the Use of Functions as Data

In SKILL++ it is much easier to treat functions as data than it is in SKILL.

Assigning a Function Object to a Variable

In SKILL++, function objects are stored in variables just like other data values. You can use the familiar SKILL algebraic or conventional function call syntax to invoke a function object indirectly. For example, in SKILL++:

```
addFun = lambda( ( x y ) x+y ) => funobj:0x1e65c8
addFun( 5 6 ) => 11
```

In SKILL, the same example can be done two different ways, both of them less convenient.

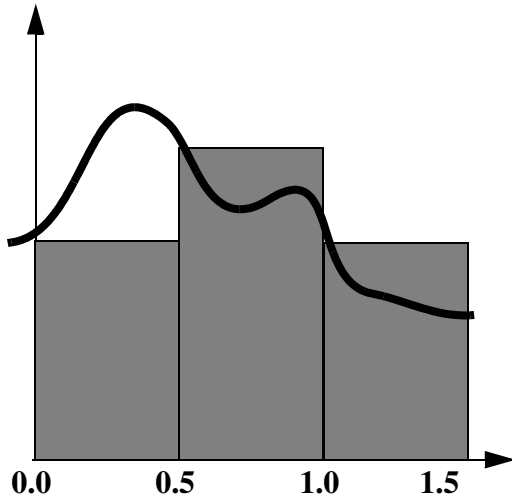
```
addFun = lambda( ( x y ) x+y )
apply( addFun list( 5 6 ) ) => 11
```

or

```
putd( 'addFun lambda( ( x y ) x+y ) )
addFun( 5 6 ) => 11
```

Passing a Function as an Argument

To pass a function as an argument in SKILL++ does not require special syntax. In SKILL the caller must quote the function name and the callee must use the `apply` function to invoke the passed function.



The `areaApproximation` function in the following example computes an approximation to the area under the curve defined by the `fun` function over the interval `( 0 1 )`. The approximation consists of adding the areas of three rectangles, each with a width of 0.5 and with heights of `fun( 0.0 )`, `fun( 0.5 )`, and `fun( 1.0 )`.

In SKILL++

```
procedure( areaApproximation( fun )
  0.5*( fun( 0.0 ) + fun( 0.5 ) + fun( 1.0 ) )
) => areaApproximation

areaApproximation( sin ) => 0.6604483
areaApproximation( cos ) => 1.208942
```

In SKILL

```
procedure( areaApproximation( fun )
  0.5*(
    apply( fun '( 0.0 ) ) +
    apply( fun '( 0.5 ) ) +
    apply( fun '( 1.0 ) )
  )
) => areaApproximation

areaApproximation( 'sin ) => 0.6604483
areaApproximation( 'cos ) => 1.208942
```


SKILL++ Closures

A SKILL++ closure is a function object containing one or more `free` variables (defined below) bound to data. Lexical scoping makes closures possible. In SKILL, dynamic scoping prevents effective use of closures, but a SKILL++ application can use closures as software building blocks.

Relationship to Free Variables

Within a segment of source code, a `free` variable is a variable whose binding you cannot determine by examining the source code. For example, consider the following source code fragment.

```
procedure( Sample( x y )
  x+y+z
)
```

By examination, `x` and `y` are not free variables because they are arguments. `z` is a free variable. In SKILL++ lexical scoping implies that the bindings of all of a function's free variables is determined at the time the function object (closure) is created. In SKILL, dynamic scoping implies that all references to free variables are determined at run time.

For example, you can embed the above definition in a `let` expression that binds `z` to 1. Only code in the same lexical scope of `z` can affect `z`'s value.

```
let( (( z 1 ))
  procedure( Sample( x y )
    x+y+z
  )
  ;;; code that invokes Sample goes here.
)
```

How SKILL++ Closures Behave

A SKILL++ application can use closures as software building blocks. The following examples increase in complexity to illustrate how SKILL++ closures behave.

Example 1

```
let( (( z 1 ))
  procedure( Sample( x y )
    x+y+z
  )
  ;;; code that invokes Sample goes here.
  Sample( 1 2 )
)
=> 4
```

Example 2

```
let( (( z 1 ))
  procedure( Sample( x y )
    x+y+z
  )
  ;; code that invokes Sample goes here.
  z = 100
  Sample( 1 2 )
)
=> 103
```

This example assigns 100 to the binding of `z`. Consequently, the `Sample` function returns 103. In this case, dynamic scoping and lexical scoping agree.

Example 3

```
let( (( z 1 ))
  procedure( Sample( x y )
    x+y+z
  ) ; procedure
  ;; code that invokes Sample goes here.
  let( (( z 100 ))
    Sample( 1 2 )
  ) ; let
) ; let
=> 4
```

This example invokes `Sample` from within a nested `let` expression. Although dynamic scoping would dictate that `z` be bound to 100, lexically it is bound to 1.

Example 4

```
procedure( CallThisFunction( fun )
  let( (( z 100 ))
    fun( 1 2 )
  )
)

let( (( z 1 ))
  procedure( Sample( x y )
    x+y+z
  ) ; procedure
  CallThisFunction( Sample )
)
=> 4
```

In this SKILL++ example, the `let` expression binds `z` to 1, creates the `Sample` function and then passes it to the `CallThisFunction` function. Whenever the `Sample` function runs, `z` is bound to 1. In particular, when the `CallThisFunction` function invokes `Sample`, `z` is bound to 1 even though `CallThisFunction` binds `z` to a different value prior to calling `Sample`.

Therefore, `Sample` has encapsulated a value for its free variable `z`. To do this in SKILL is impossible, because dynamic scoping dictates that `Sample` would see the binding of `z` to 100.

Therefore, SKILL++ allows you to build a function object that you can pass an argument, certain that its behavior will be independent of specifics of how it is ultimately called.

Example 5

```
W = let( ( ( z 1 ) )
        lambda( ( x y )
                x+y+z
              )
      )
=> funobj:0x1bc828
W( 1 2 ) => 4
```

In this example, the name `Sample` is insignificant because the `let` expression itself does not contain a call to `Sample`. Instead, the `let` expression returns the function object. This function object is a closure. The code returns a distinct closure each time the code is executed. In SKILL++, there is no way to affect the binding of `z`. The function object has effectively encapsulated the binding of `z`.

Example 6

The `makeAdder` function below creates a function object which adds its argument `x` to the variable `delta`. Each call to `makeAdder` returns a distinct closure.

```
procedure( makeAdder( delta )
          lambda( ( x ) x + delta )
        )
=> makeAdder
```

In SKILL++, you can pass 5 to `makeAdder` and assign the result to the variable `add5`. No matter how you invoke the `add5` function, its local variable `delta` is bound to 5.

```
add5 = makeAdder( 5 )=> funobj:0x1e3628
add5( 3 ) => 8

let( ( ( delta 1 ) )
    add5( 3 )
  ) => 8

let( ( ( delta 6 ) )
    add5( 3 )
  ) => 8
```

SKILL++ Environments

This section introduces the run-time data structures called `environments` that SKILL++ uses to support lexical scoping. This section covers how SKILL++ manages environments during run time. Understanding this material is important if your application use closures.

For more information, [Using SKILL and SKILL++ Together](#) covers how inspecting environments can help you debug SKILL++ programs.

The Active Environment

During the execution of your SKILL++ program, the set of all the variable bindings is called an `environment`. To accommodate the sequence of nested lexical scopes which contain the current SKILL++ statement being executed, an environment is a list of environment frames such that

- SKILL++ stores all the variables with the same lexical scope in a single environment frame
- The sequence of nested lexical scopes correspond to a list of environment frames

Consequently, each environment frame is equivalent to a two column table. The first column contains the variable names and the second column contains the current values.

The Top-Level Environment

When a SKILL++ session starts, the active environment contains only one environment frame. There are no other environment frames. All the built-in functions and global variables are in this environment. This environment is called the top-level environment. The `toplevel( 'ils )` function call uses the SKILL++ top-level environment by default. However, it is possible to call the `toplevel( 'ils )` function and pass a non-top-level environment. Consider an expression such as

```
let( (( x 3 )) x ) => 3
```

in which we insert a call to `toplevel( 'ils )`. During the interaction we attempt to retrieve the value of `x` and then set it. References to `x` affect the SKILL++ top-level.

```
ILS-<2> let( (( x 3 )) toplevel( 'ils ) x )
ILS-<3> x
*Error* eval: unbound variable - x
ILS-<3> x = 5
5
ILS-<3> resume()
3
ILS-<2>
```

Compare it with the following in which we call `toplevel` passing in the lexically enclosing (active) environment. Thus the `toplevel` function can be made to access a non-top-level environment!

```
ILS-<2> let( (( x 3 )) topLevel( 'ils theEnvironment() ) x )
ILS-<3> theEnvironment()->??
((x 3))
ILS-<3> x
3
ILS-<3> x = 5
5
ILS-<3> resume()
5
ILS-<2> x
*Error* eval: unbound variable - x
ILS-<2>
```

Creating Environments

During the execution of your program, when SKILL++ evaluates certain expressions that affect lexical scoping, SKILL++ allocates a new environment frame and adds it to the front of the active environment. When the construct exits, the environment frame is removed from the active environment.

Example 1

```
let( (( x 2 ) ( y 3 ))
      x+y
    )
```

When SKILL++ encounters a `let` expression, it allocates an environment frame and adds it to the front of the active environment.

An Environment Frame

Variable	Value
x	2
y	3

To evaluate the expression `x+y`, SKILL++ looks up `x` and `y` in the list of environment frames, starting at the front of the list. When the expression terminates, SKILL++ removes it from the active environment. The environment frame remains in memory as long as there are references to the environment frame.

In this simple case, there are none, so the frame is discarded, which means it's garbage and therefore liable to be garbage collected.

Example 2

```
let( ( ( x 2 ) ( y 3 ) )  
    let( ( ( u 4 ) ( v 5 ) ( x 6 ) )  
        u*v+x*y  
    )  
)
```

At the time SKILL++ is ready to evaluate the expression $u * v + x * y$, there are two environment frames at the front of the active environment.

Environment Frame for the Outermost let

Variable	Value
x	2
y	3

Environment Frame for the Innermost let

Variable	Value
u	4
v	5
x	6

To determine a variable's location is a straight-forward look up through the list of environment frames. Notice that x occurs in both environment frames. The value 6 is the first found at the time the expression $u * v + x * y$ is evaluated.

Functions and Environments

When you create a function, SKILL++ allocates a function object with a link to the environment that was active at the time the function object was created.

When you call a function, SKILL++ makes the function object's environment active (again) and allocates a new environment frame to hold the arguments. For example,

```
procedure( example( u v )  
    let( ( ( x 2 ) ( y 3 ) )  
        x*y + u*v  
    )  
)  
example( 4 5 )
```

allocates the following:

An Environment Frame

Variable	Value
u	4
v	5

and adds to the front of the active environment, which is the environment saved when the `example` function was first created.

When the function returns, SKILL++ removes the environment frame holding the arguments from the active environment and, in this case, the environment frame becomes garbage. It then restores the environment that was active before the function call.

Persistent Environments

The `makeAdder` example below shows a function which allocates a function object and then returns it. The returned function object contains a reference to the environment that was active at the time the function object was created. This environment contains an environment frame that holds the argument to the original function call.

Therefore, whenever you subsequently call the returned function object, it can refer the local variables and arguments of the original function even though the original function has returned.

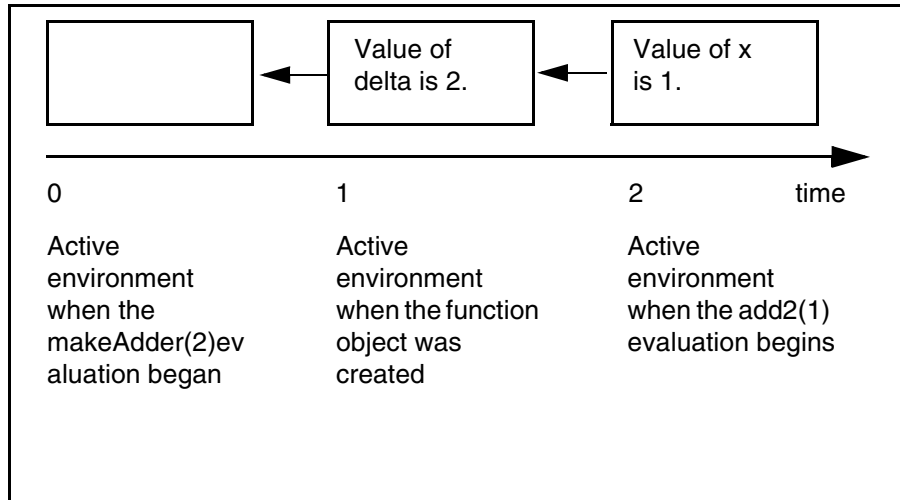
This capability gives SKILL++ the power to build robust software components that can be reused. Full understanding of this capability is the basis for advanced SKILL++ programming.

```
procedure( makeAdder( delta)
  lambda( ( x ) x + delta )
)
=> makeAdder
add2 = makeAdder(2) => funobj:0x1e6628
add2( 1 ) => 3
```

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About SKILL++ and SKILL

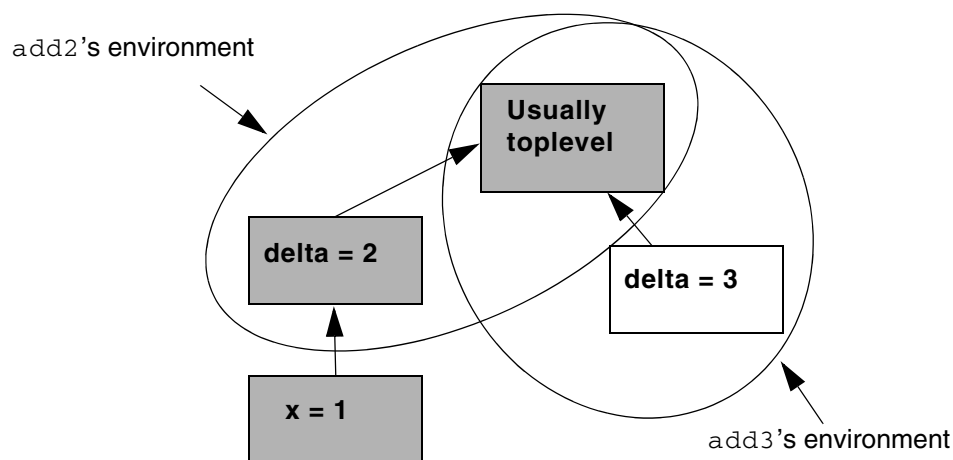
The function object that `makeAdder` returns is within the lexical scope of the `delta` argument.



Calling `makeAdder` again returns another function object.

```
add3 = makeAdder(3) => funobj:0x1e6638
```

The figure below shows several environments. The encircled environments belong to the `add2` and `add3` functions. The gray environment is the active environment at the time `add2(1)` at its entry point. The other environment belongs to the `add3` function, which becomes active only if `add3` is called.



Using SKILL++

This chapter deals with the pragmatics of writing programs in the Cadence® SKILL++ language.

About SKILL++ and SKILL provides an overview of the differences between the Cadence SKILL language and SKILL++.

Using SKILL and SKILL++ Together focuses on the key areas in which SKILL++ semantics differ from SKILL semantics.

SKILL++ Object System describes a system that allows for object-oriented interfaces based on classes and generic functions composed of methods specialized on those classes.

Related Topics

Declaring Local Variables in SKILL++

Sequencing and Iteration

Software Engineering with SKILL++

SKILL++ Packages

Declaring Local Variables in SKILL++

SKILL++ provides three binding constructs to declare local variables together with initialization expressions. The syntax for `let`, `letseq` and `letrec` is identical but they differ in the order of evaluation of the initialization expressions and in the scope of the local variables.

Syntax Template for `let`, `letseq` and `letrec`

```
let(  
  ( ( s_var1 g_initExp1 ) ( s_var2 g_initExp2 )... )  
  g_bodyExp1
```

```
g_bodyExp2
...
)
```

Using let

Each local variable has the body of the `let` expression as its lexical scope. The order of evaluation of the initialization expressions and the binding sequence is unspecified. You should avoid cross-references between variables in a `let` expression.



Caution

The initialization expression bound to one local variable should not refer to any of the other local variables.

Example 1

```
let( ( ( x 2 ) ( y 3 ) )
      x*y
    ) => 6
```

Example 2

```
let( ( ( x 2 ) ( y 3 ) )
      let( (( z 4 ))
            x + y + z
          ) ; let
    ) ; let
=> 9
```

Example 3

```
let( ( ( x 2 ) ( y 3 ) )
      let( (( x 7 ) ( z x+y ) )
            z*x
          )
    ) => 35
```

Because the initialization expressions are outside of the scope of the `let`, `z` is bound to `2+3`, instead of `7+3`.

Example 4

```
let( ( ( x 2 ) ( y 3 ) )
      let( (( x 7 ) ( foo lambda( ( z ) x + y + z ) ) )
            foo( 5 )
          )
    )
```

```
)  
=> 10
```

This example shows that the initialization expressions are also outside the scope of the `let`. Specifically, the occurrence of `x` in the body of `foo` is in the scope of the outer `let`.

Using `letseq`

Use `letseq` to control the order of evaluation of the initialization expressions and the binding sequence of the local variables. Evaluation proceeds from left to right. The scope of each variable includes the remaining initialization expressions and the body of `letseq`. It is equivalent to a corresponding sequence of nested `let` expressions.

Example 1

```
letseq( ( ( x 1 ) ( y x+1 ) )  
        y  
      )  
=> 2
```

The code above is a more convenient equivalent to the code below in which you control the sequence explicitly by the nesting.

```
let( ( ( x 1 ) )  
     let( ( ( y x+1 ) )  
          y  
        )  
      )
```

Example 2

```
let( ( ( x 2 ) ( y 3 ) )  
     letseq( (( x 7 ) ( z x+y ) )  
             z*x  
           )  
      ) => 70
```

This example is identical to “[Example 3](#)” on page 314 except that the inner `let` is replaced with `letseq`.

Example 3

```
let( ( ( x 2 ) ( y 3 ) )  
     letseq( (( x 7 ) ( foo lambda( ( z ) x + y + z ) ) )  
             foo( 5 )  
           )  
      )  
=> 15
```

This example is identical to “[Example 4](#)” on page 314 except that the inner `let` is replaced with `letseq`.

Using letrec

Unlike `let` and `letseq`, each variable’s scope is the entire `letrec` expression. In particular, each variable’s scope includes all of the initialization expressions. Each initialization expression can refer to the other local variables with the following restriction: each initialization expression must be executable without accessing the other variables. This restriction is met when each initialization expression is a `lambda` expression. Therefore, use `letrec` to declare mutually recursive local functions.

Example 1

```
letrec(
  ( ;; variable list
    ( f
      lambda( ( n )
        if( n > 0 then n*f(n-1) else 1
          ) ; if
        ) ; lambda
      ) ; f
    ) ; variable list
  f( 5 )
) ; letrec
=> 120
```

This example declares a single recursive local function. The `f` function computes the factorial of its argument. The `letrec` expression returns the factorial of 5.

Example 2

```
procedure( trParity( x )
  letrec(
    (
      ( isEven
        lambda( (x)
          x == 0 || isOdd( x-1 )
        ) ; lambda
      ) ; isEven
      ( isOdd
        lambda( (x)
          x > 0 && isEven(x-1)) ; lambda
        ) ; isOdd
      ) ;
    if( isEven( x ) then 'even else 'odd )
  ) ; letrec
) ; procedure
```

The `trParity` function returns the symbol `even`, if its argument is even, and returns the symbol `odd` otherwise. `trParity` relies on two mutually recursive local functions `isEven` and `isOdd`.

Using procedure to Declare Local Functions

As an alternative to using `letrec` to define local functions, you can use the `procedure` syntax.

Example 1

This example uses the `procedure` construct to declare a local function instead of using `letrec`.

```
procedure( trParity( x )
  procedure( isEven(x)
    x == 0 || isOdd( x-1 )
  )
  procedure( isOdd(x)
    x > 0 && isEven(x-1)
  )
  if( isEven( x ) then 'even else 'odd )
) ; procedure
```

Example 2

```
procedure( makeGauge( tolerance )
  let( ( ( iteration 0 ) ( previous 0.0 ) test )
    procedure( performTest( value )
      ++iteration
      test = ( abs( value - previous ) <= tolerance )
      previous = value
      when( test list( iteration value ) )
    ) ; procedure
    performTest
  ) ; let
) ; procedure

G = makeGauge( .1 )
=> funobj:0x322b28
G(2) => nil                ;; first iteration
G(3)=> nil                ;; second
G(3.01) = >               ;; third iteration
( 3 3.01 )
```

The `makeGauge` function declares the local `performTest` function and returns it. This function object is the gauge. Passing a value to the gauge compares it to the previous value passed then updates the previous value. The gauge returns a list of the iteration count and the value, or `nil`. The function object has access to the local variables `iteration`, `previous`, and `test`, as well as access to the argument `tolerance`. Notice that using

a gauge object can simplify your code by isolating variables used only for the tracking of successive values.

For another `makeGauge` example, see [“Example 4: Using a Gauge When Computing the Area Under a Curve”](#) on page 321.

Example 3

```
procedure( trPartition( nList )
  procedure( loop( numbers nonneg neg )
    cond(
      ( !numbers list( nonneg neg ) )
      ( car( numbers ) > 0
        loop(
          cdr( numbers )
          cons( car( numbers ) nonneg ) ; ppush on nonneg
          neg
        ) ; loop
      )
      ( car( numbers ) < 0
        loop(
          cdr( numbers )
          nonneg
          cons( car( numbers ) neg ) ; ppush on neg
        ) ; loop
      )
    ) ; cond
  ) ; procedure
  loop( nList nil nil )
) ; procedure

trPartition( '( 3 -2 1 6 5 ) ) => ((5 6 1 3) (-2))
```

In this example, the `trPartition` function separates a list of integers into a list of non-negative elements and negative elements. The local `loop` function is recursive.

Sequencing and Iteration

The following sequencing and iteration functions are provided in SKILL++:

- Use `begin` to construct a single expression from one or more expressions.
- Use `do` to iteratively execute one or more expressions.
- Use a named `let` construct to extend the `let` construct with a recursive iteration capability.

Using begin

Use `begin` to construct a single expression from one or more expressions. The expressions are evaluated from left to right. The return value of the `begin` expression is the return value of the last expression in the sequence.

The `begin` function is equivalent to the `progn` function. The `progn` function is used to implement the `{ }` syntax. Use the `begin` function to write SKILL++-compliant code.

Example 1

```
ILS-1> begin(  
    x = 0  
    printf( "Value of x: %d\n" ++x )  
    printf( "Value of x: %d\n" ++x )  
    x  
    ) ; begin  
Value of x: 1  
Value of x: 2  
2
```

This example shows a transcript using the `begin` function.

Example 2

```
ILS-1> { x = 0  
    printf( "Value of x: %d\n" ++x )  
    printf( "Value of x: %d\n" ++x )  
    x }  
Value of x: 1  
Value of x: 2  
2
```

This example uses the `{ }` braces to group the same expressions.

Using do

Use `do` to iteratively execute one or more expressions. The `do` expression allows multiple loop variables with arbitrary variable initializations and step expressions. You can specify

- One or more loop variables, including an initialization expression and a step expression for each variable.
- A termination condition that is evaluated before the body expressions are executed.
- One or more termination expressions that are evaluated upon termination to determine a return value.

Syntax Template for do Expressions

```
do( (
  ( s_var1 g_initExp1 [g_stepExp1] )
  ( s_var2 g_initExp2 [g_stepExp2] )...)
  ( g_terminationExp g_terminationExp1 ...)
  g_loopExp1 g_loopExp2 ...)
=> g_value
```

A `do` expression evaluates in two phases: the initialization phase and the iteration phase.

The initialization expressions `g_initExp1`, `g_initExp2`, ... are evaluated in an unspecified order and the results bound to the local variables `var1`, `var2`, ...

The iteration phase is a sequence of steps going around the loop zero or more times with the exit determined by the termination condition.

1. Each iteration begins by evaluating the termination condition.
2. If the termination condition evaluates to a non-nil value, the `do` expression exits with a return value computed as follows:
3. The termination expressions `terminationExp1`, `terminationExp2`, ... are evaluated in order. The value of the last termination condition is returned as the value of the `do` expression.
4. Otherwise, the `do` expression continues with the next iteration as follows.
5. The loop body expressions `g_loopExp1`, `g_loopExp2`, ... are evaluated in order.
6. The step expressions `g_stepExp1`, `g_stepExp2`, ..., if given, are evaluated in an unspecified order.
7. The local variables `var1`, `var2`, ... are bound to the above results. Reiterate from step one.

Example 1

```
procedure( sumList( L )
  do(
    (
      ( tail L cdr( tail ))
      ( sum 0 sum + car( tail ))
    )
    ( !tail sum )
  )
) ; procedure

sumList( nil ) => 0
sumList( '( 1 2 3 4 5 6 7 ) ) => 28
```


Example 2

By definition, the sum of the integers 1, ..., N is the Nth triangular number. The following example finds the first triangular number greater than a given limit.

```
procedure( trTriangularNumber( limit )
do(
  (
    ( i 0 i+1 )          ;; start loop variables
    ( sum 0 )            ;; no update expression
                        ;; same as ( sum 0 sum )
    )                    ;; end loop variables
    ( sum > limit         ;; test
      sum                 ;; return result
    )
    sum = sum+i ;; body
  ) ; do
) ; procedure

trTriangularNumber( 4 ) => 6
trTriangularNumber( 5 ) => 6
trTriangularNumber( 6 ) => 10
```

Example 3

```
procedure( approximateArea( dx fun lower upper )
do(
  ( ; loop variables
    ( sum
      0.0          ;; initial value
      sum+fun(x)   ;; update
    )
    ( x
      lower        ;; initial value
      x+dx          ;; update expression
    )
  ) ; end loop vars
  ( x >= upper ;; exit test
    dx*sum    ;; return value
  )
  ;; no loop expressions
  ;; all work is in the update expression for sum
) ; do
) ; procedure

approximateArea( .001 lambda( ( x ) 1 ) 0.0 1.0 ) => 1
approximateArea( .001 lambda( ( x ) x ) 0.0 1.0 ) => .4995
```

The function `approximateArea` computes an approximation to the area under the graph of the `fun` function over the interval from `lower` to `upper`. It sums the values `fun(x)`, `fun(x+dx)`, `fun(x+dx+dx)` ...

Example 4: Using a Gauge When Computing the Area Under a Curve

```
procedure( makeGauge( tolerance )
  let( ( ( iteration 0 ) ( previous 0.0 ) test )
```

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Using SKILL++

```
lambda( ( value )
  ++iteration
  test = ( abs( value - previous ) <= tolerance )
  previous = value
  when( test list( iteration value ) )
  ) ; lambda
) ; let
) ; procedure

procedure( computeArea( fun lower upper tolerance )
  let( ((gauge makeGauge( tolerance )) result )
    do(
      ( ; loop variables
        (
          dx
          1.0*(upper-lower)/2 ;;; initial value
          dx/2 ;;; update expression
        )
      ) ; end loop variables
      ( result =
        gauge( approximateArea( dx fun lower upper ) )
        result
      )
      nil ;;; empty body
    ) ; do
  ) ; let
) ; procedure

computeArea( lambda( ( x ) 1 ) 0 1 .00001 ) => ( 2 1.0 )
computeArea( lambda( ( x ) x ) 0 1 .00001 ) => ( 16 0.4999924 )
pi = 3.1415
computeArea( sin 0 pi/2 .00001 ) =>(18 0.9999507)
```

The `computeArea` function invokes `approximateArea` iteratively until two successive results fall within the given tolerance. The `dx` loop variable is initialized to $1.0 * (\text{upper} - \text{lower}) / 2$ and updated to `dx/2`. This example uses a gauge to hide the details of comparing successive results. The source for the `makeGauge` function is replicated for your convenience. See [Example 2](#) on page 317 in the "Using procedure to Declare Local Functions" section for a discussion of `makeGauge` and gauges in general.

Using a Named let

The named `let` construct extends the `let` construct with a recursive iteration capability. Besides the name you provide in front of the list of the local variables, the named `let` has the same syntax and semantics as the ordinary `let` except you can recursively invoke the named `let` expression from within its own body, passing new values for the local variables.

Syntax Template for Named let

```
let(
  s_name
  ( ( s_var1 g_initExp1 ) ( s_var2 g_initExp2 )... )
  g_bodyExp1
```

```
g_bodyExp2
...
)
```

Example 1

```
let(
  factorial
  (( n 5 ))
  if( n > 1
    then
      factorial( n-1)*n
    else
      1
  )
) ; let => 120
```

This example computes the factorial of 5 with a named `let` expression. Compare the example above with the following

```
let( (( n 5 ))
  procedure( factorial( n )
    if( n > 1
      then
        n*factorial( n-1)
      else
        1
    ) ; if
  ) ; procedure
  factorial( n )
) ; let => 120
```

and with the following

```
letrec(
  ( ;;; variable list
    ( n 5 )
    ( factorial
      lambda( ( n )
        if( n > 0 then n*factorial(n-1) else 1 ) ; if
      ) ; lambda
    ) ; f
  ) ; variable list
  factorial( n )
) ; letrec => 120
```

Example 2

```
let( loop      ;;; name for the let
  (           ;;; start let variables
    ( numbers '( 3 -2 1 6 -5 ))
    ( nonneg nil )
    ( neg nil )
  )           ;;; end of let variables
  cond(
    ( !numbers ;;; loop termination test and return result
      list( nonneg neg )
    )
  )
```

```

    )
    ( car( numbers ) > 0 ;;; found a non-negative number
      loop( ;; recurse
        cdr( numbers )
        cons( car( numbers ) nonneg )
        neg
      ) ; loop
    )
    ( car( numbers ) < 0 ;;; found a negative number
      loop( ;; recurse
        cdr( numbers )
        nonneg
        cons( car( numbers ) neg )
      ) ; loop
    )
  ) ; cond
) ;;; loop let
=> ((613) (-5 -2))
```

This example separates an initial list of integers into a list of the negative integers and a list of the non-negative integers. Compare this example with the `trPartition` function in “[Example 3](#)” on page 318 which explicitly relies on a local recursive function.

Software Engineering with SKILL++

SKILL++ supports several modern software engineering methodologies, such as

- Procedural packages
- Modules
- Object-oriented programming with classes (see [SKILL++ Object System](#))

SKILL++ also facilitates information hiding. `Information hiding` refers to using private functions and private data which are not accessible to other parts of your application. Information hiding promotes reusability and robustness because your implementation is easier to change with no adverse effect on the clients of the module.

SKILL++ Packages

A `package` is a collection of functions and data. Functions within a package can share private data and private functions that are not accessible outside the package. Packages are a hallmark of modern software engineering.

SKILL++ facilitates two approaches to packages.

- You can explicitly represent the package as an collection of function objects and data. Clients of the package use the arrow (->) operator to retrieve the package functions. Different packages can have functions with the same name.
- You might want to reimplement a collection of SKILL functions as a SKILL++ package. Informal SKILL packages have no opportunity to hide private functions or data. Reimplementing a SKILL package in SKILL++ provides the opportunity. Usually, you want to do this in a way that clients do not need to change their calling syntax. In this case, you do not need to represent the collection as a data structure. Instead, the package exports some of its function objects and hides the remainder.

The Stack Package

```
Stack = let( ()
  procedure( getContents( aStack )
    aStack->contents
  ) ; procedure
  procedure( setContents( aStack aList )
    aStack->contents = aList
  ) ; procedure
  procedure( ppush( aStack aValue )
    setContents(
      aStack
      cons(
        aValue
        getContents( aStack )
      ) ; cons
    )
  ) ; procedure
  procedure( ppop( aStack )
    letseq( (
      ( contents getContents( aStack ) )
      ( v car( contents ) )
    )
    setContents( aStack cdr( contents ) )
    v
  ) ; letseq
  ) ; procedure
  procedure( new( initialContents )
    list( nil 'contents initialContents )
  ) ; procedure
  list( nil 'ppop ppop 'ppush ppush 'new new )
) ; let

=> ( nil
  ppop funobj:0x1c9b38
  ppush funobj:0x1c9b28
  new funobj:0x1c9b48 )
```

Using the Stack Package

```
S = Stack->new( '( 1 2 3 4 ) ) => (nil contents ( 1 2 3 4 ))
Stack->ppop( S ) => 1
Stack->ppush( S 1 ) => (1 2 3 4)
```

Comments

The Stack package is represented by a disembodied property list. Alternate representations such as a defstruct are possible. The only requirement is that the package data structure obey the `->` protocol.

Only the `ppush`, `ppop`, and `new` function are visible to the clients of the package.

The `ppush` and `ppop` functions use the `getContents` and `setContents` functions. If you choose a different representation for a stack, you only need to change the `new`, `getContents`, and `setContents` functions. The `getContents` and `setContents` functions are hidden to protect the abstract behavior of a stack.

Retrofitting a SKILL API as a SKILL++ Package

```
define( stackPush nil )
define( stackPop nil )
define( stackNew nil )
let( ()
  procedure( getContents( aStack )
    aStack->contents
  ) ; procedure
  procedure( setContents( aStack aList )
    aStack->contents = aList
  ) ; procedure
  procedure( ppush( aStack aValue )
    setContents(
      aStack
      cons(
        aValue
        getContents( aStack )
      ) ; cons
    )
  ) ; procedure
  procedure( ppop( aStack )
    letseq( (
      ( contents getContents( aStack ) )
      ( v car( contents ) )
    )
    setContents( aStack cdr( contents ) )
    v
  ) ; letseq
  ) ; procedure
  procedure( new( initialContents )
    list( nil 'contents initialContents )
  ) ; procedure
  stackPush = ppush
  stackPop = ppop
  stackNew = new
nil
) ; let
```

Using the `stackNew`, `stackPop`, and `stackPush` Functions

```
S = stackNew( '( 1 2 3 4 ) ) => (nil contents (1 2 3 4))
stackPop( S ) => 1
stackPush( S 5 ) => (5 2 3 4)
```

Comments

This example assumes `stackNew`, `stackPop`, and `stackPush` are the names of the functions to be exported from the `stack` package. As is customary, the package prefix `stack` informally indicates the functions that compose a package.

- The `getContents` and `setContents` functions are local functions invisible to clients.
- Using the `define` forms for `stackNew`, `stackPop`, and `stackPush` is not strictly necessary. Using the `define` form alerts the reader to those functions which the ensuing `let` expression assigns a value to an exported API.

SKILL++ Modules

You can structure a SKILL++ module around a creation function, which the client invokes to allocate one or more instances of the module. The client passes an instance to a procedural interface.

The `makeStack` and `makeContainer` functions in the following examples are creation functions in the following sense: when you call `makeStack` it “creates” a stack instance. The stack instance is a function object whose internals can only be manipulated (outside of the debugger) by the `ppushStack` and `popStack` functions.

The creation function has

- Arguments
- Local variables
- Several local functions that can access the arguments to the creation function and can communicate between themselves through the local variables

The creation function returns one of the following, depending on the implementation:

- A single local function object
- A data structure containing several of the local function objects
- A single function object which dispatches control to the appropriate local functions

Stack Module Example

A `stack` is a well-known data structure that allows the client to push a data value onto it and to pop a data value from it.

The Procedural Interface

The following table summarizes the procedural interface functions to the sample stack module.

Action	Function Call	Return Value
Allocate a stack.	<code>makeStack(aList)</code>	Function object
Push a value onto the stack.	<code>pushStack(aStack aValue)</code>	A list of the stack contents
Pop a value from the stack.	<code>popStack(aStack)</code>	A popped value

The variable `aStack` is assumed to contain a stack object allocated by calling the `makeStack` function.

Allocating a Stack

```
S = makeStack( '( 1 2 3 4 ) ) => funobj:0x1e36d8
```

Popping a Value

```
popStack( S ) => 1  
popStack( S ) => 2
```

Pushing a Value on the Stack

```
pushStack( S 5 ) => (5 3 4)
```

Returns a list of stack contents at this point.

Implementing the `makeStack` Function

The `makeStack` function returns a function object. This function object is an instance of the stack module. In turn, this function object returns one of several functions local to the `makeStack` function.

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```
procedure( makeStack( initialContents )
  let( (( theStack initialContents ))

    procedure( ppush( value )
      theStack = cons( value theStack )
    )

    procedure( ppop( )
      let( (( v car( theStack ) ))
        theStack = cdr( theStack )
        v
      )
    )

    lambda( ( msg)                ;;;; return a function object
      case( msg
        ( ( ppush ) ppush )
        ( ( ppop ) ppop )
        ( t nil )
      )
    ) ; let
  ) ; procedure
```

Implementing the pushStack and popStack Functions

The variable `aStack` contains a function object.

- When `aStack` is called, it returns the appropriate local function `ppush` and `ppop`.
- The `ppush` and `ppop` functions are within the lexical scope of the local variable `theStack`.

Notice the syntactic convenience of calling the stack object indirectly through the `fun` variable.

```
procedure( pushStack( aStack aValue )
  let( ( ( fun aStack( 'ppush ) ) )
    fun( aValue )          ;;;; retrieve local ppush function
                          ;;;; call it
    )
  )

procedure( popStack( aStack )
  let( ( ( fun aStack( 'ppop ) ) )
    fun()                  ;;;; retrieve local ppop function
                          ;;;; call it
    )
  )
```

The Container Module

`Containers` are like variables with an important difference. You can reset a container to the original value that you provided when you created the container.

The Procedural Interface

The following table summarizes the procedural interface to the sample container module. This interface relies on the availability of several function objects in the container instance's data structure. The arrow ($\rightarrow$) operator is used to retrieve the interface functions.

Action	Function Call	Return Value
Allocate a container with an initial value.	<code>aContainer = makeContainer(aValue)</code>	A disembodied property list representing the container instance.
Return the container's current value.	<code>aContainer->get()</code>	Current value in <i>aContainer</i>
Store a new value in the container.	<code>aContainer->set(bValue)</code>	The container's new value.
Reset the container to the initial value	<code>aContainer->reset()</code>	The container's original value.

Implementing the makeContainer Function

- The `makeContainer` function returns a disembodied property list containing the local functions as property values.
- The three functions `resetValue`, `setValue`, and `getValue` are local but are accessible through `makeContainer`'s return value.
- Unlike the stack module example, there are no global functions in the procedural interface.

```

procedure( makeContainer( initialValue )
  let( ( (value initialValue))          ;;; initialize the container
    procedure( resetValue()             ;;; reset value
      value = initialValue
    )
    procedure( setValue( newValue )      ;;; store new value
      value = newValue
    )
    procedure( getValue( )               ;;; return current value
      value
    )

    resetValue()
    list( nil                             ;;; the return value
      'set setValue
      'get getValue
      'reset resetValue
    )
  )

```

```
    ) ; let  
  ) ; procedure
```

Allocating Container Instances

```
x = makeContainer( 0 )  
=> (nil  
    set funobj:0x1e38f8  
    get funobj:0x1e3908  
    reset funobj:0x1e38e8 )
```

This example allocates a container instance with initial value 0.

```
y = makeContainer( 2 )  
=> (nil  
    set funobj:0x1e3928  
    get funobj:0x1e3938  
    reset funobj:0x1e3918 )
```

This example allocates a container instance with an initial value of 2. Notice that the returned value contains different function objects.

Retrieving Container Values

```
x->get() + y->get() => 2
```

This example retrieves the values of the two containers and adds them. Notice the conventional function call syntax accepts an arrow ($\rightarrow$) operator expression in place of a function name to access member functions.

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Using SKILL and SKILL++ Together

This chapter discusses the pragmatics of developing programs in the Cadence® SKILL++ language. You should be familiar with both SKILL and SKILL++, specifically the material in [“About SKILL++ and SKILL”](#) on page 293 and [“Using SKILL++”](#) on page 313.

Developing a SKILL++ application involves the same basic tasks as for developing SKILL applications. Because most viable applications will involve tightly integrated SKILL and SKILL++ components, there are several more factors to consider:

- **Selecting an interactive language**

When entering a language expression into the command interpreter, you need to choose the appropriate language mode.

- **Partitioning an application into a SKILL portion and a SKILL++ portion**

You are free to implement your application as a heterogeneous collection of source code files. You need to choose a file extension accordingly.

- **Cross-calling between SKILL and SKILL++**

In general, SKILL++ functions and SKILL functions can transparently call one another. However, a few families of SKILL functions can operate differently when called from SKILL than when called from SKILL++. You need to be able to identify such SKILL functions, adjust your expectations, and exercise caution, when calling them from SKILL++.

- **Debugging a SKILL++ program**

In a hybrid application, errors can occur in either SKILL functions or SKILL++ functions. Displaying the SKILL stack will reveal SKILL++ environments and SKILL++ function objects which you will want to examine.

- **Communicating between SKILL and SKILL++**

Data allocated with one language is accessible from the other language. For example, you can allocate a list in a SKILL++ function and retrieve data from it in a SKILL function. Both languages use the same print representations for data.

The phrase ‘from within a SKILL program’ means

- from a SKILL interactive loop
- from within SKILL source code, outside of a function definition
- from within a SKILL function

Similar definitions apply ‘from within a SKILL++ program’.

For more information, see the following sections:

- [Selecting an Interactive Language](#) on page 335
- [Partitioning Your Source Code](#) on page 336
- [Cross-Calling Guidelines](#) on page 336
- [Redefining Functions](#) on page 338
- [Sharing Global Variables](#) on page 338
- [Debugging SKILL++ Applications](#) on page 340

Selecting an Interactive Language

You can use SKILL or SKILL++ for interactive work. Both languages support a read-eval-print loop in which you repeat the following steps:

1. You type a language expression.
2. The system parses, compiles, and evaluates the expression in accordance with either SKILL or SKILL++ languages syntax and semantics.
3. The system displays the result using the print representation appropriate to the result's data type.

Starting an Interactive Loop (toplevel)

You can call the `toplevel` function to start an interactive loop with either SKILL or SKILL++. SKILL is the default.

- To select the SKILL language, type

```
toplevel( 'il )
```

- To select the SKILL++ language and the SKILL++ top-level environment, type

```
toplevel( 'ils )
```

- To select the SKILL++ language and the environment to be made active during the interactive loop, pass the environment object as the second argument:

```
toplevel( 'ils envobj( 0x1e00b4 ) )
```

In this example, the environment object is retrieved from the print representation.

Exiting the Interactive Loop (resume)

Use the `resume` function to exit the interactive loop, returning a specific value. This value is the return value of the `toplevel` function. The following example is a transcript of a brief session, including prompts.

```
> R = toplevel( 'ils )
ILS-<2> resume( 1 )
1
> R
1                                     ;;;return value of the toplevel function.
```

Partitioning Your Source Code

You are free to implement your application as a heterogeneous collection of source code files. The `load` and `loadi` functions select the language to apply to the source code based on the file extension.

```
FileA.il  
FileB.il  
FileC.ils  
FileD.ils  
FileE.ils
```

Functions defined in each file can call functions defined in the other files without regard to the language in which the functions are written. The syntax for function calls is the same regardless of whether the function called is a SKILL function or a SKILL++ function. Specifically,

- SKILL++ functions are visible to the SKILL portions of your application
- SKILL functions are automatically visible to the SKILL++ portions of your application

You may call SKILL application procedural interface functions from a SKILL++ program. Most applications continue to rely heavily on SKILL functions.

Cross-Calling Guidelines

Several key semantic differences between SKILL and SKILL++ dictate certain guidelines you should follow when calling SKILL functions from SKILL++, including the following:

- All SKILL++ environments, other than the top-level environment, are invisible to SKILL
- All SKILL++ local variables are invisible to SKILL

You should avoid calling SKILL functions that call

- The `eval`, `symeval`, or `evalstring` functions
- The `set` function

You should avoid calling `nlambda` SKILL functions.

Avoid Calling SKILL Functions That Call `eval`, `symeval`, or `evalstring`

When you call the one-argument version of `eval`, `symeval`, or `evalstring` functions from a SKILL function, you are using dynamic scoping. Any symbol or expression which you

pass to a SKILL function will probably evaluate to a different result than it would have in the SKILL++ caller.

In general, to determine whether a SKILL function calls any of these functions, you should consult the reference documentation.

Avoid Calling nlambda Functions

The nlambda category of SKILL functions are highly likely to call the `eval` or `symeval` functions.

A SKILL `nlambda` function receives all of its argument expressions unevaluated in a list. Such a function usually evaluates one or more of the arguments. The `addVars` SKILL function adds the values of its arguments.

```
nprocedure( addVars( args )
  let( (( sum 0 ))
    foreach( arg args
      sum = sum+eval(arg)
    ) ; foreach
    sum
  )
)

let( ((x 1) (y 2) (z 3 ))
  addVars( x y z )
)

=> 6
```

When called from SKILL++, the `eval` function uses dynamic scoping to resolve the variable references. In this case, the variable `x` was unbound.

```
ILS-<2> let( ((x 1) (y 2) (z 3 ))
  addVars( x y z )
)
*WARNING* (addVars): calling NLambda from Scheme code -
  addVars(x y z)
*Error* eval: unbound variable - x
ILS-<2>
```

If necessary, reimplement the SKILL `nlambda` function as a SKILL or SKILL++ macro, using `defmacro`. For example,

```
defmacro( addVars ( @rest args )
  `let( (( sum 0 ))
    foreach( arg list( ,@args )
      sum = sum + arg
    ) ; foreach
    sum
  )
)
```

Use the set Function with Care

Avoid calling SKILL functions that in turn call the `set` function. Usually such SKILL functions store values in other SKILL variables. If you call such a function from SKILL++ and pass a quoted local variable, the SKILL function will not store the value in the SKILL++ local variable. Instead, the value goes into the SKILL variable of the same name.

The following `SetMyArg` SKILL function behaves differently when called from SKILL++ than when called from SKILL.

SetMyArg Called from SKILL

```
> procedure( SetMyArg( aSymbol aValue )
    set( aSymbol aValue )
    ) ; procedure
SetMyArg
> let( ( ( x 3 ) )
    SetMyArg( 'x 5 )
    x
    ) ; let
5
```

SetMyArg Called from SKILL++

```
> toplevel 'ils
ILS-<2> let( (( x 3 ))
    SetMyArg( 'x 5 )
    x
    ) ; let
3
```

Redefining Functions

During a single session, you are warned when you redefine a SKILL function to be a SKILL++ function, or visa versa.

You are only likely to encounter this when doing interactive work and are confused about which language “owns” the interaction.

Sharing Global Variables

It is usually desirable to avoid relying on global variables. However, it is sometimes necessary or expedient for the SKILL++ and SKILL portions of your application to communicate through global variables.

Using importSkillVar

Before the SKILL and SKILL++ portions of your application can share a global variable, you must first call the `importSkillVar` function. The SKILL++ global variable and the SKILL global variable will then be bound to the same location.

For example, consider the following interaction with a SKILL top level.

```
> delta = 2
2
> procedure( adder( y )
    delta+y
) ;
adder
>
```

To set the value of the `delta` variable from within a SKILL++ program, you should first

```
importSkillVar( delta )
```

as the following sample interaction with a SKILL++ top level shows.

```
> toplevel 'ils
ILS-<2>delta
*Error* eval: unbound variable - delta
ILS-<2> importSkillVar( delta )
ILS-<2> delta
2
ILS-<2> adder( 4 )
6
```

Note: You do not need to import `delta` just to call `adder` from within SKILL++ code.

How importSkillVar Works

Although understanding this level of detail is not necessary to effectively use `importSkillVar`, this section is provided for expert users.

In SKILL++, a variable is bound to a memory location called the variable's binding. The familiar operation of “storing a value in a variable” stores the value in the variable's binding. SKILL++ variable bindings are organized into environment frames. The SKILL++ top-level environment contains all the variable bindings initially available at system start up.

Normally, all global (top-level) SKILL++ variables are bound to the function slot of the SKILL symbol with the same name as the variable. For example, the variable `foo` is bound to the function slot of the symbol `foo`. Consequently, in SKILL++, when you retrieve the value of a SKILL variable, you are getting the contents of the symbol's function slot.

The `importSkillVar` function directs the compiler to instead bind a SKILL++ global variable in the top-level environment to the value slot of the symbol with the same name.

Informally, you can use `importSkillVar` to enable access to a SKILL symbol's current value binding from within SKILL++.

Evaluating an Expression with SKILL Semantics

As an advanced programmer, you might find that separating closely related SKILL code and SKILL++ code into different files is distracting or otherwise not convenient. For example, suppose that in the middle of a SKILL++ source code file you want to declare a SKILL function that refers to a SKILL variable.

Using the previous example

```
delta = 5
procedure( adder( y )
    delta+y )
```

The `inSkill` macro below allows you to splice SKILL language source code into a SKILL++ source code file. You can use the `inSkill` macro as shown.

```
inSkill(
    delta = 5
    procedure( adder( y )
        delta+y ) ; procedure
)
```

Debugging SKILL++ Applications

This section addresses common tasks that arise when debugging hybrid SKILL and SKILL++ applications.

Examining the Source Code for a Function Object

Use the `pp` SKILL function to display the source code for a global function. The `pp` function expects that its argument is a symbol. It retrieves the function object stored in the function slot of the symbol you pass. To use `pp` to display the source code for a function object, store the function object in an unused global.

In SKILL++

```
G4 = funobj( 0x1e3628 )
pp( G4 )
```

In SKILL

```
putd( 'G4 ) = funobj( 0x1e3628 )
pp( G4 )
```

The `pp` function pretty-prints the function object stored in the function slot of the symbol. The `pp` function uses the global symbol to name the function object. This is only seriously misleading if the function object is recursive.

Pretty-Printing Package Functions

Use the `pp` function as explained above to pretty-print package functions.

```
MathPackage = let( ()
  procedure( add( x y ) x+y )
  procedure( mult( x y ) x*y )
  list( nil 'add add 'mult mult )
)
=> (nil add funobj:0x1c9c48 mult funobj:0x1c9c58)

ILS-1> Q = MathPackage->add
funobj:0x1c9c48
ILS-1> pp( Q )
procedure( Q(x y)
  (x + y)
)
```

Inspecting Environments

A significant SKILL++ application is likely to include many function objects, each with its own separate environment. While debugging, you may need to interactively examine or set a local variable in an environment other than the active environment.

You can

- Retrieve the active environment
- Inspect an environment with the `->` operator
- Retrieve the environment of a function object

Retrieving the Active Environment

The `theEnvironment` function returns the enclosing lexical environment when you call it from within SKILL++ code.

Example 1

```
Z = let( (( x 3 ))
  theEnvironment()
) ; let
=> envobj:0x1e0060
```

This example returns the environment that the `let` expression establishes. The value of `z` is an environment in which `x` is bound to 3. Each time you execute the above expression, it returns a different environment object.

Example 2

```
W = let( (( r 3 ) ( y 4 ))
        let( (( z 5 ) ( v 6 ))
              theEnvironment()
            )
      )
```

This example returns the environment that the nested `let` expressions establish.

Testing Variables in an Environment (`boundp`)

Use the `boundp` function to determine whether a variable is bound in an environment. The optional second argument should be a SKILL++ environment.

```
boundp( 'b W ) => nil
boundp( 'r W ) => t
```

Using the `->` Operator with Environments

You can use the `->` operator against an environment to read and write variables bound in the environment.

```
W->z => 5
W->v = 100
```

Alternatively, you can use the `symeval` function to retrieve the value of a variable relative to an environment.

```
symeval( 'r W ) => 3
```

Alternatively, you can use the `set` function to set the value of a variable in an environment.

```
set( 'r 200 W ) => 200
```

Using the `->??` Operator with Environments

Use the `->??` operator to dump out the environment as a list of association lists with one association list for each environment frame.

```
W->?? => ( ((z 5) (v 6)) ((r 3) (y 4)) )
```

Evaluating an Expression in an Environment (eval)

Use the `eval` function to evaluate an expression in a given lexical environment.

```
eval( '( z+v ) W ) => 11
eval( '( z = 100 ) W ) => 100
eval( '( z+v ) W ) => 106
```

Examining Closures

As function objects, closures have both source code and data. For example, consider the following closure generated when the `makeAdder` function is called.

```
procedure( makeAdder( delta )
  lambda( ( x ) x + delta )
)
=> makeAdder
add5 = makeAdder( 5 )
=> funobj:0x1fe668
```

Examining the Source Code

Use the `pp` function as explained above to examine the source code for the closure.

```
ILS-1> pp( add5 )
procedure( add5(x)
  (x + delta)
)
nil
```

Examining the Environment

Install the SKILL Debugger and use the `theEnvironment` function to retrieve the environment for the function object. Use the `->??` operator to examine the environment.

```
theEnvironment( funobj( 0x1fe668 ) )->??
=> (((delta 5)))
```

See the `makeStack` example in [“Implementing the makeStack Function”](#) on page 328

```
S = makeStack( '( 1 2 3 ) ) => funobj:0x1e3758
E = theEnvironment( S ) => envobj:0x1e00b4
E->push => funobj:0x1e3738
E->initialContents => (1 2 3)
```

General SKILL Debugger Commands

Tracing (tracef)

You can only use the `tracef` function to trace SKILL functions or SKILL++ functions defined in the top-level SKILL++ environment.

Setting Breakpoints

You can only set breakpoints at SKILL functions or SKILL++ functions defined in the top-level SKILL++ environment.

Calling the break Function

You can insert a call to the `break` function but that requires redefining the function that calls the `break` functions. If called from SKILL++, the enclosing lexical environment is the active environment during the debugger session.

```
ILS-<2> MathPackage = let( ()
  procedure( add( x y ) break() x+y )
  procedure( mult( x y ) x*y )
  list( nil 'add add 'mult mult )
)
(nil add funobj:0x1c9ca8 mult funobj:0x1c9cb8)
ILS-<2> MathPackage->add( 3 4 )
<<< Break >>> on explicit 'break' request
SKILL Debugger: type 'help debug' for a list of commands or debugQuit to leave.
ILS-<3> theEnvironment()->??
(( (x 3)
  (y 4)
  )
  ((add funobj:0x1c9ca8)
   (mult funobj:0x1c9cb8)
  )
)
ILS-<3>
```

Examining the Stack (stacktrace)

During the execution of both SKILL and SKILL++ function calls, use the `stacktrace` function to examine the SKILL stack. The stack will probably contain several function object and environment object references. You can use the techniques discussed above to display source code for a function object and to examine an environment.

For example, at the break point in the previous example, the `break` function passes the active environment to the break handler.

Cadence SKILL Language User Guide

Using SKILL and SKILL++ Together

Cadence SKILL Language User Guide

Using SKILL and SKILL++ Together

SKILL++ Object System

To gain benefits from object-oriented programming, the Cadence® SKILL language requires extensions beyond lexical scoping and persistent environments.

The Cadence SKILL++ Object System allows for object-oriented interfaces based on classes and generic functions composed of methods specialized on those classes. A class can inherit attributes and functionality from another class known as its superclass. SKILL++ class hierarchies result from this inheritance relationship.

To attain the maximum benefit from the SKILL++ Object System, you should only use it with lexical scoping, because lexical scoping magnifies the power of the interfaces you can develop with the SKILL++ Object System.

You do not need to be familiar with another object-oriented programming language or system to understand or use the SKILL++ Object System. However, if you are familiar with the Common Lisp Object System (CLOS), you can apply your experience of CLOS in learning the SKILL++ Object System as the SKILL++ Object System is modelled after a subset of the Common Lisp Object System.

For more information, see the following sections:

- [Basic Concepts](#) on page 347
- [Class Hierarchy](#) on page 362
- [Browsing the Class Hierarchy](#) on page 364
- [Advanced Concepts](#) on page 366

Basic Concepts

The following items are central concepts of the SKILL++ Object System:

- [Classes and Instances](#) on page 348
- [Generic Functions and Methods](#) on page 348

- [Subclasses and Superclasses](#) on page 349

For more information, see the following sections:

- [Defining a Class \(defclass\)](#) on page 350
- [Instantiating a Class \(makeInstance\)](#) on page 353
- [Initializing an Instance \(initializeInstance\)](#) on page 353
- [Reading and Writing Instance Slots](#) on page 354
- [Defining a Generic Function \(defgeneric\)](#) on page 355
- [Defining a Method \(defmethod\)](#) on page 356

Classes and Instances

A class is a data structure template. A specific application of the template is termed an instance. All instances of a class have the same slots. SKILL++ Object System provides the following functions:

- `defclass` function to create a class
- `makeInstance` function to create an instance of a class
- `initializeInstance` function to initialize a newly created instance

Generic Functions and Methods

A `generic function` is a collection of function objects. Each element in the collection is called a `method`. Each method corresponds to a class. When you call a generic function, you pass an instance as the first argument. The SKILL++ Object System uses the class of the first argument to determine which methods to evaluate.

To distinguish them from SKILL++ Object System generic functions, SKILL functions are called `simple functions`. The SKILL++ Object System provides the following functions.

- `defgeneric` function to declare a generic function
- `defmethod` function to declare a method

Subclasses and Superclasses

SKILL++ Object System supports both single and multiple inheritance. In single inheritance, one class B can inherit structure slots and methods from another class A. You can describe the relationship between the class A and class B as follows:

- B is a subclass of A
- A is a superclass of B

In multiple inheritance, class B can inherit structure slots and methods from multiple classes. For example, class A and class C. In this case, the relationship between class A, B, and C is as follows:

- B is a subclass of A and C
- A is a superclass of B
- C is a superclass of B

Class Precedence List

All inheritance decisions are governed by the class precedence list, which is an ordered list of a given class and its superclasses. The following rules determine the precedence order of classes:

- In single inheritance: A class is always more specific than its superclass. So the order of precedence flows from left to right. For example,

```
defclass(A ())  
defclass(B (A))  
; order of precedence is from B to A
```

- In multiple inheritance: For a given class, superclasses listed on the left are more specific than those listed on the right. So the order of precedence is from the superclass on the left to the superclass on the right. For example,

```
defclass(A () ())  
defclass(B () ())  
defclass(X (A B) ()) ; in class X, class A should come before class B  
defclass(Y (B A) ()) ; in class Y, class B should come before class A  
; as per the above rule, the following code results in an error:  
defclass(M (X Y) ())
```

Defining a Class (defclass)

The domain of geometric objects provides good examples for using object oriented programming. Use the `defclass` function to define a class. You specify the superclass, if any, and all the slots of the class.

```
defclass( GeometricObject
  ()                ;;; superclass
  ()                ;;; list of slot descriptions
) ; defclass
```

This example defines the `GeometricObject` class. Defining the `GeometricObject` class allows the subsequent definition of default behavior of all geometric objects. It has no slots. Because no superclass is specified, the superclass is the `standardObject` class.

```
defclass( Triangle
  ( GeometricObject ) ;;; superclass
  (
    ( x )            ;;; x slot description
    ( y )            ;;; y slot description
    ( z )            ;;; z slot description
  )
) ; defClass
```

This example defines the `Triangle` class. It declares that

- The `Triangle` class is a subclass of the `GeometricObject` class
- Each instance shall have three slots named `x`, `y`, and `z`

Slot Options

Slot options, also known as slot specifiers, govern how you initialize the slot as well as your access to the slot.

Slot Option	Value	Meaning
<code>@initarg</code>	symbol	Defines a keyword argument for the <code>makeInstance</code> function. Note: You can supply more than one <code>@initarg</code> for a given slot. For more information on the order of precedence used when multiple <code>@initargs</code> are supplied, see Rules of Initialization on page 352.
<code>@initform</code>	expression	Defines an expression which initializes the slot.

Slot Option	Value	Meaning
@reader	symbol	Defines a generic function with this name. The function returns the value of the slot.
@writer	symbol	Defines a generic function with this name. The function accepts a single argument which becomes the new slot value.

Example 1

```
defclass( Triangle
  ( GeometricObject )
  (
    ( x
      @initarg x
    )
    ( y
      @initarg y
    )
    ( z
      @initarg z
    )
  )
) ; defClass
```

Example 2

```
defclass( Circle
  ( GeometricObject )
  (
    ( r @initarg r )
  )
) ; defClass
```

Inheritance of Slots

The following rules govern the inheritance of slots in subclasses:

- If a subclass is inherited from a superclass, it also inherits the slots of the superclass.
- If a subclass is inherited from multiple superclasses, which have slots with the same name, then only one slot of the given name is inherited. For example:

```
defclass( Z1 () ((a @initform 1) (b @initform 1)))
defclass( Z2 () ((b @initform 2) (a @initform 2) (zz)))
defclass( Z3 () ((b @initform 3) (a @initform 3) (zz)))
defclass( Z4 () ((b @initform 4)))
defclass( Z5 (Z1 Z2 Z3 Z4) ())
defclass( Z6 (Z4 Z3 Z2 Z1) ())
z5 = makeInstance('Z5 )
```

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```
z5->?
=> (a b zz) ;; slots a, b, and zz are not duplicated
```

Note: If you define a class with two slots that have the same name, SKILL creates the class but also issues a warning.

```
defclass(A () ((slotA) (slotB) (slotA @initform 42)))
*WARNING* duplicate slot slotA
```

Similarly, an error is raised if you define a class in which two @reader or @writer slot options have the same name. For example:

```
defclass(A () ((aa @reader getA) (ff @reader getA)))
*Error* defclass: slots (aa ff) cannot use the same name for @reader - getA

defclass(B () ((aa @reader getB) (dd @writer getB)))
*Error* defclass: slot aa cannot use the same name for @reader and @writer - getB
```

Rules of Initialization

- If a subclass is inherited from a superclass, it also inherits the slots and methods of the superclass. When you instantiate such a subclass, the initialization arguments (@initargs) of the subclass have precedence over the initialization arguments of the superclass. For example:

```
defclass( A1 ()
  ((slot @initform 1 @initarg a)))
defclass( B1 (A1)
  ((slot @initform 2 @initarg b)))
(slotValue (makeInstance 'B1 ?b 3) 'slot) => 3
(slotValue (makeInstance 'B1 ?a 3) 'slot) => 3
(slotValue (makeInstance 'B1 ?b 3 ?a 4) 'slot) => 3
(slotValue (makeInstance 'B1 ?a 3 ?b 4) 'slot) => 4
```

For more information on instantiating classes, see [Instantiating a Class \(makeInstance\)](#).

- If two or more @initargs initialize the same slot, then the order of precedence of the initialization arguments is from left to right. For example:

```
defclass( C1 () ((slot1)
  (slot3 @initarg s3 @reader rs3)
  (slot4 @initarg s4 @writer ws4 @initarg (s4a 'slot34))))
(slotValue (makeInstance 'C1 ) 'slot4 ) => slot34
(slotValue (makeInstance 'C1 ?s4a 5 ) 'slot4 ) => 5
(slotValue (makeInstance 'C1 ?s4 5 ) 'slot4 ) => 5
(slotValue (makeInstance 'C1 ?s4 5 ?s4a 6 ) 'slot4 ) => 5
(slotValue (makeInstance 'C1 ?s4a 5 ?s4 6 ) 'slot4 ) => 6
```


Instantiating a Class (makeInstance)

Use the `makeInstance` function to instantiate a class. The first argument designates the class you are instantiating. Subsequent keyword arguments initialize the instance's slots. The `makeInstance` function returns the newly allocated instance of the class.

```
procedure( makeTriangle( x y z )
  if( x<y+z && y<z+x && z<x+y
    then
      makeInstance( 'Triangle
        ?x 1.0*x
        ?y 1.0*y
        ?z 1.0*z
      )
    else
      error(
        "%n %n %n fail triangle inequality test\n"
        x y z
      )
  ) ; if
) ; procedure
```

```
exampleTriangle = makeTriangle( 3 4 5 ) => stdobj:0x1e6030
```

The print representation for a SKILL++ Object System instance consists of `stdobj :` followed by a hexadecimal number.

Note: `makeInstance` function does not check for invalid `@initargs`. If your code contains invalid `@initargs`, `makeInstance` accepts the `@initargs` as additional `@rest` options.

Initializing an Instance (initializeInstance)

The `initializeInstance` function is called by `makeInstance` to initialize a newly created instance. You can define methods for `initializeInstance` to specify the actions that need to be taken when the instance is initialized. You can also use `initializeInstance` to add initialization parameters in addition to those defined by `@initform`.

```
defclass(A ()
  (
    (x @initarg x @initform 1)
    (y @initarg y @initform 2)
    (product)
  )
)

defmethod( initializeInstance @after ((obj A) @key product @rest args)
  if(product then
    obj->product = product
```

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```
else
  obj->product = obj->x * obj->y
)
printf("initializeInstance : A : was called with args - obj == '%L'
  product == '%L' rest == '%L'\n" obj product args)
printf("  object initialized to: %L\n" obj->??)
)

makeInstance('A)
  initializeInstance : A : was called with args - obj == 'stdobj@0x2d61020'
product == 'nil' rest == 'nil'
  object initialized to: (x 1 y 2 product 2)
=> stdobj@0x2d61020

makeInstance('A ?x 5 ?y 10)
  initializeInstance : A : was called with args - obj == 'stdobj@0x2d61038'
product == 'nil' rest == ' (?x 5 ?y 10)'
  object initialized to: (x 5 y 10 product 50)
=> stdobj@0x2d61038

makeInstance('A ?product 30)
  initializeInstance : A : was called with args - obj == 'stdobj@0x2d61050'
product == '30' rest == 'nil'
  object initialized to: (x 1 y 2 product 30)
=> stdobj@0x2d61050
```

Reading and Writing Instance Slots

You can use the arrow operator to read a slot's value.

```
exampleTriangle->x => 3.0
exampleTriangle->y => 4.0
exampleTriangle->z => 5.0
```

You can use the `->` operator on the left-side of an assignment statement.

```
exampleTriangle->x = 3.5
```

The `->??` expression returns a list of the slots and their values.

Another approach is to use the `@reader` and `@writer` slot options to define generic functions for reading and writing slots when you define the class.

```
defclass( Triangle
  ( GeometricObject )
  (
    ( x
```

```

        @initarg      x
        @reader        get_x
        @writer        set_x
    )
    ( y
        @initarg      y
        @reader        get_y
        @writer        set_y
    )
    ( z
        @initarg      z
        @reader        get_z
        @writer        set_z
    )
) ; defClass

```

```

exampleTriangle = makeTriangle( 3 4 5 ) => stdobj:0x1e603c
get_y( exampleTriangle ) => 4.0
set_x( exampleTriangle 3.5 ) => 3.5

```

Defining a Generic Function (defgeneric)

Use the `defgeneric` function to define a generic function. The body of the generic function defines the default method for the generic function.

```

defgeneric( Perimeter ( geometricObject )
    error( "Subclass responsibility\n" )
) ; defgeneric

```

This example indicates that relevant subclasses of the `geometricObject` class, such the `polygon` class, should have a `Perimeter` method. Although not strictly necessary to do so, defining a generic function before defining any methods for it has two advantages:

- You can specify a default method

Using the `defgeneric` function gives you control over the default method. When you invoke a generic function that has no default method, the SKILL++ Object System raises an error. The following example illustrates calling a generic function which does not have a method defined for the specific argument.

```

Perimeter( 3 )
*Error* (Default-method) generic:Perimeter class:fixnum

```

- You can document the template argument list

In the absence of a generic function, the first method you define automatically declares the generic function. The method's argument list becomes the template argument list.

You can also use the `defgeneric` function to associate a *proxy* class, called a *generic function class*, with the generic function. A *proxy* class is useful when defining customSpecializer methods for a particular class, or for defining dependency protocol where the methods are specialized on a particular generic function class. The basic need for an

application specific *proxy* class is to be able to differentiate one group of generic functions from others in an application specific way.

By default, all generic functions are associated with `class:ilGenericFunction`. So, to be able to use a *proxy* class you need to inherit the class from `class:ilGenericFunction` as shown in the following example.

Example 1

```
defclass(niGF (ilGenericFunction) ()) ; generic class 'niGF
defgeneric(niTest (x y) ?genericFunctionClass niGF) ; generic function niTest is
associated with class:niGF
```

In this example, `niGF` is the *proxy* class for the generic function, `niTest`. The instance of this class is created when either a generic function is defined (at `defgeneric` time) or when this class is accessed for the first time.

Note: This lazy creation of the *proxy* object is especially true for generic functions loaded from a context.

To retrieve a *proxy* instance from the generic function object, use the `getGFproxy` function as shown in the example below. The instance of the proxy object retrieved is stored in property list of generic function symbol.

Example 2

```
getGFproxy('niTest)
=> stdobj@0x83c0018
classOf(getGFproxy('niTest))
=> class:niGF
classOf(getGFproxy('printself)) ;; class of standard generic function (printself)
=> class:ilGenericFunction ;; default
getGFproxy('abc)
=> nil ;; non-existing generic function
```

In addition, you can inherit from a generic function *proxy* class by using the `defclass` function, which is in turn inherited from `class:ilGenericFunction` (or optionally from any other `StandardObject` class).

Defining a Method (defmethod)

Use the `defmethod` function to define a method. You do not need to define the generic function before you define a method for it. When you invoke a generic function, the SKILL++

Object System chooses the method to run based on the class of the first argument you pass to the function.

- If you use conventional syntax `defmethod( ... )` in place of the LISP syntax `( defmethod ... )`, use white space to separate the method name from the argument list.
- You must specify the class of the method's first argument to the `defmethod` function. The first argument to `defmethod` uses the following syntax:

```
( s_arg1 s_class )
```

Example 1

```
defmethod( Perimeter (( triangle Triangle ))
  let( (
    (x triangle->x)
    (y triangle->y)
    (z triangle->z)
  )
    x+y+z
  ) ; let
) ; defmethod
```

```
Perimeter( exampleTriangle ) => 12.0
```

This example defines a method named `Perimeter`. It is specialized on the `Triangle` class.

Example 2

```
defmethod( Perimeter (( c Circle ))
  2*c->r*3.1415
) ; defmethod
```

This example defines a `Circle` class and defines the `Perimeter` method for the `Circle` class.

You can specify additional optional arguments while defining a method of a generic function by using the `@rest` option. For example:

```
defgeneric( myTest (x @rest _args))
; The following derived methods use additional @key and @optional arguments:
defmethod( myTest (x) 1)
=> t
defmethod( myTest ((x string) @key (z 2)) 3)
=> t
defmethod( myTest ((x number) @optional a b c) 4)
=> t
```

The eqv Specializer

While defining a method using `defmethod`, you can use the `eqv` specializer to specialize the method on objects other than classes (for example, some value of its arguments).

When `eqv` is encountered in a method declaration, the value of the argument of the method is compared to the `eqv` value. If a match is found, the method is executed. For example,

```
defgeneric( factorial (x))
defmethod( factorial ((x fixnum)) ;; #1
  (times x (factorial (sub1 x))))
defmethod( factorial ((x (eqv 0)) ;; #2
  1)
; method #2 is applicable if the argument is eqv to 0
```

Defining Method Combinations (@before, @after, and @around)

Once you have defined a generic function, you can combine it with methods that execute before or after the normal implementation. There are three kinds of additional or auxiliary methods that you can use with `defmethod`: `@before`, `@after`, and `@around` methods.

A standard method combination will have `defmethod` as the primary method and a method qualifier (`@before`, `@after`, `@around`) between the name of the method and the parameter list.

```
defmethod( mymethod @before ((x number)) )
defmethod( mymethod @after ((x fixnum)) )
defmethod( mymethod @around ((x number) y))
```

All the applicable methods are evaluated and partitioned into separate lists according to their qualifiers. A diagrammatic representation of the order in which the applicable methods are invoked is given below:

```
(@around methods | if exist . . . @before methods | if exist . . . primary
method(s) | if exist . . . @after method(s))
```

The detailed order in which the applicable methods are invoked is as follows:

- The `@around` methods, if defined, are executed first.

If an `@around` method invokes `callNextMethod`, execute the next most specific `@around` method, with all its `callNextMethod` methods.

Note: If no applicable primary method is defined but an `@around` method exists, an

error occurs.

- If there are no `@around` methods or if an `@around` method invokes `callNextMethod` but there are no further `@around` methods to call then proceed as follows:

- The `@before` methods are executed thereafter, with the most-specific method invoked first. All `@before` methods are called before any of the primary methods.

If you use a `callNextMethod` in a `@before` method, an error returns.

- Then, the most-specific primary methods are called.

You can use a `callNextMethod` inside the body of a primary method to call the next most-specific primary method.

- The `@after` methods are called after the primary methods but in the reverse order, with the least-specific method called first.

If you use a `callNextMethod` in an `@after` method, an error returns.

Note: Calling a generic function that has no primary method, but only `@before` or `@after` method qualifiers, results in an error.

If there is an error in a method, the execution returns to the nearest toplevel.

Example 1

```
defmethod( mymethod (( x fixnum)) ;; # 1 - primary method
... )
defmethod( mymethod @before (( x number )) ;; # 2
... )
defmethod( mymethod @before (( x fixnum )) ;; # 3
... )
defmethod( mymethod @after (( x fixnum )) ;; #4
... )
defmethod( mymethod @around (( x fixnum)) ;; # 5 - primary method
. . .
callNextMethod()
. . .)
```

When `mymethod` is invoked (with a `fixnum` argument), the methods are called in the following order:

#5 -> #3 -> #2 -> #1 -> #4

Example 2

```
defmethod(aTest @around ((x number) y)
  /* 1 : around method*/
  callNextMethod())
defmethod(aTest @before ((x number) y)
  /* 2 : before method */
  ...)
defmethod(aTest ((x number) y)
  /* 3 : a primary method */
  callNextMethod())
defmethod(aTest @after ((x number) y)
  /* 4 : after method */
  ...)
defmethod(aTest @around ((x systemObject) y)
  /* 5 : another @around method */
  callNextMethod())
aTest(1)
=> #1 -> #5 -> #2 -> #3 -> #4 (calling order)
```

Multi-Method Dispatch

SKILL++ supports multi-method dispatch. With multi-method dispatch, all arguments of a method are treated equally and the method to be applied is decided at runtime, based on the dynamically-determined types of arguments.

It means that you can define more than one method with the same name in your code. When the method call is made, instead of one parameter specializer determining the method to be applied, it is determined by multiple parameter specializers.

Example: Single-Dispatch

```
;; body of method1
;; method1 specialized on its first argument only (obj)
defmethod ( method1 ((obj string) x y)
) ; end of method1(string)
```

Example: Multiple-Dispatch

```
;; body of method2
;; method2 specialized on 3 arguments:
```



```
;; obj of type string
;; x of type number
;; y of class "classY"
defmethod ( method2 ((obj string) (x number) (y classY))
) ; end of method2
```

Method Specificity

In case, all applicable methods have the arguments of the same type, the methods are sorted on the order of specificity. So, the most-specific primary method is called first; other methods can then be called from the primary method by using the `callNextMethod`. For example,

```
defmethod( test ((x number) (y string))
printf("test number/string\n")
callNextMethod()
)

defmethod( test ((x fixnum) (y string))
printf("test fixnum/string\n")
callNextMethod()
)

defmethod( test ((x number) (y primitiveObject))
printf("test number/primitiveObject\n")
callNextMethod()
)

defmethod( test ((x fixnum) (y primitiveObject))
printf("test fixnum/primitiveObject\n")
callNextMethod()
)

defmethod( test ((x t) (y t))
printf("test t/t\n")
)
```

The class precedence list is used in determining the method specificity:

```
t -> systemObject -> primitiveObject -> string
t -> systemObject -> primitiveObject -> number -> fixnum
```

So, for `test(1 "test")` the order of method calls is:

```
; all the applicable methods are called according to the class precedence of
arguments:
=> test fixnum/string
=> test fixnum/primitiveObject
=> test number/string
=> test number/primitiveObject
=> test t/t
```

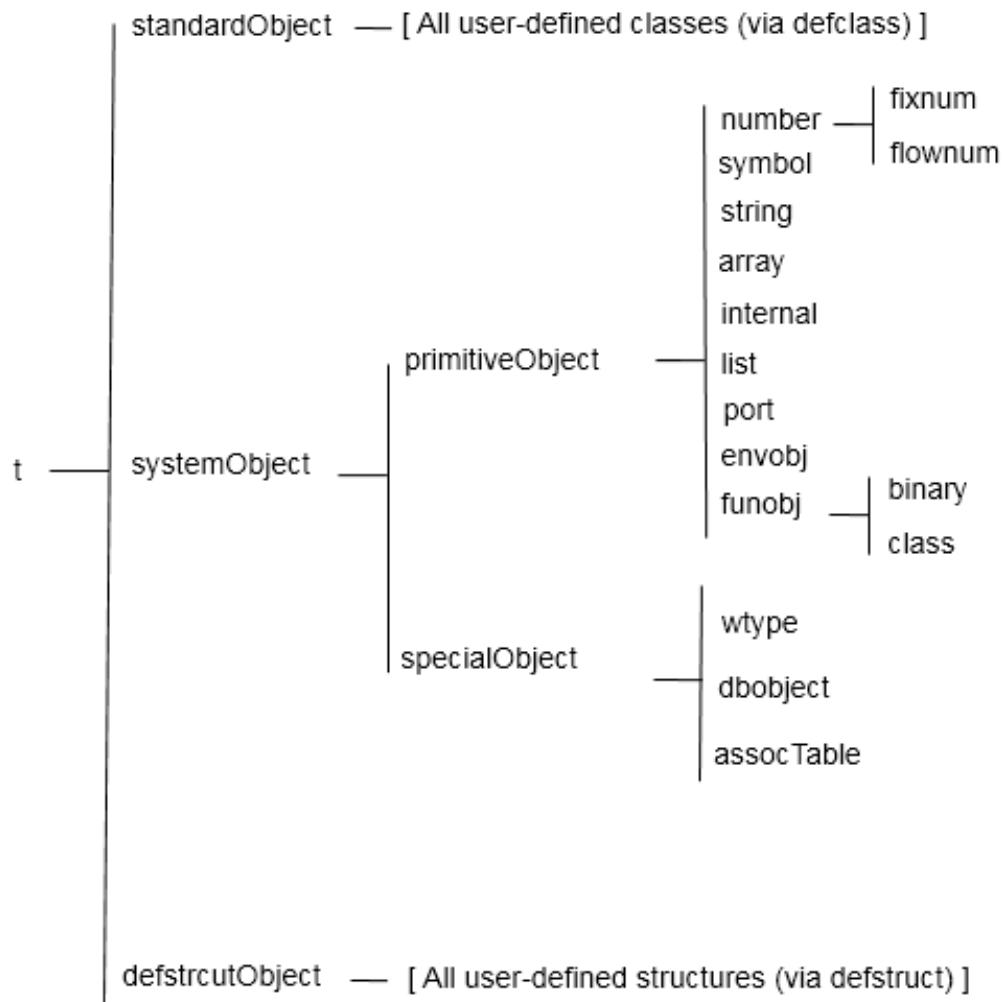
For `test(1.0 "test")`, the order of method calls is:

```
;three applicable methods are called according to the class precedence of
arguments:
=> test number/string
```

```
=> test number/primitiveObject
=> test t/t
```

Class Hierarchy

The diagram below is a horizontal view of the SKILL++ Object System class hierarchy.



- `t` is the superclass of all classes. Class `t` has two immediate subclasses, `standardObject` and `systemObject`.
- `standardObject` is the superclass of all classes you define with `defclass` function. This is the primary portion of the class hierarchy that you can extend.
- `systemObject` is the superclass of `primitiveObject` and `specialObject`. No subclasses of `systemObject` can be used with `defclass`.

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- `primitiveObject` is the superclass of all SKILL built-in classes.
- `specialObject` is the superclass of all classes corresponding to the C-level registrable “user-types.”
- `defstructObject` is a superclass of all structures defined by `defstruct` function (You can create such a class with `addDefstructClass` function).

This example shows how you can list all of the subclasses. Run this program to see what the class hierarchy is at any given time.

```
procedure( getDirectSubclasses( classObject )
  foreach( mapcar c subclassesOf( classObject )
    className( c )
  ) ; foreach
) ; procedure

procedure( getAllSubclasses( classObject )
  let( ( direct )
    direct = getDirectSubclasses( classObject )
    cons(
      className( classObject )
      direct && foreach( mapcar c direct
        getAllSubclasses( findClass( c ) )
      ) ; foreach
    ) ; cons
  ) ; let
) ; procedure

getAllSubclasses( findClass( 't ) ) =>

(t
  (standardObject
    (GeometricObject
      (Triangle)
      (Point)
    )
  )
  (systemObject
    (primitiveObject
      list()
      (port)
      (funobj)
      (array)
      (string)
      (symbol)
      (number
        (fixnum)
        (flonum)
      )
    )
  )
  ( specialObject
    (other)
    (assocTable)
  )
)
```

Browsing the Class Hierarchy

The SKILL++ Object System provides a number of functions for browsing the class hierarchy:

- [Getting the Class Object from the Class Name](#) on page 364
- [Getting the Class Name from the Class Object](#) on page 364
- [Getting the Class of an Instance](#) on page 365
- [Getting the Class of the Environment Object \(envObj\)](#) on page 365
- [Getting the Superclasses of an Instance](#) on page 365
- [Checking if an Object Is an Instance of a Class](#) on page 365
- [Checking if One Class Is a Subclass of Another](#) on page 366

Examples in these sections refer to the following code:

```
defclass( GeometricObject
  ()      ;; superclass
  ()      ;; list of slot descriptions
) ; defclass

defclass( Triangle
  ( GeometricObject ) ;; superclass
  (
    ( x @initarg x ) ;; x slot description
    ( y @initarg y ) ;; y slot description
    ( z @initarg z ) ;; z slot description
  )
) ; defClass

exampleTriangle = makeTriangle( 3 4 5 ) => stdobj:0x1e6030
```

Getting the Class Object from the Class Name

Use the `findClass` function to get the class object from its name. Use a SKILL symbol to represent the class name.

```
findClass( 'Triangle ) => funobj:0x1cb2d8
```

Getting the Class Name from the Class Object

Use the `className` function to get the class symbol. The term `class symbol` refers to the symbol used to represent the class name. The SKILL++ Object System uses a SKILL symbol to represent the class name.

```
className( findClass( 'Triangle ) ) => Triangle
```

Getting the Class of an Instance

Use the `classOf` function to get the class of an instance.

```
className( classOf( exampleTriangle ) ) => Triangle
```

Getting the Class of the Environment Object (envObj)

Use the `classOf` function to get the class of the environment object `envObj`.

```
z = let( (( x 3 ))
  theEnvironment()
) ; let
=> envobj:0x1e0060
classOf(z)
=> class:envobj
```

Getting the Superclasses of an Instance

Use the `superclassesOf` function to get the superclasses of a class. The function returns a list of class objects.

```
L = superclassesOf( classOf( exampleTriangle ) )
foreach( mapcar classObject L
  className( classObject )
) ; foreach
=> (Triangle GeometricObject standardObject t)
```

Checking if an Object Is an Instance of a Class

Use the `classp` function to check if an object is an instance of a class. You can pass either the class symbol or the class object as the second argument.

Example 1

```
classp( exampleTriangle 'Triangle ) => t
classp( 5 'fixnum ) => t
classp( 5 'Triangle ) => nil
```

5 is a `fixnum`. 5 is not an instance of `Triangle`.

Example 2

```
classp( exampleTriangle 'GeometricObject ) => t
classp( exampleTriangle 'standardObject ) => t
classp( exampleTriangle t ) => t
```

This example illustrates that `classp` returns `t` for all superclasses of the class of an instance. `Triangle` is a subclass of `GeometricObject`. `GeometricObject` is a subclass of `standardObject`. `standardObject` is a subclass of `t`.

Checking if One Class Is a Subclass of Another

Use the `subclassp` function to determine whether one class is a subclass of another.

Example 1

```
subclassp(
    findClass( 'Triangle )
    findClass( 'GeometricObject )
) => t
```

`Triangle` is a subclass of `GeometricObject`.

Example 2

```
subclassp(
    findClass( 'Triangle )
    findClass( t )
) => t
```

`Triangle` is a subclass of `t`.

Example 3

```
subclassp(
    findClass( 'Triangle )
    findClass( 'fixnum )
) => nil
```

`Triangle` is not a subclass of `fixnum`.

Advanced Concepts

This section covers the following advanced aspects of the SKILL++ Object System:

■ Method Argument Restrictions

- [Applying a Generic Function](#) on page 368
- [Incremental Development](#) on page 369
- [Methods and Slots](#) on page 370
- [Sharing Private Functions and Data Between Methods](#) on page 370
- [Using Custom Specializers in SKILL++ Methods](#) on page 371

Method Argument Restrictions

Method argument lists have the following restrictions:

Aspect	Restriction
Number of arguments	The methods of a generic function can have additional <code>@optional</code> arguments.
<code>@rest</code> arguments	All methods of a generic function must take <code>@rest</code> arguments if any of the methods take <code>@rest</code> arguments.
Keyword arguments	Each method of a generic function must ■ Take <code>@rest</code> arguments if any of the methods take <code>@rest</code> arguments ■ Allow a superset of the keyword arguments specified in the <code>defgeneric</code> declaration <code>@rest</code> picks up all keyword arguments that have no matching keyword in the formal argument list. Different methods may have different default forms for the optional arguments and may accept different set of keywords.

Example

```
(defmethod test ((x class1) (y class2) @key a @rest _rest) ... )
(defmethod test ((x class3) (y class2) @key b @rest _rest) ... )
(defmethod test ((x class1) y @rest _rest) ... )
(defmethod test (x y @rest _rest) ... )
```

In the example above, the method `test()` can be called with or without `@key` arguments, provided at least two required arguments are passed to `test()`.

Applying a Generic Function

When you apply a generic function to some arguments, the SKILL++ Object System performs the following actions to complete the function call. This process is called *method dispatching*. The SKILL++ Object System

1. Retrieves the methods of the generic function.
2. Determines the class of the first argument to the generic function.

Based on the class of the first argument passed to the generic function, the SKILL++ Object System finds

- ☐ No applicable methods

SKILL++ Object System calls the default method for the generic function if one exists. Otherwise it signals an error.

- ☐ Only one applicable method
- ☐ More than one applicable method

This situation occurs when you have methods specialized on one or more superclasses of the first argument's class.

3. Determines applicable methods by examining the method's class specializer.

A method is applicable if it specialized on the class of the first argument or a superclass of the class of the first argument.

4. Sorts the applicable methods according to the chain of superclasses of the first argument's class.

- ☐ The first method in the ordering is the most specific method.
- ☐ The last method in the ordering is the least specific method.

5. Calls the first method.

You can invoke the `callNextMethod` function from within a method to access the next applicable method in the ordering. For example:

```
defgeneric( describe (obj) () )
defclass( GeometricObject () () ) ; no slots or superclasses

defclass( Point
  ( GeometricObject )
  (
    ( name      @initarg name )
    ( x         @initarg x );;; x-coordinate
    ( y         @initarg y );;; y-coordinate
  )
)
```



```
) ; defclass

defmethod( describe (( object GeometricObject ))
  className( classOf( object ))
) ; defmethod

defmethod( describe (( p Point ))
  sprintf( nil "%s %s @ %n:%n"
    callNextMethod( p )
    p->name
    p->x
    p->y
  )
) ; the most specific method

aPoint = makeInstance( 'Point ?name "A" ?x 1 ?y 0 )
describe( aPoint )
=> "Point A @ 1:0"
```

In the example, the `describe` generic function has two methods that are applicable to the argument `aPoint`:

- The method specialized on the `Point` class
- The method specialized on the `GeometricObject` class

The method specializing on the `Point` class is the more specific method, therefore the SKILL++ Object System applies the most specific method to the argument.

Incremental Development

In the SKILL++ environment , you can redefine SKILL++ functions incrementally. You should observe the following guidelines when redefining SKILL++ Object System elements of your application:

- **Redefining methods**

During development, you can expect to redefine methods about as frequently as you redefine procedures. You can redefine a method as long as the redefined method's argument list continues to conform to the generic function.

- **Redefining generic functions**

You need to redefine a generic function to change the generic function's default method or argument list. Such need occurs infrequently. When you redefine a generic function, the SKILL++ Object System discards all existing methods for the generic function.

- **Redefining classes**

You need to redefine a class when you want to

- ☐ Change the superclass
- ☐ Add or remove a slot
- ☐ Add or remove a slot option

If you need to redefine a class, you should exit the SKILL++ environment and reload your application. A frequent need to redefine classes probably indicates that you should analyze your application before further programming.

Methods and Slots

Methods are usually more expensive to use compared to slots but they offer data hiding and safety. Consider whether the `Triangle's Area` method should access a slot containing the (precomputed) area or whether the area should be computed. The nature of your application dictates your final decision.

Computing the area may be costly if, for example, the area of triangles is used often. In such a situation, it would be more advantageous to add a slot for area to the triangle class. But then we would have to add `@writer` methods for the sides of a triangle to recalculate the area when the length of a side changes.

Sharing Private Functions and Data Between Methods

Using lexical scoping with the SKILL++ Object System allows all methods specialized on a class to share private functions and data.

The methods for a class might need access to data, such as an association table, that is shared between all instances of the class. Slots you specify in the `defclass` declaration are allocated within each instance of the class.

The methods for a class might all rely on certain helper functions which you need to make private.

Using the following template as a guide achieves both goals.

```
defgeneric( Fun1 ( obj ... ) ... )
defgeneric( Fun2 ( obj ... ) ... )
defclass( Example ( ... ) ( ... ) )
let(
  (
    ( classVar1 ... )           ;data shared between all
    ( classVar2 ... ) ... )    ;instances of the class
  )
  procedure( HelpFun1( ... ) ;private helper functions
  procedure( HelpFun2( ... )
  ...
```

```
defmethod( Fun1 (( obj Example) ... )
defmethod( Fun2 (( obj Example ) ... )
...
); let
```

Using Custom Specializers in SKILL++ Methods

In the SKILL++ object system you can define generic functions which are a collection of methods specialized on the class of one or more arguments passed to the generic function. This allows you to simply implement polymorphism in SKILL - in other words you can have multiple implementations of the same function which are focused on different objects, and this can make the code easier to maintain and develop as it avoids having to have a large `cond` or `case` function call within your code to switch between different objects; in addition the method dispatch will call the most applicable method first, with potentially that method being able to call the next most applicable method and so on.

The basic approach allows you to specialize on the class of the object passed to the function. For example:

```
defgeneric(CCScomputeArea (obj) error("Unrecognized object %L\n" obj)
)
defmethod(CCScomputeArea ((obj CCSrect))
; calculate area of a rectangle object
)
defmethod(CCScomputeArea ((obj CCScircle))
; calculate area of a circle
)
defmethod(CCScolor ((obj CCSshape))
; return the color of a shape (CCSrect and CCScircle are subclasses of CCSshape)
)
```

The `CCSshape`, `CCScircle`, and `CCSrect` are class names here that would have been defined elsewhere. This is an incomplete example just showing the methods so as to show the principle.

Sometimes you might have an object but want to further distinguish which method to call based on the value of a slot of that object. You can then use method dispatch to separately perform some work based on a slot or attribute value. One example might be that you have a database object (`dbobject`) in Virtuoso, which could represent a figure in the design (e.g. an instance, or a shape of some sort). The `dbobjects` in Virtuoso do have distinct objects for each kind of object - the `objType` attribute specifies whether the object is a "rect", a "path", a "net", a "cellView" and so on. However, because these are not instances of SKILL++ classes, there is limited ability to write methods specializing on those specific objects. You can write a method using a class name corresponding to the type of the object:

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```
defmethod(CCScomputeArea ((obj dbobject))
; handle all the different kinds of database object within this method and compute
the area appropriately
)
```

Normally, you cannot further refine the method to specialize on the object solely if it is a *rect*, or solely if it is a *pathSeg*, for example. This is because there are no subclasses of *dbobject* and so, by default, you cannot specialize on the *objType* of the database object.

What you would like to do is to use polymorphism to specialize on the *objType* of the database object in order to compute the area and perimeter of a variety of figure types.

Introduction

In addition to specializing on classes, SKILL++ allows specializing either on the value of a basic type, using the built-in *eqv* specializer, or by defining custom specializers to match specific objects. In order to enable these custom specializers, it is necessary to define a class plus three different methods to support the capability.

Before using custom specializers, you should consider whether this is the right mechanism for what you are trying to achieve. The danger is that you can end up hiding the logic of your program in the method dispatch, and ending up with methods specialized on particular values distributed across multiple files and this can make it hard to understand the flow of the program. You may be better off using a *case* or *cond* function within your code to group all the logic about different values in one place. One of the key benefits of declaring a method specialized on a class (the standard mechanism) is that the method also applies to subclasses too, unless a more specific method is defined. With custom specializers, the method can only be applied to specific matching instances, and if your custom specializers are complex they can end up with an order dependency which is very hard to reason about. It is possible to resolve any order dependency by defining the *ilSpecMoreSpecificp* method to determine the order, but again this adds additional complexity which may be unobvious to a reader of the code (this article does not use *ilSpecMoreSpecificp*).

Defining Methods Using Non-class Specializers

First of all, let us describe how you would declare a method using these specializers. As was seen above, the argument for a method normally is of the form *(obj className)* to make it specific to an instance of that class (or any of the children of that class). In order to use the custom specializers (or the *eqv* specializer), you would use the form *(obj (specializerSymbol arg1 [arg2 ... argN]))*. The *specializerSymbol* can be either *eqv* or one you define yourself - as will be outlined below. If *eqv* is used, then a single argument is given, and then the object will be matched against that value. So for example:

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```
defmethod(CCStranlate ((str string)) strcat("EN: " str)) defmethod(CCStranlate
((str (eqv "hello")) "bonjour") CCStranlate("goodbye") => "EN: goodbye"
CCStranlate("hello") => "bonjour"
```

With the *eqv* specialization, it only makes sense to give a single argument (the other arguments are ignored), but with custom specializers you can specify multiple arguments to aid the specificity of the match. Note that with the *eqv* specialization, the argument is evaluated when the *defmethod* is defined (and so you must quote symbols if the argument is a symbol), but for custom specializers they are unevaluated and can only be literal values.

Using Custom Specializers With *defmethod*

For custom specializers, you need to define a class that is used to compare an argument to see if it matches the specialization, plus three methods: *ilGenerateSpecializer*, *ilEquivalentSpecializers* and *ilArgMatchesSpecializer*. The following example shows how this is done for a specialization designed to match the *objType* of the database object. This means that you can then use methods similar to the one given below:

```
defmethod(abFigArea ((obj (abDbFig "rect"))))
  abAreaOfBBox(obj~>bBox)
)
defmethod(abFigArea ((obj (abDbFig "donut"))))
  let((inner outer)
    outer=abFigMaths.PI*obj~>outerRadius**2
    inner=abFigMaths.PI*obj~>innerRadius**2
    outer-inner
  )
)
```

Defining a class for the specialization

In this case, the symbol *abDbFig* has been used for the specialization. The *ilGenerateSpecializer* method takes care of recognizing this symbol and generating an instance of the specialization class. The specialization class in this case is defined as follows:

```
defclass(abDbFigSpecializer ())
(
  (objType @initarg objType)
)
)
```

The *objType* slot in the class is used to store the value of *objType* that the method is trying to match against.

ilGenerateSpecializer

The `ilGenerateSpecializer` method is called whenever a custom specializer is used in a `defmethod` as in the above example. It can either return the code (the SKILL expression) needed to generate the instance, or can directly return the instance of the specializer class. Three arguments are specified for it:

```
ilGenerateSpecializer(g_genericFunctionObj s_specSymbol l_argList)
```

Argument	Description
<code>g_genericFunctionObj</code>	An instance of the <code>ilGenericFunction</code> class or a proxy class derived from it. Using proxy classes is one way of associating custom specializers with generic functions. The SKILL Language User Guide describes how to define proxy classes to define a group of generic functions to use common specializers. A cut-down example of using a proxy class is given at the bottom of this article to compare with the preferred method.
<code>s_specSymbol</code>	The symbol used as the first entry in <code>defmethod</code> to trigger the custom specializer (<code>abDbId</code> in the example above). Note that this does not strictly need to be a symbol.
<code>l_argList</code>	The remaining arguments to the specializer - ("rect") in the example above. If more than one argument is given, this is a list of all the arguments.

An example of `ilGenerateSpecializer` using an `eqv` specializer to match the expected symbol. Since the generic function object is not used, it is prefixed with an underscore to prevent any impact on the SKILL Lint score and to indicate that it is unused.

```
defmethod(ilGenerateSpecializer (_gf (_spec (eqv 'abDbFig)) args)  
makeInstance('abDbFigSpecializer ?objType car(args)))
```

Another way of implementing this is to return the code needed to generate the instance:

```
defmethod(ilGenerateSpecializer (_gf (_spec (eqv 'abDbFig)) args)  
`makeInstance('abDbFigSpecializer ?objType ,car(args))  
)
```

ilEquivalentSpecializer

The `ilEquivalentSpecializer` method is called whenever more than one method is defined for a generic function. It is intended to identify when identical custom specializers

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have been specified in the case of method redefinition, so that the new method replaces the previously defined one, rather than adding a new method. Having two methods with identical specializers would lead to errors during dispatch, which is why this method must be defined. The arguments signature is:

```
ilEquivalentSpecializer(g_genericFunctionObj g_specObj1 g_specObj2j)
```

Argument	Description
<code>g_genericFunctionObj</code>	An instance of the <code>ilGenericFunction</code> class or a proxy class derived from it. (See <code>ilGenerateSpecializer</code> above)
<code>g_specObj1</code>	An instance of the <code>specializer</code> class to be compared.
<code>g_specObj2</code>	The second instance of the <code>specializer</code> class to be compared

An example of `ilEquivalentSpecializer` to compare that the `objType` is identical for two instances, needed for method redefinition:

```
defmethod(ilEquivalentSpecializers (_gf (spec1 abDbFigSpecializer)
    (spec2 abDbFigSpecializer))
spec1->objType==spec2->objType
)
```

ilArgMatchesSpecializer

Finally the `ilArgMatchesSpecializer` method is needed to determine whether a argument passed to the generic function at run time matches a particular `specializer` class, so that it then knows which method should be called. This has this argument signature `ilArgMatchesSpecializer`.

```
(g_genericFuncObj g_specObj g_argument)
```

Argument	Description
<code>g_genericFunctionObj</code>	An instance of the <code>ilGenericFunction</code> class or a proxy class derived from it. (See <code>ilGenerateSpecializer</code> above)
<code>g_specObj</code>	An instance of the <code>specializer</code> class being compared against.
<code>g_argument</code>	The argument to the method that is being compared.

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The method should check the type/class of the argument being compared to prevent errors if a candidate argument doesn't have the slots being accessed within the `ilArgMatchesSpecializer`. This could be done by adding a class specializer on the third argument, or including the type/class check as part of the check within the method itself.

For example:

```
defmethod(ilArgMatchesSpecializer (_gf (spec abDbFigSpecializer) (arg dboject))
  spec->objType==arg->objType
)
```

Programming Examples

See the following sections for programming examples:

- [List Manipulation](#) on page 378
- [Symbol Manipulation](#) on page 378
- [Sorting a List of Points](#) on page 379
- [Computing the Center of a Bounding Box](#) on page 380
- [Computing the Area of a Bounding Box](#) on page 381
- [Computing a Bounding Box Centered at a Point](#) on page 381
- [Computing the Union of Several Bounding Boxes](#) on page 382
- [Computing the Intersection of Bounding Boxes](#) on page 382
- [Prime Factorizations](#) on page 383
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- [Counting Values in a List](#) on page 389
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- [Regular Expression Pattern Matching](#) on page 391
- [Geometric Constructions](#) on page 392
- [Examples of How to use Custom Specializers in SKILL++ Methods](#) on page 405

List Manipulation

A list is a linear sequence of Cadence® SKILL language data objects. The elements of a list can have any data type, including symbols or other lists. The printed presentation for a SKILL list uses a matching pair of parentheses to enclose the printed representations of the list elements. The `trListIntersection` and `trListUnion` functions illustrate

- Documenting at the function level
- Using the `setof` function
- Using the `member` function

The `trListUnion` function also illustrates the `nconc` function, which destroys all but its last argument. In this case, the first argument is a new, anonymous list created by the `setof` function.

```
procedure( trListIntersection( list1 list2 )
  setof( element list1
    member( element list2 )
  ) ; setof
) ; procedure

procedure( trListUnion( list1 list2 )
  nconc(
    setof( element list2
      !member( element list1 )
    ) ; setof
    list1
  ) ; nconc
) ; procedure

trListIntersection( `(a b c) `(b c d)) => (b c)
trListUnion( list(1 2 3) list(3 4 5 6)) => ( 4 5 6 1 2 3)
```

Symbol Manipulation

A symbol is the primary data structure within SKILL. A SKILL symbol has four data slots: the name, the value, the function definition, and the property list. Except for the name slot, all slots can be empty.

The `trReplaceSymbolsWithValues` function makes a copy of an arbitrary SKILL expression, in which all references to a symbol are replaced by the symbol's value.

```
a = "one" b = "two" c = "three"
testCase = '( 1 2 ( a b ) )
trReplaceSymbolsWithValues( testCase ) => (1 2 ("one" "two"))

testCase = '(1 ( a ( c ) ) b )
trReplaceSymbolsWithValues( testCase ) => (1 ("one" ("three")) "two")
```

The `trReplaceSymbolsWithValues` illustrates

- Using recursion to process an arbitrary SKILL expression, making a copy of the expression in which all references to a symbol are replaced by the symbol's value.
- How the `cond` function handles several possibilities.

The `listp` function determines whether the expression is a list.

The `symbolp` function determines whether the expression is a list.

The `symeval` function retrieves the value of the expression, provided it is a symbol.

In the general case, the `cond` function recursively descends into the `car` of the expression and the `cdr` of the expression and builds a list from the results.

```
procedure( trReplaceSymbolsWithValues( expression )
  cond(
    ( null( expression ) nil )
    ( symbolp( expression ) symeval( expression ) )
    ( listp( expression )
      cons(
        trReplaceSymbolsWithValues( car( expression ) )
        trReplaceSymbolsWithValues( cdr( expression ) )
      )
    )
    ( t expression )
  ) ; cond
) ; procedure

x = 5
a = 1
trReplaceSymbolsWithValues( `(x a))
=> (5 1)
```

Sorting a List of Points

The `trPointLowerLeftp` function indicates whether `pt1` is located to the lower left of `pt2`. This function illustrates

- Documenting at the function level
- Using the `xCoord` and `yCoord` functions

Note: The `xCoord` and `yCoord` functions are aliases for the `car` and `cadr` functions.

- Using the `cond` function

```
procedure( trPointLowerLeftp( pt1 pt2 )
  let( ( pt1x pt2x pt1y pt2y )
    pt1x = xCoord( pt1 )
    pt2x = xCoord( pt2 )
    cond(
```

```
        ( pt1x < pt2x t )
        ( pt1x == pt2x
          pt1y = yCoord( pt1 )
          pt2y = yCoord( pt2 )
          pt1y < pt2y )
        ( t nil )
      ) ; cond
    ) ; let
  ) ; procedure
trPointLowerLeftp( 0:0 100:100)
=> t
trPointLowerLeftp( 100:100 0:0)
=> nil
```

The `trSortPointList` function returns a list of points sorted destructively and illustrates

- Documenting at the function level

- Using the sort function

```
procedure( trSortPointList( aPointList )
  sort( aPointList 'trPointLowerLeftp )
); procedure
x = list(100:0 100:100 0:0 50:50 0:100)
trSortPointList( x )
=>
(( 0 0)
 (0 100)
 (50 50)
 (100 0)
 (100 100))
```

Computing the Center of a Bounding Box

The `trBBoxCenter` function returns the point at the center of a bounding box and illustrates

- Documenting at the function level

- Using the `xCoord` and `yCoord` function

- Using the `lowerLeft` and `upperRight` function

- Using the colon (:) operator to build a point

```
procedure( trBBoxCenter( bBox )
  let( ( llx lly urx ury )
    ury = yCoord( upperRight( bBox ) )
    urx = xCoord( upperRight( bBox ) )

    llx = xCoord( lowerLeft( bBox ) )
    lly = yCoord( lowerLeft( bBox ) )
    ( urx + llx )/2 : ( ury + lly )/2
  ) ; let
) ; procedure
```

```
trBBoxCenter( list(0:0 100:100))  
=> (50 50)
```

Computing the Area of a Bounding Box

The `trBBoxArea` function returns the area of a bounding box and illustrates

- Documenting at the function level
- Using the `xCoord` and `yCoord` functions
- Using the `lowerLeft` and `upperRight` functions
- Using parentheses to change priority of arithmetic operations

```
procedure( trBBoxArea( bBox )  
  let( ( llx lly urx ury )  
    urx = xCoord( upperRight( bBox ) )  
    ury = yCoord( upperRight( bBox ) )  
    llx = xCoord( lowerLeft( bBox ) )  
    lly = yCoord( lowerLeft( bBox ) )  
    ( ury - lly ) * ( urx - llx )  
  ) ; let  
) ; procedure
```

Computing a Bounding Box Centered at a Point

The `trDot` function returns bounding box coordinates with a given point as its center and illustrates

- Using `@key` to declare keyword arguments
- Establishing default values for a keyword argument
- Documenting at the function level
- Using the `let` function
- Using the `xCoord` and `yCoord` functions
- Building a bounding box with the `list` function and colon (`:`) operator

```
procedure( trDot( aPoint @key ( deltaX 1 ) ( deltaY 1 ) )  
  let( ( llx lly urx ury aPointX aPointY )  
    aPointX = xCoord( aPoint )  
    aPointY = yCoord( aPoint )  
    llx = aPointX - deltaX  
    urx = aPointX + deltaX  
    lly = aPointY - deltaY  
    ury = aPointY + deltaY  
    list( llx:lly urx:ury )  
  )
```

```
    ) ; let
  ) ; procedure
  trDot( 100:100 ?deltaX 50 ?deltaY 50)
  => ((50 50) (150 150))
```

Computing the Union of Several Bounding Boxes

The `trBBoxUnion` function returns the smallest bounding box coordinates containing all the boxes in a given list and illustrates

- Using `foreach( mapcar ... )`
- Using the `apply` function with the `min` and `max` functions
- Using the `list` function and the colon (`:`) operator to construct a bounding box
- Documenting at the function level

```
procedure( trBBoxUnion( bBoxList )
  let( ( llxList llyList
        urxList uryList
        minllx minlly
        maxurx maxury
      )
    llxList = foreach( mapcar bBox bBoxList
      xCoord( lowerLeft( bBox ) ) )
    llyList = foreach( mapcar bBox bBoxList
      yCoord( lowerLeft( bBox ) ) )
    urxList = foreach( mapcar bBox bBoxList
      xCoord( upperRight( bBox ) ) )
    uryList = foreach( mapcar bBox bBoxList
      yCoord( upperRight( bBox ) ) )
    minllx = apply( 'min llxList )
    minlly = apply( 'min llyList )
    maxurx = apply( 'max urxList )
    maxury = apply( 'max uryList )
    list( minllx:minlly maxurx:maxury )
  ) ; let
) ; procedure
trBBoxUnion( list( list(0:0 100:100) list(50:50 150:150)) )
=> ((0 0) (150 150))
```

Computing the Intersection of Bounding Boxes

The `trBBoxIntersection` function illustrates

- Using `foreach( mapcar ... )`
- Using the `apply` function with the `min` and `max` functions
- Using the `cond` function

■ Using the `list` function and colon (`:`) operator to construct a bounding box

```

procedure( trBBoxIntersection( bBoxList )
  let( ( llxList llyList
        urxList uryList
        maxllx  maxlly
        minurx  minury )
    llxList = foreach( mapcar bBox bBoxList
                      xCoord( lowerLeft( bBox ) ) )
    llyList = foreach( mapcar bBox bBoxList
                      yCoord( lowerLeft( bBox ) ) )
    urxList = foreach( mapcar bBox bBoxList
                      xCoord( upperRight( bBox ) ) )
    uryList = foreach( mapcar bBox bBoxList
                      yCoord( upperRight( bBox ) ) )
    minurx = apply( 'min urxList )
    minury = apply( 'min uryList )
    maxllx = apply( 'max llxList )
    maxlly = apply( 'max llyList )
    cond(
      ( maxllx >= minurx nil )
      ( maxlly >= minury nil )
      ( t
        list( maxllx:maxllx minurx:minury ) )
    ) ; cond
  ) ; let
) ; procedure

trBBoxIntersection( list( list(0:0 100:100) list(50:50 150:150)) ) =>
( (50 50) (100 100) )

```

Prime Factorizations

A prime factorization of an integer is a list of pairs and is an example of an association list. The first element of each pair is a prime number that divides the number and the second element is the exponent to which the prime is to be raised. Each such pair is termed a prime-exponent pair.

```

pf1 = '( ( 2 3 ) ( 3 5 ) )
pf2 = '( ( 3 2 ) ( 5 2 ) ( 7 3 ) )

pf3 = '( ( 3 2 ) ( 5 4 ) ( 7 2 ) )
pf4 = '( ( 3 6 ) ( 7 3 ) ( 11 2 ) ( 5 1 ) )

```

The `assoc` function is used to determine whether a prime number occurs in a prime factorization. It returns either the prime-exponent pair or `nil`. For example:

```

assoc( 2 pf1 ) => ( 2 3 )
assoc( 7 pf2 ) => ( 7 3 )

```

Evaluating a Prime Factorization

To evaluate the prime factorization means to perform the arithmetic operations implied:

- For each prime-exponent pair, raise the prime to the corresponding exponent
- Multiply the resulting list of integers together

For example, evaluating the prime factorization

```
( ( 3 2 ) ( 5 2 ) ( 7 3 ) )
```

is equivalent to evaluating

```
3**2 * 5**2 * 7**3
```

The `trTimes` function multiplies a list of numbers together. It handles two cases that the `times` function does not handle.

The `trTimes` function illustrates

- Using an `@rest` argument to collect an arbitrary number of arguments into a list.
- Using the `apply` function. In the general case, we use the `apply` function to invoke the normal `times` function
- Using the `cond` function
- The `null` function tests for an empty list. The `onep` function tests whether a number is one

```
procedure( trTimes( @rest args )
  cond(
    ( null( args ) 1 )
    ( onep( length( args ) ) car( args ) )
    ( t apply( 'times args ) )
  ) ; cond
) ; procedure
```

The `trEvalPF` function evaluates the prime factorizations. For example:

```
pf1 = '( ( 2 3 ) ( 3 5 ) )
pf2 = '( ( 3 2 ) ( 5 2 ) ( 7 3 ) )
trEvalPF( pf1 ) => 1944
trEvalPF( pf2 ) => 77175
```

The `trEvalPF` function illustrates

- Using the `apply` function with the `trTimes` function above
- Using the `foreach( mapcar ... )`
- Using the `car` and `cadr` functions

```
procedure( trEvalPF( pf )
  apply( 'trTimes
    foreach( mapcar pePair pf
      car( pePair )**cadr( pePair )
    ) ; foreach
```



```
    ) ; apply  
  ) ; procedure
```

Computing the Prime Factorization

The `trLargestExp` function returns the largest `x` such that `divisor ** x <= number` and illustrates

- Documenting at the function level
- Using the `preincrement` operator
- Using the `let` expression to initialize a local variable to a non-nil value
- Using the `while` function

```
procedure( trLargestExp( number divisor )  
  let( ( ( exp 0 ) )  
    while( zerop( mod( number divisor ) )  
      ++exp  
      number = number / divisor  
    ) ; while  
    exp  
  ) ; let  
  ) ; procedure
```

The `trPF` function returns the prime factorization a number. For example:

```
trPF( 1003 )      => ((59 1) (17 1))  
trPF( 10003 )     => ((1429 1) (7 1))  
trPF( 100003 )    => ((100003 1))  
trPF( 123456 )    => ((643 1) (3 1) (2 6))
```

The `trPF` function illustrates

- Documenting at the function level
- Using the `postincrement` operator
- Using the `let` expression to initialize a local variable to a non-nil value
- Using the `while` function
- Using parentheses to control operator precedence
- Using `let` to initialize a local variables to non-nil values
- Using a `while` loop
- Using `when` with an embedded assignment expression
- Using `trLargestExp`

```
procedure( trPF( aNumber )
  let( ( ; locals
    ( divisor 2 )
    result
    exp
    ( num aNumber )
  )
  while( num > 1
    when( ( exp = trLargestExp( num divisor )) > 0
      result = cons( list( divisor exp ) result )
      num = num / divisor ** exp
    ) ; when
    divisor++      ;;; try next divisor
  ) ; while
  result
) ; let
) ; procedure
```

Multiplying Two Prime Factorizations

The `trPFMult` function returns the prime factorization of the product of two prime factorizations, `pf1` and `pf2`. The `trPFMult` function uses the following algorithm to construct the resultant prime factorization.

1. Those prime-exponent pairs whose primes occur in only one of the prime factorizations lists are carried across unaltered into the resultant prime factorization.
2. For those primes that have entries in both prime factorizations, a prime-exponent pair using the prime is included with an exponent equal to the sum of prime's exponent in either prime factorization.

The `trPFMult` function illustrates

- Using the `setof` function and the `assoc` function to build two prime factorizations

The list `pf1Notpf2` contains all the prime-exponent pairs in `pf1` whose prime does not occur in `pf2`.

The list `pf2Notpf1` contains all the prime-exponent pairs in `pf2` whose prime does not occur in `pf1`.

- Using `foreach( mapcan ... )` to manage the lists of primes found in both lists

The list `bothpf1Andpf2` contains prime-exponent pairs whose primes are found in both `pf1` and `pf2`. The exponent is the sum of the exponent for the prime in `pf1` and `pf2`.

- Using the backquote (‘) and comma (,) operators
- Using the `nconc` function to destructively append the three lists `pf1Notpf2`, `pf2Notpf1`, and `bothpf1Andpf2`

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```
trPFMult( pf1 pf2 ) => ( ( 2 3 ) ( 5 2 ) ( 7 3 ) ( 3 7 ) )
procedure( trPFMult( pf1 pf2 )
  let( ( pf1Notpf2 pf2Notpf1 bothpf1Andpf2 pePair2 )
    pf1Notpf2 = setof( pePair1 pf1
      !assoc( car( pePair1 ) pf2 )
    )
    pf2Notpf1 = setof( pePair2 pf2
      !assoc( car( pePair2 ) pf1 )
    )
    bothpf1Andpf2 = foreach( mapcan pePair1 pf1
      when( pePair2 = assoc( car( pePair1 ) pf2 )
        ;; build a list containing a single prime-exponent pair.
        ;; The mapcan option to the foreach function
        ;; destructively appends these lists together.
        '( (
          ,car( pePair1 )
          , (cadr( pePair1 )+cadr( pePair2 ))
        ) )
      ) ; when
    ) ; foreach
    ;; destructively append the three lists together.
    nconc( pf1Notpf2 pf2Notpf1 bothpf1Andpf2 )
  ) ; let
) ; procedure
```

Using Prime Factorizations to Compute the GCD

The `trPFGCD` function returns the prime factorization of the greatest common denominator (GCD) of two prime factorizations. This function illustrates

- Using `foreach( mapcan ... )` to merge two association lists.

Primes found in both association lists are given an exponent equal to the minimum of the exponents in both the association lists.

- Using the backquote (```) and comma (`,`) operators.

```
procedure( trPFGCD( pf1 pf2 )
  let( ( pePair2 )
    foreach( mapcan pePair1 pf1
      pePair2 = assoc( car( pePair1 ) pf2 )
      when( pePair2
        ;; build a list containing a single prime-exponent pair.
        ;; The mapcan option to the foreach function
        ;; destructively appends these lists together.
        '( (
          ,car( pePair1 )
          ,min( cadr( pePair1 ) cadr( pePair2 ) )
        ) )
      ) ; when
    ) ; foreach
  ) ; let
) ; procedure
```

The `trGCD` function illustrates finding the greatest common denominator (GCD) of two numbers by

- Finding their prime factorizations
- Manipulating the prime factorizations to find the prime factorization of the GCD
- Evaluating the prime factorization

```
procedure( trGCD( num1 num2 )
  trEvalPF( trPFGCD( trPF( num1 ) trPF( num2 ) ) )
) ; procedure
```

Fibonacci Function

This example illustrates a recursive implementation of the Fibonacci function, implemented directly from the mathematical definition.

```
procedure( fibonacci(n)
  if( (n == 1 || n == 2) then 1
  else fibonacci(n-1) + fibonacci(n-2)
))
fibonacci(3)          => 2
fibonacci(6)          => 8
```

The same example implemented in SKILL using LISP syntax looks like the following:

```
(defun fibonacci (n)
  (cond
    ((or (equal n 1) (equal n 2)) 1)
    (t (plus (fibonacci (difference n 1))
              (fibonacci (difference n 2)))))
  )
)
```

Factorial Function

This is the recursive implementation of the factorial function

```
procedure( factorial( n )
  if( zerop( n ) then
    1
  else
    n*factorial( n-1)
  ) ; if
) ; procedure
```

This is an iterative implementation

```
procedure( factorial( n )
  let( ( ( f 1 ) )
    for( i 1 n
```

```
        f = f*i
      ) ; for
    f ;;; return the value of f
  ) ; let
) ; procedure
```

Exponential Function

This function computes e to the power x by summing terms of the power series expansion of the mathematical function. It uses the factorial function.

```
procedure( e( x )
  let( ((sum 1.0))
    for( n 1 10
      sum = sum + (1.0/factorial(n)) * x**n
    ) ; for
    sum ;;; return the value of sum.
  ) ; let
) ; procedure
```

To get a sense of the accuracy of this implementation of the e function, observe

```
e( log( 10 ) ) => 9.999702 ;; should be 10.0
```

Counting Values in a List

The `trCountValues` function tallies the number of times each distinct value occurs as a top-level element of a list. It prints a report and returns an association list that pairs each unique value with its count. Two implementations are presented. The results are equivalent except for ordering.

- The first implementation of `trCountValues` produces

```
trCountValues( '( 1 2 a b 2 b 3 ) ) =>
((a 1) (b 2) (3 1) (2 2) (1 1))
```

and prints

```
1 occurred 1 times.
2 occurred 2 times.
3 occurred 1 times.
b occurred 2 times.
a occurred 1 times.
```

- The second implementation of `trCountValues` produces

```
trCountValues( '( 1 2 a b 2 b 3 ) ) =>
((3 1) (b 2) (a 1) (2 2) (1 1))
```

and prints

```
3 occurred 1 times.
b occurred 2 times.
a occurred 1 times.
```

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```
2 occurred 2 times.  
1 occurred 1 times.
```

The first implementation of the `trCountValues` function illustrates

- Using an association table to maintain the counts

Each element in the list is used as an index into the table.

- Using the `tableToList` function to build an association list for an association table

```
procedure( trCountValues( aList )  
  let( ( ( countTable makeTable( "ValueTable" 0 ) ) )  
  
    foreach( element aList  
      countTable[ element ] = countTable[ element ] + 1  
    ) ; foreach  
  
    foreach( key countTable  
      printf( " %L occurred %d times.\n"  
        key countTable[ key ] )  
    ) ; foreach  
  
    tableToList( countTable ) ;; convert to assoc list  
  ) ; let  
) ; procedure
```

The second implementation of the `trCountValues` function illustrates

- Using an association list to maintain the counts
- Using the `assoc` function to retrieve an entry, if any, for an element
- Using the `rplaca` function to update the count for an entry

```
procedure( trCountValues( aList )  
  let( ( countAssocList countEntry count )  
  
    foreach( element aList  
      countEntry = assoc( element countAssocList )  
      if( countEntry  
        then ;; update the count for this element  
          count = cadr( countEntry )  
          rplaca( cdr( countEntry ) ++count )  
        else ;; add an new entry for this element  
          countAssocList = cons(  
            list( element 1 ) ;; new entry  
            countAssocList )  
      ) ; if  
    ) ; foreach  
  
    foreach( entry countAssocList  
      printf( " %L occurred %d times.\n"  
        car( entry ) ;; the element  
        cadr( entry ) ;; the count  
      )  
    ) ; foreach  
  
    countAssocList ;; return value
```

```
    ) ; let  
  ) ; procedure
```

Counting Characters in a String

The `trCountCharacters` function counts the occurrences of characters in a string.

The `trCountCharacters` illustrates

- Using the `parseString` function to construct a list of characters that occur in a string
- Using either implementation of the `trCountValues` function to count the occurrences of the single-character strings

```
procedure( trCountCharacters( aString )  
  trCountValues( parseString( aString "" ) )  
  ) ; procedure
```

Regular Expression Pattern Matching

The following functions take the regular expression pattern matching functions `rexMatchp` and `rexMatchList` provided by SKILL and build two new functions `shMatchp` and `shMatchList`, which provide a simple shell-filename-like pattern matching facility.

The rules:

- Target strings should be single words
- A pattern is to match the entire target words
- Special characters in a pattern:

*	Matches any string, including the null string
?	Matches any single character
[...]	Same as in the original <code>rex</code> package
.	Matches <code>.</code> literally

The function `sh2ed` is used to build a regular expression that is passed to the `rex` functions.

```
(defun sh2ed (s)  
  (let ((sh_chars (parseString s ""))  
        (ed_chars (tconc nil "^")))  
    (while sh_chars  
      (case- (car sh_chars)
```

```
( "*" (tconc ed_chars ".") (tconc ed_chars "*"))
( "?" (tconc ed_chars ".") )
( "." (tconc ed_chars "\\") (tconc ed_chars ".") )
( t (tconc ed_chars (car sh_chars)))
(setq sh_chars (cdr sh_chars))
(tconc ed_chars "$")
(buildString (car ed_chars) ""))

(defun shMatchp (pattern target)
  (rexMatchp (sh2ed pattern) target))

(defun shMatchList (pattern targets)
  (rexMatchList (sh2ed pattern) targets))

shMatchp("*.out" "a.out")      => t
shMatchp("test.?" "test.il")   => t
shMatchp("*.?" "test.out")     => nil

shMatchList("*test.?" '("ALUtest.1" "data.in" "MEMtest.13" "test.5"))=>
("ALUtest.1" "test.5")
```

Geometric Constructions

Here is an extensive example of a SKILL++ Object Layer application.

Application Domain

A geometric construction is a collection of points and lines you build up from an initial collection of points. The initial collection of points are called *free* points. You can add points and lines to the collection through various familiar constraints. You can constrain

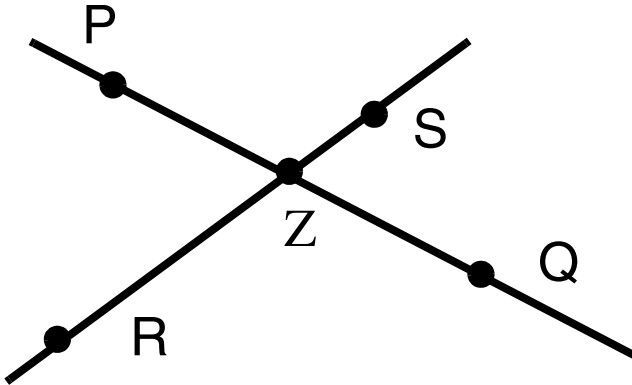
- A point to lie on two intersecting lines
- A line to pass through two points
- A line to pass through a point and to be parallel to another line
- A line to pass through a point and to be perpendicular to another line

When you move any one of the free points, the application propagates the change to all the constrained points and lines

Example

1. You specify the free points P and Q.
2. You construct the line PQ passing through P and Q.
3. You specify the free points R and S.

4. You construct the line RS passing through R and S.
5. You construct the intersection point Z of the line PQ and the line RS.



When you move any of the points P, Q, R, or S the lines PQ and RS and the point Z move accordingly.

Implementation

The implementation uses the SKILL++ Object System to define several classes and generic functions. The following sections discuss

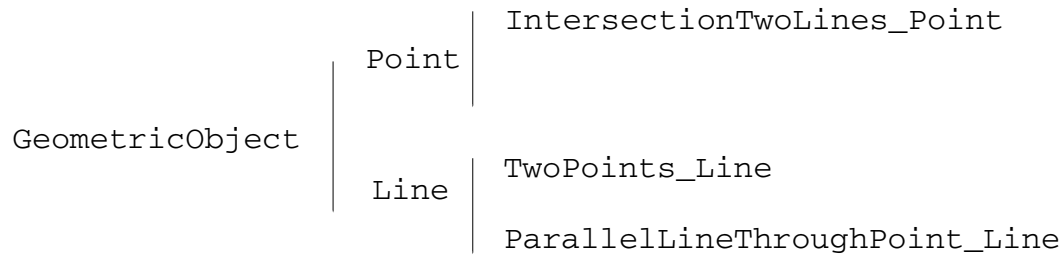
- The class hierarchy
- The generic functions
- The source code
- Several examples

To focus on SKILL++ language issues, the implementation does not address graphics. Instead, you non-graphically

1. Call a SKILL function repeatedly to specify several free points.
2. Call other SKILL functions to construct the dependent points and lines.
3. Enter a SKILL expression to change the coordinates of one of the free points.
4. Call a SKILL function to propagate the change through the constrained points and lines.
5. Call a SKILL function to describe one of the constrained points or lines.

Classes

The implementation uses the SKILL++ Object System to define several classes in the following class hierarchy.



GeometricObject Class

The `GeometricObject` class represents all the objects in the construction. It defines the `constraints` slot. This slot lists all the other objects which need to be notified when the object updates.

Point Class

The `Point` class represents a point with slots `x` and `y`.

Line Class

The `Line` class represents a line with the slots `A`, `B`, and `C`. These are the coefficients in the line's equation

$$Ax + By + C = 0$$

IntersectionTwoLines_Point Class

The `IntersectionTwoLines_Point` class is a subclass of the `Point` class and represents a point that lies on two intersecting lines. It includes two slots that store the lines.

TwoPoints_Line Class

The `TwoPoints_Line` class is a subclass of the `Line` class and represents a line passing through two points. The class defines two slots to store the points.

ParallelLineThroughPoint_Line Class

The `ParallelLineThroughPoint_Line` class is a subclass of the `Line` class and represents a line that passes through a point parallel to another line.

To specify a free point, you instantiate the `Point` class. To specify a constrained point or line, you instantiate the associated subclass.

Generic Functions

The implementation uses several generic functions.

- The `Update` function propagates changes. It updates the coordinates or equations of an object and call itself recursively for each dependent object.
- The `Describe` function prints a description of an object and calls itself recursively for each dependent object.
- The `Validate` function verifies that a point or line satisfies its constraints.
- The `Connect` function adds an object to another object's list of constraints.

The `defgeneric` declarations for these generic functions each declare a default method that raises an error.

Describing the Methods by Class

The following tables describe, for each generic function, all the methods by class. In the following tables,

- Earlier rows are at the same level or higher in the class hierarchy.
- The etc. rows refer to other constraint subclasses of `Point` or `Line` that you can add to the implementation to extend its functionality.

Update Methods

Class	Action
GeometricObject	Call <code>Update</code> for each of the dependent objects.
Point	No method for this class.
Line	No method for this class.
IntersectionTwoLines_Point	Based on the two lines, recompute the coordinate of the point. Call the next <code>Update</code> method.
TwoPoints_Line	Based on the two points, recompute the equation of the line. Call the next <code>Update</code> method.
etc.	

Describe Methods

Class	Method description
GeometricObject	Raise an error since there is no meaningful description at level.
Point	Print a description of the point's coordinates.
Line	Print a description of the line's equation.
IntersectionTwoLines_Point	Call the next <code>Describe</code> method to display the coordinates. Then call <code>Describe</code> for each of the two lines that define this point.
TwoPoints_Line	Call the next <code>Describe</code> method to display the equation. Then call <code>Describe</code> for each of the two point that define this line.
etc.	

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Validate Methods

Class	Method description
GeometricObject	Raise an error since there is no meaningful validation to perform at this level of an object.
Point	No method for this class
Line	No method for this class
IntersectionTwoLines_Point	Verify that the point's coordinates satisfy the equations of the two lines.
TwoPoints_Line	Verify that the two point's coordinates satisfy the line's equation.
etc.	

Connect Methods

Connect generic Function

Class	Method description
GeometricObject	Add a dependent object to the list of dependent objects.
Point	No method for this class.
Line	No method for this class.
IntersectionTwoLines_Point	No method for this class.
TwoPoints_Line	No method for this class.
etc.	

Source Code

```
;;; toplevel( 'ils )

defgeneric( Connect ( obj constraint )
  error( "Connect is a subclass responsibility\n" )
) ; defgeneric

defgeneric( Update ( obj )
  error( "Update is a subclass responsibility\n" )
) ; defgeneric

defgeneric( Validate ( obj )
```

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```
error( "Validate is a subclass responsibility\n" )
) ; defgeneric

defgeneric( Describe ( obj )
error( "Describe is a subclass responsibility\n" )
) ; defgeneric

;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;
;;;; GeometricObject
;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;

defclass( GeometricObject
(
(
( constraints
@initform nil
)
)
) ; defclass

defmethod( Connect (( obj GeometricObject ) constraint )
when( !member( constraint obj->constraints )
obj->constraints = cons( constraint obj->constraints )
) ; when
) ; defmethod

defmethod( Update (( obj GeometricObject ))
printf( "Updating constraints %L for %L\n"
obj->constraints obj )
Validate( obj )
foreach( constraint obj->constraints
Update( constraint )
) ; foreach
t
) ; defmethod

;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;
;;;; Point
;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;

defclass( Point ( GeometricObject )
(
( name @initarg name )
( x @initarg x );;; x-coordinate
( y @initarg y );;; y-coordinate
)
) ; defclass

defmethod( Describe (( obj Point ))
printf( "%s at %n:%n\n"
className( classOf( obj )) obj->x obj->y )
) ; defmethod

defmethod( Validate ((obj Point))
t
)
```

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```
////////////////////////////////////
//////////////////////////////////// IntersectionTwoLines_Point
////////////////////////////////////
defclass( IntersectionTwoLines_Point ( Point )
(
    ( L1 @initarg L1 )
    ( L2 @initarg L2 )
)
) ; defclass

defmethod( Describe (( obj IntersectionTwoLines_Point ))
callNextMethod( obj );; generic point description
printf( "...intersection of\n")
Describe( obj->L1 )
Describe( obj->L2 )
) ;defmethod

procedure( make_IntersectionTwoLines_Point( line1 line2 )
let( ( point )
point = makeInstance( 'IntersectionTwoLines_Point
?L1 line1
?L2 line2
)
Update( point )
Connect( line1 point )
Connect( line2 point )
point
) ; let
) ; procedure

defmethod( Validate (( obj IntersectionTwoLines_Point ))
let( ( A1 B1 C1 A2 B2 C2 x y )
A1 = obj->L1->A
B1 = obj->L1->B
C1 = obj->L1->C
A2 = obj->L2->A
B2 = obj->L2->B
C2 = obj->L2->C
x = obj->x
y = obj->y
when( A1*x+B1*y+C1 != 0.0 || A2*x+B2*y+C2 != 0.0
error( "Invalid IntersectionTwoLines_Point\n" )
) ; when
t
) ; let
) ; defmethod

defmethod( Update (( obj IntersectionTwoLines_Point ))
;; check to see if my two lines have values ...
printf( "Figure out my x & y from lines %L %L\n"
obj->L1 obj->L2 )
let( ( A1 B1 C1 A2 B2 C2 det )
A1 = obj->L1->A
B1 = obj->L1->B
C1 = obj->L1->C
A2 = obj->L2->A
B2 = obj->L2->B
C2 = obj->L2->C
```

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```
det = A1*B2-A2*B1
when( det == 0
    error( "Can not intersect two parallel lines\n" )
)
obj->x = ((-C1)*B2-(-C2)*B1)*1.0/det
obj->y = (A1*(-C2)-A2*(-C1))*1.0/det
) ; let
callNextMethod( obj )
) ; defmethod

;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;
;;;;;;;;;;;;;;;;;; Line
;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;

defclass( Line ( GeometricObject )
(
    ( A )
    ( B )
    ( C )
)
) ; defclass

defmethod( Describe (( obj Line ))
    printf( "%s %nx+%ny+%n=0\n"
        className( classOf( obj ))
        obj->A obj->B obj->C
    )
) ; defmethod

;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;
;;;;;;;;;;;;;;;;;; TwoPoints_Line
;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;

defclass( TwoPoints_Line ( Line )
(
    ( P1 @initarg P1 )
    ( P2 @initarg P2 )
)
) ; defclass

defmethod( Describe (( obj TwoPoints_Line ))
    callNextMethod( obj )
    printf( "...containing\n" )
    Describe( obj->P1 )
    Describe( obj->P2 )
) ; defmethod

procedure( make_TwoPoints_Line( p1 p2 )
    let( ( line )
        line = makeInstance( 'TwoPoints_Line
            ?P1 p1 ?P2 p2 )
        Update( line )
        Connect( p1 line )
        Connect( p2 line )
        line
    ) ; let
) ; procedure
```


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```
defmethod( Validate (( obj TwoPoints_Line ))
  let( (x1 y1 x2 y2 A B C)
    x1 = obj->P1->x
    x2 = obj->P2->x
    y1 = obj->P1->y
    y2 = obj->P2->y
    A = obj->A
    B = obj->B
    C = obj->C
    if( A*x1+B*y1+C != 0.0 then
      error( "Invalid TwoPoints_Line\n" ))
    if( A*x2+B*y2+C != 0.0 then
      error( "Invalid TwoPoints_Line\n" ))
    t
  ) ; let
) ; defmethod

defmethod( Update (( obj TwoPoints_Line ))
  let( (x1 y1 x2 y2 m b)
    x1 = obj->P1->x
    x2 = obj->P2->x
    y1 = obj->P1->y
    y2 = obj->P2->y
    if( x2-x1 != 0
      then
        m = (y2-y1)*1.0/(x2-x1)
        b = y2-m*x2
        obj->A = -m
        obj->B = 1
        obj->C = -b
      else
        obj->A = 1.0
        obj->B = 0.0
        obj->C = -x1
    ) ; if
  ) ; let
  callNextMethod( obj )
) ; defmethod

;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;
;;;;;;;;;;;;;;;; ParallelLineThroughPoint_Line
;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;

defclass( ParallelLineThroughPoint_Line ( Line )
  (
    ( P @initarg P)
    ( L @initarg L)
  )
) ; defclass

defmethod( Validate (( obj ParallelLineThroughPoint_Line ))
  let( (x1 y1 A B C LA LB LC)
    x1 = obj->P->x
    y1 = obj->P->y
    LA = obj->L->A
    LB = obj->L->B
```

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```
LC = obj->L->C
A  = obj->A
B  = obj->B
C  = obj->C
when( A*LB-LA*B != 0.0 || A*x1+B*y1+C != 0
      error( "Invalid ParallelLineThroughPoint_Line\n" ))
t
) ; let
) ; defmethod

defmethod( Describe (( obj ParallelLineThroughPoint_Line ))
  callNextMethod( obj )
  printf( "...Containing\n" )
  Describe( obj->P )
  printf( "...Parallel to\n" )
  Describe( obj->L )
) ; defmethod

procedure( make_ParallelLineThroughPoint_Line( point line )
  let( ( parallel_line )
    parallel_line = makeInstance(
      'ParallelLineThroughPoint_Line
      ?P point
      ?L line
    )
    Update( parallel_line )
    Connect( point parallel_line )
    Connect( line parallel_line )
    parallel_line
  ) ; let
) ; procedure

defmethod( Update (( obj ParallelLineThroughPoint_Line ))
  let( ( A B C x1 y1 )
    A = obj->L->A
    B = obj->L->B
    C = obj->L->C
    x1 = obj->P->x
    y1 = obj->P->y
    obj->A = A
    obj->B = B
    obj->C = -(A*x1+B*y1)
  ) ; let
  callNextMethod( obj )
) ; defmethod
```

Example 1

This example

- Creates a point P and describes it
- Creates another point R and describes it
- Constructs the line L that passes through the points P and R and describe it

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- Changes the x coordinate of P to 1 and calls the `Update` function to propagate the changed coordinates of P

In this version of the application, it is your responsibility to call the `Update` function after changing the coordinates of a free point. You should not use the `->` operator to change the coordinates of a non-free point.

- Describes the line L to verify that it has been updated

```
P = makeInstance( 'Point ?x 0 ?y 4 )
stdobj:0x36f018

Describe( P )
Point at 0:4
t

R = makeInstance( 'Point ?x 3 ?y 0 )
stdobj:0x36f024

Describe( R )
Point at 3:0
t

L = make_TwoPoints_Line( P R )
Updating constraints nil for stdobj:0x36f030
stdobj:0x36f030

Describe( L )
TwoPoints_Line 1.333333x+1y+-4.000000=0
...containing
Point at 0:4
Point at 3:0
t

P->x = 1
1

Update( P )
Updating constraints (stdobj:0x36f030) for stdobj:0x36f018
Updating constraints nil for stdobj:0x36f030
t

Describe( L )
TwoPoints_Line 2.000000x+1y+-6.000000=0
...containing
Point at 1:4
Point at 3:0
t
```

Example 2

This example

- Creates four points P, Q, S, and R
- Constructs the line PQ that passes through the points P and Q

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- Constructs the line SR that passes through the points S and R
- Constructs the point Z that is on both the lines PQ and SR and describes the point Z
- Changes the y coordinate of the point S to 4 and updates the point S
- Describes the point Z to verify that it has been updated

```
P = makeInstance( 'Point ?x 0 ?y 4 )
stdobj:0x36f090

Q = makeInstance( 'Point ?x 0 ?y -4 )
stdobj:0x36f09c

S = makeInstance( 'Point ?x -3 ?y 0 )
stdobj:0x36f0a8

R = makeInstance( 'Point ?x 3 ?y 0 )
stdobj:0x36f0b4

PQ = make_TwoPoints_Line( P Q )
Updating constraints nil for stdobj:0x36f0c0
stdobj:0x36f0c0

SR = make_TwoPoints_Line( S R )
Updating constraints nil for stdobj:0x36f0cc
stdobj:0x36f0cc

Z = make_IntersectionTwoLines_Point( PQ SR )
Figure out my x & y from lines stdobj:0x36f0c0 stdobj:0x36f0cc
Updating constraints nil for stdobj:0x36f0d8
stdobj:0x36f0d8

Describe( Z )
IntersectionTwoLines_Point at 0.000000:0.000000
TwoPoints_Line 1.000000x+0.000000y+0=0
...containing
Point at 0:4
Point at 0:-4
TwoPoints_Line -0.000000x+1y+-0.000000=0
...containing
Point at -3:0
Point at 3:0
t

S->y = 4
4

Update( S )
Updating constraints (stdobj:0x36f0cc) for stdobj:0x36f0a8
Updating constraints (stdobj:0x36f0d8) for stdobj:0x36f0cc
Figure out my x & y from lines stdobj:0x36f0c0 stdobj:0x36f0cc
Updating constraints nil for stdobj:0x36f0d8
t

Describe( Z )
IntersectionTwoLines_Point at 0.000000:2.000000
TwoPoints_Line 1.000000x+0.000000y+0=0
...containing
Point at 0:4
Point at 0:-4
TwoPoints_Line 0.666667x+1y+-2.000000=0
```

```
...containing
Point at -3:4
Point at 3:0
t
```

Extending the Implementation

Consider the following extensions:

- Protecting the implementation from inconsistent access

You can use the `->` operator to change coordinates of any point, even a constrained point. You can extend the implementation to allow the user to change only coordinates of free points.

- Adding graphics

You can extend the implementation to associate a database object with each point or line and update the database object at appropriate times. You should have to affect only the `Point` and `Line` classes.

Examples of How to use Custom Specializers in SKILL++ Methods

An example of using SKILL++ custom specializers to allow defining a method specialized on the database object type. Standard SKILL++ methods are specialized on the class of an object passed to the generic function, but this allows extending it to further divide based on some attribute of a non-SKILL++ object.

This example implements `abFigArea(obj)` that calculates the area of various types of database object, You can see the individual methods further down the code. It implements calculations for most (not guaranteed to be complete) figure object types in Virtuoso. It also includes `abFigPerimeter(obj)` to show an example of a second generic function using the same specializers.

To call:

```
abFigArea(dbId) => area (in micron^2)
abFigPerimeter(dbId) => length of perimeter (in microns)
```

Limitations: does not currently compute the perimeter of an arc (It is difficult to compute the length for elliptic arcs).

```
*****
SCCS Info: @(#) abAreaMethod.ils 12/17/20.11:01:03 1.2
```

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```
*/
;-----
; First define a class to store the information needed to match an
; object against the custom specializer
;-----
defclass(abDbFigSpecializer ()
  (
    (objType @initarg objType)
  )
)
;-----
; This method is called when a defmethod using a custom specializer
; is used. This uses an eqv specializer, so if the defmethod is
; of this form: defmethod(methodName ((obj (abDbFig "ellipse")))) ...
; then it will create an instance of the specializer class to
; store the database objType in.
; ilGenerateSpecializer is passed the generic function object, the symbol
; which is the first term of the specializer in the defmethod, and then
; a list of the arguments (all literal; no evaluation is performed)
;-----
defmethod(ilGenerateSpecializer (_gf (_spec (eqv 'abDbFig)) args)
  makeInstance('abDbFigSpecializer ?objType car(args))
)
;-----
; This method is used when the generic function is called. It
; tries against each specializer object that there is a method
; defined for, and this method determines whether the specializer
; matches the objType of the database object passed in.
; The method is passed the generic function object, the object for the
; specializer being checked against, and the actual object for the argument
; to the method.
; Note that the method should check that the argument is the expected
; object type as well as checking the values match - this avoids situations
; where retrieving (in this case) ->objType fails because the object has no
; slot objType.
;-----
defmethod(ilArgMatchesSpecializer (_gf (spec abDbFigSpecializer) (arg dbobject))
  spec->objType==arg->objType
)
;-----
; Must define a method to check whether two specializer objects are
```

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```
; equivalent. Failure to do this will lead to errors if you redefine
; methods using the custom specializer
;-----
defmethod(ilEquivalentSpecializers (_gf (spec1 abDbFigSpecializer)
    (spec2 abDbFigSpecializer))
    spec1->objType==spec2->objType
)
/*****
*
*          abAreaOfBBox (bBox)
*
*      A utility function to calculate the area of a bBox. Several
* objects are rectangular, so having a common function makes this simpler
*
*****/
defun(abAreaOfBBox (bBox)
    destructuringBind(((llx lly) (urx ury)) bBox
        (urx-llx)*(ury-lly)
    )
)
/*****
*
*          abAreaOfPolygon(pointList)
*
*      A utility function to calculate the area of a polygon.
*      Effectively integrates around the polygon.
*
*****/
defun(abAreaOfPolygon (pointList)
    let((sum 0) firstPt lastPt dx)
        firstPt=lastPt=car(pointList)
        foreach(point cdr(pointList)
            dx=xCoord(point)-xCoord(lastPt)
            sum=sum+dx*(yCoord(point)+yCoord(lastPt))
            lastPt=point
        )
        sum=sum+((xCoord(firstPt)-xCoord(lastPt))*
            (yCoord(firstPt)+yCoord(lastPt)))
        0.5*abs(sum)
    )
)
```

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```
/******
*
*          abPerimeterOfBBox(bBox)
*
*   A utility function to calculate the perimeter of a bBox
*
******/
defun(abPerimeterOfBBox (bBox)
  destructuringBind(((llx lly) (urx ury)) bBox
    2*(urx-llx+ury-lly)
  )
)
/******
*
*          abPerimeterOfPolygon(pointList @optional (closed t))
*
*   Utility function to compute the perimeter of a polygon pointList.
*   By default this is a closed polygon, but if the second argument
*   is passed as nil, will find the length of a line with that pointList.
*
******/
defun(abPerimeterOfPolygon (pointList @optional (closed t))
  let(((sum 0) firstPt lastPt)
    firstPt=lastPt=car(pointList)
    foreach(point cdr(pointList)
      sum=sum+sqrt(
        (xCoord(lastPt)-xCoord(point))**2 +
        (yCoord(lastPt)-yCoord(point))**2
      )
      lastPt=point
    )
    when(closed
      sum=sum+sqrt(
        (xCoord(lastPt)-xCoord(firstPt))**2 +
        (yCoord(lastPt)-yCoord(firstPt))**2
      )
    )
    sum
  )
)
;-----
```


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```
; Some of the methods need PI, so this associates the math
; constants with the abFigMaths symbol (so can use abFigMaths.PI
; for example)
;-----
defMathConstants('abFigMaths)
/*****
*
*                               abFigArea(obj)
*
* The generic function - pass in a database object and compute the
* area. The generic function handles anything that isn't recognised and
* tries to give a vaguely useful error message
*
*****/
defgeneric(abFigArea (obj)
    let((objType)
        objType=
            if(dbIsId(obj) then
                strcat(type(obj) " - " obj~>objType)
            else
                type(obj)
        )
        error("abFigArea: unrecognized object %L (%s)\n" obj objType)
    )
)
/*****
*
*                               abFigPerimeter(obj)
*
* The generic function for perimeter - pass in a database object and
* compute the length of the perimeter. The generic function simply
* errors gracefully when passed an unrecognised object.
*
*****/
defgeneric(abFigPerimeter (obj)
    let((objType)
        objType=
            if(dbIsId(obj) then
                strcat(type(obj) " - " obj~>objType)
            else
                type(obj)
    )
)
```

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```
        )
        error("abFigPerimeter: unrecognized object %L (%s)\n" obj objType)
    )
)
;-----
; Now all the methods specialized on each objType.
; First, the abFigArea methods
;-----
;-----
; Just use the bBox of these objects to compute the area
;-----
defmethod(abFigArea ((obj (abDbFig "rect"))))
    abAreaOfBBox(obj~>bBox)
)
defmethod(abFigArea ((obj (abDbFig "inst"))))
    abAreaOfBBox(obj~>bBox)
)
defmethod(abFigArea ((obj (abDbFig "mosaic"))))
    abAreaOfBBox(obj~>bBox)
)
defmethod(abFigArea ((obj (abDbFig "figGroup"))))
    abAreaOfBBox(obj~>bBox)
)
defmethod(abFigArea ((obj (abDbFig "stdVia"))))
    abAreaOfBBox(obj~>bBox)
)
defmethod(abFigArea ((obj (abDbFig "customVia"))))
    abAreaOfBBox(obj~>bBox)
)
;-----
; Polygons, boundaries, blockages and halos have a point list
; that we can compute the area from
;-----
defmethod(abFigArea ((obj (abDbFig "polygon"))))
    abAreaOfPolygon(obj~>points)
)
defmethod(abFigArea ((obj (abDbFig "areaBlockage"))))
    abAreaOfPolygon(obj~>points)
)
defmethod(abFigArea ((obj (abDbFig "layerBlockage"))))
    abAreaOfPolygon(obj~>points)
```

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```
)
defmethod(abFigArea ((obj (abDbFig "PRBoundary"))))
    abAreaOfPolygon(obj~>points)
)
defmethod(abFigArea ((obj (abDbFig "areaBoundary"))))
    abAreaOfPolygon(obj~>points)
)
defmethod(abFigArea ((obj (abDbFig "snapBoundary"))))
    abAreaOfPolygon(obj~>points)
)
defmethod(abFigArea ((obj (abDbFig "clusterBoundary"))))
    abAreaOfPolygon(obj~>points)
)
defmethod(abFigArea ((obj (abDbFig "areaHalo"))))
    abAreaOfPolygon(obj~>points)
)
defmethod(abFigArea ((obj (abDbFig "layerHalo"))))
    abAreaOfPolygon(obj~>points)
)
;-----
; Few individual objects which are more interesting
;-----
defmethod(abFigArea ((obj (abDbFig "donut"))))
    let((inner outer)
        outer=abFigMaths.PI*obj~>outerRadius**2
        inner=abFigMaths.PI*obj~>innerRadius**2
        outer-inner
    )
)
defmethod(abFigArea ((obj (abDbFig "ellipse"))))
    ; pi*a*b where a and b are the radii of the ellipse
    destructuringBind(((llx lly) (urx ury)) obj~>bBox
        abFigMaths.PI*(urx-llx)*(ury-lly)/4.0
    )
)
defmethod(abFigArea ((obj (abDbFig "path"))))
    abAreaOfPolygon(dbGetPathBoundary(obj))
)
defmethod(abFigArea ((obj (abDbFig "pathSeg"))))
    abAreaOfPolygon(obj~>boundary)
)
```

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```
;-----  
; Finally a bunch of objects with no area  
;-----  
defmethod(abFigArea ((_obj (abDbFig "line")))  
    0.0  
)  
defmethod(abFigArea ((_obj (abDbFig "dot")))  
    0.0  
)  
defmethod(abFigArea ((_obj (abDbFig "arc")))  
    0.0  
)  
defmethod(abFigArea ((_obj (abDbFig "ruler")))  
    0.0  
)  
;-----  
; could use the bBox, but area doesn't really make sense for labels  
;-----  
defmethod(abFigArea ((_obj (abDbFig "label")))  
    0.0  
)  
defmethod(abFigArea ((_obj (abDbFig "textDisplay")))  
    0.0  
)  
;-----  
; Now the abFigPerimeter methods  
;-----  
;-----  
; Just use the bBox of these objects to compute the perimeter  
;-----  
defmethod(abFigPerimeter ((obj (abDbFig "rect"))  
    abPerimeterOfBBox(obj~>bBox)  
)  
defmethod(abFigPerimeter ((obj (abDbFig "inst"))  
    abPerimeterOfBBox(obj~>bBox)  
)  
defmethod(abFigPerimeter ((obj (abDbFig "mosaic"))  
    abPerimeterOfBBox(obj~>bBox)  
)  
defmethod(abFigPerimeter ((obj (abDbFig "figGroup"))  
    abPerimeterOfBBox(obj~>bBox)
```

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```
)
defmethod(abFigPerimeter ((obj (abDbFig "stdVia"))))
    abPerimeterOfBBox(obj~>bBox)
)
defmethod(abFigPerimeter ((obj (abDbFig "customVia"))))
    abPerimeterOfBBox(obj~>bBox)
)
;-----
; Polygons, boundaries, blockages and halos have a point list
; that we can compute the perimeter from
;-----
defmethod(abFigPerimeter ((obj (abDbFig "polygon"))))
    abPerimeterOfPolygon(obj~>points)
)
defmethod(abFigPerimeter ((obj (abDbFig "areaBlockage"))))
    abPerimeterOfPolygon(obj~>points)
)
defmethod(abFigPerimeter ((obj (abDbFig "layerBlockage"))))
    abPerimeterOfPolygon(obj~>points)
)
defmethod(abFigPerimeter ((obj (abDbFig "PRBoundary"))))
    abPerimeterOfPolygon(obj~>points)
)
defmethod(abFigPerimeter ((obj (abDbFig "areaBoundary"))))
    abPerimeterOfPolygon(obj~>points)
)
defmethod(abFigPerimeter ((obj (abDbFig "snapBoundary"))))
    abPerimeterOfPolygon(obj~>points)
)
defmethod(abFigPerimeter ((obj (abDbFig "clusterBoundary"))))
    abPerimeterOfPolygon(obj~>points)
)
defmethod(abFigPerimeter ((obj (abDbFig "areaHalo"))))
    abPerimeterOfPolygon(obj~>points)
)
defmethod(abFigPerimeter ((obj (abDbFig "layerHalo"))))
    abPerimeterOfPolygon(obj~>points)
)
;-----
; Few individual objects which are more interesting
;-----
```

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```
defmethod(abFigPerimeter ((obj (abDbFig "donut"))))
  let((inner outer)
    outer=2*abFigMaths.PI*obj~>outerRadius
    inner=2*abFigMaths.PI*obj~>innerRadius
    outer+inner
  )
)

defmethod(abFigPerimeter ((obj (abDbFig "ellipse"))))
  ;-----
  ; there is no exact formula for the perimeter of an ellipse
  ; so instead use Ramanujan's approximation
  ;-----
  destructuringBind(((llx lly) (urx ury)) obj~>bBox
    let((a (urx-llx)/2.0) (b (ury-lly)/2.0) h)
      h=(a-b)**2/(a+b)**2
      abFigMaths.PI*(a+b)*(1+3*h/(10+sqrt(4-3*h)))
    )
  )
)

defmethod(abFigPerimeter ((obj (abDbFig "path"))))
  abPerimeterOfPolygon(dbGetPathBoundary(obj))
)

defmethod(abFigPerimeter ((obj (abDbFig "pathSeg"))))
  abPerimeterOfPolygon(obj~>boundary)
)

;-----
; Lines and rulers - the perimeter is just one side
; I guess it could be double this, maybe?
;-----

defmethod(abFigPerimeter ((obj (abDbFig "line"))))
  abPerimeterOfPolygon(obj~>points nil)
)

defmethod(abFigPerimeter ((obj (abDbFig "ruler"))))
  abPerimeterOfPolygon(obj~>points nil)
)

;-----
; This is left to an exercise for the reader (mainly because it is
; pretty difficult - there is no exact formula for this)
;-----

defmethod(abFigPerimeter ((_obj (abDbFig "arc"))))
  0.0
```

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```
)
;-----
; Finally a bunch of objects with no perimeter
;-----
defmethod(abFigPerimeter ((_obj (abDbFig "dot"))))
    0.0
)
defmethod(abFigPerimeter ((_obj (abDbFig "label"))))
    0.0
)
defmethod(abFigPerimeter ((_obj (abDbFig "textDisplay"))))
    0.0
)
```

Simple Code Example Using a Generic Function Proxy

This example (not as complete as the `abFigArea/abFigPerimeter` example) uses a different strategy of defining a proxy for the `ilGenericFunction` and using that to specialize the custom specializer methods. This proxy class must be specified via the `?genericFunctionClass` argument to `defgeneric` when defining a generic function. The disadvantage of this approach is that it could disguise how the method are actually being specialized - in this case just a string representing the `objType` is used to distinguish each method, and that may not be as clear that something special is happening. Therefore, it is probably best to use the approach above (if using custom specializers at all).

```
;-----
; define a proxy for the generic function object
; Any generic functions defined using this will then trigger the
; right custom specializer methods
;-----
defclass(CCSproxyGF (ilGenericFunction)
    ()
)
;-----
; A class to hold the specializer details
;-----
defclass(CCSdbobjectSpecializer ())
    (
        (objType @initarg objType)
    )
)
defmethod(ilGenerateSpecializer ((gf CCSproxyGF) spec _args)
```

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```
;-----  
; Don't need any arguments - just use the specializer (which can  
; be a string or a symbol)  
;-----  
makeInstance('CCSdbobjectSpecializer ?objType get_string(spec))  
)  
defmethod(ilEquivalentSpecializers ((gf CCSproxyGF)  
  (spec1 CCSdbobjectSpecializer) (spec2 CCSdbobjectSpecializer))  
  spec1->objType==spec2->objType  
)  
defmethod(ilArgMatchesSpecializer ((gf CCSproxyGF) (spec CCSdbobjectSpecializer)  
  (arg dbobject))  
  spec->objType==arg->objType  
)  
;-----  
; Note that the generic function is defined indicating that it uses a  
; different class other than ilGenericFunction. This means that the  
; custom specializer methods above will be used  
;-----  
defgeneric(CCSgetFullObjType (obj) ?genericFunctionClass CCSproxyGF  
  when(dbobjectp(obj)  
    obj->objType  
  )  
)  
;-----  
; Methods using this approach. Note that this is not terribly transparent  
; that they are dealing with database objects  
;-----  
defmethod(CCSgetFullObjType ((obj ("rect")))  
  "rectangle"  
)  
defmethod(CCSgetFullObjType ((obj ("inst")))  
  "instance"  
)
```